

Foundation Course in Managerial Economics

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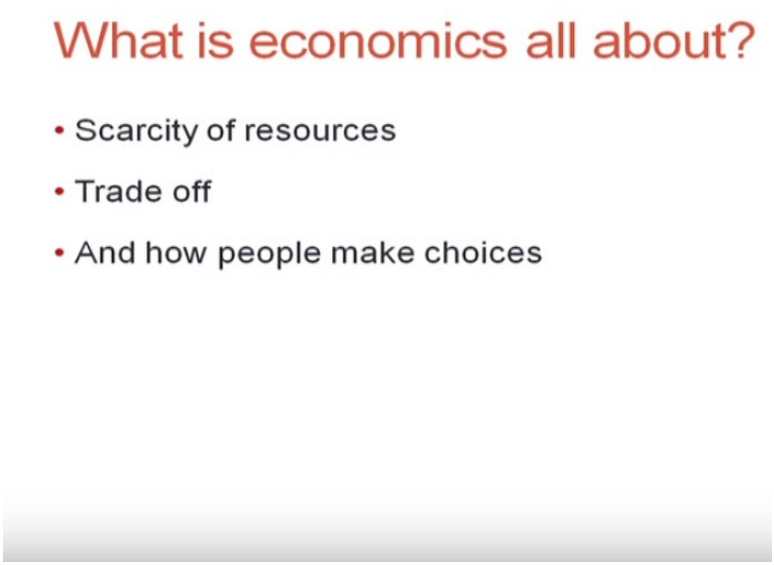
Lecture - 01

Principles of Economics

Welcome to the foundation course in Managerial Economics. I am Dr. Barnali Nag from Vinod Gupta School of Management, IIT Kharagpur. This course is about, we are primarily going to focus on microeconomic theories, which I used in decision making in different organizations.

We are basically going to go through different tools and concepts of economic theory to see how the different organizations, people in the organizations, the stake holders, the government, the consumers, the producers, firms, banks, how they interact with each other. So they make decisions at different, their own levels, and these decisions interact with each other in the market and result in certain kinds of market outcomes. So we are going to see how people make decisions and how are choices made in the economy.

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So, but this is about managerial economics. Now what is economics all about? Economics is about, it is actually many of us might think that economics is about how banks function, how investment is made, how prices are determined. Yes, but more importantly economics is about how people make decisions.

It is about people actually and how people, be it a homemaker, a student, an entrepreneur, a policy maker who is in the government making policies, how they make decisions. So this is about, so economics is more about how people make decisions, but why is it complicated? Why is it complicated?

It is complicated because everyone is faced with scarcity of resources. There is scarcity of resources in the economy and be it a person who is running a household, be it a student who is basically surviving on some stipend or some help from his home, or be it a government who is surviving on taxes, be it a nation which has its own endowments of natural resources.

So everyone is faced with some amount of scarcity and why are we using the term scarcity? We are using the term scarcity because wants and needs are unlimited and resources are limited. So that is the reason that scarcity is a problem because everyone has his, everyone, a person would like to has innumerable wants and desires but he has limited income.

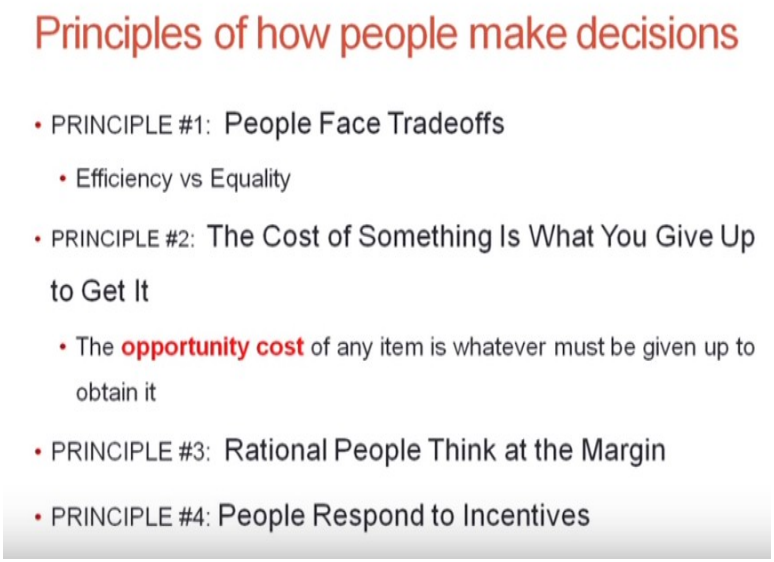
Similarly, a government would like to do everything. It would like to eradicate poverty; it would like to take care of its environment. It would like to focus on its economic growth and boost its producers to improve production. So everyone has lot of goals in mind, but resources are limited.

So scarcity of resources is a problem and that leads to choices. Basically you have to make a trade off. You cannot have everything together and how people make choices that is the question that we are trying to understand through different kinds of economic theories.

So before I move on to the various lectures, we are going to discuss lot of economic theories and concepts and we are going to study lot of models later, but in the first module of this lecture let me focus on certain principles, some fundamental principles of economics.

There are 10 principles of economics which has been laid down beautifully by Professor Gregory Mankiw whose book also I have recommended for this course and this is about principles of how people make decisions, how people interact with each other, and how different nations how they interact and how what are the principles which guide the nations, their macroeconomic decisions.

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So the first set of principles of how people make decisions as I earlier already said, it is principle number one is people face tradeoffs, people face tradeoffs as I already said there is scarcity of

resources and you have lot of goals and you have to make a choice. So people make tradeoffs, people face tradeoffs rather and they have to choose between two, three kinds of options.

One of the very important tradeoff in economics that which is a huge challenge for all economists is a tradeoff between efficiency and equality. I will give an example, I am not going to discuss, this is a very vast area of research and a huge problem, but I am not going to focus too much on it, but I will, it is worth a mention so I am going to just give a small example.

Say for example the government decides that everyone should be should have a minimum standard of living, should have some minimum income. Now, so it decides where is the government going to give this money to the poor from? So the government is basically going to tax the rich.

Now when they tax the rich it is this incentive for the rich to work hard. So indirectly when the government is trying to improve the equality in the economy by taking the money away from the rich and distributing in the poor what the government is doing is it is basically reducing the efficiency in the economy right? So efficiency versus equality is a very important and very challenging tradeoff that every economy faces.

Second principle is the cost of something is what you give up to get it, what you give up to get it and this brings us to a very important concept in economics which we are going to come across frequently later in the course, it is called the opportunity cost. The opportunity cost of any item is whatever must be given up to get it.

Say for example a student decides after completing his engineering that I am going to instead of taking the job that has been offered through the campus interview, I am going to pursue my pursue further education. What is the opportunity cost of the further education? The opportunity cost is basically the salary or the income that the student is foregoing by taking up the option of doing higher education.

So this is what opportunity cost is. So basically nothing comes for free. It is every choice has a opportunity cost. Third principle is rational people think at the margin. What does this mean? This is, this is also very important principle because later whenever we develop any kind of models and any kind of decision making, we are going to use this principle repeatedly.

Say for example, again taking the example of the same student who is going to say he decides that I am going to do a 1-year MBA course, 1 year management education course for which he has to pay some fee. Now the decision is going to depend on only the cost that he is going to incur during this 1 year of education against the income that he is foregoing for this 1 year of education. He is not going to look at any other of its expenses or his incomes in the past.

Or say another example, say a person is hiring people, a person a producer he is a manufacturer he is hiring people. Now one more person whether he should hire that person or not will depend on how much salary or how much wage he has to pay to the person against how much output he is going to get from the person. So it is always at the margin. He does not think about the in totality what his production is, what his cost is, no. He is basically thinking at the margin. This is what we all of us do.

So the fourth principle is people respond to incentives. This is easy to understand. Say for example the government decides that people need to smoke less. So the government basically imposes a tax on smoking cigarettes. There is a tax on cigarettes, and smoking, the amount of smoking comes down.

So this is a negative incentive to smoke. Similarly you could have positive incentives also. Say the government wants people to save more. So it has lot of, it offers a lot of tax benefits for saving, saving more. So people respond to incentives and this is true everywhere, in all walks of life right?

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Now second is principles of how people interact. So people not only they make decisions for themselves they also interact with others, others in the sense they interact with other consumers, other producers, firms, when they go and go to look for job they interact with the employers. So there is always interaction across people, different people interact and they these interactions influence their decisions.

So these principles are going to guide us to understand how people interact with each other. So one of the principles is trade can make everyone better off. Now when I say trade can make everyone better off, I do not necessarily mean trade between countries. When I say trade it is basically exchange of goods for money across different individuals also.

What this basically means is if I am good at producing something and another person is good at producing something else, it makes sense for us to exchange the goods that we are good at making instead of trying to make everything myself okay?

So principle 5 is trade can make everyone better off and this is so true for countries also like the countries they do not have to if what whatever they are good at making they basically manufacture that and export those commodities and they import commodities which other countries are good at making. So as a result what is happening? We are getting cheaper products from other countries and other countries are being able to utilize the efficient production or the quality of production that we are able to offer.

So the next principle is markets are usually a good way to organize economic activity. Here 2 things are important. One is what do I mean by organize economic activity and the second is the word usually. So that means it is not always the case that markets are good way to organize economic activity, but mostly they are. So what do you mean by organize economic activity?

Now organize economic activity means basically what to produce in the economy? How much to produce of anything? How to produce it and who is going to buy it? As I said in the beginning that there are unlimited needs and wants in the economy and various things are demanded in the economy right from fundamental basic stuff like food, clothes, and shelter to something as fancy as having a birthday party on moon.

So there is maybe it is possible to have at least some consumers at some part of the world who would like to have something like as fancy as this, but whether it gets produced in the economy or not that gets easily decided in the market. So market is a good way to organize economic activity and it decides, but how does it do so? How does the market do so?

So basically people can go to the market with their demands. They can go to the market looking for stuff they would like to buy and they have certain willingness to pay. They would like to pay a certain price for every stuff. So they can go to the market and express their needs and wants and the firms can decide that whether it is it makes sense for me to produce this these stuff and sell in the market or not and if they do so then there is the stuff gets produced in the market.

So this is how the market basically resolves the problem of what gets produced in the market, but how does it do so? It does by the demand. There is a demand for a certain product, people are willing to pay a certain price for it and price acts as a signal in the market. So when price acts as a signal in the market the basically the producer decides that if I am able to provide this stuff at this price and if he does so then that stuff gets produced in the market and this is how the price is acting as signal in the market.

They are like the invisible hand, the term coined by Adam Smith which says that basically the market guides people where they are basically trying to satisfy their own interest. People are satisfying their own interest, the consumer is trying to meet his own interest. Producer is trying to meet his own interest of maximizing profit and the people and the product that are being

demanded in the market they get produced in the market. So this is how the market are, is usually a good way to organize economic activity.

But, now the second, the next principle basically talks about it is the continuation of the previous principle where we used the term usually. Now, markets are usually a good way but not always and when the markets are not good way to organize economic activity, governments can sometimes improve market outcomes. So when do governments improve market outcomes? Basically governments improve market outcomes when markets fail. What do I mean by markets failing? When do markets fail?

Markets fail under lot of situations like externalities, public goods, the government's attempt to have more equality in the economy. These are certain situations where the markets may fail and what do I mean by externality, what is externality? Externality basically means that a certain, all activities do not have do not always it is not that when say for let me explain through example.

Say for example a tannery. It is producing, it is tanning leather and it is polluting the nearby stream, water stream. Now the people who are buying the products from this tannery, they are not affected by maybe may or may not be affected by this pollution. Similarly tannery itself it is it will be charging the price depending price of tannery depending on how much cost it is incurring.

It is not going to take into account the fact that it is polluting the water stream nearby. So who pays for it? So basically an activity is causing or affecting people or stakeholders who are not directly involved in this economic activity. These are not people who are either producing the leather. These are not people who are consuming the leather nevertheless they are getting affected. These are maybe people staying nearby who are using the water from the stream. So there is a cost to the economy.

There is someone has to incur that cost and who is going to incur that cost? Who is going to impose that cost on this activity, the government. So the government can impose a certain so the government can come and say to the tannery that look you are polluting the nearby stream and so you have to pay certain taxes extra with which we will clean the stream or the government might say that you have to directly pickup incur the cost of cleaning this stream.

So this is where the government has to intervene in the market. Another example is public goods. Say for example keeping the air clean. Who is going to do it? It is a public good. Public good is something from which you cannot exclude anyone. So if an organization says that look I am going to clean the air, air nearby and you pay for it. Can it exclude people who are not paying for it? No it cannot. So these are examples of public good where the government needs to step in and it needs to take the initiative to produce because here you cannot exclude people from consuming these goods and another example is say the, say the government wants more equality.

Say for example in the labour market or where people are getting hired, people are willing to work for a very low wage which is not acceptable to the government or which is not acceptable to the society in general because it is inhuman. So there the government needs to step in and say that look you have to pay a wage which is minimum, the minimum amount should be this amount. So these are situations where the government can intervene and these are situation where the markets fail.

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Now the third set of principles is principle of how the economy works as a whole, basically how the economy works as a whole. This, these principles are more applicable in macroeconomics, which we are not going to focus on, which is beyond the scope of this course, but nevertheless I

am going to go through the through these principles. One is that a country's standard of living depends on its ability to produce goods and services.

So how well an economy does or what is the amount of economic growth in a for a country solely depends on how the country is able to produce out of its limited resources. Every country has certain amount of say certain amount of natural endowment of resources, some amount of labour, some amount of capital stock, say for example machineries, equipment, infrastructure that a country has and given all these all this setup the economy basically produces.

Now how well the economy does and how the economy growth can be rapid in a country depends on how efficiently the country is able to produce out of its limited resources. So and that would basically depend on the productivity of these countries. So if a country is able to improve the productivity of its economic activities, then it is able to do well. Another principle is that prices rise when the government prints too much money. Price rise is what?

Price rise is basically inflation. So inflation happens when the government prints too much money. That means there is too much money in the economy which is chasing too little goods. You are not producing enough but government has printed lot of money. So lot of money is chasing little amount of goods so the prices of these goods go up.

So this is the general principle that prices rise when the government prints too much money and that the last but not the least the principle that society faces a short-run tradeoff between inflation and unemployment. So it has been seen very often that various policies of the government which try to address unemployment or unemployment is basically what?

When economic activity is falling when the not enough is getting produced in the economy then the organizations are going to lay off workers. So when they lay off workers unemployment is going to rise. So basically there the what does the government do to target improvement in employment, in the economy? Basically they try to boost economic activity in the short run.

So when the government tries to boost economic activity through various policies it has been seen that in the short run they cause inflation or price rises and it is easy to understand because from principle 8 we already know that you cannot increase your output or economic activity from your resources unless and until you improve your productivity which is not possible in the

short run and if you are not able to increase your productivity other ways of trying to increase output in the short run is done at the cost of inflation.

So these are the 10 principles of how the economy works, how people interact, how the different stakeholders they make decisions in the economy and later we are going to study various economic models to see how managerial decisions are made and we are going to use these principles often.

We are going to come back and revisit these principles while we discuss the different kinds of different models and quickly since I have gone through these 10 principles another thing that I would like to emphasize is that this course is on primarily on microeconomics and we are calling it managerial economics because this is about decision making process of individuals or managers who are who need to understand how consumers make decisions, how governments make decisions, how the producers they themselves need to make decisions in different kinds of market structures. So this is more about microeconomics where the individual is involved.

So the first 2 types of principles are more relevant here but nevertheless we are going to keep on referring to the first 7 principles that we have discussed here and macroeconomics is more about understanding the overall economy. Macroeconomics is more about understanding the how the economy in totality functions.

The different so we are going to primarily focus on microeconomics and the another thing that I would like to emphasize is we are going to develop various models. These models say for example we would like to see how is price determined in the market. We would like to see what kind of markets, what are the different kinds of market structures.

So there are various products that are sold. So each we are going to see these products, the nature of these products determine what kind of market they have. Say for example they are I am going to explain them later but like say they say monopoly or oligopoly or competitive market a monopolistic competition. So to understand each of these market structures we are going to develop models. Now what are these models? These models are basically they try to replicate the reality and we are going to make certain assumptions to make it very simple to understand the basic functioning of these markets.

So they do not necessarily mean that they take into account all the problems associated with this market but it is like if you are building an airplane you would like to first see how the plane functions, how it takes off from the ground and flies in the, so you are not going to look at the small intricacies how the seats are laid and how the window shutter works and so similarly we are going to exclude some, many of the details of a market, yet try to understand what is the fundamental functioning of these models. So this is what we are going to do and in the next module I am going to start with the supply demand framework. Thank you.

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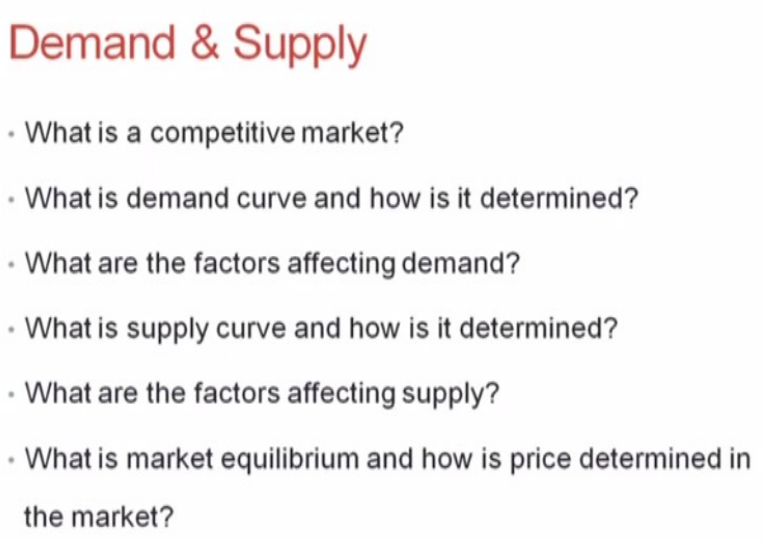
Lecture - 02

Demand - Supply Framework

Welcome to the next module of foundation course in Managerial Economics. We entered the previous module talking about the various principles of economics and I said that I am going to discuss a lot of models and let us start the course with a very fundamental and very powerful framework used in economics which is called the demand supply framework. It is a very simple framework yet it is very powerful because it can be used to understand lot of changes in the market.

Lot of market interactions in the market and be it market for goods, services or be it market for even foreign exchange, be it market for investment, savings, funds in the funds market loanable funds market. So any kind of market this demand supply framework, very simple framework of demand and supply is able to explain a lot of changes in these markets.

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So before going to the going into the actual framework, let me show what are the kinds of questions we are trying to address here. Now first I am going to briefly introduce a concept called competitive market. This is an assumption that we are going to take while explaining this framework of demand and supply.

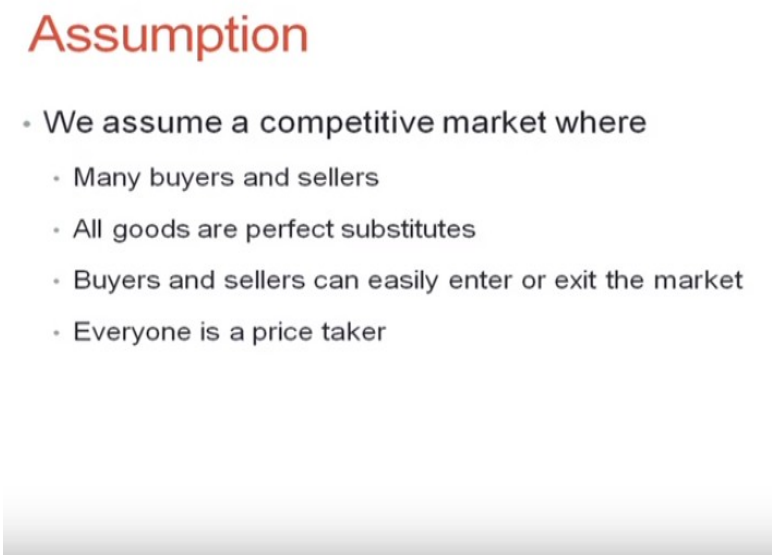
Competitive markets or perfect competition we are going to do in lot of details later, but I am going to briefly introduce the concept of competitive market here and because this is assumption that we are going to take and we are going to also try to understand what is demand and or what is demand curve and how is it determined. Basically we are trying to understand demand and demand curve is something a relationship that a mathematical or a graphical relationship that we are going to use to understand demand and how is it determined?

Then we are going to ask what are the factors that affect demand? Similar to the demand curve we are going to develop something called the supply curve and we are going to try to understand where does this supply curve come from. What is this supply curve and what are its major determinants? What are the factors which affect supply? And finally once we have the demand curve and the supply curve in the market we are going to be able to understand that there is a equilibrium in the market where demand equals supply.

Basically, what is being demanded is getting produced. So how does that happen? So that is the market equilibrium and that is also going to answer the question that how is price determined in

the market. So we said in the previous module we discussed that price acts as a signal in the market right? So when price acts as a signal in the market, where does this price come from? So this is what we are going to see in this first week's lecture of demand and supply.

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As I said we are going to take make certain assumptions. So when we discussed the framework of demand and supply we are going to assume a competitive market here. What does a competitive market mean? Competitive market does not, it is actually contrary to what we might think that in a competitive market everyone tries to be better than the other and everyone is trying to basically survive in the market and outwit its other rivals but that is not the case at all in the case of a competitive market where basically the assumption is there are many buyers and sellers.

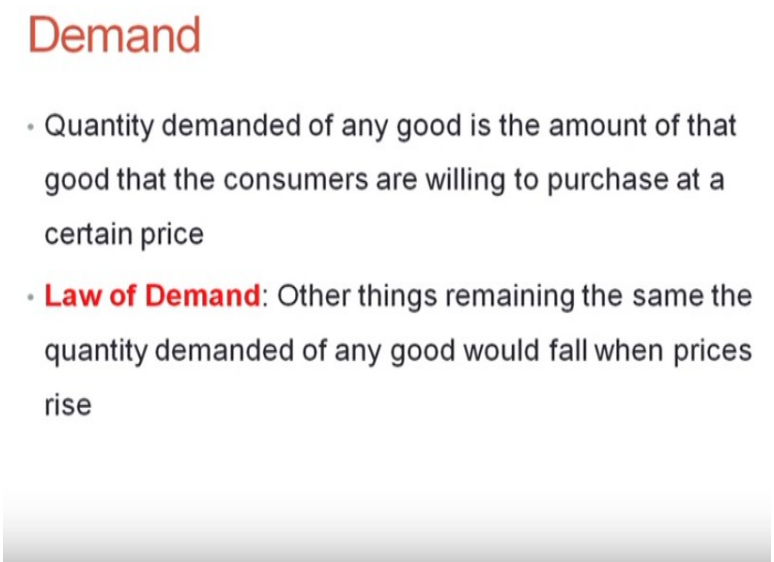
There are so many buyers and sellers that it does not matter. It does not, no buyer or no seller can affect each other's decision. No one can affect each other. It does not matter if a buyer exits the market, no one is going to notice. Similarly, if a seller enters the market no one is going to notice in the sense that it is not going to affect the prices or it is not going to affect the output that is being sold in the market. So buyers, there are infinite numbers of if not infinite, a very large number of buyers and sellers and so no single buyer or no single seller is important in the market in any way.

All goods are perfect substitutes. Again contrary to our common notion of the term competition that we use in the layman's language the competition term we use where we are sort of one might think that we are competing to better one's product than the others but in case of a competitive market all products are perfect substitutes. What one seller is selling is no way different than what the other seller is selling. So to the buyer it does not matter who buys the product, from whom he buys the product from.

So and thirdly the buyers and sellers can easily enter or exit the market. So when it does not make sense, for somehow if he feels that the cost is too much, going up too much, he can easily quit the market and no one is going to notice also because it does not affect the market. Similarly, a buyer can anytime, a buyer can enter the market, he can exit the market without affecting the market.

So a result of all these 3, first 3 assumptions is that everyone is a price taker. So price gets determined in the market, everyone looks at that price and if the price is acceptable to them they buy or they sell. So these are the assumptions behind competitive market and this is the assumption that we are going to take and now I am going to discuss about demand.

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Now what is demand? Quantity demanded of any good is the amount of that good that the consumers are willing to purchase at a certain price. So every consumer has a demand, every consumer has a demand. He basically expresses demand for a certain product like I am going to purchase this many units of pizza if the price is this.

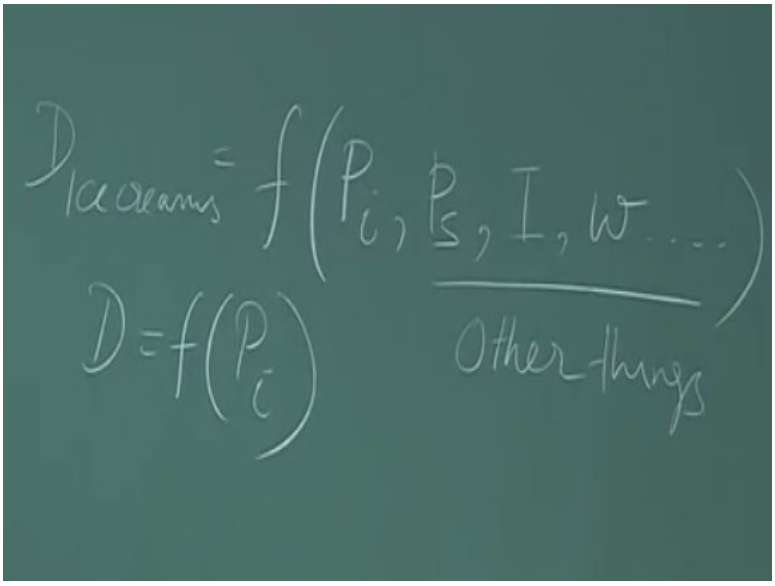
I am going to purchase so many garments if the price is this. I am going to make so many phone calls if the per unit of per phone call it costs me this much. So these are the different willingness to pay which is there in the minds of the consumer. So this is represented in the demand curve in the market.

And that brings us to something which we call the law of demand. Law of demand because this is something which is most often than not observed in the market that if the price goes down the demand for the product goes up and similarly if the price goes up the demand for the product goes down. So let me show through an illustration.

Say for example we are talking about a market where there is demand for market for ice-creams and there is a demand for ice-creams. So where does this demand come from in the market. Firstly, the demand comes from the individuals, one every buyer is going to go to the market and express his demand and every buyer has his own preference own taste everything and then he expresses his demand for ice-cream in the market.

So at a certain price what I am going to demand would not be the same as or may not be the same as the amount that someone else is going to demand and before actually deriving the demand curve let me show where does this demand come from. Say for example how many units of ice-cream I would like to buy?

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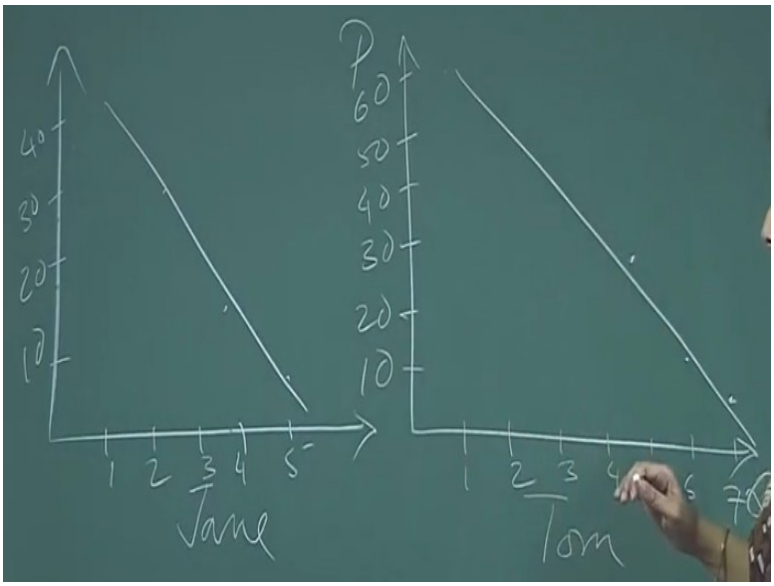
That demand function if I write it as a demand function say for example the demand for ice-creams would be function of certain variables that means certain my decision to buy the ice-cream is dependent on lot of things. Firstly, the price of the ice-cream. Second what? Price of something which is similar to ice-cream.

Say for example frozen desserts that are available. If I am, I have gone to the market and I am looking for something cold, something frozen, I would look at ice-creams, I would look at frozen desserts. I would probably look at cold drinks also. So price of substitute products. That is similar products which can meet the similar kind of need that the ice-cream meets.

Thirdly, what if I am rich? I am going to buy more probably, so my income. Then what else? Say for example if it is summer I might buy more ice-creams. When it is winter I buy less or the temperature outside, what is the day temperature outside like, so maybe weather. So one can imagine lot of factors on which my demand for ice-cream is dependent on.

So out of these the demand curve that I have defined here that the quantity demanded of any good, amount that the consumers are willing to purchase. So that demand curve for ice-cream is basically a function of only the price of ice-cream. Other things remaining the same. So these are the other things. So in the law of demand where I have already showed that other things remaining the same, the quantity demanded of any good. So other things, these are the other things that I was talking about. So these are the other things.

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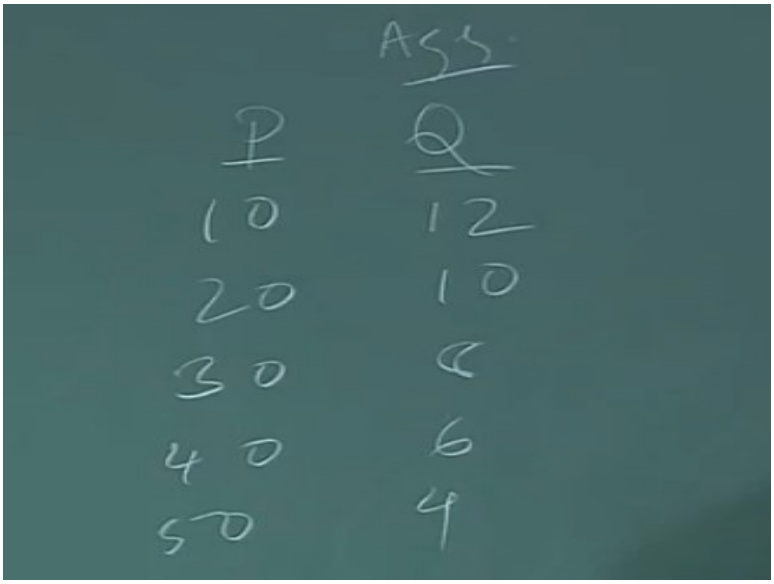
Now, so now basically I am going to, what I am looking for is a relationship between the quantity of ice-cream that I am I would like to buy and the price of ice-cream. Now one more thing that I would like to mention here is we are always going to stick to this framework of price and quantity where quantity is in the x axis and price is on the y axis. This is something which are which we are going to uniformly use during the entire course and this is something which is commonly used where price is always on the y axis and quantity on the x axis.

So now say I am trying to draw the demand curve of a particular individual. Say a person called Tom. Now Tom has say the ice-cream could be of various prices, 10, 20, 30, 40, 50, 60 and there are various quantities. Now Tom knows that he decides that I am going to buy say 1, 2, 3, 4, 5, 6, 7. So I am going to buy 7 units of ice-cream if the price is 10 rupees.

I am going to buy 6 units of ice-cream if it is 20. I am going to buy 30 units if it is sorry I am going to buy 5 units if it is 30 rupees and this is how he follows the law of demand and this is how he expresses his demand in the market. So this is his demand curve for ice-cream. Similarly, there could be another person, there could be another person say Jane who is not so much fond of ice-cream as Tom and she is basically she also expresses her demand for ice-cream and when the price is 10, 10 rupees she purchases see she purchases 5 units of ice-cream.

When the price goes up, even less; when it goes up, even less, and this is her demand curve. So each individual comes to the market which with their demand curve and all the demand gets, all the demand that aggregates in the market and we get something called the market demand curve. So how is the market demand curve arrived at?

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So basically market demand curve say market demand curve is going to so for the market demand curve at price of 10 rupees, price of 10 rupees, market demand is 5 units for Jane and 7 units for Tom. So it is 12 units; 20, 30, 40, 50. So say we aggregate the quantities demanded by Jane and Tom at each of the prices and this is the aggregate quantity that we get in the market.

So say it is 12, 10, 8, 6, 4. So this is the market demand curve. This is the market demand schedule that we arrive at and here one thing I would just like to point out is I am taking the example of 2 people aggregating it and showing the market demand curve but going by our assumption of perfect competition in a perfectly competitive market there cannot be 2 buyers, 2 buyers is too less.

So they if in a market there are only 2 buyers they can actually influence the market. But this is just for simplicity that I am showing it. Actually, there can be whole lot of buyers and this is how the market demand curve is arrived at. You basically aggregate the quantities for each of the

prices and you get the market demand curve. So this was the law of demand and in the next module I am going to talk about the different determinants of demand.

So this is the this was the like okay I have cleaned the board now but I showed the different other variables which affect demand and we are going to see how changes in those variables basically affect the demand schedule. So this is what we are going to do in the next module. Thank you.

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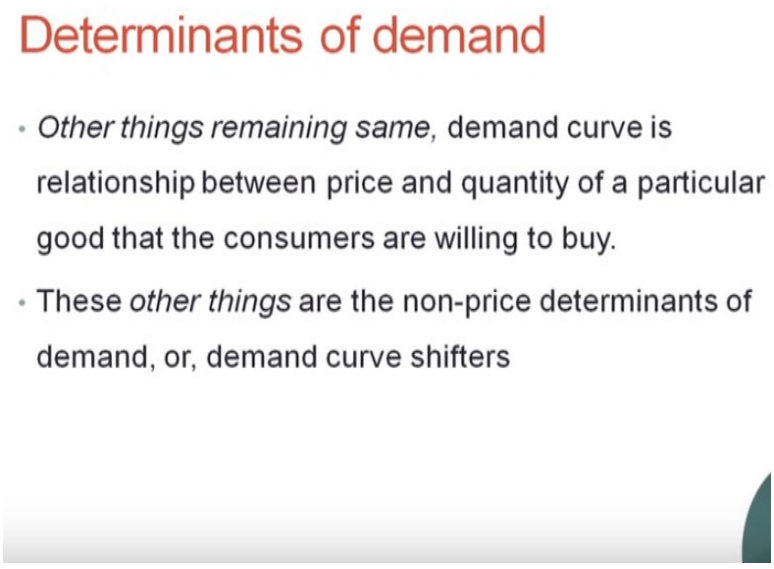
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Lecture - 03

Determinants of Demand

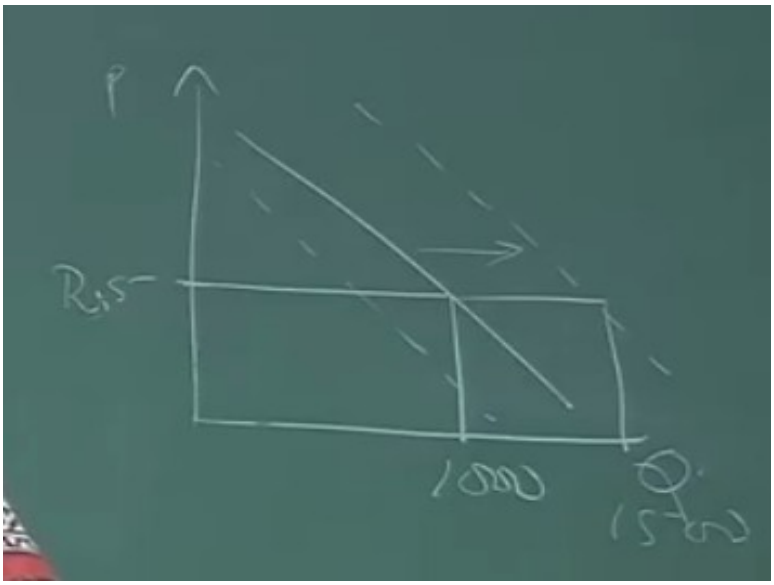
Welcome to the third module of the of this week's lecture on Demand and Supply. We started with the demand supply framework and we discussed how demand is determined in the market and we have not yet talked about the supply side. In this module I am going to discuss about the price and non-price determinants of demand. So I showed in the previous module that demand is a function of the price of a good and also a lot of factors which could influence demand.

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So what are the determinants of demand? Now other things remaining same demand curve is relationship between price and quantity of a particular good that the consumers are willing to buy. Now other things remaining same, this is italicized, because as I had mentioned earlier that there are lot of factors which influence demand and the demand curve that I drew for say for Tom, the relationship between price and quantity, price and quantity of ice-creams.

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So this, this was the demand curve for either the individual Tom or we can imagine this to be the demand curve for the market of ice-creams. Other things remaining same means that the price of other products say for example frozen desserts or cold drinks or other alternatives of ice-cream that the consumer might consider, those prices are same; people's tastes are same, the weather is same outside.

So given all the variables that can influence demand being same how changes in price is going to change my demand for a certain good say in this example the demand for ice-creams. So if the price of ice-cream falls, I am going to demand more of it. If the price of ice-cream increases I am going to demand less of it. So that is the demand of ice-creams.

Now, focusing on the other things. These other things are the non-price determinants of demand or demand curve shifters. So say for example if any of the variables that I talked about in the

which determine demand, any of the variables they change, then the demand curve is going to basically change right or left. That is for the same price now the consumer is going to demand either more of that good or less of that good depending on which variable has changed. So let us see some examples.

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So these are the non-price determinants of demand. Number one is the number of buyers. Say let us take an example of a college campus where there is a book shop. Now if the college campus the college has increased the number of seats for students this year and more number of students have joined the college the demand for books is going to go up. So basically the demand curve shifts to the right if the number of students has gone up in the college campus this year. So the number of buyers increasing increases the market demand curve.

Second is income. As I already said that if someone has more of income he is he or she is probably going to demand more of a good. If I am rich, probably I can afford more ice-creams, more pizzas, more of the good stuff and if I am not so rich I am going to demand less. So if my income goes up, I have a big promotion, I am going to probably demand more of some goods.

Now there is a difference. There is something called normal good and there is something called inferior good. Now, what is normal good? Normal good is, the demand for normal good

increases with income. So this is what I said the good stuff about. It is the all those things that I would ideally like to consume more but I could not because I had less income.

So when my income goes up I am going to start consuming more of these goods. So in case of normal goods again my demand curve is going to shift to the right if my income goes up but there are certain goods which are called inferior goods. These are goods where if my income goes up I would like to probably like to consume less of these goods or maybe I am probably shifting to other kind of goods.

A good example could be the quality of rice that I consume. So if my income goes up, I am going to consume probably better quality rice and the existing quality of rice that I am currently consuming, probably I will stop consuming those rice. So in that case if my income goes up my demand curve is going to shift to the left. So for the same price I am going to consume less of that product. So these are called inferior goods.

Third is the prices of related goods. Related goods means one could be it could be a substitute. As I said in case of ice-creams the price of frozen dessert, kulfi, cool drinks. So these are probably substitute goods and the demand increases with increase in price of substitute. Lot of substitute products one can imagine say for example say tea and coffee beverages.

So imagine in the railway platform there is a person who is selling tea and the price, going price is 5 Rs and that 5 Rs he is probably selling 1000 cups of tea, 1000 cups of tea. Now say there is a coffee outlet somewhere and suddenly that person decides I am going to charge a, he was earlier charging a certain price and he decides I am going to charge much more okay. People are really fond of coffee and I am going to charge a much higher price for coffee.

So in that case, lot of people who were earlier buying coffee they are going to now shift to the tea seller and at the same price that the tea seller was selling his tea at Rs 5 now he is going to sell instead of 1000 units he is going to sell more of those units say 1500 okay. So at the same price his demand increases. So his demand shifts to the right.

On the other hand, there are certain kinds of products which have compliments. They are these are called compliments. Say for example say a person who is selling coffee or say in the say for example compliments is say a printer and the ribbon that you use in the printer. They are

complimentary goods. So if the price of ribbon goes up, the demand for printer probably is going to go down.

So or if the more importantly say if the price of printer goes up the demand for ribbons is probably going to go down. So these are in case of complimentary goods your the price the way you respond to a price is the same that you respond to the same good and in case of substitute goods your response is going to be in the opposite direction.

Say for example if the price of the substitute good goes up your demand for the good goes up. If the price of the complimentary good goes up your demand for this good goes down. So this was about price of related goods. Another is taste and preference. Taste and preference. Say people have become very environment conscious and they decide we are no longer going to drive petrol driven or diesel driven cars. So what is going to happen?

The demand for these petrol driven or diesel driven cars is going to go down and maybe the demand for say battery operated cars are going to go up or say for example people are turning into vegetarians. So the demand for non-veg goods or non-veg items are going to go down and when demand goes down for the same price the demand curve is going to shift to the left and in case of the battery operated car for the same price people are going to demand more of the good. So the demand curve shifts to the right.

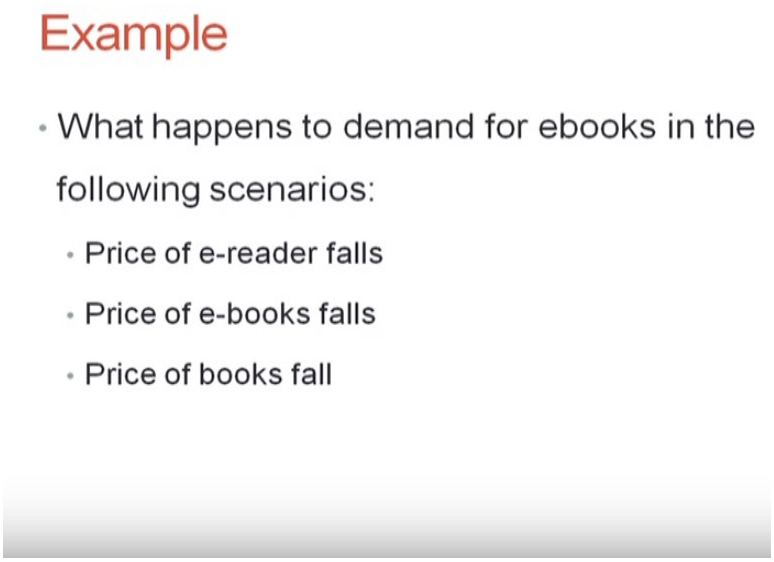
Another determinant of demand is expectations. If people are expecting say prices to rise, people are expecting say some festival is coming up and you expect the price of sugar to go up or price of some other fruits and vegetables generally to go up. So you might like to buy more of it and hoard it in your fridge. So your expectation about the price in the future makes you demand more right now.

Say there is some disturbance in the middle-east and you speculate that the price of oil is going to go up. So you might like to fill up your tank right now because you know that probably in the next 15 days, 1 month prices are going to really shoot up. So this is how expectation influences your demand.

So these are the non-price determinants of demand and the what is price determinant of demand is the price itself. So if the price changes your demand, your movement is going to be along the

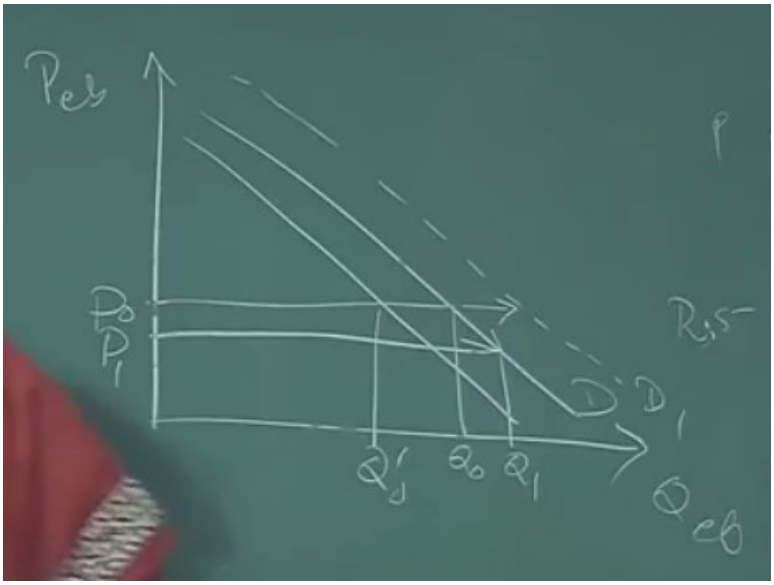
demand curve and if the non-price determinants they change in the economy the demand curve is basically going to shift to the right, right or the left.

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So let us take an example. What happens to demand for e-books in the following scenarios? So there is a demand for e-books. So let us say this is the demand for e-books and these are the prices of e-books and this is the demand curve.

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So what happens to demand for e-books in the following scenarios? Say price of e-reader falls. So what happens when the price of e-reader falls? What is e-reader basically? E-reader is what you use the e-reader to read the e-books. So if the price of e-reader falls, lot of people are going to start buying the e-readers.

Say for example the demand for kindle goes up in the and you start purchasing kindle online on Amazon and when the price of e-reader falls you buy more of e-readers, the people have started buying more of e-readers and so when they buy e-readers they have to buy e-books also to read on those e-readers. So the demand for e-books also go up.

So for the same price of e-books, if this is the average price of e-books, for the same price of e-books demand goes up. So demand curve shifts to the right. So demand curve shifts to the right. So at every price, at every price, now the number of e-books that is demanded is more than what was there earlier.

Second, say price of e-books fall. So what happens in that case? Do the does the demand curve shift? Does the demand curve shift to the right or left, no. The demand curve does not shift because the price has changed. So when price of e-books fall, so if this was the existing price and price has now fallen, movement is basically along the demand curve and when movement is along the demand curve and this is the new price P1 and this is the new quantity Q1 which is more than the earlier quantity.

So when the price of e-books e-book falls movement is along the demand curve and the total quantity demanded goes up. Price of books fall, say price of paperbacks. The price of paperbacks fall. Then what happens? What are books? Books are basically substitute of e-books. Books are substitute of e-books.

So when price of books fall the demand for e-books is going to go down. People would like to maybe continue to buy more of paperbacks or books. So the demand goes down. So for the same price of e-books now people are demanding less of the e-books. People are demanding less of the e-books. So this is what happens in case of when price of books fall.

So this summarizes how the what are the price and non-price determinants of demand are and we have seen that if the price changes the output changes along the demand curve and all other

determinants of demand when they change the demand curve basically shifts to the right or the left. So the quantity demanded can change in response to both price and non-price determinants of demand.

So price is also a determinant of demand and other factors are also determinants of demand but only the non-price determinants shift the demand curve and the price determinant basically leads to movement along the demand curve. In the next module we are going to talk about, we are going to develop the supply curve, we are going to see what are the determinants of supply curve and we are going to see what are the shifters of supply curve. Thank you.

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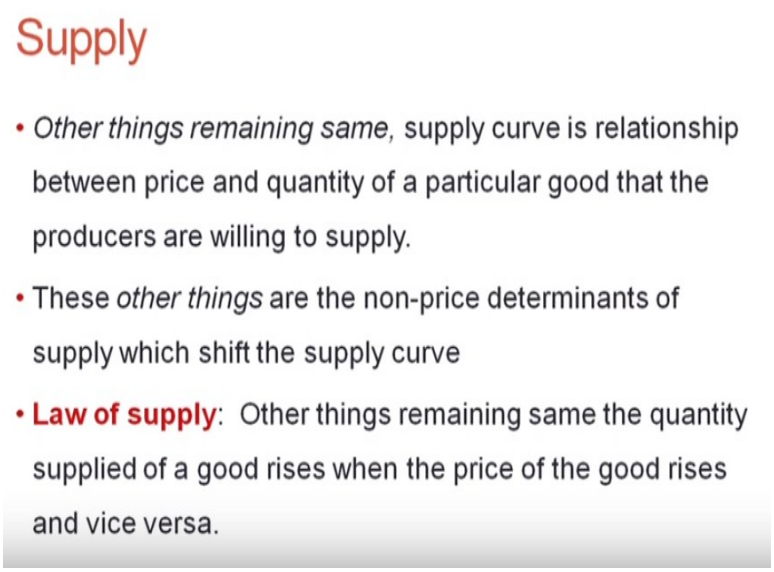
Lecture - 04

Supply Curve, Determinants of Supply Curve

Welcome to the next module of the first week of our discussion, the introductory course the foundation course in managerial economics and we started with introduction of the demand supply framework in the previous lectures and we developed the demand curve.

We said the demand the individual demand curve basically fallows the law of demand where when prices fall people demand for more goods. Now today and we also looked at different determinants of demand, different shifts, what shifts the demand curve. Similarly, today we are going to look at the supply curve and its determinants.

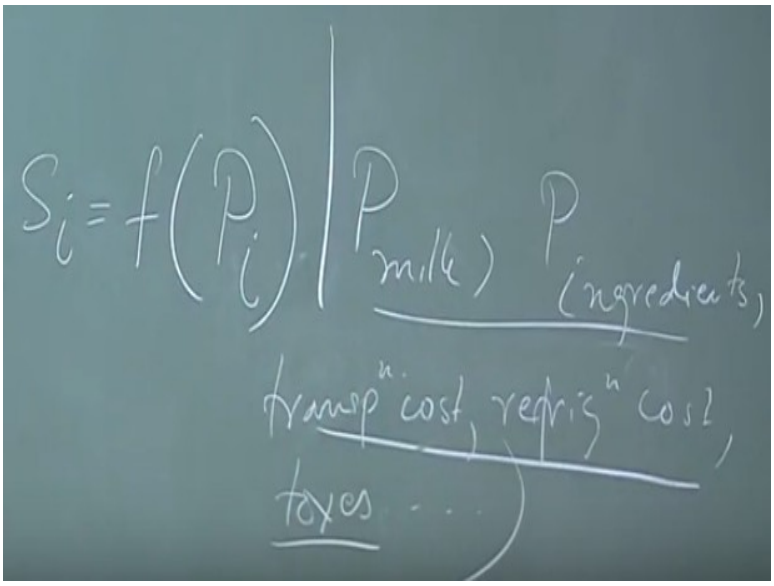
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So what is supply? Supply is basically other things remaining same. Supply curve is relationship between price and quantity of a particular good that the producers are willing to supply. So again very similar to the if you have followed the last previous classes on demand you will be able to relate more to the supply curve that we are going to develop right now.

Basically supply what when a producer decides how much to supply. Is he just dependent on the price of the good? No. It is dependent on the lot of factors. Say for example the supply of we started with the market for ice-creams when we discussed demand. So a producer who is supplying ice-creams, what is his supply dependent on? It is dependent on the price of ice-cream and it is also dependent on other factors say price of ice-cream.

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This is of, supply of ice-cream is a function of price of ice-cream and what else? Say for example milk is required for producing ice-cream. So if the price of milk goes up then the it is a problem to the suppliers, their cost goes up. So their willingness to supply more of ice-cream goes down. So supply is also a function of price of say milk. Price of say other ingredients, other ingredients are there in ice-cream.

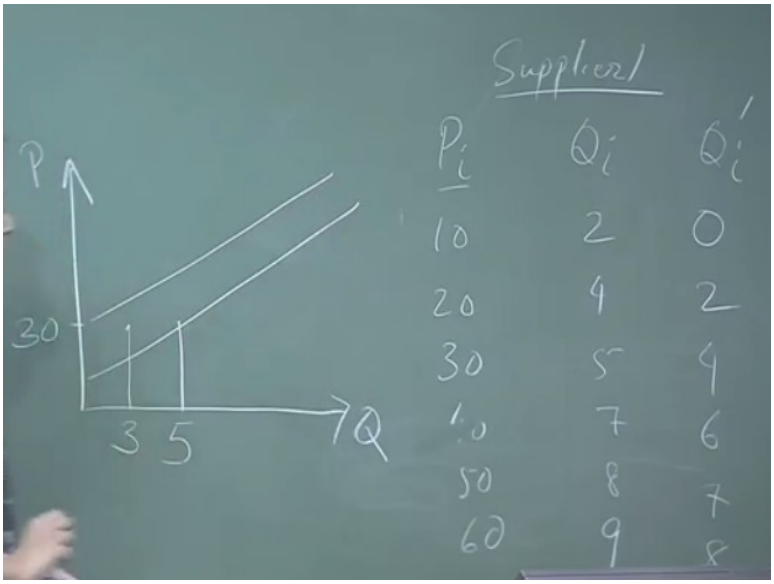
Then maybe supply of ice-cream is dependent on the cost of transportation, transportation cost, refrigeration cost and maybe taxes. So one can imagine lot of things which influence the supply of any product but what the supply curve that we are trying to develop here that is a relationship between the price of the commodity and the supply of the commodity.

When I say other things remaining same, I am assuming all these other components or other variables, they are fixed. So supply curve or the relationship the supply function that we are developing here is the relationship between the price of the variable, price of the product, and the supply of the product or the quantity supplied of the product, other things remaining constant.

So this is what we do and so these other things are the non-price determinants of supply which shift the demand curve. Now let us draw the supply curve. Now supply curve similar to the demand curve, supply curve also follows the law of supply.

What is the law of supply? It says that other things remaining same the quantity supplied of a good rises when the price of the good rises and vice versa. It is kind of makes sense. It makes it is kind of understandable that when the price is going to increase, the producer is more interested in selling the product. So that is the law of supply.

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Say for example for 1 person let us assume this is the price of ice-cream and this is the quantity that a particular supplier, supplier 1. So this is his supply condition when the price is 10, 20, 30, 40, 50, and 60 rupees. He is willing to supply say, he is willing to say supply say 2000 kgs of ice-cream okay. So he is supplying to the market. So this is in 1000 kgs maybe. So when if the price goes up he supplies 4, still goes up it is 5, 7, 8, 9.

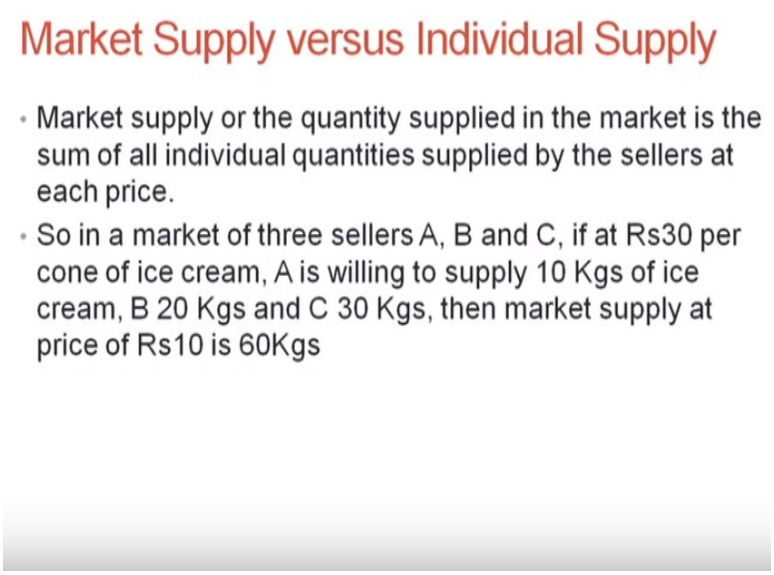
So this is his supply schedule. So this is his supply schedule. So they are positively related. As price goes up so does the quantity that he is willing to supply. So it is a positively sloping supply curve, is a positively sloping curve. So supply curve is a positively sloping curve. So this is the supply curve of one single individual. How do we find out the supply curve of the entire market because that is what we are aiming to do.

We are trying to find out the market conditions, the equilibrium price in the market from the individual demand and supply curve. So how do we get the market supply curve is the same way

that we found out the market demand curve. What we see is basically, so this is one person. This is his willingness to supply. There could be another supplier in the market, say another person whose supply schedule looks something like this. He is maybe he is a less efficient supplier.

So if the price is 10, he is able to supply only 0 units and when price is he cannot supply unless and until the price is more than 10. Say at 20, he is willing to supply 2, 4, 6, 7, 8 okay. So this could be his supply. Similarly, there are n number of lot of suppliers in the market and at each price we sum up all the supply that the producers are willing to do and we say that is the market supply curve. Say at 10 rupees the aggregate of all the quantities is gives us the market supply at that price. Similarly, at 20 the aggregate of all the quantities that gives us the supply and that way we get the market supply curve.

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So market supply or the quantity supplied in the market is the sum of all individual quantity supplied by the sellers at each price. So in a market of say 3 sellers A, B, and C if at Rs 30 per cone of ice-cream, A is willing to supply 10 kgs of ice-cream, B is willing to supply 20 kgs, C 30 kgs, then the market supply at price of Rs 10 is 60 kgs.

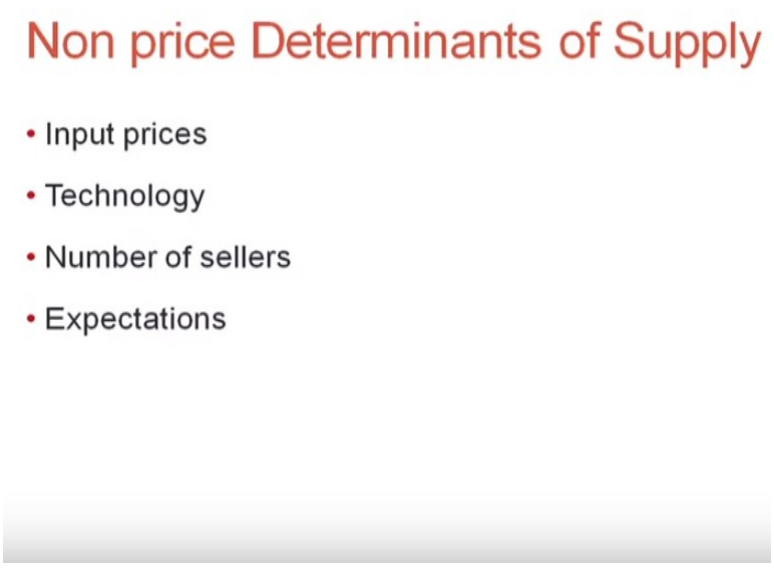
So again just to remember that at the very beginning we have assumed this is a perfectly competitive market which means that the number of suppliers is much more than 2 or 3. So 2 or

3 suppliers in the market actually very strictly speaking it violates the assumption of competition, but this is just to explain how the supply curve is the market supply curve is determined. So in a perfectly competitive market obviously the number of suppliers is going to be very high, but the supply curve is determined in this way.

So again, so again coming back to the supply function or the supply condition we are interested in knowing, what affects supply? So this is the relationship between price and quantity in the market so obviously if the price is going to move up or down similarly supply is going to move up or down. So that is the price determinant of supply. That is price determining supply.

But there are other factors like the supply function we wrote in the beginning and there were lot of variables there, lot of things which were influencing supply and we assumed them to be same when we developed this supply curve. Now what happens if any of those components, those variables they change? Then what happens to supply? That is what we are interested in knowing also and they are called the non-price determinants of supply.

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So there can be various non-price determinants of supply. Say for example input prices as I said the prices of ingredients, price of milk, even the transportation cost, all these are costs to the supplier. So any price goes up, then the supplier is willing to supply less. So if input prices are

going up so at the same price say the supplier was say at when price was 30 rupees, the supplier was willing to sell 5000 kgs of ice-cream.

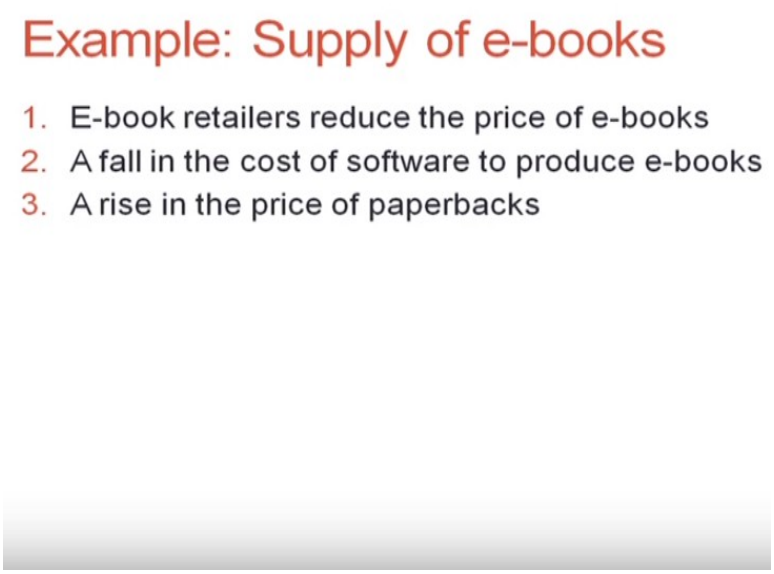
Now when costs go up when it is very difficult for the supplier to buy the ingredients, buy the milk, pay the transportation charge etc., he will be willing to supply less than 5000 kgs of ice-cream. So at the same price at 30, now probably he is supplying 3000 kgs. So in that case for and that happens at each price. At each price, the supplier is now willing to supply less than what he was supplying earlier. What does that mean? That means that the supply curve is shifting to the left. So supply curve has shifted to the left. This is what happens.

Again, another determinant is technology. Now the suppliers they use technology to produce. Now imagine there the same ice-cream producer he gets hold of a machine which produces ice-cream quicker and more efficiently. He needs maybe less electricity. He is able to produce more amount of ice-cream in lesser time. In that case what happens is at the same price he is now willing to supply more because the cost has gone down for him.

So in that case maybe he is willing to supply at 3 at 30 at 30 rupees he is willing to supply 7 units. So in that case for every price his supply curve has shifted to the right. Another non price determinant of supply is number of sellers. This is kind of obvious. If the quantity, if the number of sellers here goes up this total summation is going to go up and the supply curve is going to shift to the right and if the number of some sellers say they are quitting the market, so in that case the supply curve is going to shift to the left.

Similar to the demand curve, the supply curve is also influenced by expectations. So if the, it may not happen so much in the case of ice-cream, I am not aware, but say price of oil. If the price of oil if the market is kind of speculating that the price of oil is really going to go up in the future then the oil companies might like to hoard the oil. They might like to increase their inventory and reduce their supply today so that they are able to sell the oil at a future point of time at a higher price. So this is what expectation does to supply.

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Now let us take a example and finish off this module of understanding the supply curve. Say, so let us take the example of supply of e-books. What happens if the e-book retailers reduce the price of e-books? So we have discussed about lot of determinants of supply. Now which determinant is getting affected here?

Now here the price of e-book itself is changing. So what happens is there is no movement of the supply curve. The movement is along the supply curve. So if this is the supply of e-books then if the price of e-books is reducing so is the supply of e-books, the supply of e-books is going to come down.

What happens if a fall in the cost of software to produce e-books happens? If the, so it is basically becoming cheaper to produce the e-books. So when it is cheaper to produce the e-books the same thing is going to happen, the supply cure is now going to shift to the right. So basically at each price the supplier is able to supply more number of e-books.

And third is a situation where a rise in the price of paperbacks happen. So what are paperbacks? We discussed in the last lecture also, paperbacks are basically substitutes of e-books, the books that you generally get in the market and they are basically the e-book market is replacing the market for paperbacks.

So now if the if there is a rise in the price of paperbacks is it going to affect the supply curve? No. It is not going to affect the supply curve because it is going to affect the demand curve of the e-books because the paperback is a substitute of the e-books and price of e-books falling or rising is going to if the rise if there is a rise in the price of paperbacks the demand for e-books is going to increase and nothing is going to happen to the supply curve. So nothing is going to happen to the supply curves but obviously if there is a rise in the price of paperbacks that is affecting the demand supply framework in the e-books market but that we are going to discuss in a later module.

So that ends our discussion on supply curve. We have developed the supply curve. We know what are the determinants. We know how the supply curve shifts and in the following module we are going to discuss on the we are going to use both the demand curve supply curve bring them together in the market and see how they represent the market together and how that determines the equilibrium price and quantity in the market. Thank you.

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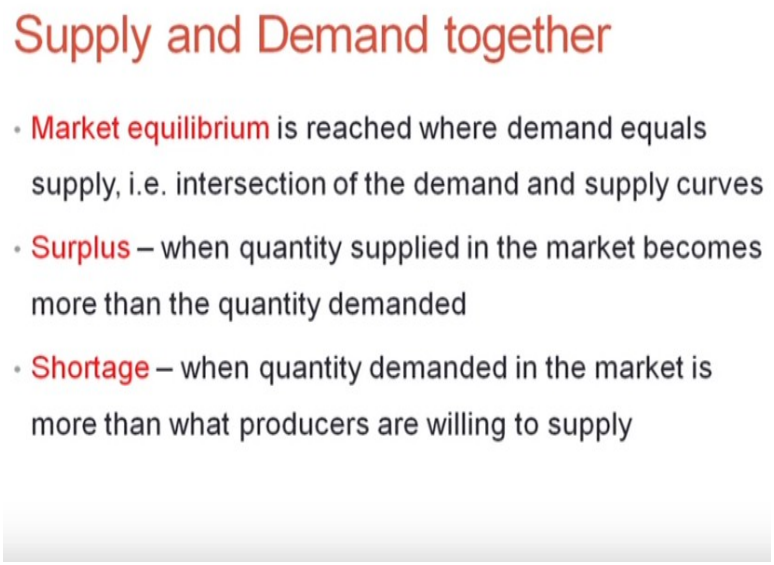
Lecture - 05

Demand Supply Equilibrium

Okay continuing with the previous modules where we developed the demand curve and the supply curve and we discussed about the various determinants of demand, determinants of supply now we are going to see how the demand curve and the supply curve together they can be used in a various ways like they can be used to see if there is any disturbance in the market, how is that resolved, what happens in the market? The basically the demand curve and supply curve they give us the market equilibrium.

We are going to see how that happens and then we are going to see that if that equilibrium is disturbed because of any disturbance either to the consumers who are the people who are demanding the goods or the producers who are the people who are supplying the goods how the market equilibrium is disturbed and does the, does the market again go to reach another equilibrium? How does that happen? So this is what we are going to see in the in this module.

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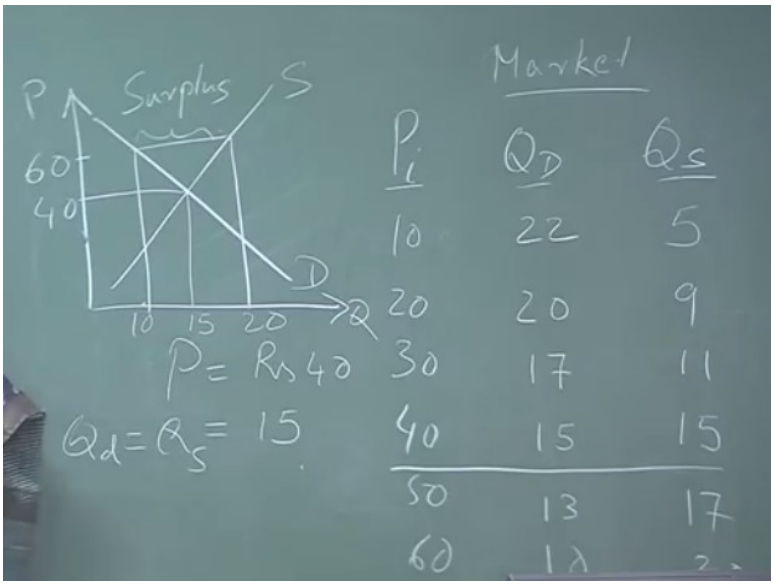


So first we are going to bring the supply and demand together in the market now. So there is a market equilibrium which is reached where demand equals supply that is the only point where which is the intersection of the demand and the supply curves. So basically what we are saying here is there is a bunch of people, the consumers who are looking for certain good.

So they have their demand, they are coming to the market looking for the goods and they have in their mind some idea about not some actually they have a definite idea about their the prices that they would be willing to pay for a certain good of certain quality and quantity. So certain quantity of good how much they are willing to pay.

So this is there in the minds of all the people who are coming to the market and similarly there are the suppliers who are bringing their product to the market and they know that there are various quantities they will be able to supply to the market given a set of prices. So they meet in the market and wherever the demand matches the supply that is the equilibrium. So let us although it is a very simple concept but still nevertheless let us try to draw it here. Let the demand and supply schedules of the so we started with the demand for ice-cream and supply of ice-cream.

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So let us say for example again, the price of ice-cream can be 10, 20, 30, 40, 50, and 60 and this is the market. So we are no longer drawing an individual demand or supply curve. It is not there is a single person coming to the single supplier and a single consumer in the market. There are whole lot of suppliers and whole lot of consumers in the market so this is the equilibrium is in the market.

So in the market this is the demand curve and this is the supply curve. So demand curve is if the price is 10 then quantity demanded is say 22, it is 20, 17, 15, 13, and 10. So the price if the price is very low the market the price is around Rs 10 per cone of ice-cream then the market is going to demand say 20000 kgs of ice-cream and how much is the entire the whole lot of suppliers who are there in the market. How much are they willing to supply?

So they are willing to supply only 5000 kgs. It does not make sense for them to produce more than that. So they are producing they are supplying 5000 kgs. So if the price is slightly less sorry ya when the price is slightly more 20, then the supply is a little more. The price goes up, supply goes up yet again. Price goes up to 40 they are willing to supply 15000. If it is the price is even more they will supply more and so this is the supply schedule and this is the demand schedule. So where do you think the market equilibrium is?

The market equilibrium is that price and that quantity where the amount demanded by the consumers is equal to the amount supplied by the consumer. So we clearly see that that point is

here. It is the this is the market equilibrium. So this is the market equilibrium and where we get that the market equilibrium is price is Rs 40 and quantity demanded is equal to quantity supplied is equal to 15. So this is where and if we draw these schedules they will also be so this is the point 15000 kgs ice-cream supplied and demanded at a price of 40 rupees. So this is the demand curve this is the supply curve. So this is the market equilibrium which we get.

Now given the equilibrium now let us understand a couple of a couple of other concepts. One is the surplus; when quantity supplied in the market becomes more than the quantity demanded. Say for example what happens if the price goes up to say 60, due to some reason the price has gone up to 60 in the market. So if the price is 60 the suppliers would be willing to supply 20, 20000 kgs of ice-cream but that cannot be the equilibrium because at price of Rs 60 how much are the consumers willing to buy?

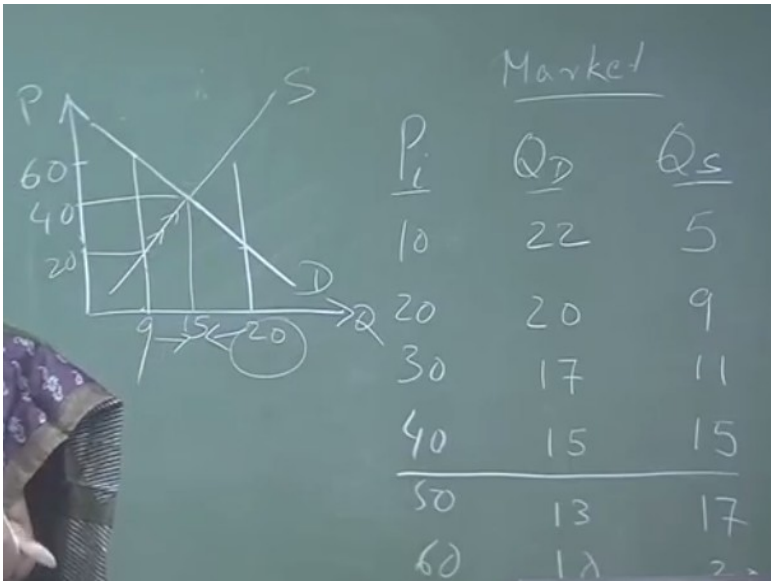
They are willing to buy only 10 units. They are willing to buy only 10 units. So there is clearly a surplus in the market. So there is clearly a surplus in the market where quantity demanded is much less than the quantity supplied. Now what happens in such a case? In such a case the producers are going to see that okay the price is high, I have, we have produced a lot of ice-cream but no one is willing to buy it.

So basically to sell the ice-cream, they will be reducing their price. So when they reduce their price gradually more people or more consumers will be entering the market showing some interest to buy the ice-creams and that will keep on happening till so this basically happens, so the price is so the quantity will gradually demanded will increase to 15 and quantity supplied will fall to 15 and the market will be back to equilibrium.

So what is happening if there is a surplus pressure the price is going to come down and when price comes down suppliers are going to reduce their supply and consumers going to raise their demand for the ice-cream and finally they again come back to the original market equilibrium.

The opposite of surplus is shortage. Now what happens when there is a shortage in the market? Say when quantity demanded in the market is more than what producers are willing to supply. So quantity demanded in the market is more than what producers are willing to supply.

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Say, now say price falls and price falls to 20 Rs, say price falls to 20 Rs. So at 20 Rs the consumers have a lot of demand for ice-creams. They would like to now, they would like to purchase 20000 kgs of ice-cream but at 20 Rs the supplier can only afford to supply 9000 kgs of ice-cream. So what happens in such a case?

In such a case the supplier is going to see that there is lot of demand in the market so they are going to increase their price or on the opposite of this the same reasoning is from the consumer side they will see that there is a shortage in the market, I want the ice-cream but it is not available. What is available in the market at this price is only 9000 kgs.

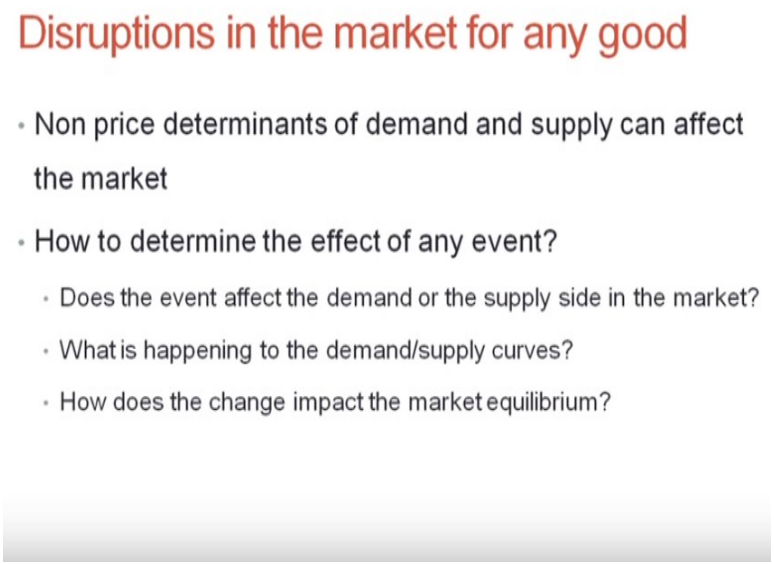
So many of the people they will be willing to pay a little more for the ice-cream. So gradually what will happen is the suppliers will increase their supply and consumers are going to reduce their demand or they are going to quit the market as they see that the price is going up. So all the people who were expressing their demand over here they are going to gradually quit the market as prices go up. As prices go up, the suppliers will be supplying more and the consumers will be demanding less till they reach the equilibrium level of 15 units.

So this is what happens in case of shortage. So what are we seeing basically? What is correcting the market? When there is any movement away from the equilibrium, there is something which

is correcting the market and the market is coming back to equilibrium and what is that? Now that is the price. Price is basically price acts as a signal in the market.

It tells the consumers that at this price what you are demanding is not to be supplied by the suppliers and to the supplier also the price tells that at this price you can only sell so much and you have to reduce the price if you want to sell more. Similarly, to the consumers it says that you have to increase the price if you want to buy more. So price acts as a signal in the market and brings the market back into equilibrium.

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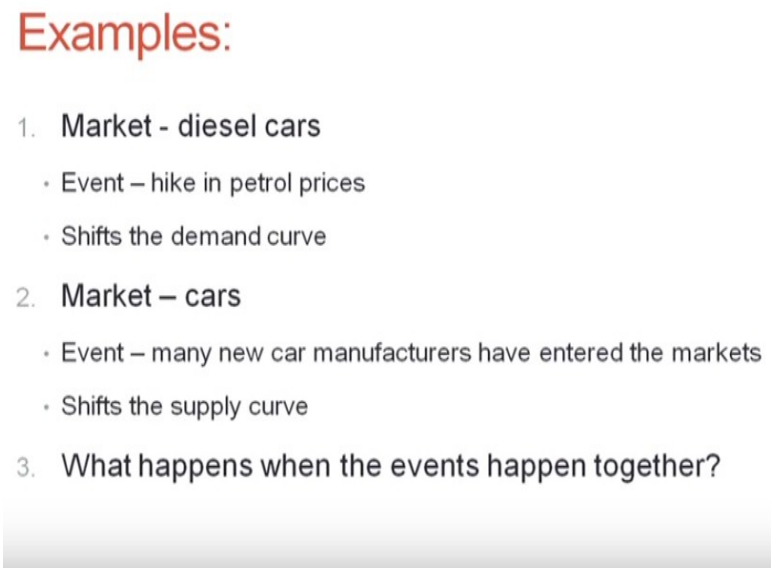
So now let us understand how exactly what happens how we can use this very simple framework to understand any disruptions in the market for any good. So how can disruptions happen in the market? Now this supply and demand shortage that the sorry the surplus and the shortage situations these situations are basically changes in the price that is the price determinant is changing and so the supply demand this equilibrium is happening in the market, but now we are going to talk about what happens any if any of these supply or demand curves any of these curves they shift in the market and what factors shift these curves. They, basically the non-price determinants of demand and supply shift the curves in the market.

So non-price determinants of demand and supply can affect the market. So whenever any determinant any disruption happens in the market how are we going to determine the effect of such an event. So basically the steps are we have to ask the questions, does the event affect the demand or the supply side in the market?

That is the first job that we need to do is we have to see that any event in the market is it affecting my demand side or is it affecting my supply side. If I have been able to figure that out then the next questions is what is happening to the demand and supply curves? Say if I have been able to determine that this event is going to affect my demand how is it affecting the demand? Is it shifting the demand curve? Is it shifting the demand curve to the right? Is it shifting it to the left? So I need to understand that.

And third is how does the change impact the market equilibrium. Once I have figured out that which of the two curves is getting affected and how it is getting affected it is easy to see what the new equilibrium is.

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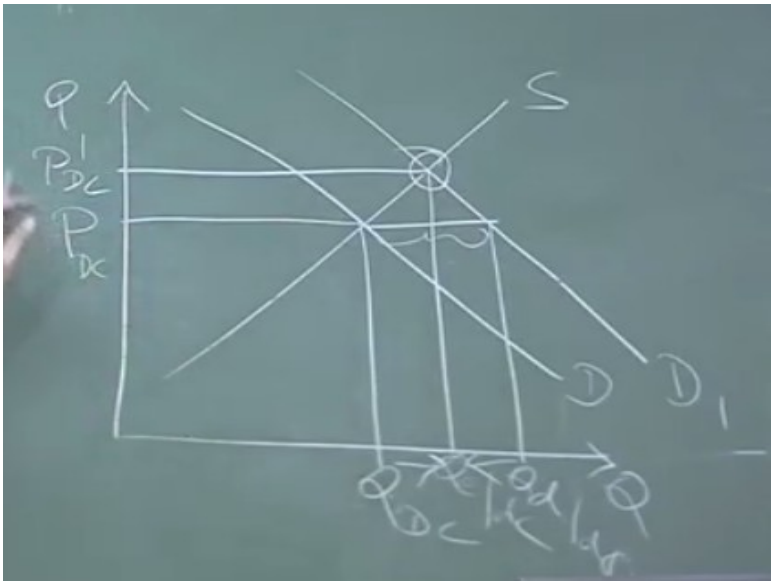


So let us try some examples and see how we can use this model to understand any changes in the market. Say we are talking about a market for diesel operated cars and a event that we are talking

about is there is hike in petrol prices. So hike in petrol prices means what? Which of my 2 factors demand and supply is going to be affected?

Now when there is a hike in petrol prices the people who are going to buy the diesel cars or the people who are going to buy the petrol cars or people who are looking for cars they will be more inclined to buy the diesel cars because the fuel of petrol cars is becoming more expensive and petrol cars are basically substitutes of diesel cars and basically using a petrol car or substitute of diesel car is becoming expensive so the consumers are going to demand more of the diesel cars.

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So let us again, so basically there is a increase in petrol prices, diesel cars the demand for diesel cars is increasing so my diesel the demand curve for diesel cars is going to shift to the right. So this is my new demand. So basically what I am saying is initially this was my equilibrium, this was the equilibrium price of car and supply of cars diesel cars say DC and the supply and now the demand has increased. So at the equilibrium level this is the existing price of diesel cars.

Now all of a sudden at since petrol prices have increased at this price of diesel cars demand for diesel car is now here. So this is the sorry this is the quantity. So this is the quantity demanded of diesel cars. This is the quantity demanded of diesel cars. But are they able to buy at this price? No they are not, because no one is supplying. How much are they supplying here? So now when

the demand curve has shifted to the right the suppliers are still not supplying so much of diesel cars.

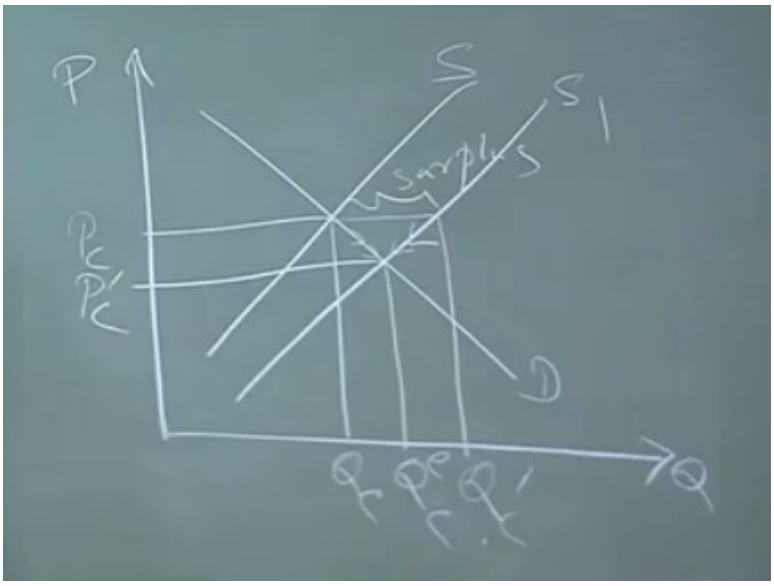
What they have observed in the market is basically demand curve has shifted up and people are willing to pay more for the cars and this is the new equilibrium where this is the new price of diesel cars and this is the new equilibrium quantity of diesel cars. So basically let us go through this once more. At equilibrium price of P DC the demand suddenly increased and quantity demanded also increased. So what situation is happening here? What is happening is a shortage, a shortage of diesel cars.

So when the supplier see that there is a shortage of the diesel cars in the market so and people are willing to buy more so they will supply more at a higher price, obviously, they cannot supply a high quantity at a lower price. So they have to follow their own supply curve and they are willing to they are telling the rest of the customers that if you are willing to pay a little more you can buy some diesel cars and the people here lot of them are going to see that the prices are increasing so they will quit the market.

So as some of them quit the market, the market basically then gradually comes back comes to a new equilibrium at this point where there is a new higher price of diesel cars and new higher quantity of diesel cars which are being sold in the market. So this is one example. So event is hike in petrol prices and this shifts the demand curve.

Now let us look at another situation. Say we are talking about generally the market for cars and the event that has happened is many new car manufacturers have entered the market. It is a some policy of the government which encourages manufacturing sector and maybe there is the economy is thriving and basically they see a lot of promise in this sector, so lot of new manufacturers have entered the market.

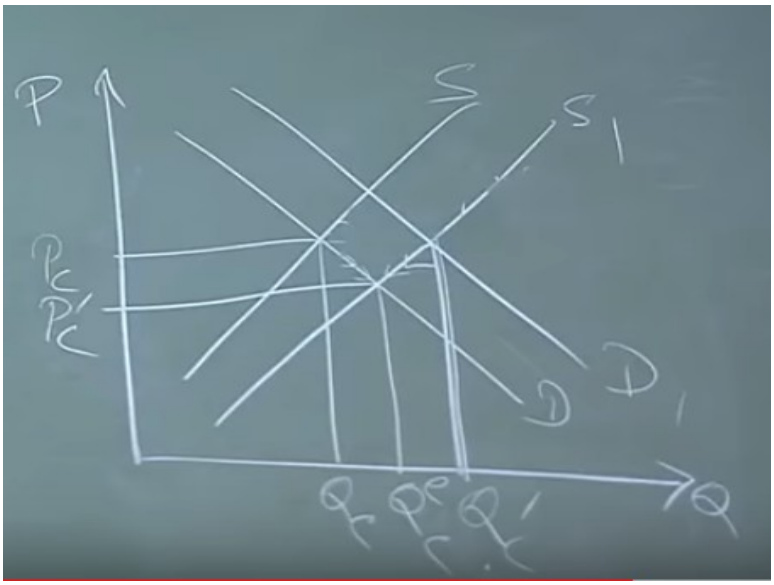
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Now what happens in that case? Which of so this event is a non-price determinant of which of the 2 factors, supply or demand? Now when the new car manufacturers have entered the market clearly the number of sellers has gone up in the market and the supply has gone up. So if this is the demand this is this was the original supply and this was the price of cars, this was the quantity of cars getting sold and now since more suppliers have entered the market supply curve has shifted to the right and at this price the equilibrium price of P c there are more suppliers to supply cars and they would be willing to sell Q c dash of number of cars. But are people willing to buy those cars? No. There is a clearly a surplus in the market.

So when there is a surplus in the market as we discussed what happens they know that there is a surplus in the market so suppliers know that they will have to reduce the price to sell the units they are willing to sell and so price goes down and as price goes down some people beyond this point, they would be willing to buy the cars. So as price goes down this is the new equilibrium for cars and this is the new price at which the cars will be available. So this is the so this is the explanation of the event. Many new car manufacturers and shifts the supply curve and what happens when the events happen together?

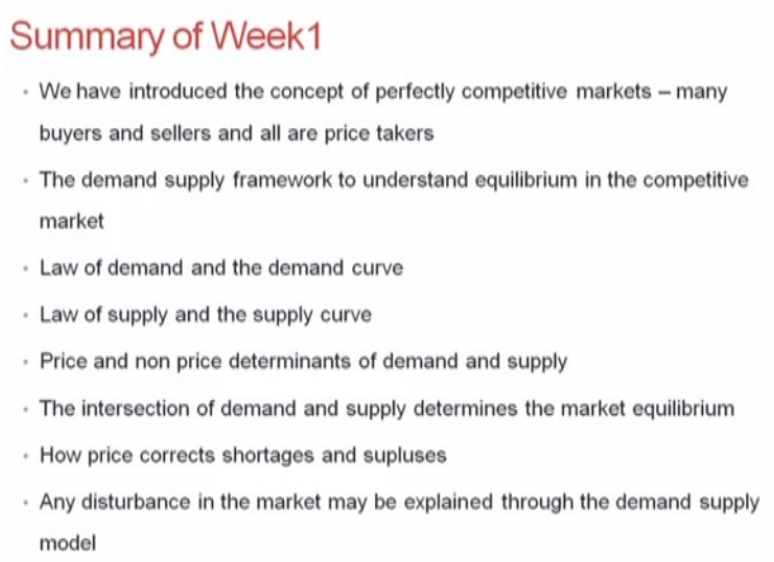
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So let us so this this is about the supply and in the previous case we saw that there was a increase in demand. So the demand shifts. So what happens? There is a intersection at this point demand and supply curve and there is a new equilibrium. So basically now there are more number of cars in the market as is so although the difference is diesel cars and cars but mind you that within cars also there are many diesel cars. So we can well assume that if the number of cars the supply of cars has increased the number of diesel cars would also increase. So this kind of situation is going to affect the market for diesel cars also.

So there is a new equilibrium and there is a new equilibrium in the diesel cars market where there is the price is the price has gone down and the output has gone up because the demand has gone up and the number of sellers in the market has also gone up. So the price has gone down and the output has gone up in the market. So this completes our discussion of the demand supply framework. We have introduced the demand supply framework and let me summarize.

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We have we started with the concept of perfectly competitive markets. We are trying to explain equilibrium in a perfectly competitive market and we started with the concept we said that there are many buyers and sellers and all are price takers and the we start we developed the demand supply framework to understand equilibrium in the competitive market. We said the demand curve comes from the law of demand. The supply curve comes from the law of supply although there are various other determinants of demand and supply.

So the intersection of demand and supply determines the market equilibrium and we saw how price corrects shortages and surpluses in the market and any disturbance in the market may be explained through the demand and supply model. It is a very simple model yet a very powerful model even later when we are going to okay we are not going to discuss about macroeconomic concepts but any market which has some price and some quantity say for example investment market the price is basically the rate of interest.

So similarly say foreign exchange market. The price is basically the exchange rate. So there also this very simple framework of demand and supply can be used to understand any kind of fluctuations in this market say if rate of interest changes what happens to saving, what happens to investment, all these could be understood.

So it is a very simple yet powerful and model in the economics course. So in the next week we are going to talk about elasticities of demand and supply. We are going to develop the we are

going to talk about elasticity, price elasticity, income elasticity etc. We are going to see how changes in prices change demand etc., and how they also help us understand various factors, the decision-making factors in case of demand and supply. Thank you.

Foundation Course in Managerial Economics

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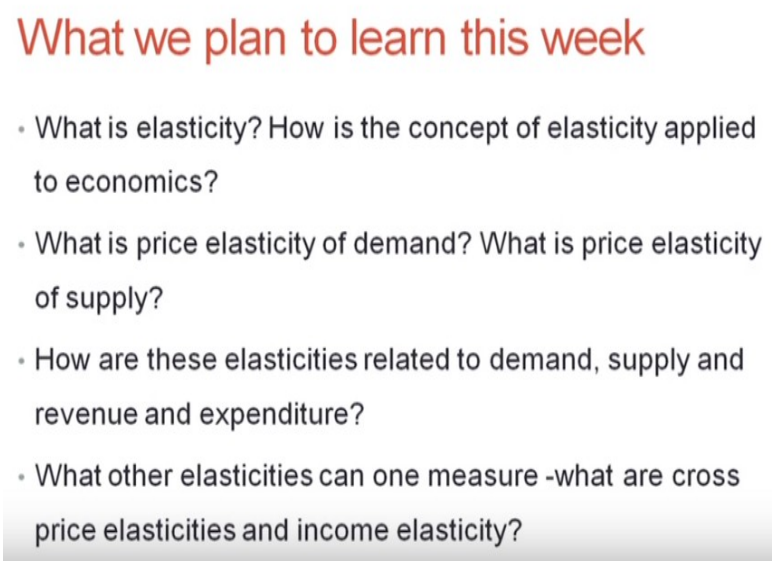
Indian Institute of Technology-Kharagpur

Lecture - 06

Elasticity of Demand and Supply

Hello and welcome back to the second week of the foundation course in Managerial Economics. Last week we developed the concept of demand and supply and we developed the concept of how demand supply equilibrium shows us what the price the market price is and what the market quantity is which basically clears the market and today we are going to extend the discussion and talk about something called elasticity of demand and elasticity of supply.

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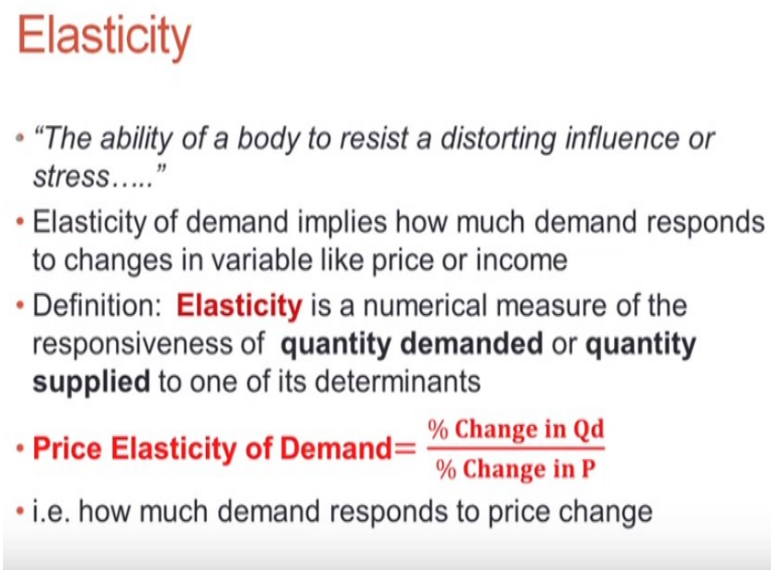


So elasticity is, so let me let me just quickly go through what we plan to learn this week. We are going to talk about what is elasticity. Most of know what elasticity is especially people with science and engineering background are very much conversant with the concept of elasticity, how it is measured etc.

And then we are going to see how is the concept of elasticity applied in economics in this framework of demand and supply and then we are going to talk about what do we mean by price elasticity of demand and what is price elasticity of supply and why is it why are we measuring these elasticity in the first place, what do we hope to achieve or understand when we are calculating the elasticities of demand and supply. What understanding or what further understanding of the market do we achieve by studying the elasticity of demand and elasticity of supply and we are going to see that elasticity actually tells us a lot about the revenue or the expenditure that the revenue that the producer is hoping to get out of producing the total number of goods or selling the total of number of goods in the market that is the revenue and we are going to see that there is a linkage between the revenue that he may expect with the elasticity of say the supply curve and similarly with the elasticity of the demand curve when the elasticity of the demand curve actually is going to tell us that how much extra units of output the consumer is likely to buy when the price of that product falls.

So just holding on to that thought let us proceed and what other elasticities can one measure so we are going to start with the concept of price elasticity of demand and later we are going to talk about price elasticity of supply but we are going to also talk about diverse types of elasticities. We are going to talk about cross price elasticity, income elasticity and once we are conversant with the idea of elasticity we are going to see that it is very easy to understand how demand or supply would respond to changes in any of say price or income or say any other determinant of demand or supply that we talked about in the last week.

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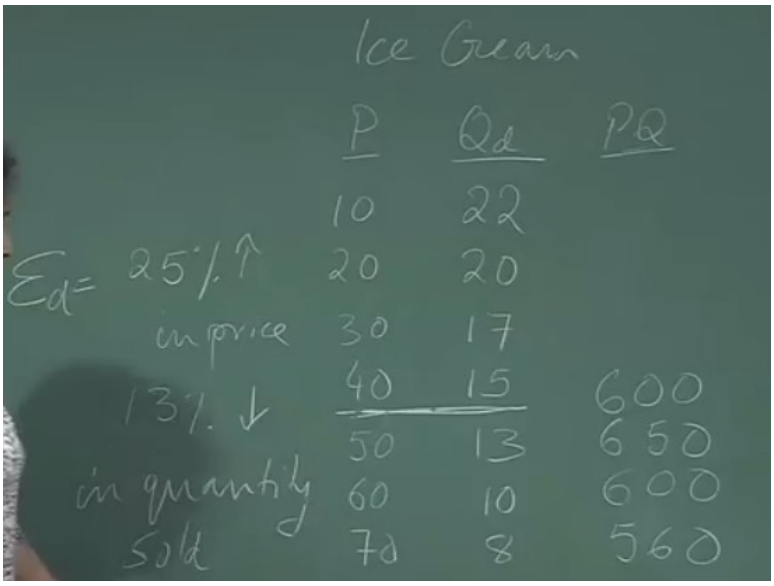


So what is elasticity? So the standard definition of elasticity is the ability of a body to resist a distorting influence or stress. That is we say that if it is in physics it is if the body can after distortion or stress if the body can come back to its original shape and size, so that is elasticity. So we are using this concept here and we are saying that elasticity of demand implies how much demand responds to changes in variables like price or income.

So how much demand is resisting an increase in price or how much distortion is happening to say demand or supply when something like any of the determinants of demand or supply say for example price when the price changes how much demand is changing and how much supply is changing in response to that price. We are trying to evaluate that. So the definition here we are using is elasticity is a numerical measure of the responsiveness of quantity demanded or quantity supplied to one of its determinants.

This is the most general definition of elasticity that we are going to use and a very specific definition of price elasticity of demand that is how much demand responds to change in price is simply given by the or it is simply measured by the ratio of percentage change in quantity demanded to percentage change in price. That is how much demand responds to price change. So we can start with an example. Let us start with an example and we are going to take exactly the same example of the ice-cream market that we introduced in the previous class.

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So say this is my demand schedule in the market, these are the prices, and these are the quantity demanded of ice-cream. So the prices were 10, 20, 30, 40, 50, 60, and 70 and against these prices the quantity demanded in the market were 22, 20, 17, 15, 13, 10, and 8. So we are assuming that this is the quantity so if the price is say 10 Rs per cone of ice-cream, the total amount that the market is going to demand is say 22000 kg of ice-cream or whatever depending on the size of the market.

So unit is not so important here. It is just enough to see that the quantity is increasing, is increasing as prices price is decreasing. So this is the demand schedule of the ice-cream. Now where does elasticity come in? Say for example, I am an ice-cream producer and I decide that somehow it does not it is not making sense for me to so earlier in the last class what was the equilibrium that we arrived at? We arrived at this equilibrium.

So say currently the market price is 40 and I am selling 15000 cones of ice-cream. So now and what is my revenue? The revenue that I am earning that this is the total amount of ice-cream, this is the total sale that I am that is happening in the market. That is 600. So now say for example, the ice-cream producers decide that 40 Rs is not enough, like my costs are going up.

Say for example my refrigeration cost as electricity bills are have gone up and refrigeration cost has gone up so I would like to increase my price from 40 to say 50 Rs. So my concern is by the law of demand I know that if I increase my price the quantity demanded in the market is going to

go down. So to the producer the concern is, is my demand is my total revenue going to continue to be this or is it going to be more or is it going to be less.

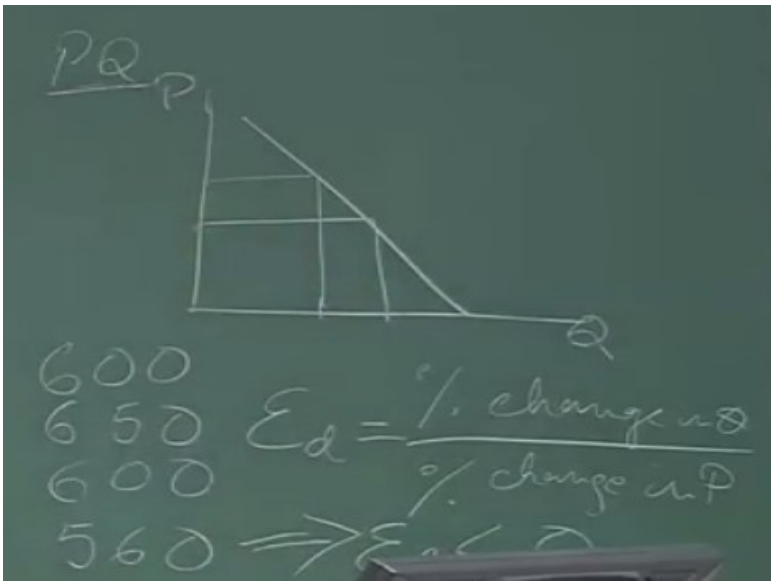
It can be more if the amount, the quantity or the response in the market to the price change is not so much as the price change. Say for example if my P goes up by a higher percentage than Q in that case I can hope to have a increase in revenue. So in that case it makes, probably makes sense for me to increase the price. So let us see what happens here.

So now I am going to see so here if I would like to increase my price from 40 to 50 I am going to see what my elasticity of demand is and as per the definition that we have shown in the slide, earlier slide, it is percentage change in quantity demanded to percentage change in price. So if I increase my price to 50 Rs then by definition my elasticity of demand against price is 25 percent increase in price and around 13 percent increase sorry it is going to be a increase in price so 13 percent fall in quantity sold.

So if my price is the amount by which the price is increasing is more than the amount by which the percentage by which quantity is increasing it makes sense for me to increase the price. So here if I increase this it to 50 I can see that here it is 650 okay. Now if I decide to increase it to 60 then what happens? Then I am seeing that it is increasing to 60 and my price has sorry my the price has increased to 60 and my quantity has fallen to 10 my but my total revenue continues to be 600.

So in that case there is the same amount of increase in price, the same amount of increase in price is leading to same amount of decline in quantity and finally if I decide that I my costs are really high and I would really like to increase the price let me increase it to 70. In that case what I see is it is 560. So when it is 560 I clearly see that there is a drop in my revenue. So having said that this is just to develop the concept of elasticity that basically elasticity what it is telling me.

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Elasticity is telling me that since P and Q move in opposite directions since we have a negatively sloping demand curve where this is P this is Q, since we have a negatively sloping demand curve every unit of price change that happens every unit of price change that happens is comes along with a change in quantity and that is going to change my revenue.

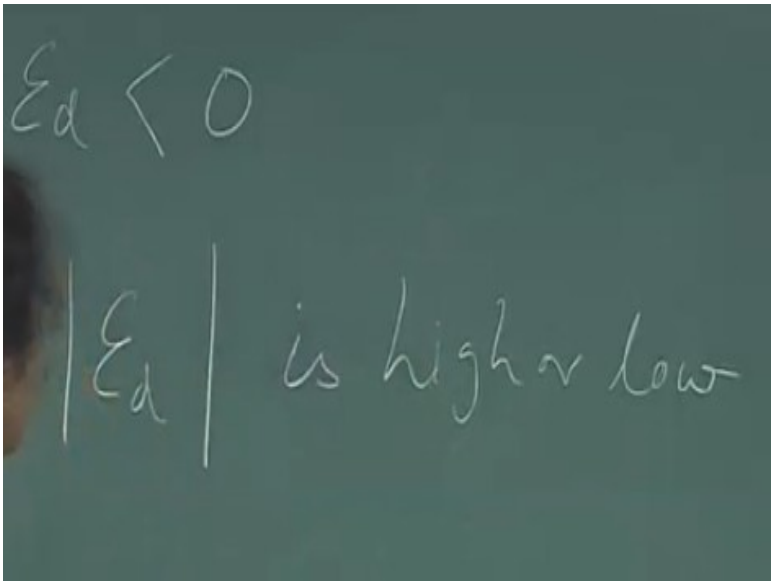
But right now we have not discussed anything about production, production function, cost, etc. How basically the producer decides if the price is to be increased or not. So this is not to say that the producer is not going to increase the price. This is not to say that since price is when he increases the price from 40 to 70 his revenue falls from 600 to 560 so the producer is not going to increase the price. No I am not saying that.

This concept this is a very simple example which I have developed to demonstrate what basically price elasticity of demand means and what it is what is its implication or usage. So along a demand curve so along a demand curve as I have already said price and quantity move in opposite directions so what do I mean by that? So basically when I say that elasticity of demand is percentage change in quantity divided by percentage change in price.

So when price is increasing or decreasing quantity moves in the opposite direction which implies that elasticity of demand price elasticity of demand is always less than 0. Price elasticity of

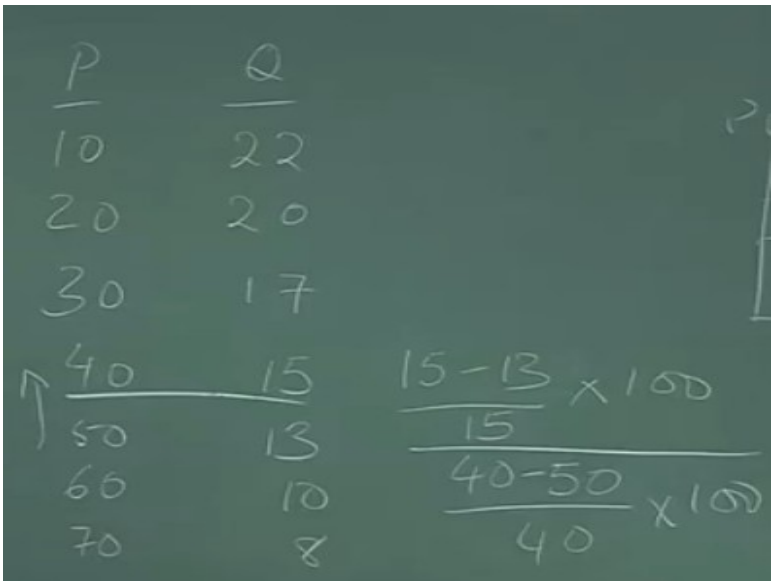
demand is always less than 0 because the demand curve has a negative slope because price and quantity moving opposite directions so elasticity of demand is always going to be less than 0 and let me clean up the board.

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So elasticity of demand is always less than 0 and henceforth whenever I say that elasticity is high or low, what I mean by that is if demand is high or low. Basically I am taking the price elasticity value in absolute terms and I am saying whether elasticity of demand is high or low. So whenever we say price elasticity of demand is high or price elasticity of demand is low what we mean by that is the absolute value of the elasticity. This elasticity is always less than 0 but what I what I mean by elasticity being high or low is the absolute value of the elasticity of demand is high or low.

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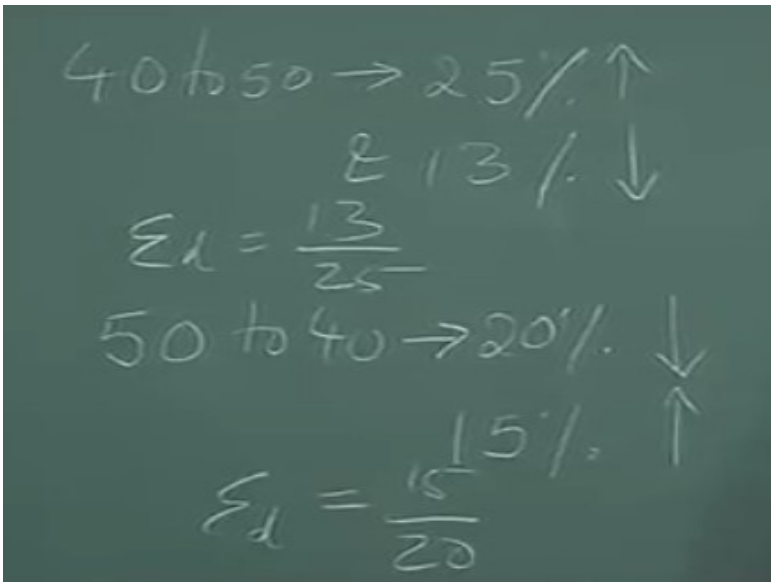
Okay another point is that going back to the same example I going back to the same example sorry I have to write it once more okay so we took the example that we are starting from this point. So equilibrium price is 40 and the quantity demanded right now in the market is 15 and we are contemplating increasing the price from 40 to 50 and we know that demand is going to decrease.

Now what is the response to price increase at this point? When I am the way that elasticity is calculated, how is it calculated? It is basically calculated when I am doing when I am trying to calculate this percentage change in quantity. So what I do is it is 40 - 50 divided by 40 into 100 I am sorry it is going to be in the this is the price increase so it will come in the denominator.

So this is how I calculate what the price elasticity of demand is when I am trying to increase the price from 40 to 50 or in other words I am trying to find out how responsive quantity is to price changes when my price is increasing from 40 to 50. Now what happens when I am trying to calculate exactly the opposite? If I am trying to calculate how is my demand going to change if my change my or if I decide to reduce my price from 50 to 40.

Is it going to be the same as increasing my price from 40 to 50? No, it is not. So in that case if I am if I am reducing the price from 50 to 40 in that case my elasticity price elasticity of demand is going to be so let me just clean this in the same way let me just clean this up and write here.

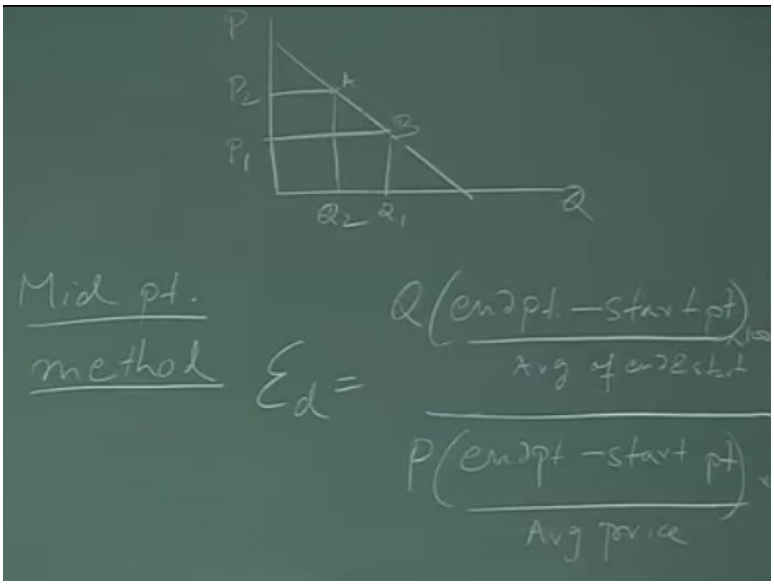
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When I am increasing from 40 to 50 price is increasing in 25 % increase in price and 13 % fall in quantity but when price falls from 50 to 40 this is a 20 % fall in price and 15 % rise in quantity and my elasticity of demand here is going to be 13 by 25 and here it is going to be 15 by 20, 15 by 20. So obviously they are not the same. So they are not the same.

So basically what that means is at what point I start calculating my or what point I start calculating my price elasticity of demand that is important in determining what the price elasticity of demand is going to be. So that kind of makes things a little uneasy or complicated so what we usually do here is we use the midpoint method.

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What we do is we use the midpoint method where we use the midpoint method where basically so if so this problem basically what we mean is if I move from A to B I get a certain elasticity price elasticity of demand which is not going to be equal to if I move from B to A. So to resolve that problem what usually is done is we take the midpoint of the 2 changes both in the numerator and denominator.

What we do is midpoint method is now I define elasticity of demand as Q endpoint minus starting point values divided by the average of the 2 points of end and starting point into 100 divided by price endpoint minus starting point divided by the average price at the 2 points. So this basically gives me the midpoint method of calculating the price elasticity of demand. So this is the midpoint method of calculating the price elasticity of demand.

And in the following lecture we are going to talk about so this is in this lecture or in this module we have developed the concept of elasticity of demand. We have seen how elasticity of demand is calculated and we have seen that basically the value of the price elasticity of demand varies or depends on the point at which it is calculated and so to resolve the problem what we have done is we have used something called the midpoint method where we are going to take the average of the beginning and the endpoint that we are looking at.

So if the price is increasing or decreasing we are going to get the same value of elasticity of demand in both the cases. Unlike in the earlier case where if your if one is moving from the point

A to B on the demand curve the price elasticity of demand that we get is different from if we move from B to A.

So this was about calculation of elasticity of demand and this is general definition of elasticity of demand and this is price elasticity of demand if it is income elasticity of demand we are going to have income in the denominator. If it is something else we are going to have the so basically whatever determinant of demand or supply that is influencing the demand or supply let us not talk about supply right away. We will be taking it up in a different module.

So to continue if we would like to see what elasticity of demand is with respect to any determinant of demand this is how we calculate. We take the quantity change in response to any determinant of demand divided by the change in the determinant of demand. So we have taken the example and we are going to work more on this so we have taken the example of price here because price elasticity of demand is very important and we are going to talk more about it.

So in the denominator here we have price but we can also imagine something like income. Say a person how much his consumption of say luxury cars in an economy. If there is increase in income how much consumption of luxury cars increases in the economy? So something like that can be calculated by having instead of P here in the denominator we can have income variable in the denominator.

So this is how elasticity of demand concept is used in economics and we are going to use more of it in the later modules and we will be developing we will be looking at relationship between the demand curves, the shapes of the demand curves, and how elasticity of demand depends on them. We are going to look at determinants of elasticity of demand etc. in the following module. Thank you.

Foundation Course in Managerial Economics

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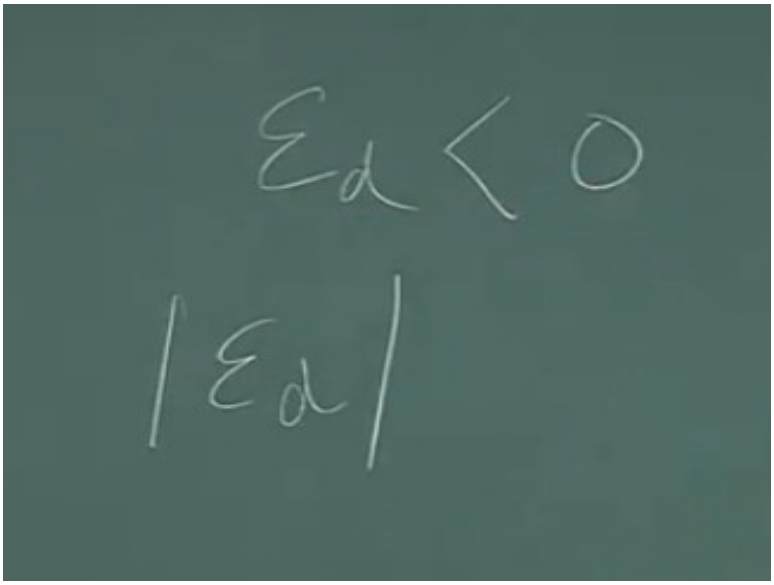
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Lecture - 07

Determinants of Elasticity of Demand

Welcome back to the next module of elasticity. We have introduced the concept of elasticity of demand and in the previous module we saw how elasticity is calculated and we said that this general definition of elasticity or the calculation it can be used to look at elasticity of demand or the responsiveness of demand to various determinants of demand. So that kind of sounds easy. So let us look at what are the determinants of demand. What determines and how if a commodity is or a product is going to be highly elastic or is it going to be going to have low elasticity.

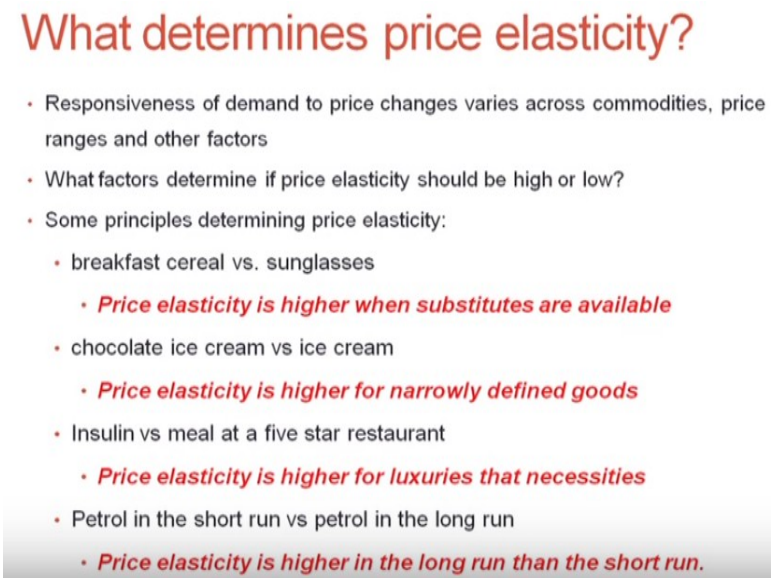
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So these are certain things we are interested to know and again repeating what I said in the previous module that elasticity of demand or price elasticity of demand is less than 0 always because of the law of demand that whenever price increases quantity is going to decrease and when price decreases the quantity demanded is going to increase.

So this is always going to be less than 0 but when I say that elasticity is high or low what I mean by that is the absolute value of the price elasticity of demand. So this is price elasticity of demand and not any elasticity. Say only price elasticity of demand is less than 0.

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So what determines price elasticity or what determines how the elasticity is going to be? What kind of elasticity can we expect for a certain product? So this is what we are trying to study here. So responsiveness of demand to price chain price changes varies across commodities, price ranges and other factors. So how much demand is going to respond to any price change that depends on a lot of factors and so we are going to discuss a few of the factors here which are which are kind of more obvious.

What factors determine if price elasticity should be high or low? So here are some principles determining price elasticity. So following are some principles determining price elasticity and I am going to explain using some examples. So as I show the examples I would actually encourage you to try to imagine before actually going to the next slide I would encourage you to imagine which of the commodities should show higher elasticity.

That is the question that we are I am going to show 2 commodities and the question that I am trying to ask is which of the 2 products should be more sensitive to price changes. This is what so the more sensitive to price change it is higher the elasticity. That is the quantity adjustment happens more in response to price changes.

So let us look at the first example. Say breakfast cereal versus sunglasses. So in case of so which one of these 2 products is going to have probably higher elasticity? So what the logic that goes on here is look at breakfast cereal. What is breakfast cereal? Breakfast cereal is a option for your breakfast. So is it that you cannot have breakfast without breakfast cereal? No, it is not so. There are whole lot of substitutes of breakfast cereal and this is just one of the choices.

And on the other hand I have sunglasses. What is sunglasses? Sunglasses are basically what you wear to protect your eyes from the sun rays especially during summer. Are there any substitutes of sunglasses? Can you so when you are going out in the sun and you would like to protect your eyes do you have any other alternative? No, you do not have any choice, if you need it you have to buy sunglasses.

So the principle that we are talking about here is price elasticity is higher when substitutes are available. What I mean to say is for breakfast cereal if the price of breakfast cereal increases people are more likely to shift to other breakfast choices. So when people move to other breakfast choices the little bit increase in price of breakfast cereal is going to lead to a huge amount of decline in quantity of breakfast cereal demanded in the market because people have other choices.

But what happens in case of sunglasses? So say for example in the summertime if there is an increase in sunglasses probably the sale of sunglasses is not going to go down too much if it is really required by the people who are wearing it. So they are not going to demand less sunglasses because there are no choices, there are they have no alternatives to sunglasses.

So second example is chocolate ice-cream versus ice-cream. So what is the logic here? Which one has higher price elasticity and which one has lower price elasticity? I am sure many of you have already figured out here that chocolate ice-cream is one type of ice-cream. So there are various alternatives of chocolate ice-cream. So if the price of chocolate ice-cream goes up many of the people are likely to shift to other flavours of ice-cream, that is possible.

So but if and exact substitute of ice-cream is not really available in the market. So that is a that is a broader concept of product. So ice-cream is a more broad concept of product and so the elasticity of demand of ice-cream in general is likely to be less than the elasticity of demand for a particular flavor of ice-cream. Same holds true for lot of other things.

Say imagine that say blue jeans the price elasticity of demand of blue jeans in particular is probably going to be higher than overall clothing in general or even jeans in general. So the more

specific a product is or more highly defined or narrowly defined a good is more narrowly defined a good is the higher is its price elasticity of demand.

My third example is insulin versus meal at a 5 star restaurant. So I am very sure that by now you have kind of figured out how to find out the elasticity of demand if it is going to be high or low. You know very well and you can imagine that any product if the price is going to increase or decrease how am I going to respond to it and that kind of gives you a clue about what is the price elasticity of demand.

So in this case I am sure you have figured out that insulin has a very low price elasticity of demand because it is a it is absolutely necessary for people who buy insulin. No one buys insulin for pleasure. They buy because it is a lifesaving drug. It is required for people who have diabetes. So insulin is required for them and they cannot go without insulin and if the price of insulin goes up they are not going to reduce their demand for insulin.

On the other hand, meal at a 5 star restaurant. It is a luxury commodity. It is a luxury service or product or whatever you call it. So meal at a 5 star restaurant if the price really goes up a person is going to or the people are going to cut back on visits to these restaurants his visit to the 5 star hotels for meals and so the price elasticity of demand is much higher in case of 5 star meal at 5 star restaurant.

So price elasticity is higher for luxuries than necessities. The price elasticity is higher for luxuries than necessities. Another example is petrol in the short run versus petrol in the long run. What does this mean? This means say the government increases the price of petrol suddenly. Suddenly there is a price hike in petrol. How much people are going to respond to the price hike? Is the respond response to the price hike going to be more in the near future or is it going to be more in the longer future or the distant future?

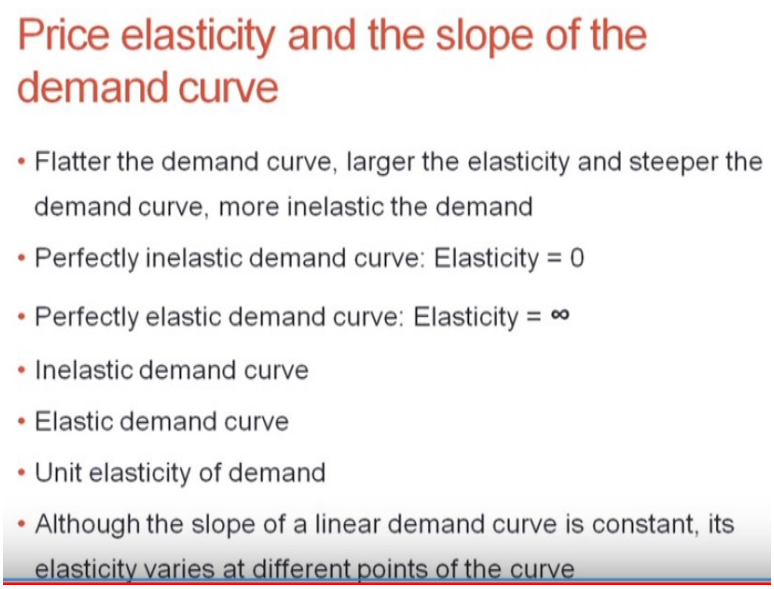
In all likelihood response to something like this is going to be slower in the short run and much more effective in the long run or much more of the the government's policy of increasing the petrol prices to probably cut down on the consumption of petrol that is going to make itself felt in the market more in the long run. Why because in the short run it is not easy for people to adjust their demand because say the demand for petrol it depends on what kind of cars they are driving. What is the way of travelling to work every day?

So all these are quite sticky situations. You cannot change your petrol car overnight because price of petrol has increased. But in the long run probably you may plan to buy a diesel car. So in

that case in the long run the economy is going to adjust to a higher price of petrol. But in the short run probably a price hike is only going to lead to more revenue for the people selling petrol than a reduction in consumption of petrol.

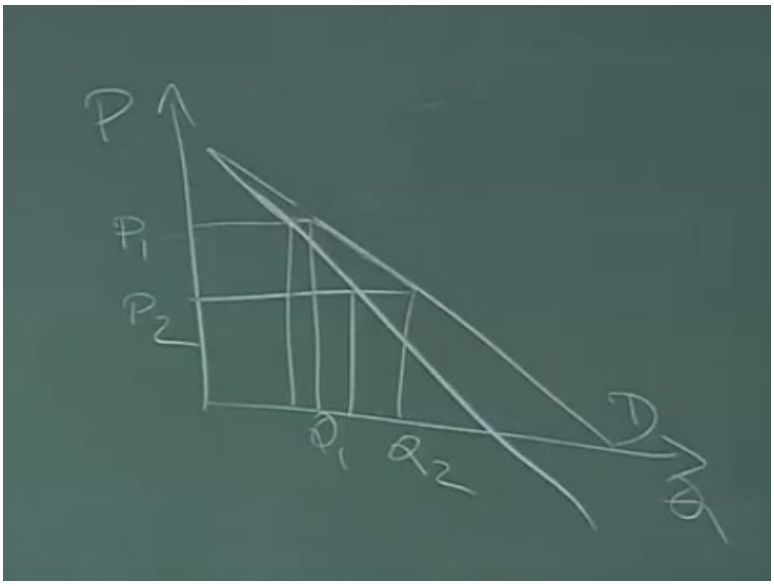
So price elasticity is higher in the long run than the short run. So in this way it is possible to imagine various situations where the elasticities would be high or low depending on the kind of determinant we are talking about, depending on the kind of situation we are talking about, depending on the nature of the product, the price ranges say for example elasticity is more if the price range is very high but if the price range is low elasticity is probably low or in other the intuition behind it is if the price is very high people are going to respond quicker to a reduction in price but if the price is too low probably people have already adjusted their consumption or further fall in price or rise in price is not going to probably change the too much of the quantity but that again depends on what kind of commodity one is selling.

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Next we are going to talk about so all this was about the price elasticity of demand etc. What we discussed in the previous slide is price elasticity of demand depending on the kind of product that one is selling in the market but every of each of these products all the products they all have a negatively sloping demand curve.

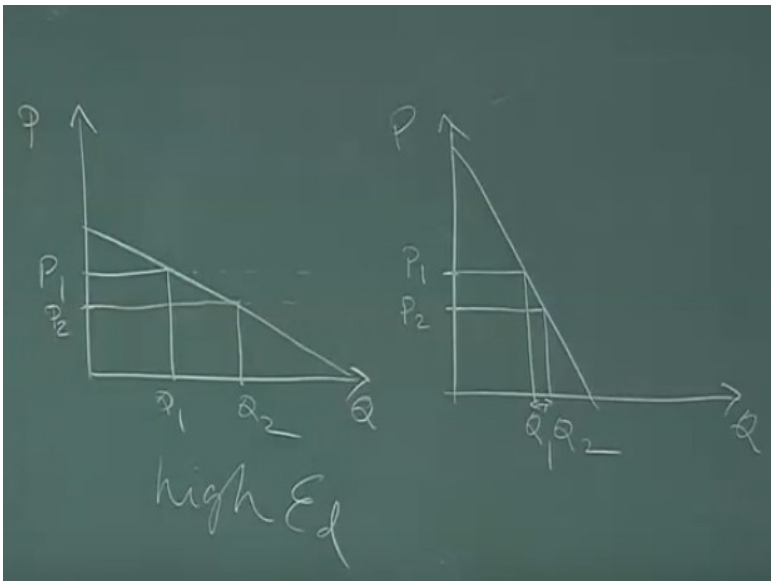
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Say all of them have a negatively sloping demand curve. Now what does this demand curve tell us about the elasticity of demand. If I draw the demand curve in a certain way is it going to tell me if the elasticity is high or low? Yes, it is. This is just any negatively sloping demand curve that I have drawn and I am going to see that obviously how much my price is changing with how much my quantity is changing with respect to price that depends on this slope of the demand curve.

So the movement from this price to this price P 1 to P 2 if there is a price fall how much my quantity is going to increase that depends on the slope of the demand curve. So if this is a steeper demand curve say ya this is a steeper demand curve so if it is a steeper demand curve so this was my initial price and this is my new quantity so the distance is lesser. So let us look at the slides. So flatter the demand curve larger the elasticity and steeper the demand curve more inelastic the demand is. Let me draw a neater picture.

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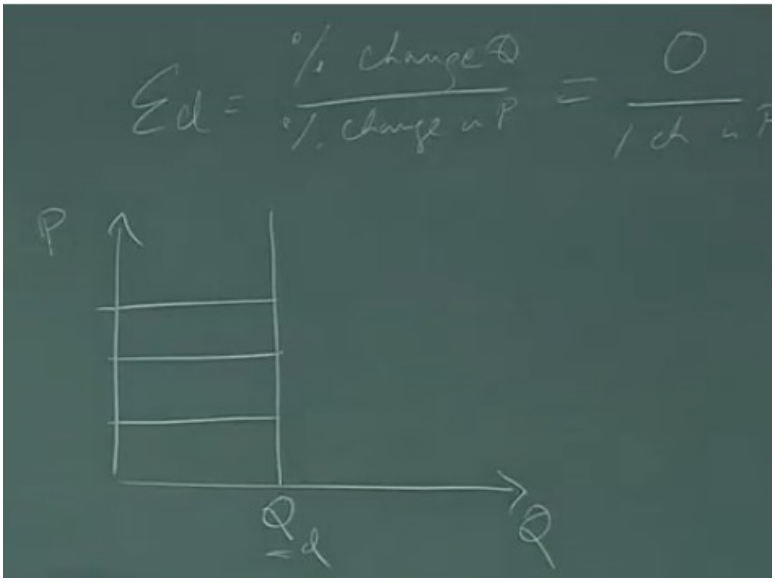
So flatter the demand curve so this is a very flat demand curve this is a very flat demand curve and this is my initial price and this is my so a little bit of fall in price is leading to a huge amount of increase in quantity so the response is so response of quantity to price is quite big. So a flat demand curve gives me a high elasticity of demand.

On the other hand I if I have a this is a very steep demand curve this is a steep demand curve and the same price change P 1, P 2. So P 1 my quantity was Q 1, P 2 my quantity is Q 2. So in case of a steep demand curve when my price falls from P 1 to P 2 my output increases. Output is going to increase. It is not that it is a inelastic demand curve so output is not responding. No, output is responding and output is increasing or not output I should not call it output this is product or the amount quantity purchased by the people by the consumer.

So the amount the quantity purchased by the consumer is increasing but by only a little bit. So the same amount of price fall is leading to a huge increase in quantity sold in the market but the same amount of price fall is leading to a very small increase in quantity sold in the market. So flatter the demand curve larger the elasticity.

Now what does a perfectly inelastic demand curve look like. What does a perfectly inelastic demand curve look like? Now a perfectly inelastic demand curve means that there is no response of quantity to prices. There is no response of quantity to change in prices.

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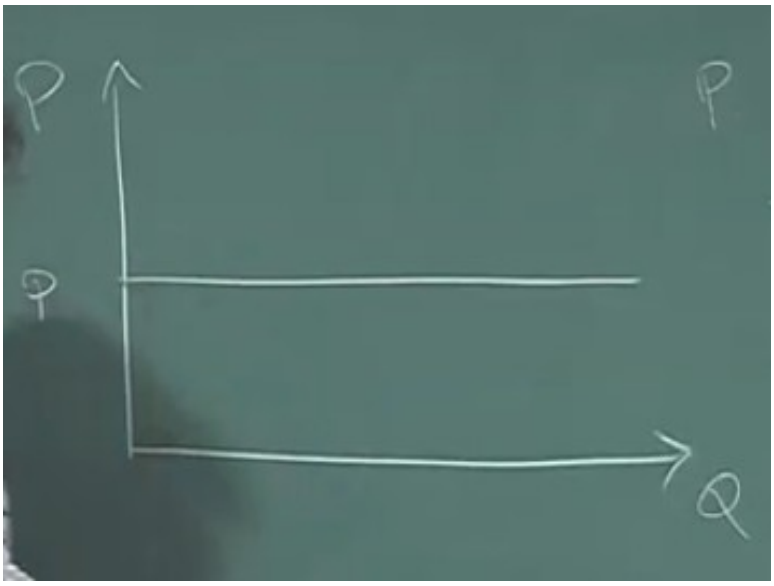
So perfectly inelastic demand curve is where perfectly inelastic demand curve basically shows that percentage change in by percentage change in P is equal to 0 by percentage change in P. So there is no change in quantity no matter what the price is. So what does the perfectly inelastic demand curve look like? It looks like this. It is a vertical demand curve.

It is a so quantity does not change. Quantity demanded does not change no matter what the prices are. So can one imagine such a demand curve in the market? Probably no. Probably there is no perfectly inelastic demand curve in the market but one example that comes to mind is say for example at some point of time when there is too much pollution in the air and there are producers or suppliers of clean fresh air.

So they are going to there is a market for air and the air that you breathe in you have to pay a price for it. So are you going to breathe more air because the price is less or are you going to breathe in less air because the price is more? No. You are going to breathe in the exactly same amount of air no matter what the price is. So this could be an example of a perfectly inelastic demand curve okay.

So next is a perfectly elastic demand curve. Perfectly elasticity demand curve means that change in quantity is infinite if there is a little bit of change in price or on the other hand that price does not change. It someone can look at it in the opposite way that no matter what the people are willing to consume price does not change in the market. So perfectly elasticity demand curve is little bit of change in price in the market your demand is going to change infinitely.

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So a perfectly elasticity demand curve is just the opposite of the perfectly inelastic demand curve and it looks like this. So it looks like this. So at this price infinite amount of quantity is sold in the market can be sold in the market and there is no price change. Or in other words if the price changes a little bit also the if the price changes little bit also and people are consuming more they completely there is an if say for example price falls then every there is the quantity increase in quantity in the market that will be demanded is going to increase infinitely and if price increases everyone is going to go out of the market so it is 0 by (()) (21:16).

So it is basically the quantity that is demanded in the market basically falls to 0 or there is infinite reduction in quantity. So what is a so before I go to the next so what is an example of this kind of demand curve?

Again this kind of demand curve is not really again not a very it is more of a theoretical concept than a real concept but we introduced a concept of perfect competition in the previous class, in the previous week and if you may remember we discussed that there are plenty of buyers and sellers or infinite number of buyers and sellers in the market and they are not price they are not price makers. They do not decide what the price is and they are not going to decide if they can increase or decrease the price and this price is basically what has been determined in the market.

So if any seller goes into a market like that a perfectly competitive market he goes there and decides that I am going to sell my product here the kind of demand curve that he is facing in a perfectly competitive market is basically a perfectly elasticity demand curve. He cannot change

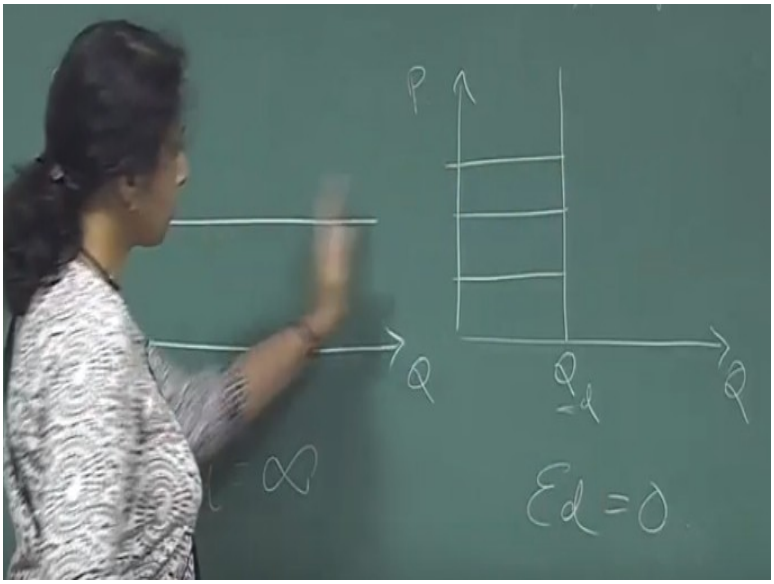
the price. He cannot change the price but he can sell as much as he wants because he sees that this is the amount that people are willing to buy at this given price.

So let me give you a little explain a little more. See imagine that a it is a huge agricultural products are basically come as close to perfectly competitive market as possible and imagine that it is a local mandi that we have seen the all the agricultural wholesale happens in those mandis and say someone is selling spinach and the rate decided in that market goes by the supplier, there are whole lot of farmers who are coming to the market to sell spinach and the demand and supply in the market that determines that this is the price of spinach say 10 Rs per bundle.

So now if any seller goes into the market it is a perfectly competitive market which means that in that mandi there are whole lot of buyers and if he offers a price which is less than P it is not possible to meet all the demand because if it is less than 10 the entire infinite amount of people in the buyers in the market will rush to him and if he sells a if he charges a higher price he is not going to be able to sell anything.

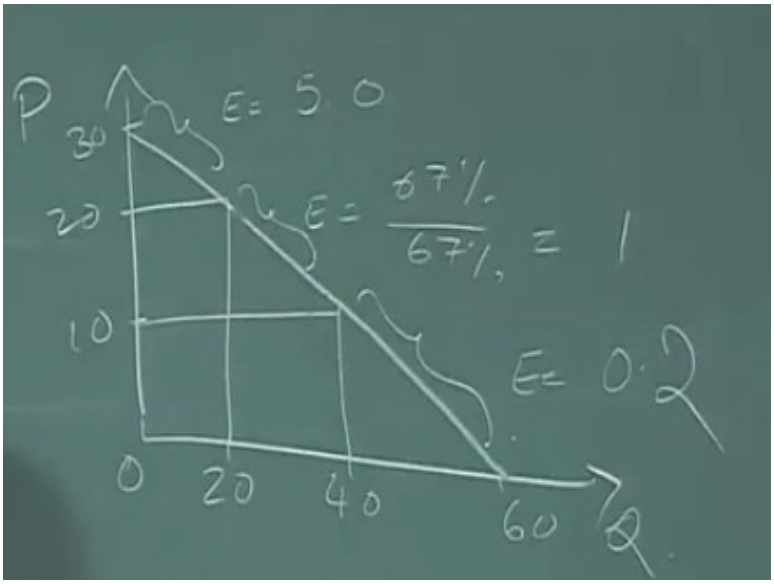
So basically he sticks to P and he can sell as much as he wants. So this is the inelastic the perfectly elasticity demand curve that a producer in a perfectly competitive market faces. We will be doing more of this when we talk about perfect competition so right now this is you can just consider this as an example and probably if it is not too clear right now it will be clearer when we talk about perfect competition in a later class.

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So elasticity in this case elasticity in this case is 0, elasticity in this case is infinite. So and we have talked about and anything in between these 2 is inelastic demand curve. Elastic demand curve we have seen that if it is a steep slope it is inelastic curve. If it is a if it has a flat slope it is a elasticity demand curve and this is, a special case of elasticity is the unit elasticity of demand which is just has one particular has a one specific case it is worth a mention.

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Say for example unit elasticity of demand is not something which is hugely important or interesting since it is a special case of elasticity of demand we are including it here. I am showing that say 20, 40, 60. Say this is the demand curve and 20, 40. So now elasticity what is the elasticity in this region when price drops from 30 to 20?

The elasticity in this region is 5.0. The elasticity in this region is 67 % by 67 % when price falls from 20 to 10 and quantity increases from 20 to 40 and this is equal to 1. So this region the is of interest because in this region elasticity is 1. Basically the same percentage change in price is going to lead to same percentage change in quantity and what is the elasticity in this region at a low price, it is 40 % by 200 % which is equal to 0.2.

So this is also illustration of something I said earlier that if you are higher up on the demand curve or where the price is too high your elasticity is probably too high. You are more sensitive to price changes and if you are in the lower part of the demand curve where price is low you are less sensitive to price changes and elasticity is low. So although the slope of a linear demand curve is constant its elasticity varies at different points of the curve.

So this is a perfect example. One should not believe that one should not mistake that since the slope of the demand curve is same every same your elasticity is going to be the same everywhere. No, that is not the case. So that ends our discussion on elasticity of demand and in the next module we are going to talk about elasticity of supply and the determinants of elasticity of supply etc. Thank you.

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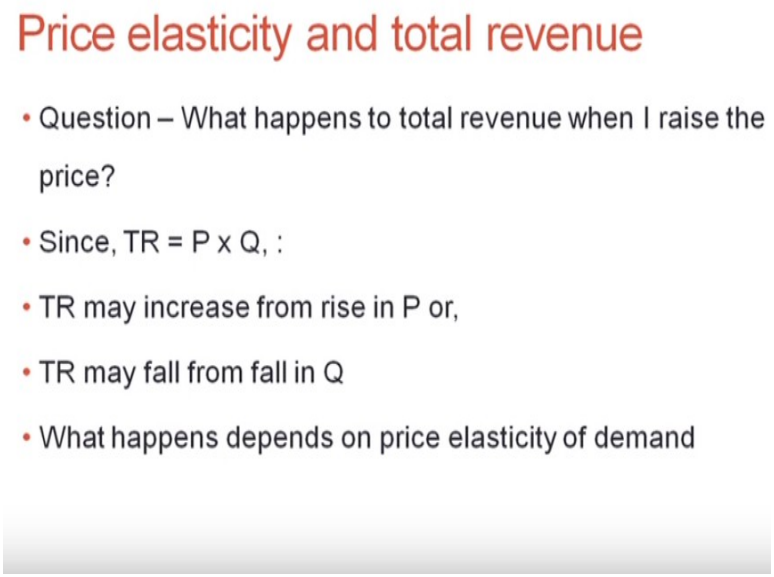
Lecture - 08

Elasticity and Revenue

Welcome back to the foundation course in managerial economics. I am Dr. Barnali Nag from VGSoM, IIT Kharagpur. We had started off with discussing about demand and supply and then we went ahead to discuss about what elasticity is and if you may remember we had started the entire discussion on elasticity with talking about what does a producer do when he is in the dilemma whether to increase his price or not because from the law of demand he knows that if he increases the price the quantity demanded is going to go down but he would like to know if the reduction in quantity is going to be compensated by the increase in price.

That is how responsive quantity demanded is to price is what he is looking for. So to continue with that discussion we are going to show here in this lecture how price elasticity is related to the revenue that a producer may expect when he is changing the price of his product.

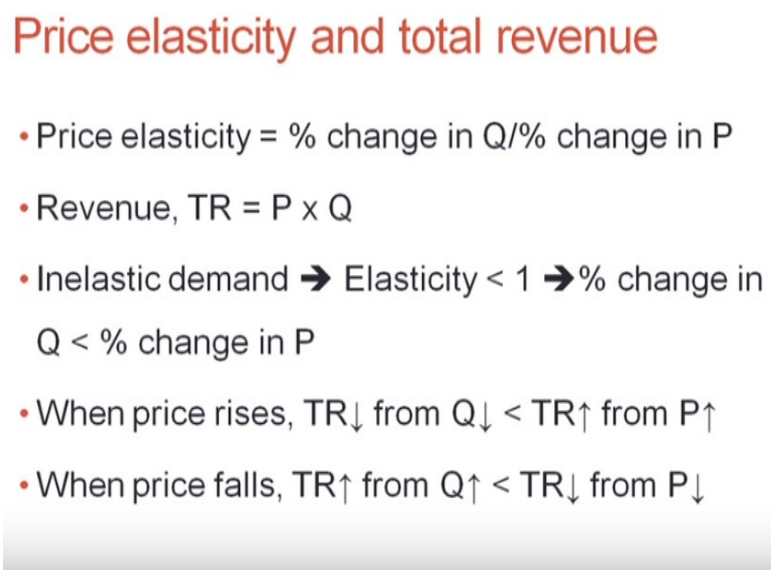
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So the question basically that we are asking is what happens to total revenue when I raise the price. Now since total revenue is equal to price multiplied by quantity total revenue may increase from rise in price or total revenue may fall from fall in quantity. So from the from this equation of total revenue equals price into quantity there are 2 components price and quantity and whenever there is any change in price quantity moves in the opposite direction. So the quantity is how much is the increase or decrease in price and related to that how much is the decrease or increase in quantity respectively the 2 cases.

So what happens depends on price elasticity of demand. Price elasticity of demand is going to tell us that what happens when price changes. Is the price affect more or the quantity affect more. So to continue with the discussion and to illustrate a little more, price elasticity and total revenue.

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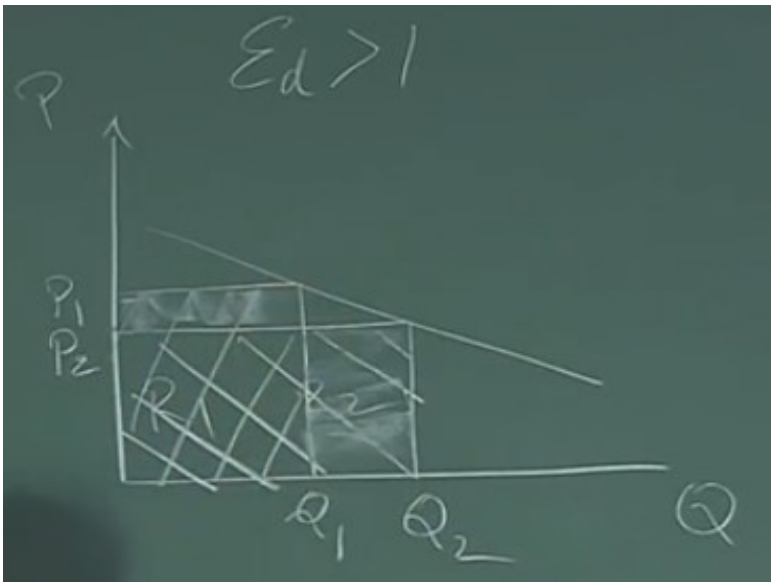


So price elasticity again going back to the definition of price elasticity is percentage change in quantity divided by the percentage change in price and revenue is P multiplied by Q. So whenever we have an inelastic demand that is elasticity is less than 1 that implies that the numerator is smaller than the denominator in the very first line where we have defined price elasticity. So elasticity less than 1 implies percentage change in quantity is less than percentage change in price.

So when price rises total revenue fall from decline in quantity and that decline is less than the total revenue the increasing impact of price increase that happens on total revenue that is when it is a inelastic demand the quantity effect or the fall in revenue from a fall in quantity is less than the increase in revenue from increase in price and similarly when price falls it is exactly the opposite effect that is revenue increase from quantity increase is actually less than revenue decrease from price decrease.

So what does that mean? That means to say that if it is an inelastic demand it makes sense to increase the price because when you are increasing the price the price increase the impact of the price increase increases total revenue and that increase is more than the fall in quantity that happens because price is risen and the fall in quantity pulls down the total revenue but that impact is less than the impact of a rise in prices to so in a nutshell it makes sense to increase the price when it is inelastic demand. Okay so let me illustrate through a diagram also. So let me illustrate through a diagram.

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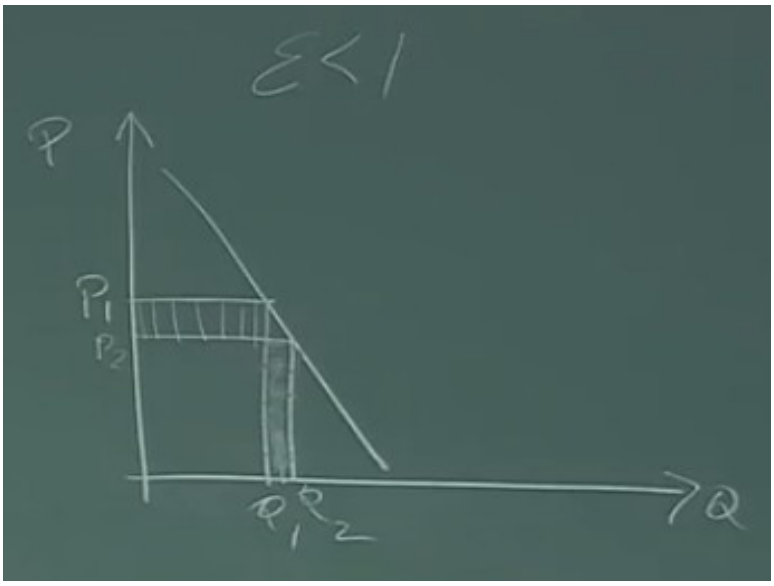


So this is the case of an elasticity demand curve. So this is a very elasticity demand curve where price elasticity is more than 1. So this is an elasticity demand curve and when price falls this is P 1 this is Q 1 and there is a fall in price the producer decides that I am going to reduce the price. So the price falls to P 2 and the quantity falls to Q 2.

Now before the price fall what is the total revenue? Total revenue is P multiplied by Q which is this area, this is the revenue 1. This is the portion that the producer was earning as revenue before there was any price change but when price fell to P 2 and quantity increased to Q 2 first of all we can see that the quantity increase here this is more than the price fall which is shown by this portion.

Also, what is the new revenue? The new revenue is this area. This is P 2 into Q 2 and this is the new revenue. So the additional amount of revenue that that the producer gets is this area and the revenue that he has foregone by reducing the prices, this area. So it looks like from the figure at least it shows that this rectangle is larger than this rectangle. So the area under this rectangle is larger than the area under this rectangle which kind of shows that if it is a flatter the slope is the more the revenue is earned when price falls, so in the case of a elasticity demand. So let me draw the case of an inelastic demand.

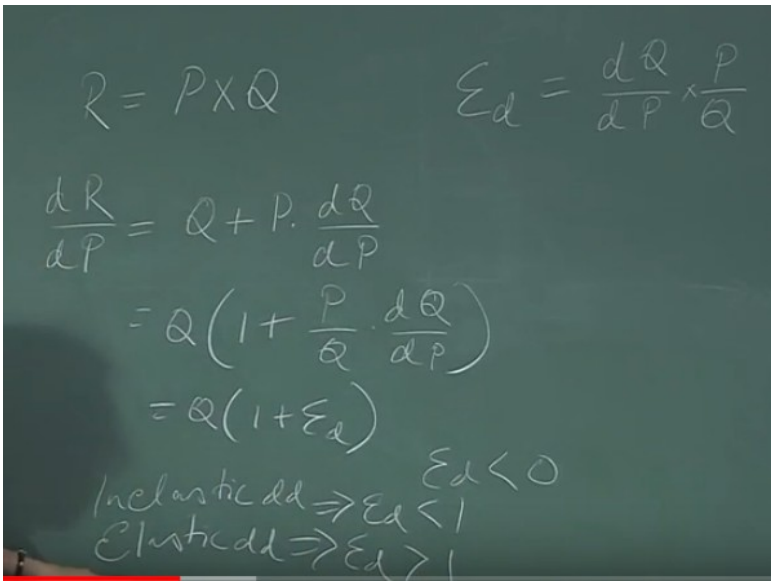
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What happens in the case of an inelastic demand? So this is P and this is a steep or elasticity is less than 1. This is inelastic demand. So in the case of inelastic demand the same price falls. This is P 1, Q 1 and price falls to price falls to P 2 and quantity increases to Q 2. Here the additional revenue that the producer gets is this very narrow rectangle and the amount that he foregoes is this upper rectangle.

So through the figures it is kind of intuitively it can be shown or graphically that the amount of revenue gained when there is a price fall through increase in revenue is much lesser in case of a inelastic demand. So this was illustration through diagram. Let me also complete the discussion by using a little bit of calculus.

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So using calculus now revenue as we wrote earlier is P multiplied by Q. Now we would like to see what happens to revenue when there is price change. So this we can write as and also let us write the definition of elasticity of demand in terms in mathematical form. This is dQ dP multiplied by P by Q.

That is change in quantity divided by percentage change in quantity divided by percentage change in price. This is elasticity of demand. So whenever there is price change then dR dP can be written as Q + P into dQ dP or taking the Q out it can be written as 1 + P by Q into dQ dP and this portion is nothing but the elasticity of demand so this is equal to 1 plus elasticity of demand.

So it is very clear from here what happens to changes change in revenue is more or less than the existing revenue is very clear from this expression. So when it is an inelastic demand and price elasticity of demand being less than curve price elasticity of demand this is always negative. So elasticity of demand is always negative and when this expression when it is an inelastic demand there is an increase in quantity.

This is a positive expression when inelastic demand implies elasticity of demand is less than 1 and elasticity demand implies elasticity of demand is more than 1. So in case of elastic demand when elasticity is more than 1 this expression becomes negative or in other words when there is a price change the revenue change happens in the opposite direction.

So what is the intuition behind it? The intuition behind it all whatever we showed graphically, through calculus, and through the by reasoning, the intuition behind this is basically that if it is a inelastic demand, say for example we took the example of say medicine, lifesaving medicine. So it is an it has a very inelastic demand. So what happens when price changes?

When price increases people are not going to cut back on consumption too much because it is a necessity. They have to buy it in any case. So that is the reason when price increases the quantity reduction or cut in quantity demand is not so much so the revenue increases in case of an inelastic demand when price increases and the opposite thing happens when price falls.

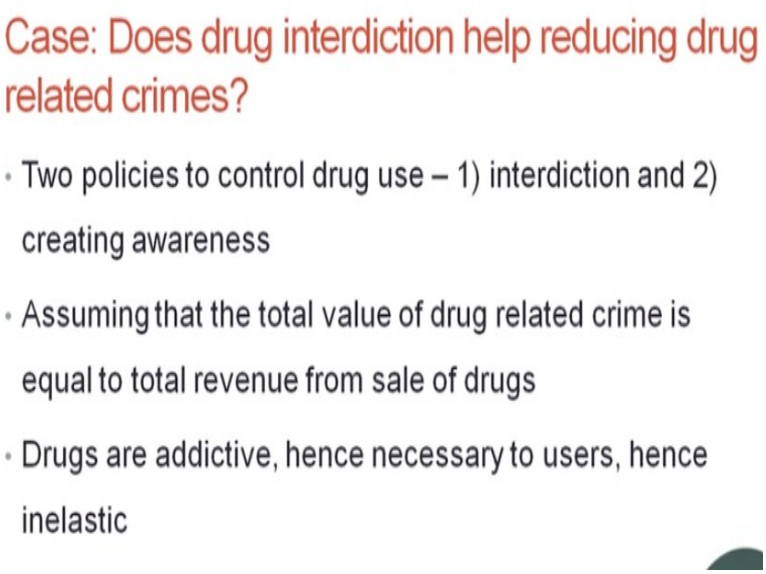
When price falls in normal circumstance or the law of demand says that when price falls consumption is going to increase but here what is happening in case of inelastic demand which is a necessity necessary good and in most cases necessary goods are something which you consume in a certain given amount. So in most of the circumstance necessary goods it is an inelastic demand where if price falls you are not going to increase your consumption too much.

So in that case the quantity effect or the effect of an increase in quantity because price has fallen that does not happen so much and so it does not make much sense to reduce the price and exactly the opposite thing happens in case of an elasticity demand.

Elasticity demand we took the example of luxury goods say and in case of luxury goods you can imagine if the price falls you would like to take the advantage and consume more but if there is a price increase it is not a necessary good and you can just as well give it up for some time or cut down your consumption or postpone your consumption. It is much more flexible and because it is flexible it is an elasticity demand.

So this was our discussion on elasticity inelastic demand and what that does to revenue and let us end the discussion with a typical case. It is it is this example is given in one of the text books that I have recommended, Principles of Economics by Mankiw and I have borrowed this example from there and this case says does drug interdiction help reducing drug related crimes.

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So you there are basically you can imagine 2 policies to control drug use. Say for example you would like to obviously you would strongly like to reduce the use of drugs among youngsters, among people, among your citizens and what can you do for that? One is you could use a very strict method of interdiction very severe punishments if you are caught with drugs and second is you could create awareness.

So we are going to take this example of elasticity, supply, demand whatever we have learnt so far to see what happens in each of these cases. Now slightly to illustrate the example we also assume that the total value of drug related crime is equal to total revenue from sale of drugs. So we are trying to quantify that what is the impact of each of these policies and how do we quantify?

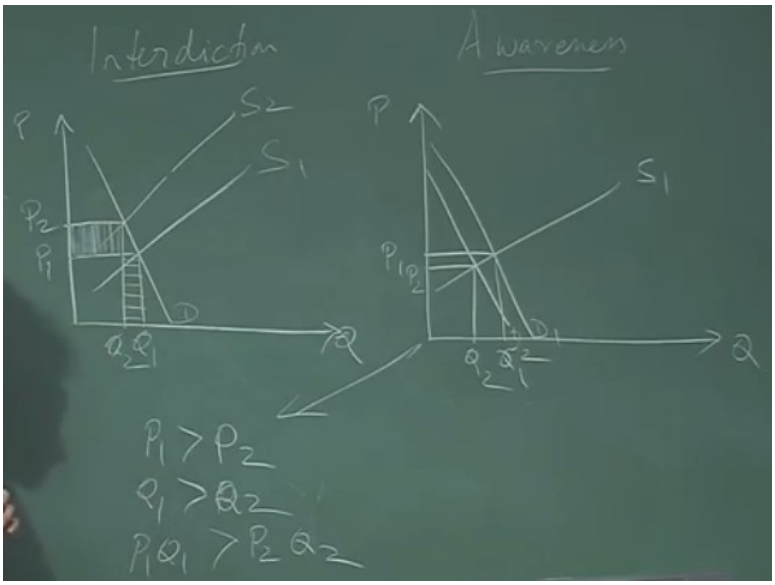
First we quantify if we are talking about drug related crimes what is the measure of that maybe we the measure we can take assume that the measure is the amount of drug that is sold or the amount of expenditure that people do on drugs by buying drugs or the amount of revenue that the drug vendors or peddlers they earn by selling the drugs and drugs are addictive hence they are necessary to users.

So they are addictive so one might assume that the consumers of drug, the people who consume the drugs are not so much price sensitive. That is no matter what the price of the drug is they would nevertheless like to buy the drugs. So the drugs we can assume that it has a very inelastic demand curve. So now let us see what happens in each of the 2 cases.

Now if the in the case of the first policy of interdiction as we discussed in the earlier classes that whenever any disturbance in the system is introduced or proposed how do we figure out that which what happens in the market. To figure that out what was the first step? The first step was that we have to find out which of the two demand and supply graphs does this impact. So here there is interdiction or there is a strong message to the people that there is severe punishment if you are caught selling drugs.

So in that case what happens? Which of the two demand and supply which of the two curves is affected. Obviously the supply curve is affected. So if there is interdiction there is a supply curve is affected. There is fear among the sellers of drugs and it probably is successful in cutting down the supply of drugs that less people are willing to sell drugs or maybe the they are more they have they are selling the same drugs probably but for them their cost has gone up. They have to be even more careful and so the price for that the cost at which they can sell the drugs that goes up and when cost goes up what happens?

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So first let us draw the demand curve for drugs. So it is a very steep demand curve where inelasticity it is a inelasticity demand curve and elasticity of demand is less than 1 and this is the demand curve for drugs and say this was the supply curve. This was the initial supply curve and

this was the initial price and quantity. This was the equilibrium initially. Now there is interdiction. So when interdiction happens what happens? The supply curve shifts to the left.

So it is costlier for the suppliers to supply the drugs. So when supply curve has shifted to the left price increases and quantity falls, quantity falls. So yes the policy is successful in reducing the amount of drugs sold to the community. So the amount falls from Q 1 to Q 2 but we have measured the drug related crime as the total revenue or the total amount of drug trading that is happening in the economy.

So we can see clearly from here that this is the revenue that is foregoing that the drug sellers are foregoing and this is the additional revenue that they are getting because drug has become more expensive and from here we can see clearly see that this is a bigger rectangle. This is a bigger rectangle and you can probably do this yourself and try to see like try to draw the demand curve steep or flat and try to do this and you will clearly see that the revenue actually increases in this case.

So the revenue actually increases in this case or in other words here the price increase that happens because supply has fallen actually leads to lesser fall in quantity. The fall in quantity is not so much because it is inelastic demand and as a result the amount of drug related crimes the way we have measured the drug related crimes that actually goes up and this is the example of interdiction.

Now what happens in the case of awareness if you are creating awareness what happens in that case? There also you have a steep demand curve D 1 and the supply curve S 1. Now in case of awareness which of the 2 curves is affected. In case of awareness where are you creating the awareness. You are basically creating the awareness among people who use the drugs.

So when awareness goes up the demand for drugs goes down. So when demand for drugs goes down the demand curve shifts to the left. So this is the new demand curve and in this case the earlier equilibrium was P 1 Q 1 and the new equilibrium is P 2 Q 2.

So not only has the quantity demanded fallen, the price has also fallen because demand has fallen and in total the PQ has fallen PQ is much less than the initial PQ. So P 1 here so P 2 is less than P 1. Q 2 is less than Q 1 and P 1 Q 1 is much more than P 2 Q 2 which shows that in case of

awareness when you create awareness let me write it here maybe the it is camera is this is blocking the camera and so in the second case in case of the awareness what is happening is P 1 is more than P 2. Q 1 is more than Q 2 and P 1 Q 1 is more than P 2 Q 2.

So the new revenue is much less than the earlier revenue. So this basically shows that the two this is a very interesting example of how elasticity can be applied and how the shift in demand and supply curves the discussion that we started with in the first case that can be illustrated through this kind of example.

So this example basically shows that if it is a inelastic demand curve and there is the so a change in the supply as against a change in demand actually creates less impact in case of a inelastic demand curve and to end the discussion another point which is worth noting is when the supply change changes when supply changes movement is along the demand curve movement is along the demand curve and when demand changes movement is along the supply curve which means to say that when supply changes the consumers feel the impact of change in prices and they respond along the demand curve and when demand changes or demand shifts the suppliers feel the price change and they respond to the price change and they do that against the or along the supply curve.

So that ends our discussion on elasticity of demand and how it can be used to explain changes in revenue and when a when a producer decides whether to change the price or not the producer needs to understand what would be the impact of a price change on the total revenue that he can expect because he knows the producer knows that the producer knows the law of demand and the producer knows that there is a negatively sloping demand curve that he is facing and whenever there is a change in price that is going to reduce the that is going to change the quantity as well.

So that ends our discussion. We are going to talk about supply in the, elasticity of supply in the next lecture. Thank you.

Foundation Course in Managerial Economics

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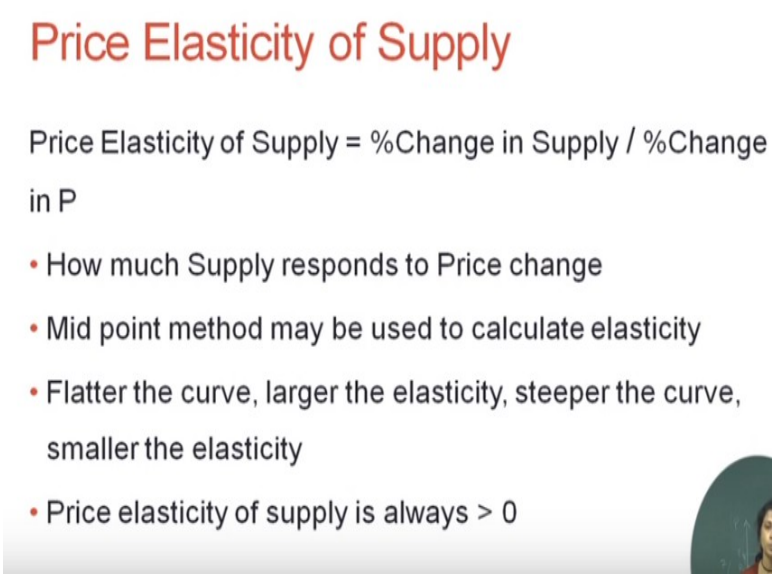
Lecture - 09

Elasticity of Supply

Welcome back to the discussion on elasticity of demand and supply. In the previous module we ended our discussion on elasticity, price elasticity of demand and we took an example of how to illustrate how a producer decides from elasticity of demand whether to changes his whether to increase or decrease his prices or not or rather to say it is not that the producer decides to change his price or not depending on elasticity it is not that a producer who has varying elastic demand he is never going to increase his price. It is not that or he is never going to reduce his price, it is not that.

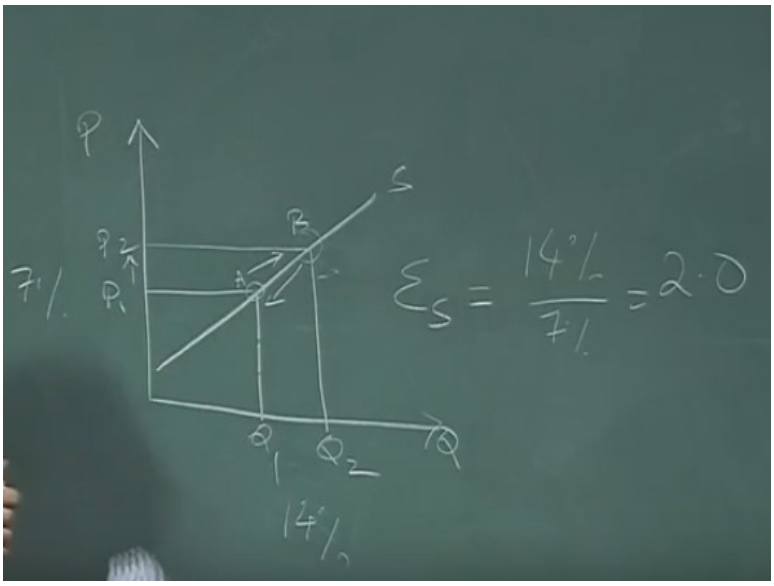
The producer's decision on whether to change the price is not solely dependent on elasticity of demand. It is dependent on a lot of things which we are going to discuss later when we talk about cost or we talk about producer theory, cost functions etc. and we are going to see that the producer's decision of how much to produce, how much to charge depends on a lot of other things than solely on the demand that he is facing but nevertheless it is the what is the impact of a price change on revenue is an important information which goes into the decision making process of a producer and this is what we discussed in the previous module and to continue with the discussion today we are going to talk about price elasticity of supply. We are going to talk about price elasticity of supply.

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And price elasticity of supply as many of you might must have already figured out would be the percentage change in supply divided by percentage change in price. So this is the percentage change in supply divided by percentage change in price or in other words how much supply responds to any price change, how much supply responds to any price change and is giving us our elasticity price elasticity of supply and like we discussed in the previous in while discussing the elasticity of demand we discussed that we talked about the midpoint method and the same midpoint method can be used to calculate elasticity of supply as well so that, let me illustrate.

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So if this is the supply curve and say this is the initial sorry this is the initial price quantity combination at which the supplier was operating and now the price changes. Now the price changes or price increases to P 2. Say price increase to P 2 and quantity also increases to Q 2; why, it follows the law of supply. By the law of supply whenever price increases it makes sense for the supplier to increase the amount of output that he can supply in the market.

So when price increases quantity also increases here. Say for example quantity increases by say 14 % when price had increased by say 7 %. So what is the elasticity of supply? So in this case elasticity of supply is equal to 14 % divided by 7 % is equal to 2 and needless to say elasticity of supply is always positive why because the slope of the supply function is positive so elasticity of supply, price elasticity of supply is always positive.

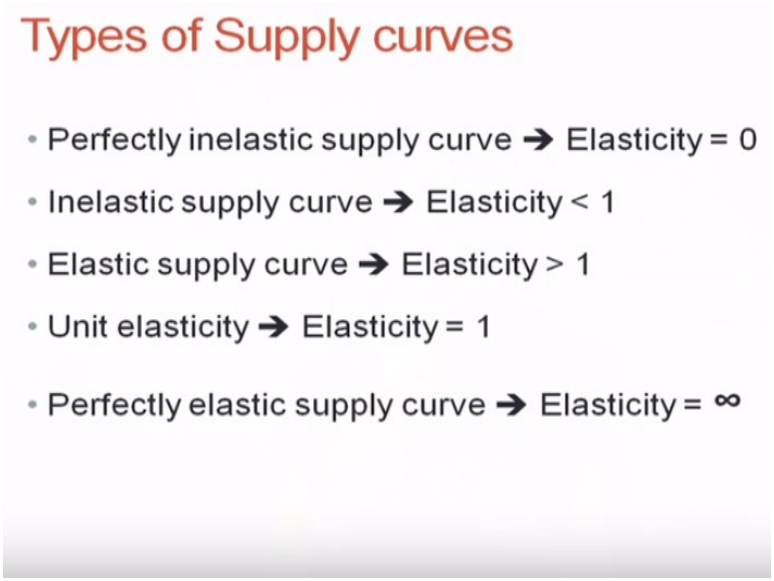
So this is price elasticity of supply and as we discussed in the case of demand also say if you are moving from point A to B or if you are moving from the point B to A if you calculate the elasticity at this point you are going to get a different number and if you calculate the elasticity from this point you are going to get a different number and to resolve that problem what we do is we use the midpoint method where we basically take the change in price divided by the average price of it at the 2 points and on the numerator we take the change in quantity divided by the average of the quantities at the 2 points and that is the midpoint method and that midpoint method is also used in case to calculate the price elasticity of supply.

So again flatter the curve, larger the elasticity; steeper the curve, smaller the elasticity. So similar to the demand the price elasticity of the demand that we have become quite conversant with so the similarly in case of the supply curve also if it is a flat supply curve the price the response of quantity to price is very high and if it is a steep supply curve the response of the quantity response against price is very small.

So that is the so that is so you have an elasticity supply curve if it is very or in other words what is the what is what is the intuition or what do we mean by a flat supply curve. What we mean by a flat supply curve is it is very or a highly elasticity supply curve is basically the it is the suppliers are highly responsive to price changes. If the price increases a little bit, the supplier can change the quantity supplied by a big amount and if in case of an inelastic supply curve the supplier is basically not too responsive to price changes.

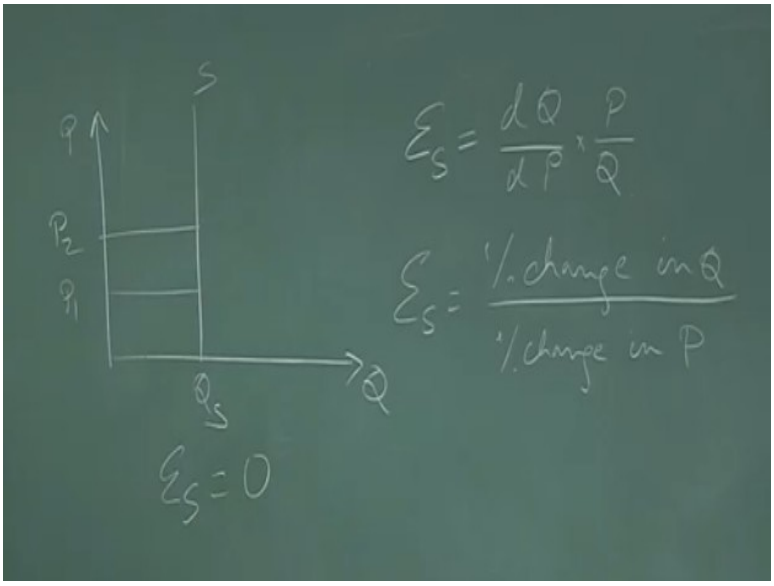
That could be because of various reasons. We are going to talk about some of them in the later slides. Basically there is a constraint or a rigidity that the supplier faces because of when the price changes a little bit the supplier is unable to change the quantity much and in that case the supplier faces an or the supplier has a very steep supply curve. And as I already said price elasticity of supply is always positive.

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Now similar to the similar to the demand curves you have different types of supply curves also and accordingly with different kinds of supply curves you have different elasticities also elasticities of supply because depending on what the slope of the what kind of slope the supply curve has accordingly the elasticity also changes of the supply curve.

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So elasticity of supply is also dQ dP into P by Q. So elasticity of supply is also defined like this or elasticity of supply is percentage change in Q divided by percentage change in P. So and in case of elasticity of supply this is the slope of the supply curve and this is positive so elasticity of supply is always positive.

Now depending on different types of supply curves you have different kinds of elasticities that will be shown by the supply curve, say perfectly inelastic supply curve. So what do we mean if you may remember we had discussed about perfectly inelastic demand curve and what did the curve look like?

In case of a perfectly inelastic demand curve the curve was a vertical line so which meant that quantity demanded was not changing no matter what the price was. Similarly, in this case where it is a perfectly inelastic supply curve elasticity is equal to 0 and I draw it here. So this is the supply curve, this is the quantity supplied and no matter what the price is the quantity supplied there is no change in quantity supplied. So there is no change in quantity supplied. So change in quantity supplied is always 0. So the inelastic supply curve always has an elasticity of elasticity of supply is equal to 0, elasticity of supply is equal to 0.

So you might like to know you might ask that what is a do we have any example of a perfectly inelasticity supply curve? Do we have an any example of perfectly inelastic supply curve where basically we are saying that the supply does not change? Supply is not changing no matter what

the price is. So one example that one could think of is say for example I have a very precious rare, say I have a painting of Picasso which I have inherited from somewhere from my great great grandparents and it is there with me and I would like to sell it.

Now I can no matter what the price of that painting is in the market, I cannot change the supply of that painting. There is there is only 1 painting or maybe 2 paintings that I have and no matter what the price is if by due to some strange reason no one is willing to pay anything and the price is 0 or if they pay me the sky then also I cannot change the supply from what I have right now. So that is a perfectly inelastic supply curve.

And so a little more realistic example realistic supply curve that you have is you have the inelastic supply curve where elasticity is less than 1 and you have the elasticity supply curve where elasticity is more than 1. So you will be able to find lot of examples where elasticity is less than 1, where elasticity is more than 1.

So elasticity less than 1 basically means that say for example there is a increase in price, you would like to increase the quantity but you have not been able to increase the quantity as much. So basically the amount by which you increase the quantity is less than the amount by which price increases and that can happen due to lot of reasons.

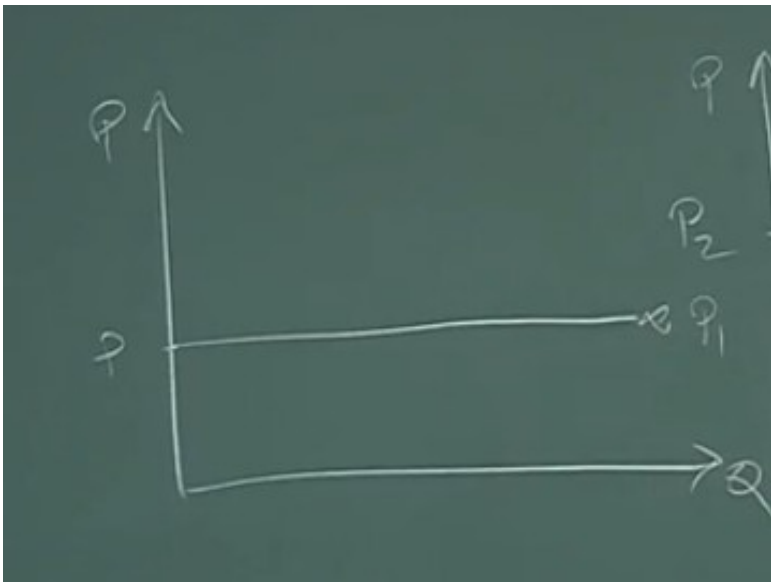
Say for example you have some say your supply depends on some natural resources that you use say for example you have a land and you produce say you are already producing rice there and the suddenly there is an increase in demand and you would like to increase and there is an increase in price in the market but you have only that much land and there is only probably if you find a little more squeeze in some more place to sow some more seeds and produce some more grains, probably you will be able to produce a little more.

But you may not be able to produce as much as you would like to given the increase in price. So that is kind of inelastic supply curve and on the other hand you have elasticity supply curve where basically you have a lot of capacity and it is possible to increase when the price increases and you may be you are operating at below capacity earlier and now because of increase in demand the prices have increased in the market and there you have enough capacity to increase

your production so that case it is a very elasticity supply curve. You have an it is a elasticity supply curve that you have.

Unit elasticity as I said in case of demand elasticity also it is a very special case where and it is although it kind of looks nice that it is unit elasticity where basically the quantity supplied increases exactly by the amount by which price increases it is actually not so much of interest it is only a special case of elasticity.

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And another extreme is you have perfectly elasticity supply curve that is perfectly elasticity supply curve which again if you may remember perfectly elasticity demand curve perfectly elasticity supply curve also looks similar where you have perfectly elasticity supply curve means if this is the price existing price, at this price supply that there is in the market is infinite. Basically at this price the supply is infinite and if the price changes a little bit the quantity supplied comes to 0. So this is the price at which supply is infinite.

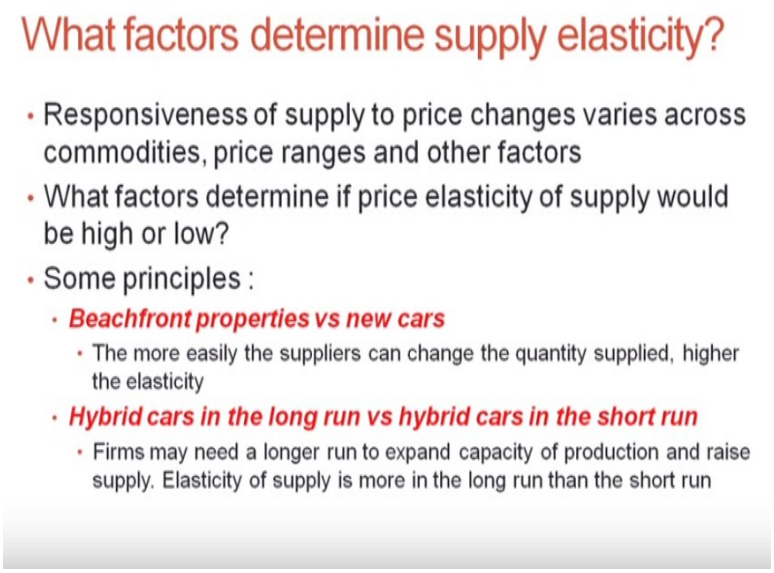
Again example again to like a perfectly inelastic supply curve perfectly inelasticity and perfectly elasticity demand curve perfectly elasticity supply curve also is not very realistic but one might imagine that perfectly elasticity supply curve is what the sellers might have in a perfectly competitive market. Say there is a perfectly competitive market and there are infinite number of

sellers like everyone is producing say grains and there is a so if everyone is producing grains the price falls to whatever is the minimum cost of production.

So no one is going to charge a price which is below that because price has is already at minimum cost of production so that is the price that everyone is charging and if anyone tries to reduce the price below that then one the in that case every everyone is going to rush to him so he is going to the consumers are going to rush to him so he has to produce more which is not possible and finally price comes back here but that is beside the point. The point is the price is not going to change from here.

Price is not going to change from here because if the price increases again the infinite amount of supply of the quantity supplied moves to here like there is again an infinite change in supply in the market. Like if the price increases so in the denominator there is a finite increase in price but in the numerator there the response will be an infinite increase in quantity because everyone is going to now whoever were selling earlier at the smaller lower price they all of them are going to now sell at this higher price. So basically at every price the quantity supplied is infinite and that is a perfectly elasticity supply curve and so the elasticity of supply is also infinite here.

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Now similar to the discussion that we had in case of demand we said that there are various factors which tell you how the demand is going to behave or how responsive is demand to price

changes. Similarly, in case of supply also there are various factors which determine whether supply is going to be elastic or is it going to be inelastic or how responsive the supplier is would depend on a lot of factors. So they are the determinants of supply elasticity.

So responsiveness of supply to price changes varies across commodities, price ranges, and other factors. So what factors determine if price elasticity of supply would be high or low? So similar to the earlier case here also we are going to use some we are going to see a few principles basically guide the supply of elasticity and let us take an example.

Say the example of beachfront properties versus new cars. So this is a market for beachfront properties. You have houses on the beach and it is a market for houses which are available on the beach, people would like to buy houses on the beach and there is a separate market which is for cars.

Now which one do you think has higher supply elasticity or in other words if the price increases in both the markets in both the markets say for example there is a boom in the economy and everyone lot of people are expecting to be rich or already have got richer and they are looking for nice houses in front of the on the sea beach and they are looking for cars.

In which of the 2 markets do you think supply is more elastic or in which of the 2 markets is it more easy to respond to the price change. That is the question that we are asking and as you may have already figured out, on the beachfront properties case there is only a limit to which the suppliers can increase the supply of beachfront properties because there is only so much of beachfront land available in the market where new property can be or new houses can be built and sold in the market.

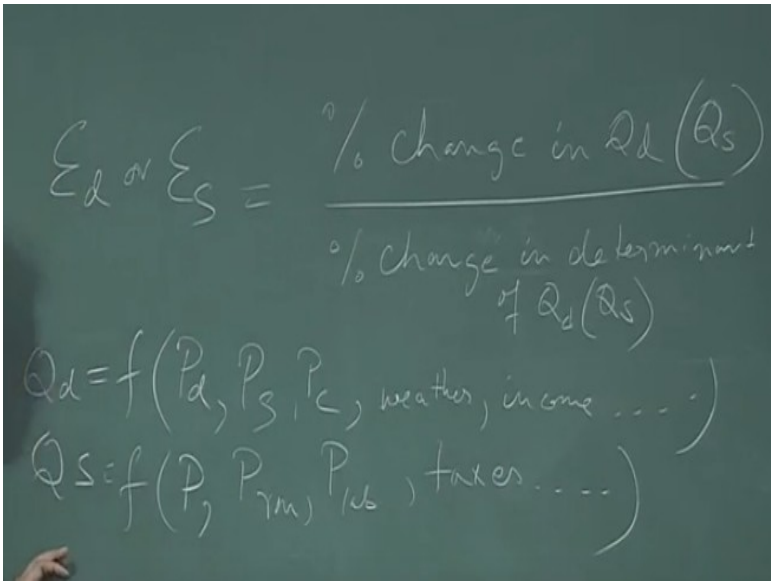
On the other hand, it is probably easier to just expand or you have built a few more plants few more factories and design a few more cars and produce new cars and sell in the market when the demand for cars go goes up in the market. So beachfront properties is something where the supplier is constrained by one of the resources, very integral or important resource which is beachfront land and but in case of new cars and in case of cars the supplier is not so much constrained by resources. So the more easily the supplier can change the quantity supplied, higher the elasticity.

Take another example which is hybrid cars in the long run versus hybrid cars in the short run. Say the economy has become very environment conscious say everyone has become very conscious and would like to have a cleaner have cleaner air cleaner environment and suddenly the demand for hybrid cars which uses less of fossil fuels the demand for these cars go up in the economy.

In that case the price of the so the demand has gone up so the price of hybrid cars has gone up and the suppliers would like to supply a lot of hybrid cars in the market. So are they able to match demand in the short run as much as they can match the demand in the long run. What do you think? So in the long in the short run the car manufacturers are only are constrained by the amount of capacity they have already created to produce the cars. But in the longer run they can set up more factories, they can expand their capacity take more loans to build new factories.

So they need some time to increase the capacity and increase the amount of cars that the economy is demanding. So they are more flexible in the longer run in production than they are in the short run in production and so their responsiveness to price increase in the short run is much less than their responsive to price increase in the longer run. So firms may need a longer run to expand capacity of production and raise supply. Elasticity of supply is more in the long run than in the short run.

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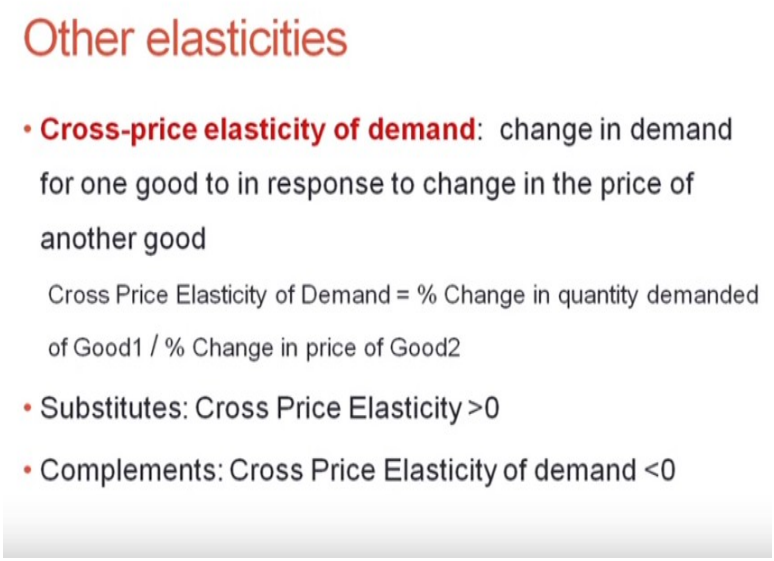


So that was about price elasticity of supply. That was about price elasticity of supply but as I said during the beginning of our lecture on elasticity we said that you can define elasticity of demand or supply as percentage elasticity of demand or elasticity of supply can be defined as percentage change in quantity demanded or quantity supplied divided by percentage change in any determinant of quantity demanded or quantity supplied.

So this is the very general definition of elasticity and this elasticity definition can also be used to find out what happens to or what is the responsiveness of quantity demanded or quantity supplied to change in any of the determinants. If you may remember we had said that quantity demanded depends on a lot of things. It is a function of the price. It is a function of lot of other goods say for example substitute goods or complementary goods or say whether income, lot of things.

Similarly, quantity supplied is a function of price, price of raw materials, price of labour, and say taxes, and lot of other things. So all these determinants any change in any of these determinants can be put in the denominator and the responsiveness can be put in the numerator and the elasticity calculated. So this can be done.

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Now so from there we are going to discuss a couple of elasticities which are very important. First is the cross-price elasticity of demand. Now what is cross-price elasticity of demand? Now this is change in demand for one good in response to change in the price of another good. So this is change in demand for one good in response to change in price of another good. So this is called cross-price elasticity of demand which is percentage change in quantity demanded of good 1 divided by percentage change in price of good 2.

So basically we are trying to see here if price of good A changes what is the impact on the quantity demanded of a good B. So how is that relevant, what happens? Why is price of some other good relevant to the quantity demanded of another good? That can happen in 2 situations, in case of substitutes.

So substitutes we have already said substitute is what substitute is basically 2 goods are substitutes when they are kind of replaceable. Say for example tea and coffee. If the price of coffee goes up you are probably going to switch to tea. Say in a railway station everyone the all of a sudden the coffee vendor decides that I am going to increase the price of coffee probably lot of people are going to shift to the tea vendors that are available around.

So there the tea and coffee are substitutes. So the cross-price elasticity or if you have the change in quantity on the numerator and change in price of change in quantity of tea consumed in response to change in price of coffee in the denominator you are going to see that the cross-price elasticity is always greater than 0 since it is a substitute.

So the law of demand behaves in an opposite way that if the price of the substitute good goes down, you are going to reduce the quantity demanded of this particular good. So if the price of coffee goes down you are going to consume less of tea and move to coffee. So the in response to so the price and quantity of substitute goods they move in the same direction. So that is the reason that substitute goods have cross-price elasticity greater than 0.

And another example is the complements. Complement is you cannot use one good without the other. Say for example in case of fountain pens is the pen and the ink, they are complements. You cannot use the pen without the ink and ink is of no use if you do not have the pen, like computers and softwares. They are all complement goods.

So in case of the complement goods if the price of the of one say the price of fountain pen goes up the demand for ink also goes down because when the price of fountain pen goes up the demand for fountain pen goes down and so the demand for ink also goes down. So in case of complements cross-price elasticity of demand is less than 0. The quantity demanded and the price they move in opposite direction just like the case of the price elasticity of demand of a single good.

That ends our discussion on elasticity. We have talked about price elasticity of demand. We have talked supply elasticity of demand. We have looked at different determinants of elasticity and we have looked at different values of elasticity ranging from 0 to infinity and we have taken some example to illustrate and finally we have talked about a more general kind of elasticities other elasticities and we have talked about the cross-price elasticity where basically we have talked about the response of one good in to against the price range in some other good. In the next class we are going to start our discussion on producer theory and cost. Thank you.

Foundation Course in Managerial Economics

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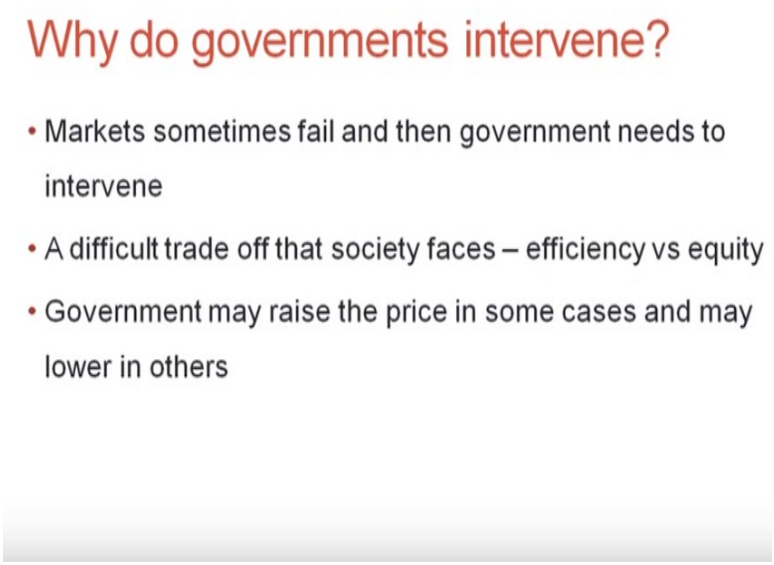
Indian Institute of Technology-Kharagpur

Lecture - 10

Demand Supply & Government Policies

Welcome back to the foundation course in managerial economics. We started with the demand and supply framework and we have been discussing about elasticity, about demand supply framework and in the very beginning of this section on demand and supply I said that this demand supply framework is a very useful tool to understand lot of changes or disruptions or disturbances in the economy and any external impact or external shock that comes to the system of the equilibrium that can be understood through the supply demand framework and in this module, today's module we are going to see or we are going to apply the supply demand framework to understand certain policies of the government. So we are going to utilize couple of modules to understand different policies of the government.

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Now in the very first lecture when we discussed about the various principles of economics you might remember that there was in one of the principles was that market is usually a good way of allocating resources. Market usually allocates the resources most efficiently in the economy and then we went on to say that there are however exceptions.

Exceptions where markets fail and we said that although market is usually a good way of doing things where basically the supply and demand they match in the market and we get equilibrium price, we get equilibrium quantity and in the process basically the allocation the very scarce resources available in the economy they get allocated into producing what is demanded most in the market and into producing what the suppliers would like to supply in the market or have the ability to supply in the market most efficiently.

So this is what we discussed and then we went on to say that markets are usually a good way but not always and when are they not the best choice to allocate resources in situations where the market fails. That is that there could be situations where the market outcome is not desirable in the economy.

Say for example say India is a very hugely populated economy with lots of people looking for jobs and very high level of poverty; many people are poor they are hungry and they are basically looking for jobs and there are not enough jobs around. So in such a situation one might imagine that many of the people would like to work at a very low amount of wage.

Now what happens? So in the framework of demand and supply the market is going to clear at a very low level of wage where many people would be willing to work and obviously the producers who are going to employ these people will be very happy to give work to these people at a very low wage.

But then the question is, is this wage acceptable? Is this wage acceptable to society, to people in general, to the government because this wage could be so low that it may not be sufficient for even to maintain a subsistence level of consumption for the people. So in these situations the market clearance that is happening, the price quantity equilibrium that is reaching through the demand supply framework of the market is not a outcome which is desirable to us.

So in such a situation the government may intervene and we also discussed about other situations of market failure where we said that there could be externalities say for example there could be some pollution happening because of some production process which is not directly harming the producer or the consumer.

So neither of the parties is actually paying any cost or assigning any cost to that negative impact of the production process but maybe that production is generally in general it is harming the environment and the other people who are neither directly involved in the production or purchase of the goods they are not happy about it. So what happens? Who takes care of that cost that the economy in general is bearing because the production process is there.

So this is again a situation of externality where basically the government steps in and government imposes a tax on the producer or the consumer and says that see this basically the principle behind the tax is that the government is trying to impose a cost on the producer and the consumer of basically engaging in a production process which is harming the other people.

So in a nutshell government basically intervenes or is supposed to intervene where the market is failing and government usually is supposed to do a good job out of it and take care of the concerns that is happening because of the market outcome if it is left to itself. So as I said market sometimes fail and then government needs to intervene and this also comes from a very difficult trade off that the society faces.

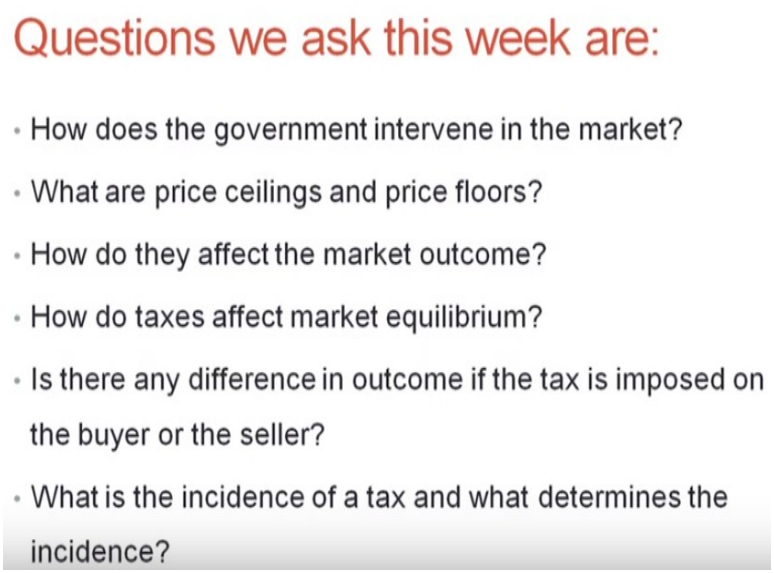
We talked about this also in the first class. We said that the very difficult tradeoff that society faces is the tradeoff between efficiency and equity. We would like everyone to have similar opportunities similar access to resources similar amount of income so we would like that to happen in a society but at the same time what happens is usually all resources go to someone who is more efficient.

Efficiency attracts the resources and they are the ones who get richer. So this is another very difficult tradeoff that the society faces and the government often and the example of the wage that I talked about is a similar example is a similar example and so government may raise the price in some cases and may lower it in others.

So basically what we are saying is since lot of outcome market outcome in the economy is not acceptable or desirable to the society in general the government may step in and in certain situations government may raise the price through various means say for example the government can actually impose a tax say by increasing the and that that eventually raises the price for a commodity or the what the government can do is it can set a price floor or a price ceiling.

So is also what the government can do and in all these cases market equilibrium is disrupted so in all these cases market equilibrium is disrupted. So obviously what the government the price that the government is setting or the tax that the government is imposing is resulting in a outcome of price and hence an outcome of output produced and purchased which is different from the price quantity equilibrium that a free market is going to arrive at.

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So I will explain through diagrams but before that so when we discussed demand and supply framework and we discussed the different policies through it the questions that we are primarily asking are how does the government intervene in the market. What does the government actually do and how does it intervene in the market?

Then already I mentioned these terms price ceilings and price floors. What are they? What are price ceilings and price floors and how do they affect the market outcome. I already said that the price ceiling, price floor, taxes which the government impose they are going to shift the market equilibrium elsewhere but how exactly does that happen and how do they affect the market outcome that is what we are going to see.

And how do taxes affect market outcome. Again taxes could be imposed either on the buyer or seller or even both so is there any difference in outcome if the taxes imposed on the buyer or the seller should the government have a preference who to tax, the buyer or the seller. What does that depend on? Does that depend on the kind of product it is taxing or is the government revenue going to be different if it taxes the 2 stakeholders differently.

So these are the questions that we are going to ask and what is the incidence of a tax and what determines the incidence? Incidence of tax means how much of the tax burden is basically shared by the buyer or the seller. How much if the government is imposing a tax what is the burden that the tax burden that the buyer actually ends up bearing or the seller actually ends up bearing. That

is the incidence of tax and some people may think that that tax the incidence of the tax should be equal to the tax. We will see so what happens.

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So these are the government interventions, the different kinds of government interventions. They can be much more complex and much more intricate government interventions but in this module we are going to discuss only about the price controls and taxes. So price controls can be of 2 types. One is the price ceiling. A legal maximum price set by the government for a particular good or service. For example, rent control or price cap on drugs.

So price ceiling is basically the government says that the price of this product cannot go up beyond a certain level. So the government basically wants the prices to not go up too high and basically the government's objective is that this particular good or service should be affordable to most of the people. So that is probably the objective of the government and that is the reason the government puts a price cap or price ceiling on it.

Say for example rent control where government says that housing should be affordable to most people. So this is a rent control. The rent cannot go up beyond a certain level or price cap on essential drug, essential medicines. So the government says that the medicine should be affordable to everyone. Most people should have access to the medicines. So the prices of the medicines cannot go too high up.

So there are numerous examples of price control price ceilings rather. The government can also impose price ceilings on essential, in most cases the price ceiling is on essential commodities where basically the objective is that it should be affordable accessible to most people and energy, fuel these are also examples where the government all over the world governments have imposed price controls with objective that energy being the fuel for growth in an economy, in fuels a lot of industries, manufacturing units, and people's houses. So that is the reason that this is a very essential fundamental commodity which should be affordable and accessible to all people.

Another example is just the opposite which is the price floor. Price floor is basically the government is saying that the price cannot fall below a certain level. Price cannot fall below a certain level and price floor is defined as a legal minimum price set by the government for a particular good or service for example minimum wage or price floor on say agricultural products, price floor on say agricultural products.

So here the government is basically trying to protect the producer. It is trying to protect the supplier. So minimum wage I gave the reason earlier when I said that I am going to again illustrate the price floor with example of the labour market and price floor on agricultural product is also a very common example whether because of technological improvement etc. basically the farmer's production or yield out of a piece of land is increasing over days but agricultural product is such a commodity which is an essential product and to relate it to the elasticity we talked about in the previous class agricultural product basically if the price falls too much if there is a whole lot of production in the economy and price rapidly falls because there is too much supply in the economy consumption of the product is not going to go up too much.

Just in the previous class we talked about price elasticity and revenue and this is just to relate to that the price elasticity of agricultural food grain for example agricultural product they are very low. Low in the sense that if the price goes down too much people are not going to raise their consumption of say wheat or food grains etc.

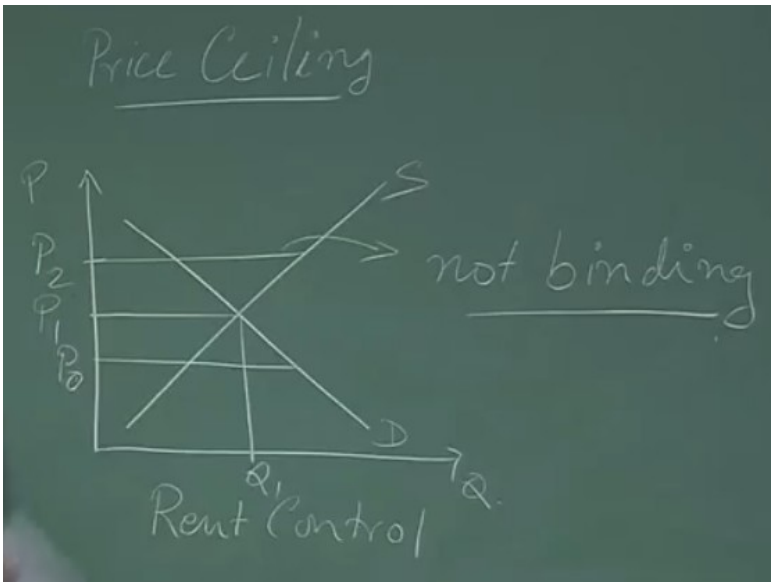
So in such a situation what happens? If there is a bumper production of crops and if it is left to the free market prices are going to go down and leading to drastic fall in income of the farmer to so to basically to protect the farmers the government can impose a price floor where it says that

okay price of wheat cannot go below this level even if the production of wheat is huge in the economy. So this is example of price floor.

Another instrument in the hands of the government to intervene in any kind of market equilibrium is taxes. Taxes it can be on buyers or it can be on sellers. For the same commodity the government can decide to tax either the seller or it can decide to tax the buyers and again taxes can be of various types and very this is a very common example that I am giving here can be a percentage of the price of a good. It can be the percentage of a price of a good or a specific amount per unit purchased or sold.

So physical quantity sold or purchased of the unit basically the government is charging a flat tax per unit of that output. So it can be of various types. So these are very broadly the definitions and examples of taxes and price controls. So let us first try to understand what happens in case of a price ceiling.

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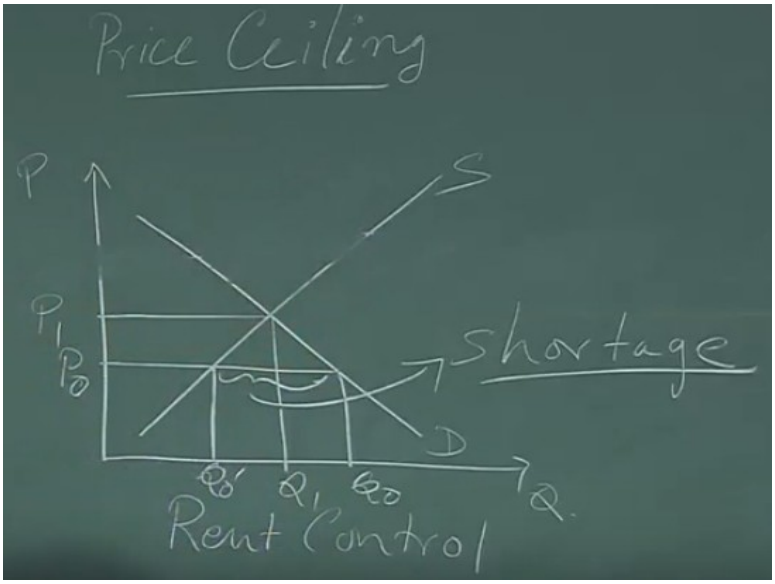
So let me say for example this is a price ceiling, say this is for rent control. Say this is example of rent control where there is an if the market is left to itself this is the demand for houses this is demand for houses for rent and this is the supply of houses which are put up on rent and this is the equilibrium price and equilibrium quantity or the number of houses that are up on rent and this is the equilibrium price.

Now what happens if there is a price ceiling? If there is a price ceiling now the market was clearing at a rent of P 1 and now the government says that this is too high or okay say for example okay the government says that the rent is too high, it cannot go up beyond this that is P 0. P 0 is the max rent that you can charge in the market if you are a home own house owner and you want to put up your house on rent you cannot charge a rent which is more than P 0 or P 0.

Now what happens in that case? But before that what happens if the government says that P 2 is the price ceiling. What happens in that case? If the price ceiling is put at a price or at a level which is higher than the market equilibrium in that case this price ceiling is not binding. This price ceiling is not binding. Why? Because the market in any case is clearing at a level which is lower than the price ceiling.

So it does not matter if the government says that the highest rent that you can charge is P 2 but no one is actually or the market is not clearing at P 2. Market is clearing at a much lower level. So no one bothers if the there is a rent control which says that it has to be maximum P 2. So we are not talking about that part of the diagram.

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What we are really interested in seeing is what happens if the price control or price ceiling is binding, if the price ceiling is binding or the price ceiling is at a level which is below the

equilibrium. If the price ceiling is at a level below the equilibrium, then what happens? Now what happens in this case is since the rent, the prevailing rent in the market is too low lot of people would like to rent houses like they will be responding along the demand curve.

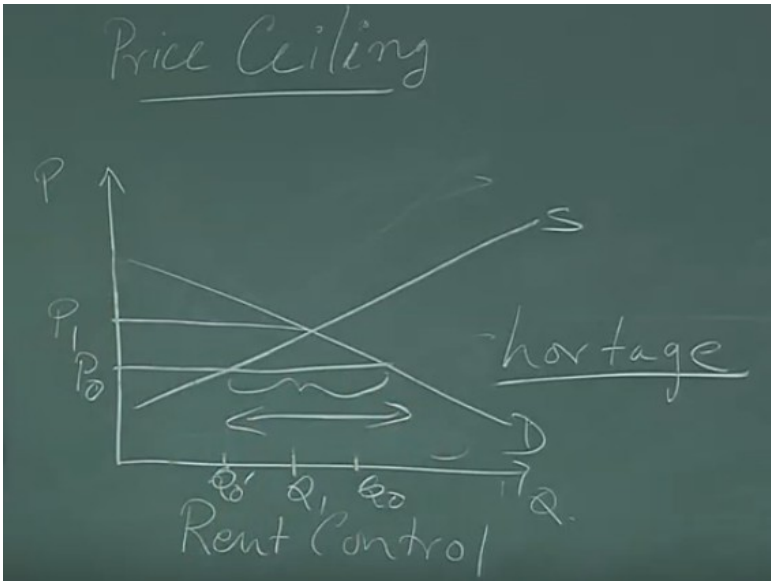
So along the demand curve if the rent is P 0, Q 0 amount of houses are demanded for rent. So this I the amount of houses which people are looking for. This is the amount of houses people are looking out for rent. But so what? Now if even if so many people are looking for rent how much is available in the market?

What will be available in the market will be as per the supply curve. So people who have put up their houses for rent in the market so they are going to only till here people who are willing to supply their houses on rent till the level of P 0, they will put up their houses up for rent in the market. So beyond these every supplier on the supply curve every supplier is seeking a price which is higher than P 0. So they will leave the market.

They will say that this this rent is too low. It is not possible for me to supply my house in the market. I do not want to put my house up for rent at this such a low level of rent so they will be out of the market and that will so basically now there are lot of people who are looking for houses in the for rent and very little people very little number of suppliers who are actually interested in renting out their houses.

So that will lead to a shortage in the market. That leads to a shortage of houses in the market. So basically what rent control is doing is it is reducing the supply in the market and it is increasing the demand in the market and thus creating a shortage in the market. So that then that is a problem. That is a problem. Okay and what is what is more interesting is this shortage is going to increase in the long run.

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In the long run what will happen is this in the long run if you think carefully you will be able to understand that in the long run the demand curve is more elasticity and the supply curve is also more elasticity. So in the long run the supply curve is more elastic and the demand curve is more elastic and as a result because of the same price ceiling the shortage is going to increase, shortage is going to increase.

Now why is the demand curve and the supply curve flatter in the long run? The demand and the supply curves are flatter in the long run because people both buyers and sellers both groups of people they get more time to adjust. They get more time to adjust to the prices and they basically adjust their preferences.

Now in the long run people who are looking for houses. They are going to be really interested in renting houses because they know that the rents have gone down because people because government has imposed a rent control and government has fixed the rent at a very low level so more number of people are going to enter the market looking for houses for rent.

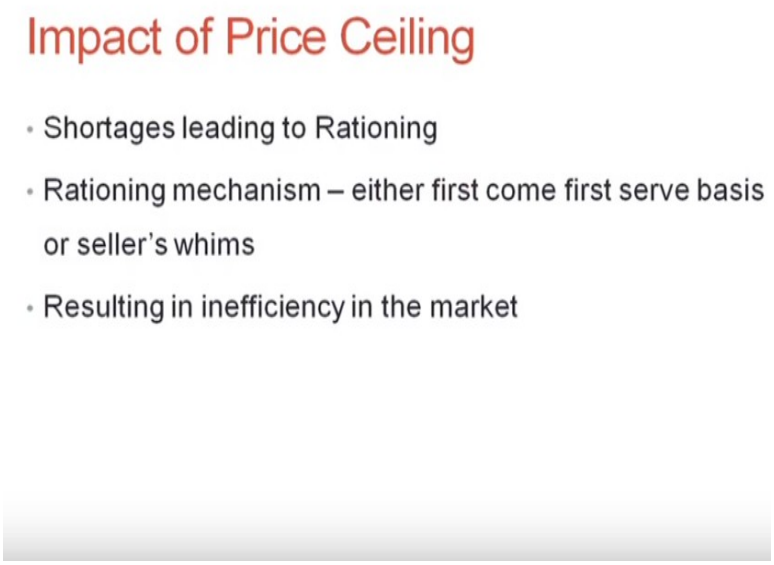
So they have a flat demand curve and in the case of the supply curve what happens is in the long run the many of the suppliers who were stuck with their houses for rent in the short run and could not take a quick decision or could not adjust their supply they gradually leave the market in the long run and more and more and more number of people who have houses which could be put up on rent they leave the market. May be they find some alternatives to their houses and they

leave the market and they basically do not want to put the houses up for rent so in the long run they have a flatter supply curve and this policy so leads to increasing shortage in the economy for houses.

So this is what happens and this example can be this the similar argument we can have in case of say energy consumption also. Say for example if the government puts a price ceiling on electricity and says that you know like the electricity prices are cannot increase more than this. what happens?

In the long run suppliers will find it less attractive to supply electricity in the market so basically they reduce their capacity and they are not too much interested to be in the supply in the production of electricity and on the other hand since prices are too low of electricity consumers are going to adjust their demand and they will shift to more and more consumption of more electricity intensive consumption behaviour. So that is what they are going to do because they know that the it is cheap to consume electricity. So this is what happens. So what are the impacts of price ceiling?

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Now so the impact of price ceiling is basically it leads to shortages and whenever there is shortage how is this fixed amount is how is this little amount of Q 0 with so many people looking for houses or so many people looking for electricity and production only limited to this

much as per the supply curve, how is that little amount of quantity to be distributed among so many people looking for the product because had it been a market equilibrium prices would have adjusted.

There would be a pressure on prices upwards and basically many people would be leaving the market. All these people would be leaving the market and prices would be going up and similarly more suppliers would be entering the market as prices would go up but that is not happening here. Here price is controlled. So there is no way price can increase so who is going to get this limited amount of production that is happening in the economy and that has to be some kind of rationing has to be there.

So how does the rationing happen? As we have seen in case of lot of rationing we are familiar in developing countries especially at the site of people queuing up in front of the say gas stations, in front of the petrol pump, there people queuing up for the to fuel their tanks because basically the price is low and too many people would like to fill up their tanks.

So that is one way of rationing and another is at seller's whims. So if I like you I am going to rent out my house to you otherwise I am not. So this is so basically it depends on seller's whims or on first come first serve basis. So this results in inefficiency in the market. Why do I say that it results in inefficiency in the market? It is because that an efficient outcome would be where all the people who are willing to pay at the market price they get the product.

But here all these people they are willing to pay a higher price but they are not able to pay a higher price and all these suppliers are willing to pay willing to produce at a higher price, they are not able to produce at a higher price so both demand is distorted and supply is controlled in the market and that is where lot of resources are wasted in the economy. That is these resources could have been put to production and that would have satisfied the demand of lot of people and lot of suppliers would be happy to sell it to the market at a slightly higher price but that is not happening. So that results in inefficiency in the market. So in the next class we are going to talk about price floors. Thank you.

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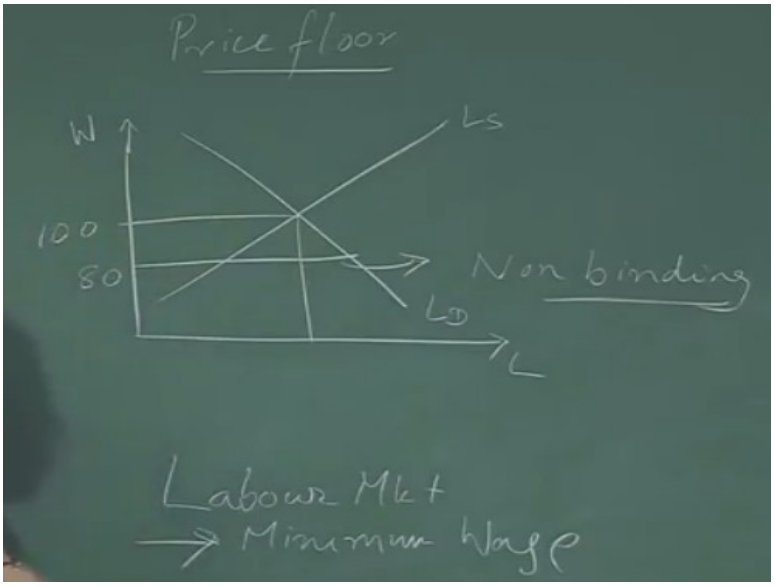
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Lecture - 11

Demand Supply & Government Policies (Contd.)

Welcome back we are now going to discuss about price floors. We ended our discussion on price ceilings and we are going to talk about now price floors. So the example that I gave in the earlier section on price floor was I talked about minimum wage. So and price floor can happen in case of agriculture products also. So let me take the example of minimum wage and let me just show what it does to the labour market.

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So this is a example of price floor and I am taking the example of a labour market where a minimum wage is imposed by the government. So as I said so this is the labour supply and this is the labour demand and this is the amount of labour. This is wage. So it is basically the same framework of prices, price of labour is nothing but wage and quantity of labour is nothing but number of people working or number of hours worked in a day etc. So it can be in any unit.

So we have on the x axis we have the labour supply or demand and this is the supply curve of labour. This is the demand curve of labour. Now as I said in the earlier class that it is possible that the economy has whole lot of people who are looking for job basically very unskilled people and with poor unskilled people who are who would like to work for a very low amount level of wage.

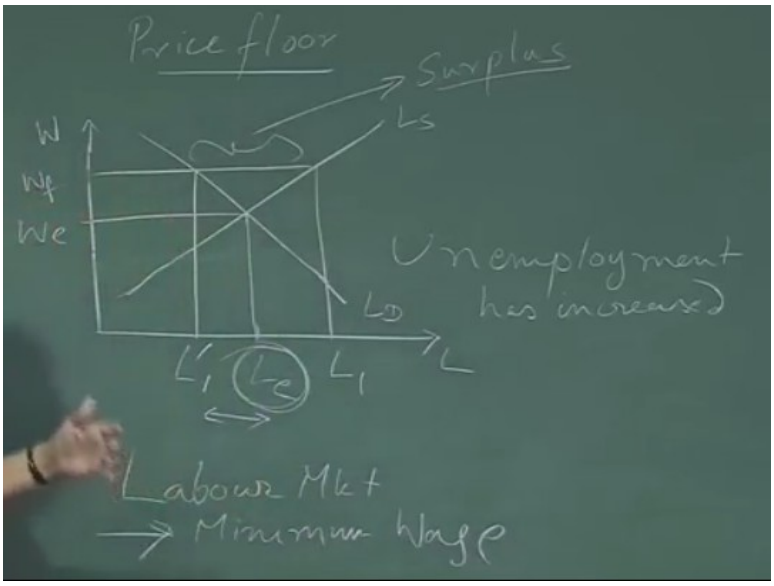
So they are all there in the market and it is a highly populated market with a huge workforce. In that case it is easy to imagine that the market is probably going to clear at a wage which is say quite low which is it is possible that the market clears say right like today in India the minimum wage is rupees 160 per day. So it is 160 per day.

So say the market clears at 100. So the market clears at 100 and the government says that see 100 is not a decent wage. It is not sufficient for a person to maintain a subsistence level of consumption at this very low wage so every employer has to pay a wage which is at least 160.

So what happens if the government decides that the wage is going to be, price floor is going to be 80, nothing. This is not a binding constraint.

So this is not this is not binding. So this price floor of 80 would be non-binding and so if the government imposes a minimum wage which is 80 and below the market clearing wage then this minimum wage is non-binding and does not disturb the equilibrium in the market. Nothing happens in the market. The market just clears at 100 because any wage above that above 80 is acceptable to the government.

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So we are not interested in this case and we are going to just stick to the above equilibrium level of prices. We are going to just stick to the above equilibrium level of prices. So this is equilibrium and let me just this is W e and the government now imposes a price floor at W f and the government says tells all employers that you cannot pay a wage which is lower than W f. Now what happens in such a situation?

It will probably it will be easier this time since we have already discussed about price ceiling so you might have already figured out that what happens in this case. Now since the wage has the market wage that can be offered has gone up this is the minimum wage. At this wage along the supply curve this many people L1 number of people or labourers would like to supply their labour. They would be very happy to work at this higher wage. So all this people now enter the job market. They are all looking for now job.

But what happens to demand, the people who are employing these people? For them the cost of employing the people have gone up. So now they have to pay a higher wage than the equilibrium they have to pay a higher wage and they are going to demand people or they are going to demand employees depending on what how much wage they have to pay.

So when they see that they have to pay W-f is the minimum that they can pay so they are going to demand ‘L1 dash’ or ‘L1 prime’ amount of labour. They are basically reducing their labour

demand. So earlier they were demanding ‘L-e’. Now they are demanding L 1 prime amount of people to work for them.

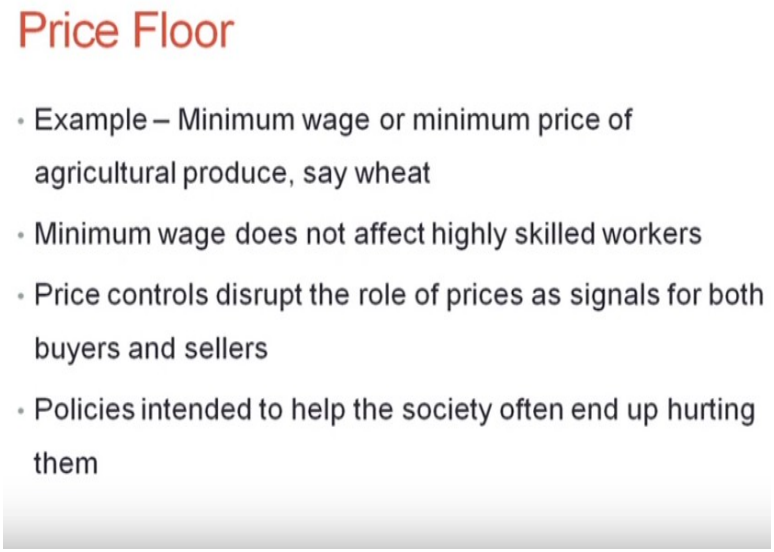
So what is happening? So basically if the market was left to itself. Market would have cleared at a lower wage and ‘L-e’ number of people would be employed but now since the government has imposed a price floor or it has imposed a minimum wage at a level which is above the equilibrium wage what has happened is the employability of people or the people looking for employees the producers looking for employees they are going to demand less number of people to work for them and on the other hand more and more people have entered the job market.

So firstly the government has created this amount of unemployment because these people could have been employed in the economy and now basically the employers are no longer employing this ‘L1 prime’ ‘L-e’ amount of people but there is also additional unemployment in the economy because now ‘L-e, L1’ amount of people have entered the labour market have entered the job market looking for jobs.

So as a result of this minimum wage unemployment has increased in the economy. Unemployment has increased in the economy because now because the producers the producers or now because the employers have to pay a higher wage they are employing less number of people so the first level of unemployment is created over here. The second level of unemployment is created because the minimum wage has gone up more people have entered the job market looking for jobs.

So they also end up being jobless and so the amount of unemployment goes up in the economy. So this is what happens so basically what is this? This is basically surplus. This is a surplus that has been created in the market. The same way that the price ceiling created a shortage in the market, price floor has created a surplus in the market. Price floor has created a surplus in the market.

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Now so coming back to the slides. So we started with example of minimum wage or minimum price of agricultural produce say wheat and we showed the impact we showed what happens in case of a price floor with the labour market and what is the how does actually labour market impact the people who are looking for jobs.

Minimum wage does not affect the highly skilled workers because highly skilled workers are probably getting a wage which is much higher than the minimum wage. So basically who are the people who lose their jobs? They are probably the people who work around who were accepting a wage level which is lower than 'W-f'.

So these are the people who are actually losing the jobs which is not a good thing because the minimum wage was probably imposed in the first place to protect this poor people who were getting a low level low amount of wage and these people are in any case many of these people are losing job because of this minimum wage requirement.

Okay and another thing that another impact of a price floor could be say for example there is a price floor on wheat say there is a price floor on wheat and the government basically is trying to protect the farmers and says that the amount of that the price of the wheat cannot be lower than a certain level.

So what happens in that case? Since there is a price floor on wheat the many more farmers would be willing to supply this product because now there is the price cannot go down any further. The price cannot go down and many farmers would like to supply. So the same way it is if we imagine this diagram for say wheat market so since the price has gone up of wheat so all these people wheat producers they will enter the market and start supplying wheat in the market.

But then what happens? When the market when the government says that I would like to protect the farmers and you can go for because of technology etc. the farmers are producing huge amount of wheat and in normal circumstance unless and until of course of there is a some you know some drought or something so in such circumstance maybe supply has contracted but in normal circumstance if there is a price flow because there is bumper production many of the farmers are willing to sell their product at a higher price and who is going to who is going to buy the products because there are no takers in the market. The demand is here.

Since the price is higher these people are people are not interested in buying more wheat. So who buys the wheat? The government's this policy will fail if the wheat is not sold because at the end of the day that is what the government is trying to do. The government is trying to protect the farmers. So what does the government do?

Government buys up this amount of surplus. So to make the policy more effective of imposing a price floor on wheat what the government has to do is it says that okay fine I am going to buy up this surplus wheat that the farmers have brought into the market but does it end there? No. So once the government purchases this wheat then this wheat has to be stored, it has to be transported, it has to be distributed.

So that also incurs a lot of cost to the government and where is that cost or where is that money going to come from for the government. So the so where is that going to come from? It is going to come from the taxes that the government raise from the people. So then it is a double problem. It is a so on the first place in the first place the government has increased the price of wheat so people are paying a higher price of wheat and secondly now the people are having to pay higher taxes to support the government to support this price floor that the government has imposed.

So basically the price control disrupts the role of prices as signals for both buyers and sellers. Now we saw in the last class that if the market is left to itself and if there is by due to whatever reason if the price if the supply is too much what happens? If there is a surplus the suppliers know that to sell this product we have to reduce the price. So they start reducing the price and as they reduce the price more and more people are buyers get interested they enter the market and they start buying.

So prices basically go down till they reach the equilibrium till they reach the equilibrium. But in case of price floor and price ceiling what is happening is the price cannot adjust in the market price cannot adjust in the market no matter there could be a shortage in the market prices are not going to go up. There could be surplus in the market prices are not going to go down and the price does not signal properly.

So basically in the long run instead of the problem resolving itself the problem becomes bigger because in the long run say in the case of the price ceiling people have adjusted the demand. They since electricity is cheaper they would now like to consume more and more electricity. So they have adjusted their habits so that they can use more and more electrical gadgets because that is the cheaper of the fuels and in the long run on the other hand to the suppliers it is less attractive to produce.

So they have shifted. They have changed their plans. They do not increase capacity because they have to sell the produce at a low price decided by the government. So the problem becomes even bigger in the long run. Unlike the free market where the prices adjust price acts as a signal and the equilibrium is reached in the long run in this case the problem actually aggravates in the long run.

So minimum wage can cause job losses and rent control similarly rent control. It can reduce the quantity or quality of affordable housing. What happens is government has introduced the policy with the very good intention of making housing affordable and accessible to people but what is happening is it does not have direct control on the suppliers.

So the suppliers lose interest and they leave the market and so the quality and quantity of housing in the market goes down and on the other hand the low prices or the low rents is a false

signal to the people looking for a housing. It is a false signal to the people that housing is cheap. So they are all adjusting their consumption habits and they are all entering the market looking for houses to rent.

So basically prices price controls they disrupt the role of prices as signals for both buyers and sellers. So basically what we are seeing is policies intended to help the society often end up hurting them. So policies intended to help the society often end up hurting them like it is people are losing jobs in case of a minimum wage and in case of surplus when there is a price floor it is the government has to incur a huge cost of for it.

However, this is not this discussion is that the scope of this discussion is limited to the demand supply framework that I have introduced and we have discussed this policy in the context of this very simple demand supply framework. That is not to say that price controls or government intervention or taxes are summarily bad. No. I am not saying that.

All that I am trying to say here is given this framework demand supply framework any disruption in the market can create some amount of inefficiency in the market but sometimes the objective of creating that inefficiency far surpasses these the inefficiency the end goal that is achieved sometimes the benefit is much higher than the cost that is we are seeing here in this very simple framework and coming back to this framework basically this teaches us how to use this framework to see how the efficiency how efficiency is arrived at in a market if it is left to itself and how prices act as a signal and help the market to come back into equilibrium. Thank you.

Foundation Course in Managerial Economics

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Lecture - 12

Taxes and Demand & Supply Framework

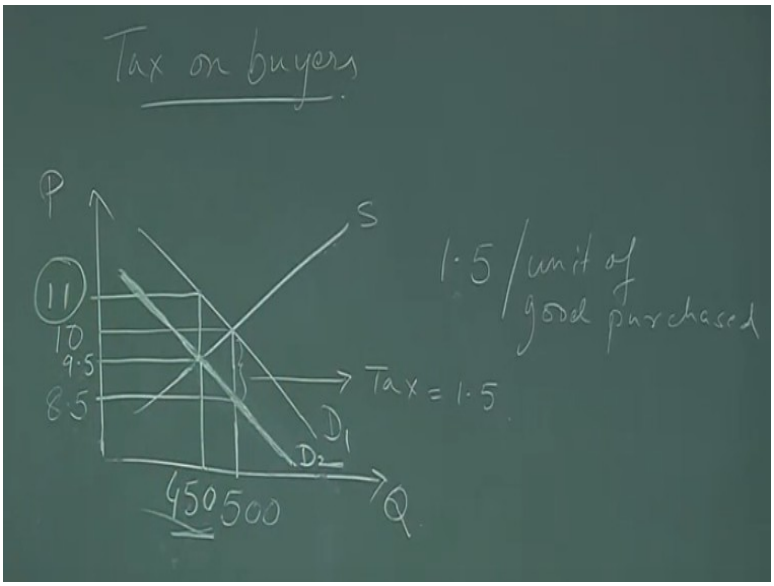
Welcome back to the discussion on demand and supply and how to use this framework to understand various policies of the government. So in the previous class we talked about the impacts. We saw how the policies can be the policies of price control and price ceilings can be analyzed through a demand supply framework and today we are going to talk about taxes.

We are going to see what happens in case of taxes how is the demand supply framework disrupted in case of taxes, what happens to equilibrium price, what happens to equilibrium quantity what is the price that the seller is getting to charge and what is the price that the buyer ends up paying. So all these are we are going to do through this demand supply framework.

So as we said in the earlier classes that tax can be either on the buyer or it can be on the seller. So first let us talk about tax on buyers. Let us see what happens if there is a tax on buyers. Now firstly what happens when there is a tax on buyers? Basically tax on buyers means for every unit of the good that this buyer tries to purchase he has to pay a tax.

So basically the product is becoming more expensive to him. So in that case basically for every unit of the product that he buys he has he is ending up paying a higher price. So his demand curve basically shifts and how does it shift, it shifts downwards. The basically the consumer is or the buyer is now demanding lesser quantity of the good at the same price.

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So let me… so this is we are taking about tax on buyers. We are talking about tax on buyers and this is the usual demand curve supply framework and this is the demand curve this is the supply curve and see there is a tax of 1.5 say rupees, Rs 1.5 per unit of the good purchased. So every unit of the good that the consumer purchases he has to pay a tax of 1.5 Rs. So he has to pay a this is the tax that he is supposed to pay per unit of good and this has been imposed by the government.

Now what happens is this is the demand and supply and say initially his he was consuming or the all the buyers in the market together were consuming 500 units at a price of say 10 Rs. They were consuming this at a price of 10 Rs. Now what happens the buyer has to pay a now the buyer has to pay this is the initial demand curve say and the buyer has to pay a tax of 1.5 per unit of goods.

So what happens is so what happens is this demand curve shifts parallelly downwards. I am sorry it does not look parallel let me let me try to draw it slightly better. So this is parallelly downwards and this is the amount of the tax equal to 1.5. Now basically what does this mean? This means that the buyers or all the consumers in the market would like to still continue to consume 500 units of the product provided that the price charged by the seller fell to 8.5 units.

So if it fell to 8.5 units then the buyers would pay 8.5 to the seller and 1.5 to the government so hence ending up paying total 10 Rs and buying 500 units. So what am I saying here basically the buyer's demand curve it still remains the same. The buyer's response or the buyer would like to continue to consume the same amount of output at the same price.

Price means how much he is paying out of its pocket. He is not bothered or he is not concerned who that money goes to. He just knows that if I take out 10 Rs from my pocket I should be able to purchase 500 units. So this is what the buyer is doing and he would like to continue to consume 500 units provided the but in that case price would have to fall to 8.5.

If the price fell to 8.5 then paying 8.5 to the producer and 1.5 to the government the buyer would be happy to shell out 10 Rs and still continue to buy 500 units. But that is not happening. That is not happening because the producer is not going to accept a price of 8.5 to sell 500 units. Hence the demand curve has shifted down basically. The demand curve has shifted down and what is the new equilibrium?

So the new equilibrium is where the new demand curve intersects with the supply curve. So the new demand curve intersects with the supply curve so along the new demand curve or let me use a different chalk to show the new demand curve. So this is the new demand curve.

So this is the new demand curve and this is the new demand curve and this demand curve basically shows that at every point on this demand curve the buyer is the price that the buyer is paying along every point of this demand curve that is the amount that is going to the seller and he knows that he is going to pay additional 1.5 to the government.

So at every point on this demand curve 1.5 of taxes has to be added to go back to the original demand curve. So this is what is happening and so now the equilibrium is at this point. This is the equilibrium and equilibrium is somewhere between 10 and 8.5 say it is at 9.5. So equilibrium is below is the equilibrium is somewhere between 10 and 8.5, say it is 9.5 and the number of units sold is less than 500. It is at say 450, this is the equilibrium.

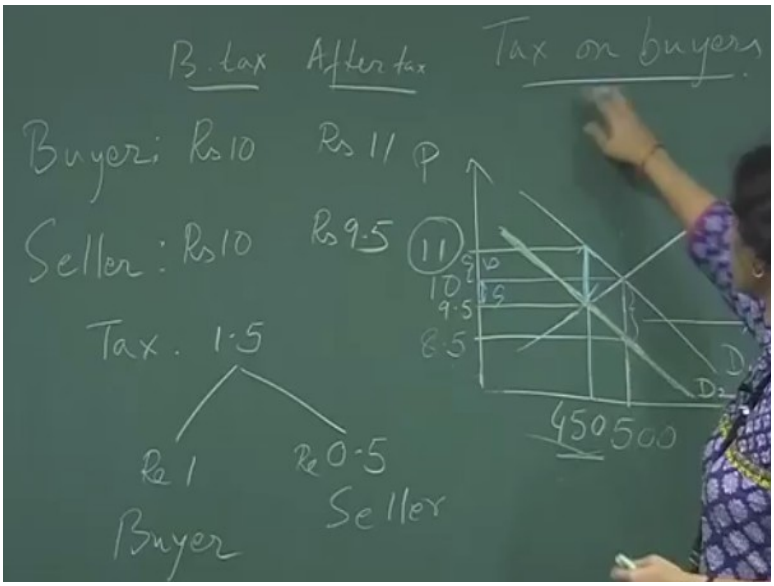
So at this point, at the new equilibrium is along the supply curve so basically the supplier is selling 450 units and receiving a price of 9.5 units and how much is the buyer paying how much

does the buyer end up shelling out. He ends up shelling out this 9.5 plus a tax of 1.5 and this is the amount that he ends up. So 9.5 plus 1.5 is 11.

So basically what we are seeing here is because a tax of 1.5 has been imposed on the buyer the buyer's demand curve shifts down by 1.5 units and every point on this curve the buyer is willing to buy a certain buy amounts at prices which he would like to pay to the supplier knowing that he has to pay additional 1.5 for every unit that he purchases.

So his equilibrium so the equilibrium the market equilibrium comes at a point where the supplier or the price that supplier receives and the price that the consumer pays to the supplier they intersect and the equilibrium is 450 here in this case and the price that the buyer is paying to the seller is 9.5 and he is paying an additional amount of 1.5 to the government so what the consumer ends up paying is 11 Rs. So he ends up paying 11 Rs. Ends up reducing his consumption from 500 to 450.

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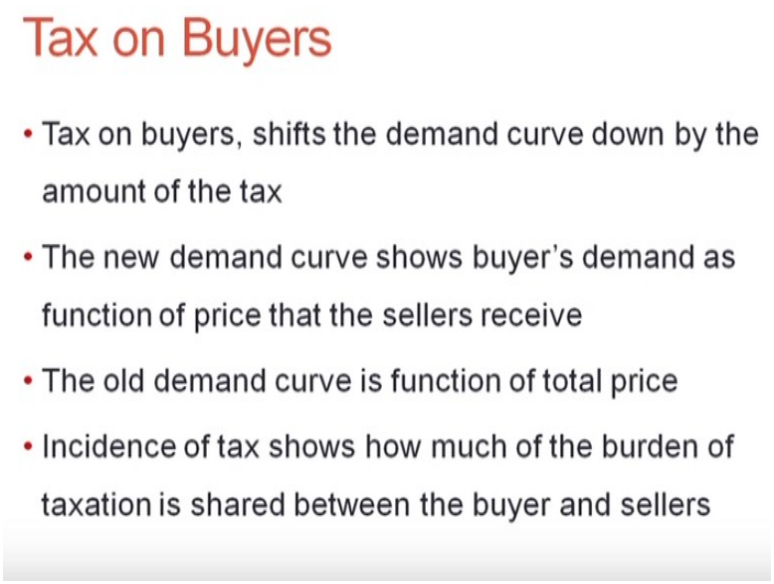
However, what is to be noted here is that government imposed a tax of 1.5. Earlier before the tax the before the tax the buyer was paying 10 Rs before tax and after tax, so 2 situations, before tax and after tax. So buyer in the before tax was paying Rs 10 to the supplier and he was buying an amount of 500 units and after tax he ends up paying Rs 11.

Whatever the seller or the seller does is before tax is he was getting the equilibrium price of Rs 10 but after tax he ends up getting Rs 9.5. He ends up getting 9.5 and the amount that is sold in the market is 450. So what do we see from here? What we see from here is so basically what we are seeing is that the tax of 1.5 is getting split up between the buyer who is paying a higher price of a price by rupee 1 and 0.5 Rs is being paid by the seller.

So what is happening basically the seller ends up selling the product at a lower price. He was earlier getting 10 Rs now he ends up charging 9.5 or slightly lower price. So this is the amount of the tax incidence that is happening on the seller because a tax of 1.5 has been imposed on the buyer. The buyer ends up paying only 1 rupee extra not 1.5 Rs extra. Although the tax is on the buyer of 1.5 per unit of good that 1.5 is getting split up that 1.5 is getting split up. So this is 1.5. This is the amount of the tax. This is the amount of the tax of 1.5 and this is this amount is being paid by the buyer and this amount is being paid by the seller.

So this amount of 1.5 of tax is basically getting split up. It creates a wedge between the buyer and the seller's prices. So the buyer ends up paying rupee 1 out of the total 1.5 amount of tax and the seller ends up getting a price 0.5 less than the original price and it is as if he is paying that amount as the tax. So this 1.5 gets split up between the buyer and the seller in the case where the tax is on the buyer. So when the tax is on the buyer this is how it works out. Next we are going to see what happens if the same tax of 1.5 per unit is imposed on the seller. Let us see what happens in that case.

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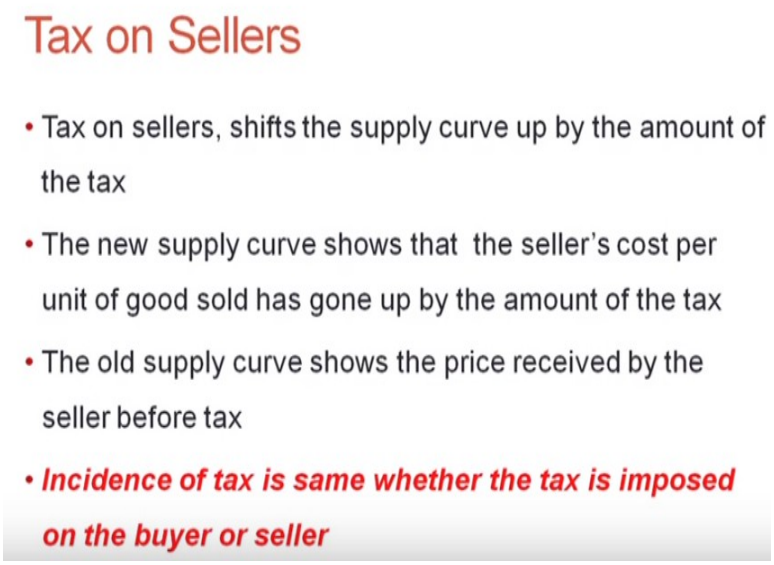
So we are going to discuss tax on seller but let me first show you the slide on tax on buyers just for completion. Now tax on buyers it shifts the demand curve down by the amount of the tax. The new demand curve shows the buyer's demand as function of price that the sellers receive. So although I have removed it the diagram from here you may remember that when the demand curve shifted downwards the new demand curve is basically the all the price on the new demand curve are the prices that the seller gets from the buyers and that does not include the taxes.

The old demand curve is function of total price. It is the price which is inclusive of taxes. So that is the price that is the actual demand curve of the consumer. Actual demand curve is what? It is basically his response to price. So his demand curve has not changed. His demand curve is still the same. The only thing that has changed is that the price along the demand curve that price is now split between the government and the seller.

So that is what we are trying to understand here. Now incidence of tax shows how much of the burden of taxation is shared between the buyers and the sellers and from this example that we took we saw that 1.5 rupee of tax is split between the buyer and the seller. The buyer pays the higher amount of 1 rupee per unit of good and the seller basically charges a price 0.5 less than the original equilibrium price and that is how he ends up paying so he is kind of so the price that the buyer is facing from the seller has gone down by 0.5.

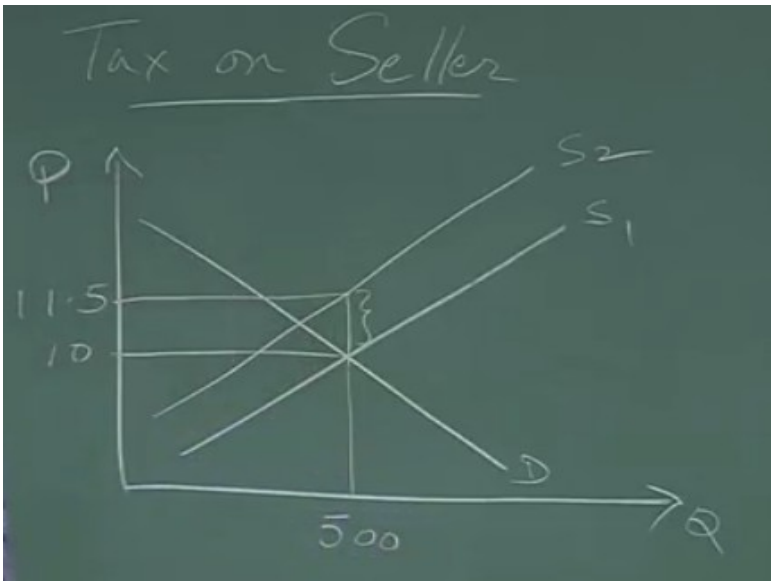
So that is the amount it can be imagined that is the tax that the seller is paying to the government. It is as if the seller has volunteered to share a part of the tax that has been imposed on the buyer. So moving on let us talk about tax on sellers. What happens if there is a tax on sellers? So if there is a tax on seller, same tax on seller what basically happens? Now when there is a tax on sellers the cost for the seller basically goes up. So every unit that the seller sells or the producer sells he is having to pay a tax. It is like a cost to him.

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So what happens is the tax on sellers shifts the supply curve up by the amount of the tax.

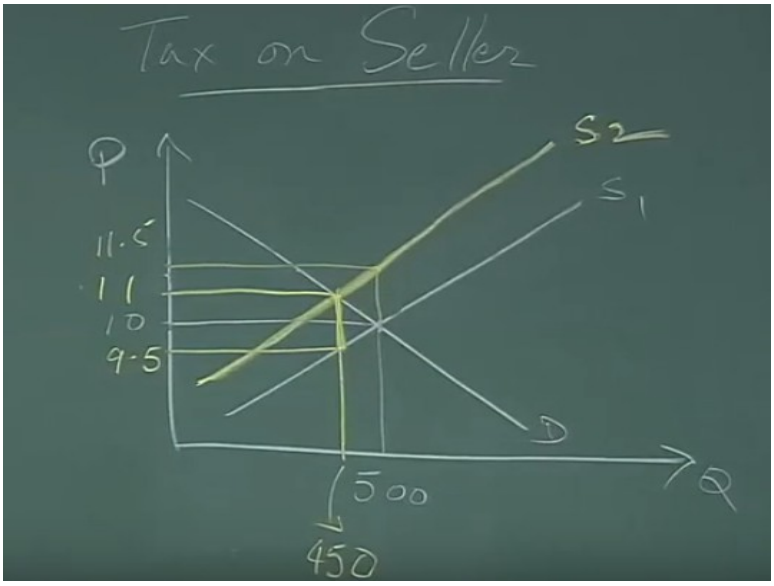
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So let me draw it here. So this is the demand curve, this is the supply curve S 1. Now 10, 500. So let us take the same equilibrium that we did earlier and now what happens is every unit that he sells he is having to pay a tax of 1.5 Rs. So every unit is costlier to sell, costlier to produce for him by 1.5 and hence the supply curve shifts to the left or upwards. So basically the supply curve shifts upwards or to the left. So this is S 2.

So to continue to sell 500 units to continue to sell 500 units the supplier would like to get a price would like to get a price over here because he has to pay now a 1.5 Rs of tax. So if this is 10 Rs he would like to get a price of 11.5 Rs by selling 500 units so that the 1.5 Rs he can pay to the government. But that is not happening. So because the consumers are not willing to pay this higher price. So let us see what happens actually.

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So let me again use a different colour for the new supply curve. So this is the new supply curve S 2. So this is the new supply curve S 2. Now what happens is now the new supply curve S 2 it intersects the demand curve but nothing has happened to the demand curve. Demand curve continues to be where it was.

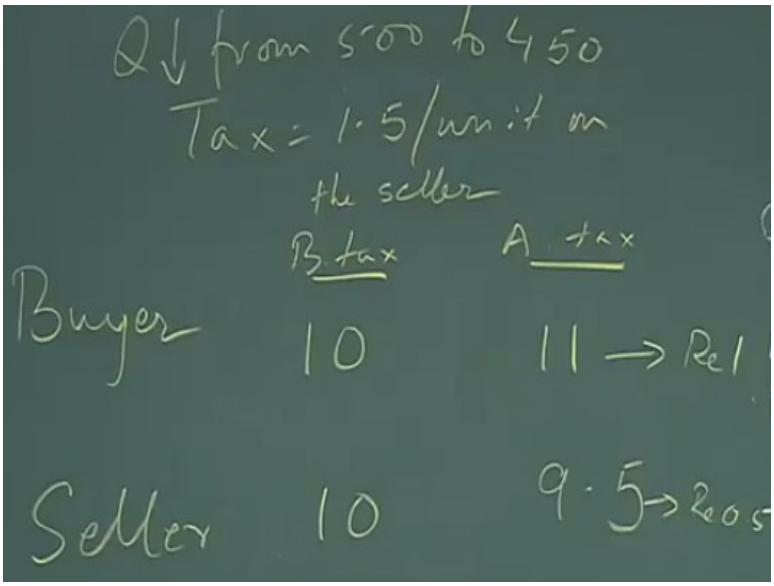
So the demand curve intersects with the new supply curve and the equilibrium is at this point, the equilibrium is at this point and it is the equilibrium is at this point which is higher than the price equilibrium price of 10 Rs and at a lower amount of quantity which is 450 units it is at 450 units and this is the amount of tax.

This is the amount of tax that has to be paid to the government. This is the amount of tax that has to be paid to the government. So basically the seller is receiving by selling 450 units. The seller is receiving a price along the original supply curve. So this is what he is getting. So this is what he is getting which is less than 10. So he is getting say 9.5 Rs per unit of the product that he is selling.

He is paying a tax of 1.5 on every product and this is the amount of the tax and this is the amount of the tax and the consumer is buying 450 units along his demand curve and this is the price that he is paying to the seller and this is 11 so which is more than 10, he is paying 11 and this was 11.5. So this is 11. So the buyer ends up paying 11 Rs to the seller. The seller receives 9.5 for

himself. The rest 1.5 he is going to pay to the government. So in this case also what is happening?

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So the quantity falls from 500 to 450 and buyer and tax is 1.5 percentage unit on the seller and this is the buyer, the seller, before tax, after tax. So before tax the buyer was paying 10 Rs. After tax buyer is paying 11 Rs. Before tax seller was receiving 10 Rs. After tax seller is receiving 9.5 Rs. So here also this 1.5 Rs of tax 1 rupee is being paid by the buyer and 0.5 rupee is being paid by the seller.

So basically what we see here is the same tax, same amount of tax of 1.5 per unit of output whether imposed on supplier or whether imposed on seller or imposed on buyer it is split up exactly in the same way in both cases. In both the cases the buyer ends up paying 1 rupee and the seller ends up paying 0.5 rupee out of this total tax and this is how they split the tax between themselves and pay the tax.

In other words, what is basically the intuition behind it? The intuition behind it is if the buyer if a tax is imposed on the buyer the seller knows that the tax has been imposed on the buyer so the buyer is basically facing a higher price. So he is going to reduce his consumption which is not good news to the seller.

So what the seller does is depending on if he sees that the tax imposition of the tax if the entire tax is to be borne by the buyer then the amount of the quantity consumed if it goes down too if it goes too much down then the revenue that comes to the seller might go down. So in that case the seller might propose to the seller might like to share a part of the tax so that the buyer basically does not have to pay 1.5 Rs per unit of the output.

So that is the intuition behind it and but another point that I wanted to make here is we have assumed a tax per unit of output a flat rate of 1.5 per unit of output because of which in our diagram the supply curve is shifting parallelly upward or the demand curve is shifting parallelly downwards.

Had it been a percentage tax a percentage imposed on the price of the good a percentage of the price per unit of the good in that case what would have happened was for higher prices the gap between the supply and between the 2 curves the new supply and old supply or the new demand and old demand, the gap would have been higher at the at higher prices and lower at lower prices.

So this is this is just something to be noted and to be careful about when you are drawing a diagram but to avoid such complexities we have taken the simpler example of a tax per unit of output. So tax on sellers is shifts the supply curve up by the amount of the tax. Now the new supply curve shows that the seller's cost per unit of good sold has gone up by the amount of the tax. The old supply curve shows the price received by the seller before tax.

That is removing the tax what exactly goes to the seller that is shown by the old supply curve and what is most interesting about this entire example and this entire exercise is that incidence of a tax is same whether the tax is imposed on the buyer or the seller. It does not matter who the tax is imposed on but the incidence is going to the same whether it is imposed on the buyer or seller.

Now in the next class we are going to so you might be curious to know that is it always that the buyer ends up paying more amount of the tax I mean how did I arrive at this at these figures of rupee 1 and 0.5. Why is it not you know like 20 paise and 1 rupee 30 paise or why is it not that seller is paying more of the tax and buyer is selling is paying less.

So that those answers we are going to get from the next part of the discussion or the next class where we are going to talk about elasticity. What is the what implication does elasticity have on this. Till then I would insist you to think about it that what is going to determine how much of the tax burden goes to the seller and how much of the tax burden actually goes to the buyer and we are also going to discuss some real life situations where the government imposes certain taxes and what is the end result of it in the next class. Thank you.

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Lecture - 13
Elasticity and Taxes

Hello and welcome back to the foundation course in Managerial Economics. We had started our discussion on how to understand the effect of different policies of the government in the demand supply framework, the very simple demand supply framework that we started with, and we are looking at the impact of different policies of the government whether it is a price control or a tax and we are trying to understand what happens in such kinds of policy interventions by the government into a free market equilibrium.

So a free market equilibrium basically gives us an intersection between the demand and supply curve and it gives us the equilibrium price in the market and the equilibrium quantity in the market which leaves both the consumers and the producer satisfied and no one is left in the market who is looking for a good at the existing price and he is not getting it.

So and we discussed that the government sometimes intervenes in the market where the market fails due to various kinds of reasons and we were discussing about taxes in the last class we discussed about we looked at what happens when a tax is imposed on a buyer or on a seller and what we saw was it does not matter whether the tax is on the buyer or the seller because in both the situations the buyer and the seller are going to distribute the tax between themselves and the amount of tax burden that the buyer bears and the amount of tax burden that the seller bears that does not change if the tax is imposed on the buyer or the seller.

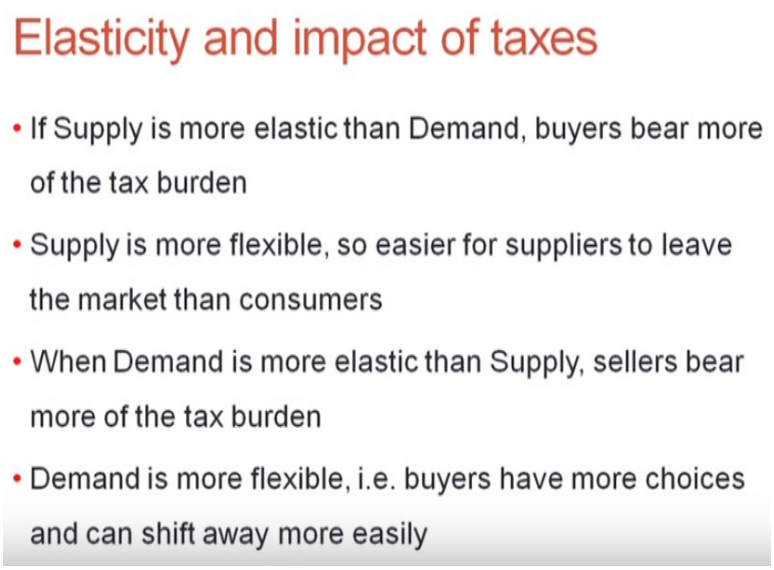
That is, it is irrespective of which party the tax is imposed on it is distributed in the same way in both cases between the buyer and the seller. So next thing and we left the discussion looking at the typical example that we took and in our example we saw that the buyer in both the cases whether the tax was on the seller or the buyer in both the cases buyer was bearing major part of the tax.

We discussed about the example where the tax was 1.5 Rs per unit of product sold and we saw that the buyer was bearing 1 rupee out of that 1.5 Rs and the seller was bearing a tax a 0.5 Rs or 50 paise out of that entire tax of 1.5 per unit of good and we saw that it did not change whether it was on the buyer or the seller.

So the question that we left are last discussion with was what determines is it that always the buyer bears the major part of the tax or what is going to determine how much part of the tax the buyer is going to bear and how much part of the tax the seller is going to bear and today we are going to see that the answer lies in our understanding of elasticity that we did a few lectures back.

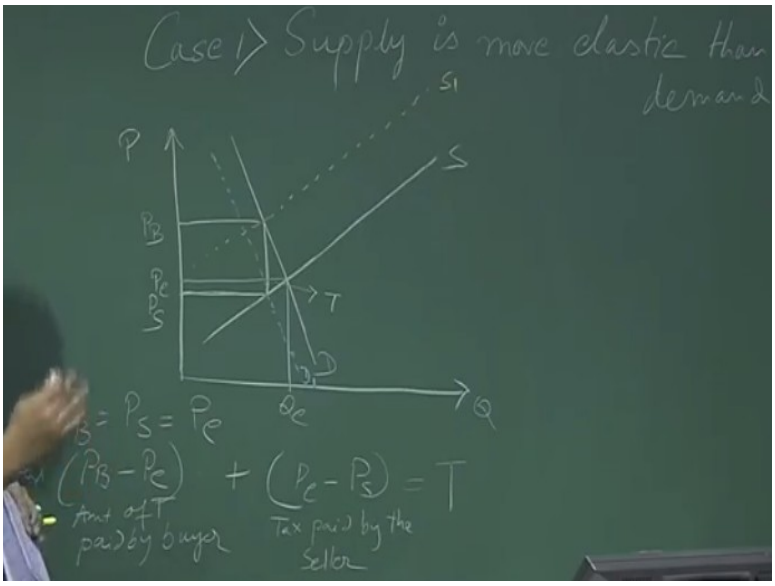
So we are going to discuss about elasticity and what impact elasticity or rather price elasticity of demand and supply has on the incidence of taxes. So we are going to talk about elasticity and impact of taxes.

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We are going to show that if supply is more elastic than demand buyers bear most of the tax burden. So if supply is more elastic than demand buyers bear most of the tax burden and let me illustrate through an example, let me show on the board why that is so.

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So we are talking about a case is one is supply is so supply is more elastic than demand, supply is more elastic than demand. So we draw a so we draw a supply curve which is more elastic means the supply curve is flatter so we draw a very flat supply curve and a steep demand curve. So supply is more elastic than demand and this is the equilibrium price equilibrium quantity so this is what happens before taxes.

Now say a tax is imposed of amount 'T' so the first thing that we can do when we are trying to find out the impact of a tax through this kind of a diagram what you can do is you can imagine a tax of T and straightaway draw the amount of the tax between the demand and the supply curve.

So this is the amount of tax that has to be paid to the government and we know that whether it is the tax is imposed on the supply or on the seller or the buyer the tax is going to be distributed in the same way in both the cases and so this tax has to be paid by the seller and the buyer and so this is the amount this is the price that the seller is going to pay on the supply curve, this is the price that the buyer is going to pay on the demand curve and so now what I am doing here is slightly different from the way I was illustrating the tax in the previous lecture.

So it is follow me through so this is the equilibrium price and the tax has been imposed so we imagine a tax of this size between the demand and the supply curve so this is T so let me colour it. So this green line is the tax of amount T. So this tax is to be split between the supplier or the seller and the buyer and the buyer now the buyer pays a higher price of P-B and the seller gets a lower price of P-S than the equilibrium price.

So this is how the tax is distributed between the two and if you are interested to know how the diagram can be completed looking at the new demand curve looking at the new supply curve etc. it is very easy to do that. If the tax is on the seller what happens the supply curve basically shifts up. So if it is on seller if it is on the seller then the supply curve is going to be a parallel shift like this. This is S1 and if it is on the buyer if the tax is on the buyer then the demand curve shifts downwards that is there is less demand now and you can imagine a this to be the new demand curve. This can be the new demand curve.

So either it is the new supply curve or it is the new demand curve. In both the cases, in both the situations the price the new prices are going to be P-B and P-S and so even without drawing the shift in the demand or the supply curve we can actually say how much of what is going to be the new price in the both cases.

So now we can see that this is how the tax is divided between the two. So this is the green line is the amount of tax that has to be split up between the two and earlier both were the price that the PB was equal to PS was earlier before tax PB was equal to PS was equal to 'Pe'. After tax PS is the price that the seller receives which is less than P-e and buyer pays a price which is higher than 'Pe' and the tax is split between the two so 'P B - P e', 'P B - P 'e is equal to 'P B – Pe' is the amount paid by the amount of tax paid by buyer plus 'Pe – PS' this is the amount of tax paid by the seller.

This is actually not paid by the seller but this is a reduction in price that the seller now receives so it is like the incidence of tax has fallen on the seller also. So the total amount that is equal to now T. So that is equal to T and so that is equal to PB - PS and that is equal to T. So T is now split up between the two.

Now coming back to our original question that how much of the tax should the buyer pay and how much of the tax should the seller pay. Now in this situation we can clearly see that when we have drawn a flat supply curve and a steep demand curve we can see that the amount of tax paid by the buyer is much higher than the amount of tax or the incidence of tax that falls on the seller.

So here we can clearly see that this is the amount of this is the amount of tax paid by buyer and this is the small amount this is the amount of tax paid by seller. This is the tax paid by seller or it is a more correct way of saying it is the incidence of tax that falls on the seller because the tax has been imposed. The seller ends up receiving a smaller price than what he was receiving earlier so it is like he has split up the amount of tax with the buyer.

So whether it is the tax has been imposed or the seller or the buyer in both the cases we see in this situation that the buyer ends up paying more major part of the tax and the seller the incidence or the burden tax burden is less on the seller. So this is what we see in the case where supply is more elastic than demand.

Now this is what we illustrated through this is what we illustrated through diagram but what is the intuition behind it. Why does it happen this way? So I am sure many of you have already figured that out or the logic would be working in your brain already that supply is more flexible. So when supply is more elastic that means what?

When supply is more elastic that means that supply is more flexible and so it is easier for suppliers to leave the market than consumers. So it is easier for suppliers to leave the market than consumers. So whenever there is tax in the market on any good whether it is on the seller or the buyer that means the market is not very friendly to either of them because the price has been increased forcibly by the government and there is a tax on the product.

So in such a situation the one of the buyer and the seller the party which is more rigid or which is more inflexible he that party has to bear the major part of the burden of the case. So in this case demand is steeper which means the people probably this is the kind of necessity necessary good or there are no substitutes or people are there are no substitute of this product so people cannot change their habits or consumption requirements so easily so even if the price increases of or the market is not very friendly still the buyer stick to the market.

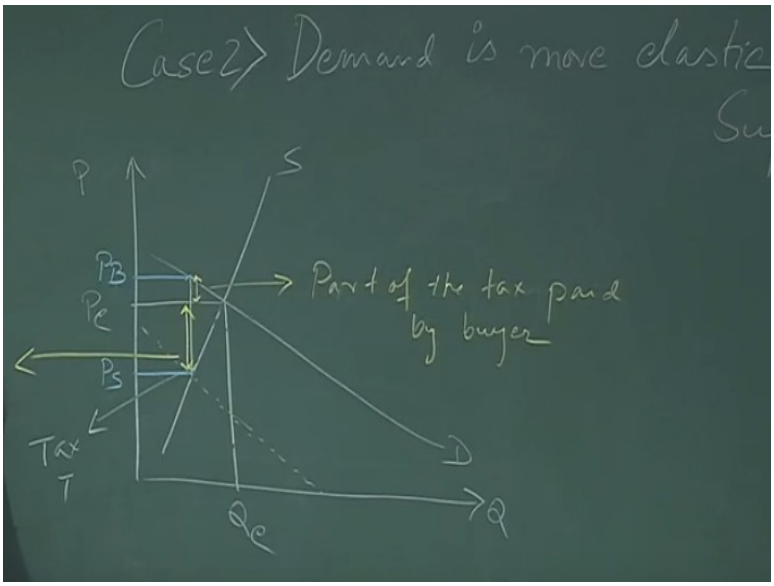
So in that case obviously who the party which party which is more flexible that is the party which can leave the market which can decide not to produce since it is not the price has the cost has gone up for the seller so the seller might just decide that I am going to produce less or I am going to really not I am not interested in paying so much of tax so I am going to produce less for unit of output I am going to pay less tax.

So in that case the supplier is more flexible so he can manage to get away with a lower by bearing a lower part of the tax and the party the buyer who has a very inelastic demand curve and cannot change his consumption requirement so easily that party has to bear the major part of the tax.

Next we are going to show that when demand is more elastic than supply, sellers bear most of the tax burden. So again it is just a matter of showing it graphically what happens or in the through diagram what happens when demand is more elastic than supply but again I am sure that the

logic has already worked in your minds that what happens when demand is more elastic than supply. So let us just draw the diagram and see what happens.

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So now we are discussing case 2 where demand is more elastic than supply. So demand is more elastic than supply that means we have a very flat demand curve and we have a steep supply curve. So demand is more elastic than supply. Again you have an equilibrium price 'Pe' and equilibrium quantity of 'Qe'.

Now again a tax of amount T imposed and so this is the almost the same amount of tax that we have been drawing so this is let this be the tax that is tax T. So this amount of tax is imposed and so immediately we draw the amount of tax between the demand and the supply curve and we get the after tax prices.

So now along the demand curve the buyer is going to pay a price of 'PB' which is higher than the equilibrium price and the along the supply curve the seller is going to receive a price of 'PS' which is less than the equilibrium price and this is the incidence of tax that is happening on the incidence of tax that is the seller is bearing.

So this is now the tax creates a wedge which is split up between the seller and the buyer and these are the prices that the price that the seller gets and this is the price that the buyer pays and the difference is basically the amount of the tax.

Now here we can clearly see that the amount of tax that the buyer is paying that is this amount this very small amount this is the amount of tax the buyer is actually paying. No matter if the say

the tax the amount of tax has been imposed on the buyer, then what happens? The demand curve basically shifts down by this amount.

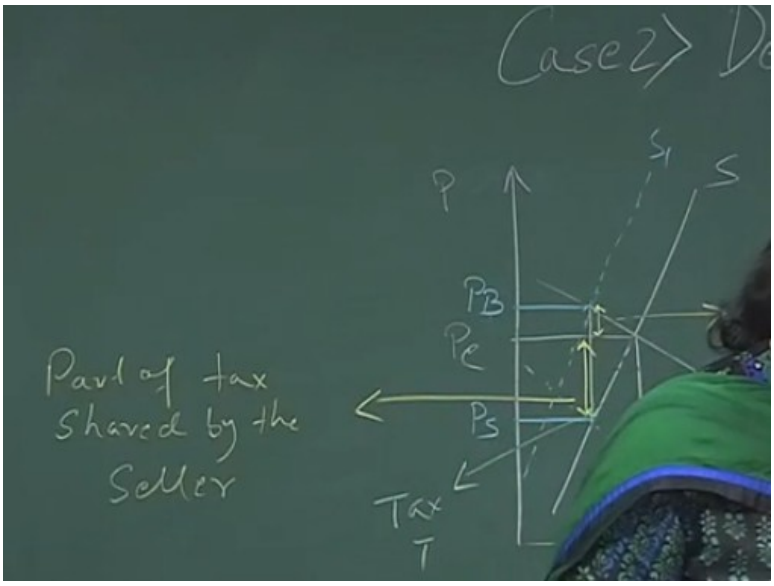
So the demand curve has made a huge shift downwards but the tax that the buyer actually ends up paying is this very small amount, this is the part of the tax paid by buyer and the seller ends up foregoing this amount of price per unit of the output. So earlier he was receiving 'Pe' now he is receiving 'PS' and this is the amount which he is which is the amount he gets less than what he was getting in during equilibrium.

So this is the amount that is part of the tax shared by the shared by the seller. So this is the part of the tax shared by the seller and in this case we clearly see that since demand is more elastic than supply what is happening is the amount of tax burden that the seller is bearing is now much more than the amount of tax burden that the buyer has to bear and again the again the reason is again the same.

Now demand is more flexible. That is buyers have more choice and can shift away more easily. So basically demand is flat or it is a very elastic demand curve means that demand is more flexible. That is probably the buyers have choices in the market and when they see it does not it is not too attractive to buy this product in the market with all these taxes they will quickly shift their demand to a probably a substitute good or probably they will stop consuming this good altogether or reduce their consumption of this good because it is not so much of necessity to them.

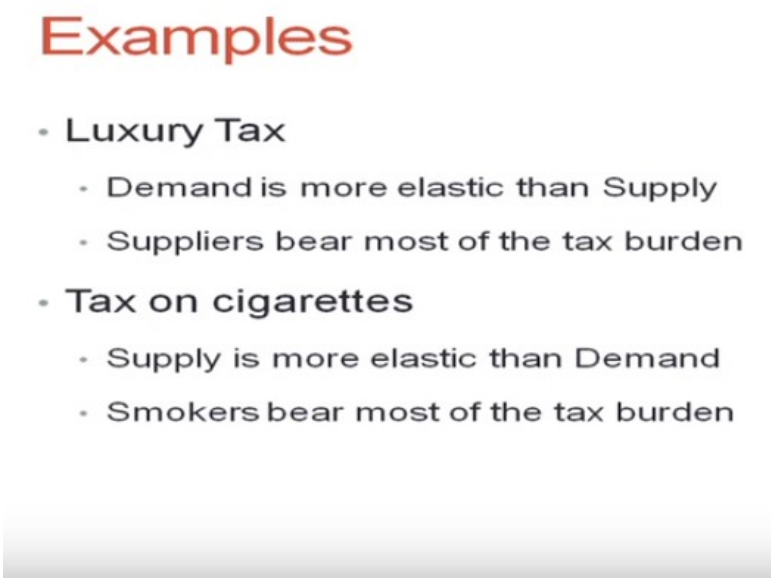
So in such a situation and we can see that the amount by which the curve shifts in each of the cases if you look at the earlier diagram also you will see that the amount by which it shifts depends on how flat or steep it is. So if the demand if this is the amount of tax the demand curve has made a huge shift that means the demand is flexible and it has made a huge shift. On the other hand, supply is steep and say if the tax had been on the supplier.

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In that case what is what would have happened? In that case supply would have shifted this to the left or to the upward and this would have been the new supply curve this would have been the new supply curve and we can clearly see that the shift in the supply curve is also very small which shows that which is basically happening because the intuition is that supply is probably inflexible at least in the short run or in the given circumstances supply does not adjust so easily so the supplier is willing to bear major part of the tax than the buyers.

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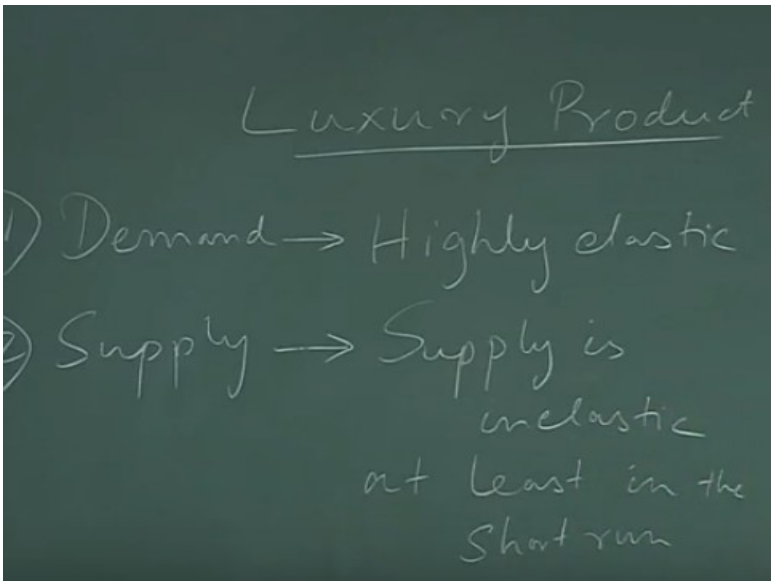


So let us discuss some examples, let us look at some examples. What happens in case of a luxury tax? What happens in case of a luxury tax? Often during the budget, we hear that the tax on luxury goods has been increased say tax on branded high-end products or taxes on room rents in

hotels or say taxes on luxury cars high-end cars the tax has gone up and sometimes even it happens that the tax is on the buyer we feel kind of happy that okay the rich are paying more taxes now and they will be paying they will have to top up more money for every unit of luxury product that they consume but does it happen that way.

What have we learnt from the exercises that we did right now? What kind of supply or demand curve does the luxury product have? Now a luxury looks at looking at demand curve luxury product is not a necessary product and sometimes luxury products have lot of lot of substitutes in the market also.

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So if the tax goes up on the luxury product that basically means that so luxury product is so in case of a luxury product first looking at demand first looking at demand is it elastic or inelastic. So luxury product is not a necessary product. It is not that people really need to consume it immediately even if the price has gone up. So people can move to other substitutes and so basically demand is highly elastic.

So demand is highly elastic. What about supply? What about supply? What about the elasticity of supply of luxury goods? Now in most cases you can imagine if the luxury good is say as I said if it is a high-end five star hotel rooms or it is high-end cars really luxurious cars, supply is going to be inelastic at least in the short run, why?

Supply is inelastic at least in the short run because that most supplying these kinds of incur a lot of cost like say for example if there is a five star hotel for the maintenance of the rooms the

hotels the taxes that they pay for having the I am not talking about the per unit tax but I am saying that the all the cost that are incurred by the hotels they are quite high and they cannot be reduced they are not so flexible in the short run.

Later we are going to talk about variable cost and fixed cost and we are going to see that in these hotels the fixed costs are very high. So basically hotels or even high-end cars the plants, the manufacturing units everything has been set.

So all of a sudden because the prices are high the supplier cannot reduce the supply or cannot all of a sudden decide that I am not going to produce any more cars or I am going to cut down on the production of cars because they already have created a enormous capacity or even if not enormous they had created certain capacity with some with some plan about production in the near future and all of a sudden he cannot just cut back on production.

So basically supply is inelastic. So supply is inelastic at least in the at least in the short run. So in such a situation this is going to happen. This is going to happen. Demand is more elastic than supply and part of the tax paid by the buyer is going to be very small and most part of the tax incidence is going to be would have to be borne by the supplier.

So is that a good policy or bad? Now since we already said that the from our understanding of elasticity that we discussed in the earlier classes now income elasticity of demand or price elasticity of demand of luxury goods is very high it is a very elastic demand. So in that case the revenue of suppliers or sellers is going to increase if they can reduce the prices.

If they cut if they reduce cost there will be enormous increase in output sold and the opposite is going to happen if they try if they raise the prices. So the government is basically discouraging this sector if it is imposing ya if it is imposing taxes on this sector's consumption.

So if it is discouraging this sector then what will happen is when the people's income grows or when the when they try to be more cost efficient or when they try to reduce their cost I mean they will not have any incentive to reduce or they will not have much incentive to grow this sector. In that case the revenue from the entire sector is going to fall.

So it at the end of the day it is not a very attractive proposition for the overall growth of the economy because this is the sector the luxury sector again going as per our understanding of elasticity that we have been discussing since the last classes since elasticity of demand for luxury products is very high it makes sense to encourage this sector to grow and be more efficiently and to reduce the prices cost and prices so that their overall revenue grows in the sector. So this was

about luxury tax and demand is more elastic than supply and suppliers bear most of the tax burden.

On the other hand, what happens when there is a tax on cigarettes? When there is a tax on cigarettes supply is more elastic than demand. So supply is more elastic than demand because cigarette is mostly addictive and to the consumers or to the smokers probably they are not going to be able to change their habits at least in the short run.

So and even unlikely to change their habits in the long run so supply is very demand is very inelastic and on the other hand supply is more elastic than demand. It is possible for a cigarette company to just start producing less cigarettes or they can just some of if they feel that it is unattractive to be in this market they can quit the market. So smokers bear most of the tax burden in this case.

So these 2 examples show us what kind of issues the policymaker needs to keep in mind when he is deciding who to impose the tax on. When I say who to it does not mean whether the buyer or the seller but on which sector so because we have already seen that imposing a tax on the buyer or the seller does not make any difference as per as incidence of tax is concerned. So we are going to discuss about efficiency of efficiency of a market equilibrium in the next class. Thank you.

Foundation Course in Managerial Economics

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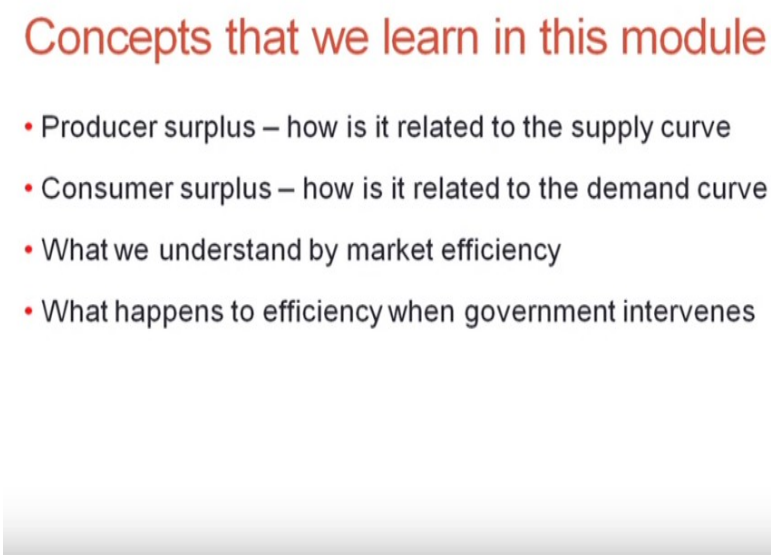
Lecture - 14

Efficiency and Market Equilibrium

Welcome back to the discussion. We have been talking about policy intervention by the government and we discussed about and everything we are discussing in the demand supply framework still and in the previous modules we looked at price regulations by the government, we looked at taxes by the government and what happens to incidence of tax, how does elasticity impact taxes etc.

We have talked, discussed about all that and in this last module of this entire session or in this entire section of the lecture on demand supply we are going to talk about something a little more theoretical. In the very first class we said that market is usually a best of way of allocating resources and today we are going to try to understand why we said so. So we are going to talk about that and we are going to say that so does that mean that the market equilibrium is the most efficient outcome in case of any market transaction. So is that the most efficient outcome?

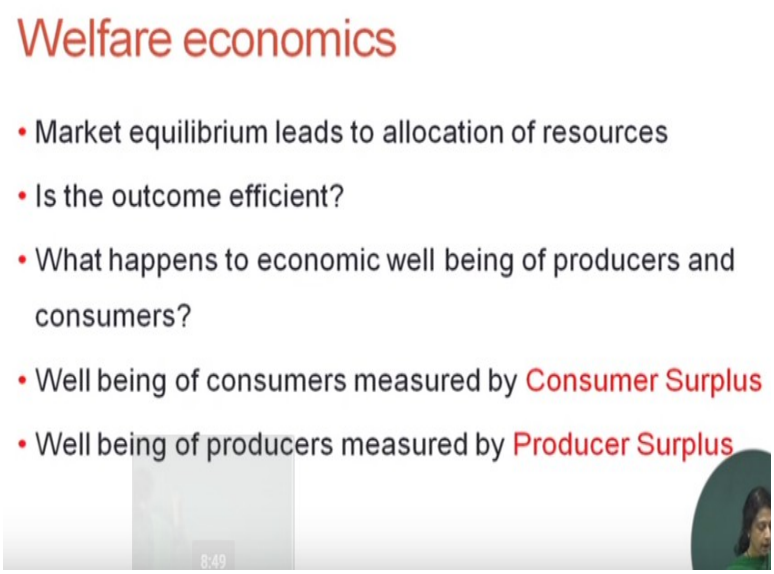
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This is what we are going to discuss and in the process the concepts we will be talking about introducing and understanding a few basic concepts which are firstly we are going to talk about something called producer surplus. So how is it related to the supply curve? So we are going to discuss about something like producer surplus and how is it related to the supply curve.

Then we are going to talk about consumer surplus, how is it related to the demand curve? Then we are going to define market efficiency or we are going to see how what is market efficiency and how producer surplus and consumer surplus they can explain market efficiency and next we are going to see what happens to efficiency when government intervenes.

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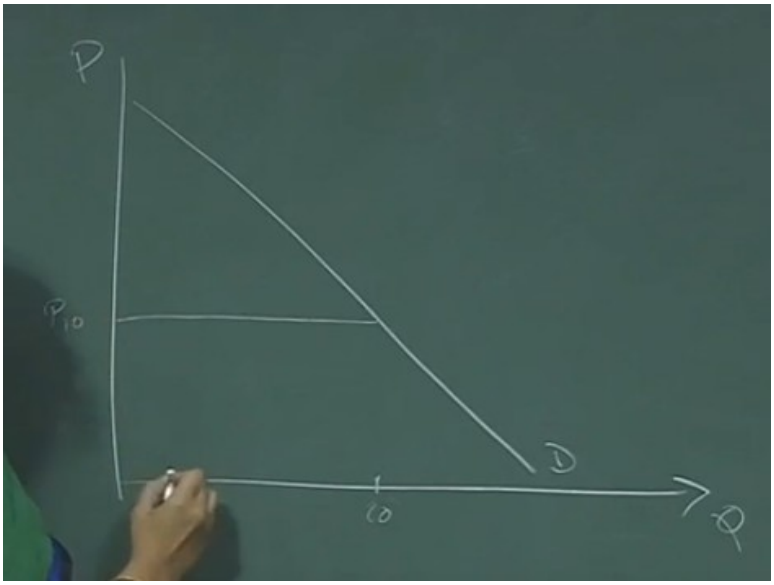
So this is all about very fundamental of an area of economics called welfare economics where we are going to basically try to define efficiency and we are going to say what is welfare economics is basically it talks about the wellbeing of all the participants in the market. So any kind of market equilibrium if it is what is it doing to the wellbeing of the producers and what is it doing to the wellbeing of the consumers.

So market equilibrium leads to allocation of resources. This we already discussed in the earlier lectures and we said that in the market equilibrium it is decided that the intersection of the demand and supply is going to give us the equilibrium price and quantity. So the quantity is decided in the market how much quantity of any good the supplier will be selling in the market and that decides how much of resources are going to be allocated and who buys this good. That is also decided through the market equilibrium people who are willing to pay the most for that good are going to buy that good.

So the market equilibrium that we get, is that an efficient outcome? That is also what we are going to see and what happens to economic wellbeing of producers and consumers and economic wellbeing of producers and consumers we define or we explain the economic wellbeing through something called the consumer surplus for the buyers and something called the producer surplus for the sellers. So let me explain.

So first let me explain what consumer surplus is. Let me start with consumer surplus and let me explain what consumer surplus is and so the consumer surplus I am going to illustrate through the demand curve.

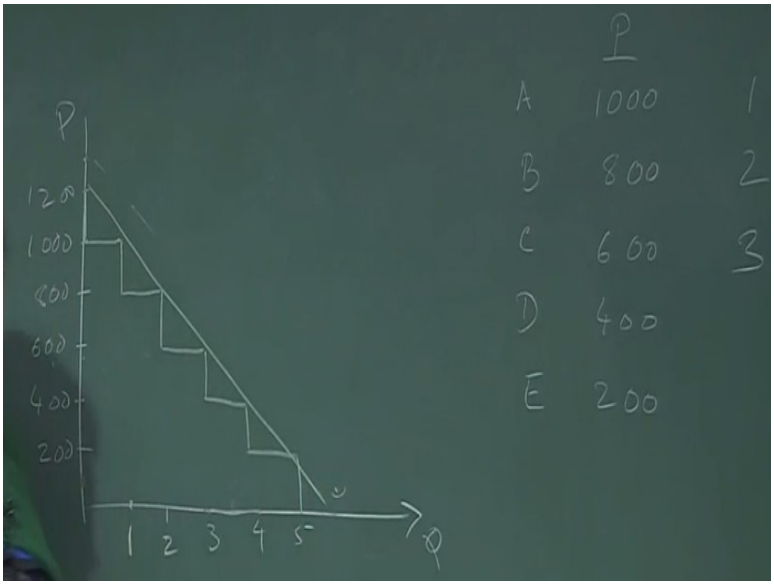
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So basically we are defining there is something called willingness to pay and what is the demand curve? What is the demand curve? Demand curve is basically every point on the demand curve every point on the demand curve so every point on the demand curve gives us shows us how much of the quantity the buyers are willing to pay.

So at so if the quantity is this much so if this is the quantity of say 10 units buyers are willing to pay an amount of say P 10. So these are the various willingness to pays of the so every point on the demand curve is the expression of willingness to pay by the consumers. So this is the willingness to pay by the consumers and let us start with a very simple economy.

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Let us start with a very simple economy where there are say 5 people A, B, C, D and E and say they are purchasing say t-shirts, t-shirts of IIT probably so they are buying t-shirts and the prices that they would be willing to pay so they have the demand curve they come to the market with some kind of demand curve.

So may be A is willing to pay a price of 1000 per t-shirt. Now B is willing to pay a little less. He would like to pay 800. C pays even less say 600 and D pays and E has a very low willingness to pay of say Rs 200. So in such a situation, in such a situation what happens in the market if the price is more than 1000 Rs? So if the price is more than 1000 Rs there are no takers. So there are basically 0 people who are willing to buy this t-shirt.

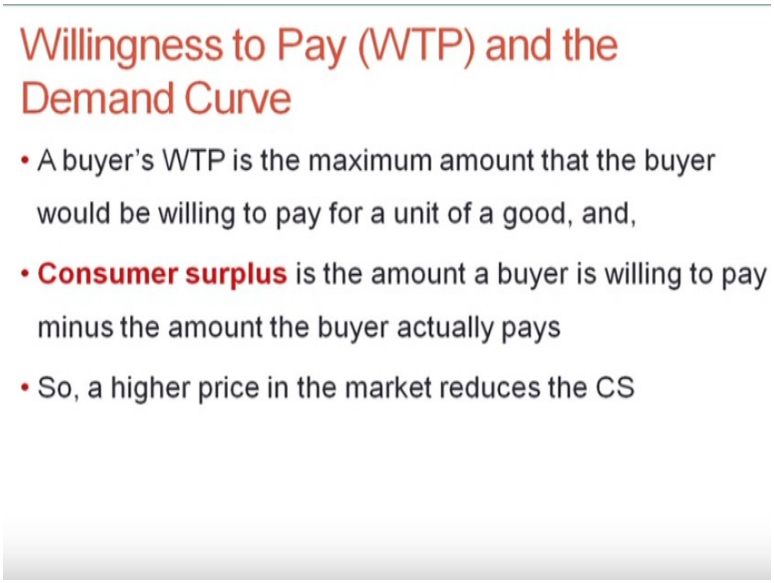
When the price is 1000 there is one person who is willing to purchase or in terms of t-shirts you can say that there is 1 t-shirt can be sold or if the price is 800 there is one more person. If the price is 800 B is willing to purchase and obviously A is also willing to purchase because his maximum willingness to pay is 1000. So here 2 t-shirts will be sold.

When the price is 600 C enters the market. He is willing to buy one, B is willing to buy one, and A is also willing to buy one. So here the price is then 3 t-shirts will be sold. So this is how we so we can draw a step diagram over here. So if the prices are 200, 400, 600, 800, 1000, and say 1200. So if the price is more than 1000, 0 people are willing to buy the t-shirt.

So when it is 100 one t-shirt will be sold; 2, 3, 4, 5. So basically it is 0 then 1 t-shirt will be sold and here 2 t-shirts will be sold 3 will be sold 4 will be sold and 5 will be sold. So basically the this is this step diagram that I have drawn from here this is this kind of looks like the demand curve. It is a negatively sloping curve or where the price when the price is falling the amount demand is increasing.

So this is a step diagram and this is how the demand curve is drawn and imagine a market where there are infinite number of buyers. Lot of plenty of buyers are there. In that case these steps will come closer to each other and gradually when the number is too high you we are going to get a smooth demand curve. We are going to get a continuous demand curve.

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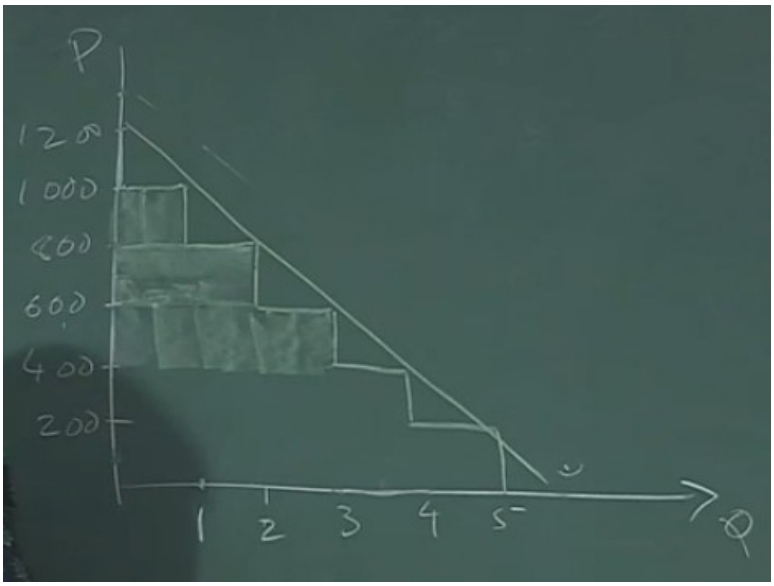


Now so coming back to the slides a buyer's willingness to pay is the maximum amount that the buyer would be willing to pay for a unit of good and consumer's surplus is the amount a buyer is willing to pay minus the amount the buyer actually pays. So let me again explain through the diagram. So basically when the second when the price is 800, B is paying a amount of 800 which is exactly equal to the amount that he would like to pay. It is not less than that.

But for A he is paying 800 so for him there is an extra benefit that he is getting by buying the t-shirt at 800. So for A there is a surplus. There is a kind of surplus of consuming this t-shirt for a

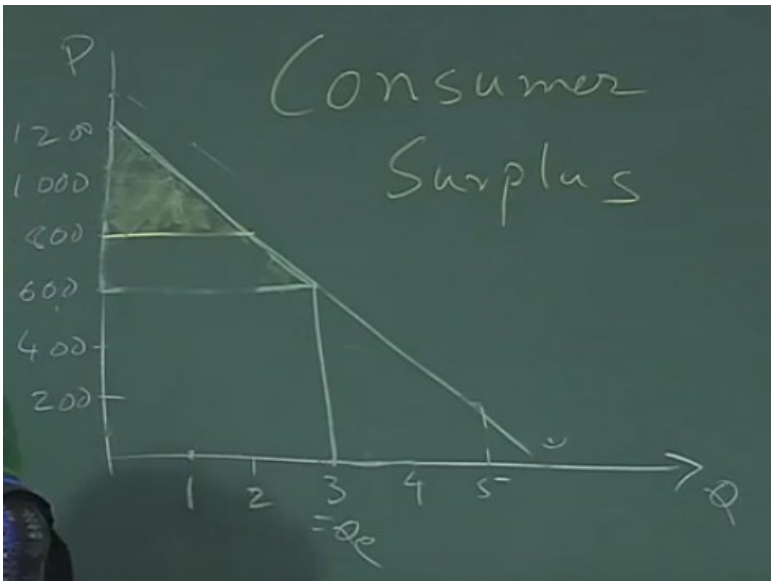
price of 800 so basically A gets a surplus of 200. So in case of when the price is 600, A gets a surplus of 400 and B gets a surplus of 200.

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So basically what we are seeing here is that all this area this is the surplus. So for A this is the surplus when he buys it for 800 Rs. For both of them this is the total area of the surplus when it is when the price is 600 this entire area of 400 this is for A and additional this amount of 200 is for B. So similarly if the price goes down even lower, then this is the amount of surplus that all the people who are entering the market or buying the product they are going to get.

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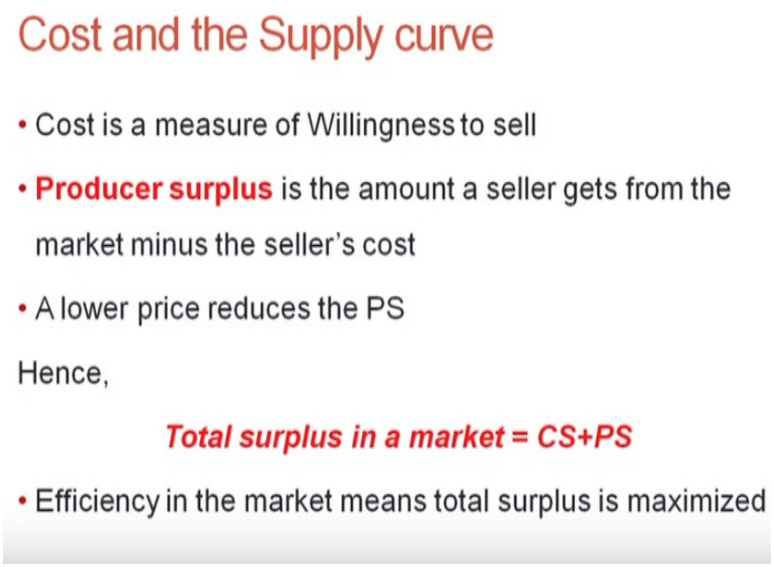
So if it is a very smooth lot of people are there and it is a smooth demand curve we can imagine if it is a smooth demand curve it is easy to imagine that if the price is 600 all the people who are buying the t-shirt and who had willingness to pay more than 600 they all have entered the market and they all are going to buy the t-shirt for 600 Rs and together they are going to get a surplus equal to this area of the triangle. So they are going to get a surplus which is equal to the area of the triangle.

So this is the market if this is the market price if the say P e is the market price if the market price P e is 600 all these people if the market price is 600, 3 t-shirts will be sold and all these people who were who had willingness to pay in this region, in this region every person if every buyer in this region has a willingness to pay which is more than 600 or equal to 600, this last person has an equal to 600. So all the people who are here they everyone is going to buy the t-shirt and for all these people this is the consumer surplus that they are going to get.

So consumer surplus is the amount the buyer is willing to pay minus the amount the buyer actually pays. So a higher price in the market reduces the consumer surplus. So obviously higher price in the market reduces the consumer. If the price now goes up to 800 if the price goes up to 800 we lose the consumer surplus in this area and consumer surplus is now less, it is only this portion. So this is the consumer surplus now. So a higher price in the market reduces the consumer surplus.

Now that was our illustration of the consumer surplus. This is consumer surplus. So this is consumer surplus and now we are going to talk about the supply curve and we are talk about something similar the surplus that a seller also gets in the market and that is called the producer surplus.

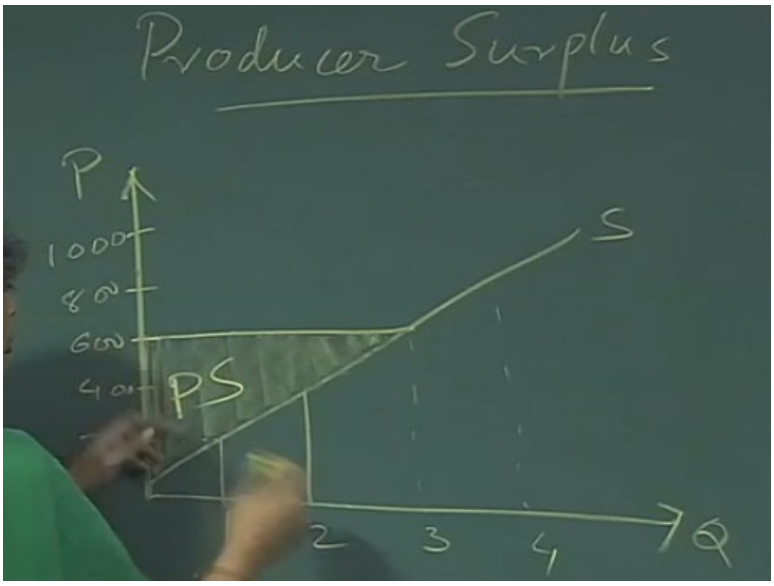
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So for the seller the cost is a measure of willingness to sell. So cost is a measure of willingness to sell. That is every seller has to incur some cost to supply a product in the market. So every seller has to incur a cost to supply a product in the market and that is the that cost is the minimum amount that he would like to charge to supply any product in the market and what is that cost? That cost is anything.

Any or the cost of all his ingredients, his labour, his opportunity cost every kind of cost that he incurs in supplying a product in the market that is the minimum price that he would accept from the market to stay in the market. If the price goes below that cost it does not make sense for he is going to make a loss and it does not make sense for him to continue to supply in the market. So basically cost is a measure of willingness to sell in the market.

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So let me draw a supply curve here. So this is we are trying to draw the producer surplus. So as I said this is the say this is the supply curve this is the supply curve of the producer and every seller every seller there are there are again like the same way that we had buyers there could be sellers in the market. So this is say 200, 400, 600, 800, and 1000. These are the prices and there is one seller in the market who can actually supply at when the price is 200.

When the price is as low as 200 there is probably one person who is willing to supply. When the price goes up 400 there one more person enters the market, he is willing to supply a unit and so he is willing to supply and the second the first person for him also now the price is more than his cost. So he is also willing to supply; so say 2, 3. So this way so as the price goes up in the market more and more sellers enter the market and they are willing to supply.

Now imagine a situation where the equilibrium price in the market that is arrived at P e is equal to 600. When P e is equal to 600 this last person the last seller who has entered the market because the price has reached 600 he does not make any profit. For him the cost is this is his willingness to pay and this is the price that he is getting. So he is good he is fine with it. He sells the product in the market. His price is equal to his equal to the willingness to sell.

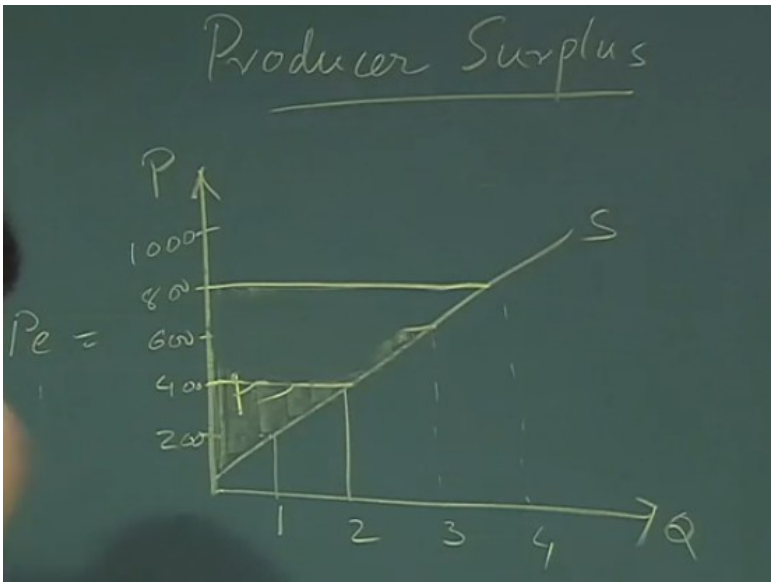
So he enters the market but every other seller in the market for whom the willingness to pay or sorry I am sorry it is not willingness to pay it is willingness to sell. So every seller in the market for whom the willingness to sell was at a price lower than 600, they all are going to sell in the

market. So they are all going to sell so imagine this first person the first seller in the market who has who operates at a very low cost.

His minimum cost is 200 or his willingness to sell is at the price of 200 that is 200 is the minimum price that he is willing to accept and because his cost is 200 now when the market price is 600 he gets a surplus. For him there is a surplus. So he gets a surplus. So the next person who is willing to for whom the willingness to sell is at a price slightly more than 200 he also gets a surplus.

So similarly every person here who have willingness to sell at a price which is lower than 600, they all get a surplus or a profit by being in the market. So they all get a surplus and this is the total amount of producer surplus or this is the total amount of surplus that the producer is getting in the market. So this is the total amount of the surplus that the producer is getting in the market and this is called the producer surplus or PS.

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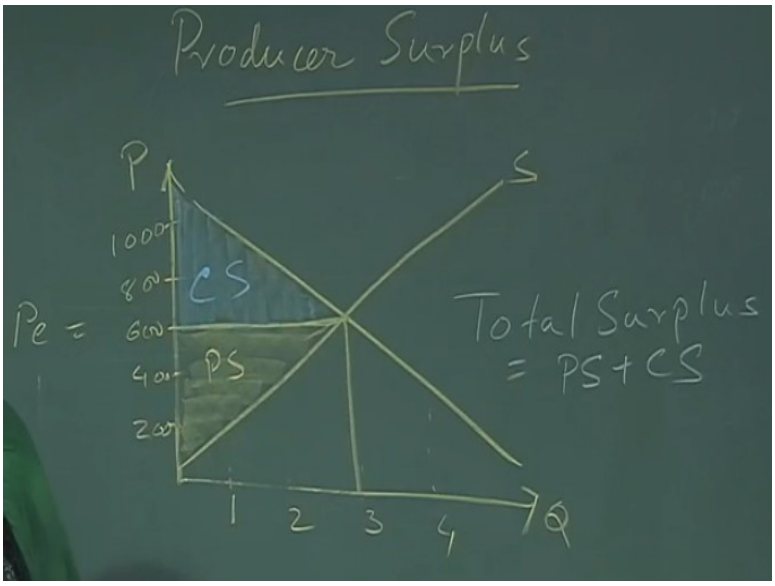


So what happens when the price the producer surplus what happens to producer surplus when the price goes up? When price goes up more sellers enter the market with more surplus say if it becomes 800 the producer surplus goes up. Now when the producer when price goes down if it is

400 this region of producer surplus is these producers all exit the market because price is now lower than their willingness to sell and so producer surplus goes down.

So basically producer surplus is the amount the seller gets from the market minus the seller's cost. A lower price reduces the producer surplus. Now hence total surplus in the market. So what is the total surplus in the market? Let us draw it neatly.

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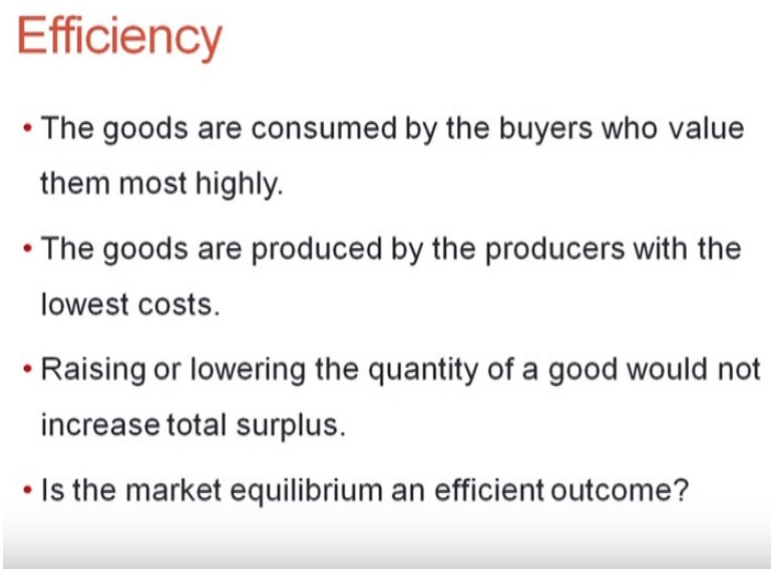
So this is the supply curve and this is the demand curve. This is equilibrium price; this is equilibrium price. Now which part is the producer surplus? As we already drew, this is the producer surplus, this is the producer surplus PS and this is the consumer surplus or CS. So when a market reaches an equilibrium they intersect it reaches an equilibrium at a price of the intersection of the two.

Both the producers and the both the producers and the consumers receive some surplus in the market and these surpluses are shown in this diagram. So this these surpluses are as shown in this diagram. So this total surplus in the market is then now what is the total so we going back to our original topic of discussion is efficiency. We were saying that if it is an efficient outcome. So we are saying that or we talked about wellbeing of the producers and suppliers in the market. So what happens to wellbeing of the consumers?

So wellbeing of the consumer is measured by this region. So the consumers wellbeing is the, this region and the producers wellbeing is measured by this region and total wellbeing of the market players or the total wellbeing of the economy is equal to this total surplus. So total surplus is equal to producer surplus plus consumer surplus. So total surplus is equal to this region of producer surplus and consumer surplus and efficiency in the market means total surplus is maximized.

So we have found a way of measuring the wellbeing of the consumer as this consumer surplus wellbeing of the producer as producer surplus and so the wellbeing of the entire market is maximized if we can show that this area is maximum in the case of the market equilibrium or in other words if we are asking the question that if the market is left to itself the outcome that happens in the market the demand supply intersection and the market equilibrium that we get is that maximizing the wellbeing of the market? If it is maximizing the wellbeing of the market then it is an efficient outcome. This is what we are saying and this is what we are trying to find out.

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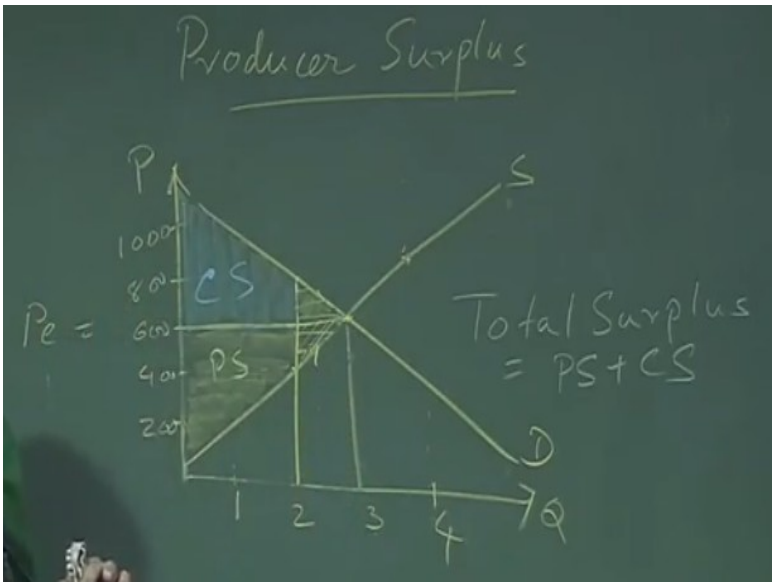
So let us try to understand what exactly we mean by efficiency. Efficiency a market outcome would be efficient if the goods are consumed by the buyers who value them most highly and the goods are produced by the producers with the lowest cost. So is that happening in this situation?

This is what we are asking that we will say that a market equilibrium or a market left to itself the equilibrium that is reached or the market outcome we are going to call this efficient if the goods are consumed by buyers who value them the most, value them the most and the goods are produced by the producers with the lowest cost.

Raising or quantity raising or lowering the quantity of a good will not increase the total surplus. So is there any way that the equilibrium that we have reached is there any way of increasing this consumer surplus or producer surplus? Any other level of output or any other level of price is that going to give us a better outcome in terms of maximizing the CS and PS?

This is what we are trying to see here and we are trying to answer the question is the market equilibrium and efficient outcome so finally we are asking that is this the most efficient outcome. Now looking at this diagram what we can see is what happens if what happens if quantity falls.

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Say for example if it is decided that due to whatever reason that instead of 3 units 2 units will be produced and so the so if 2 units are produced what happens is the suppliers would the suppliers would be charging a smaller price and the demand so and the buyers would be the buyer since they are getting less than what was available in the market so the price goes up they would be willing to purchase more at a lower price and similarly the supplier see that it makes sense to

increase the production in the market because they will be able to sell in the market so basically both of them move towards a so the producers basically start increasing they see that there is demand for the product in the market because the buyers are willing to pay a high price.

So obviously they start producing more to get a higher price so they move towards this equilibrium and similarly the consumers also do so and in both the situations and what happens to producer surplus and consumer surplus? What we see here is if they are producing if in the market the market is producing 2 units then the price the demand the producers the buyers are willing to pay is here and what and the amount of consumer surplus and producer surplus that is foregone is given by this area.

So in both the situations they see in this situation the producer see that it makes sense for them to keep on increasing a little increasing production a more little more to a greater higher price and they move towards this point so their producer surplus is enhanced and similarly consumers and as the production increases the consumers also the amount of availability in the market goes up and prices go down and more and more buyers enter the market so the consumer surplus also goes up by this amount. Similarly, the opposite is going to happen if the quantity somehow the production is say at here 4.

So if the production is here there is too much of supply at a high price and there is and the price that the buyers are willing to pay is low. So here again the same thing happens and they move towards this point and obviously since here the supply is here the looking at the producer surplus and consumer surplus point of view here the price the quantity is high and here the price is that the supplier would like to charge is at this point and there is there are no takers for this at this price.

So obviously they are going to reduce their, the production and come back to a lower price. They are going to come back to a lower price and hence and hence producer surplus gets maximized over here. So here the supply when the at these points the cost is too high for the supplier and for the buyer the willingness to pay that is the price that they are willing to pay is much lower than the cost of the suppliers. So they here there is no region where there can be any producer surplus or consumer surplus. So basically this is the region where total surplus is maximized and market equilibrium hence actually is the most efficient outcome.

In the following classes we are going to discuss the different policies of the government that we already discussed in the context of this efficiency. We are going to see what happens to producer surplus, consumer surplus etc. when there is a regulation of price when there is price regulation what happens to producer surplus, consumer surplus, or efficiency in the market. Thank you.

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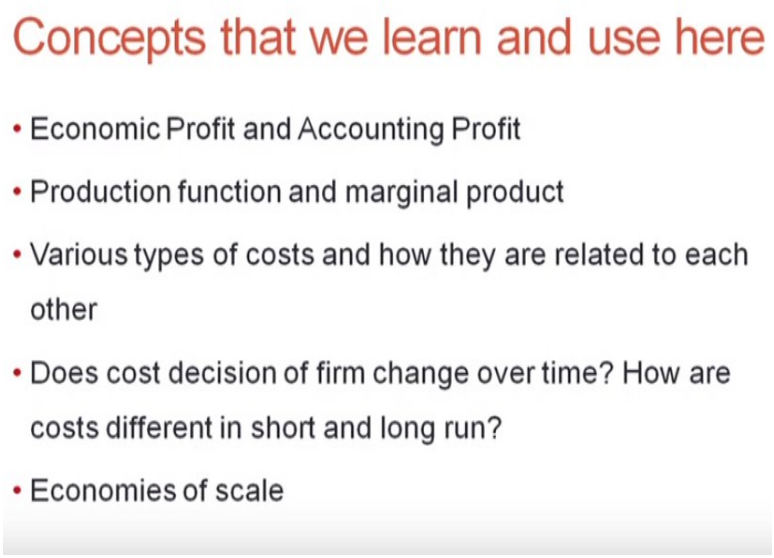
Lecture - 15

Production and Cost

Hello and welcome back to the foundation course in managerial economics. We completed our discussion on demand and supply analysis in the previous lectures and today we are going to introduce the, we are going to discuss about costs of production and production, what are cost curves.

We are basically going to define a few technical concepts which we are going to use throughout the course, later part of the course and we are going to introduce the concept of cost, the concept of production output, marginal product etc. and the discussions that we are going to have this week is going to form the foundation of most of the discussions that we will have about market structures in the consecutive weeks and we are going to see that how producers make their decision in different kinds of market structure looking at the cost curves looking at the concept of analysing cost, that we are going to do this week, how the different producer how the producers in different types of market structures they use these concepts of cost and production to decide how much output to produce, what price to charge etc. So let us start.

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So the concepts that we learn and use here are we are going to talk about something called economic profit and accounting profit. Now profit as we know it is anyone if we ask what profit is it obviously determines the difference between the revenue that we get and the cost we incur to produce that or generate that revenue.

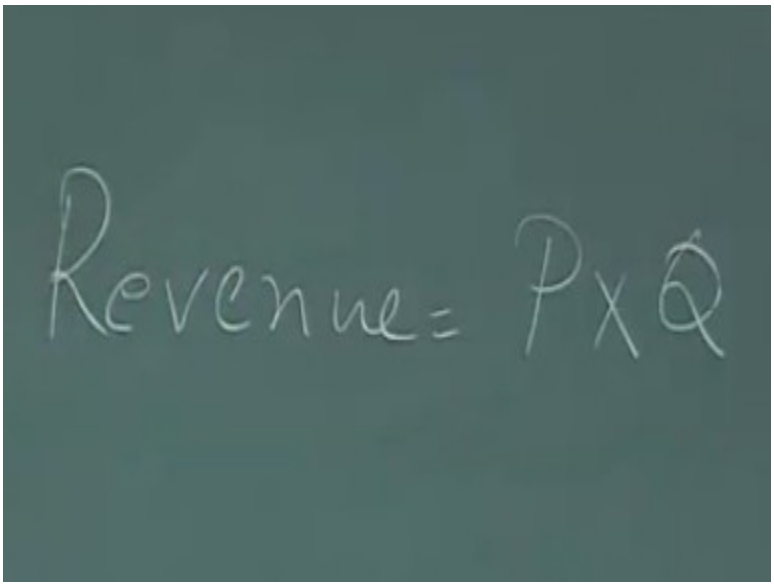
So, but yet we are going to see that there is a difference between what we are going to call economic profit and what we call accounting profit and economic profit is something that we are going to consider during the entire course. So whenever we discuss any market structure we are going to whenever we say profit we mean economic profit and not accounting profit.

Then we are going to discuss about production function about marginal product because as we discussed in the very beginning of the course that people make decisions or choices at the margin. So we are going to so this kind of dreams about our use of the concept of marginal in every sphere of discussions that we are going to have henceforth and marginal product is one of them and we are going to see what it is.

Then we are going to discuss about various types of costs and how they are related to each other. Now does cost decision of firm change over time because as we know that a firm has to make some short run decisions like a producer or a firm is deciding how much to produce in the within the say next month and the firm has to make certain long run choices long run decision.

Say for example a company needs to decide how much where does it see itself in say 2 years' time like is it going to produce the same amount of output that it is producing today or would it like to produce more or would it like to produce less so depending on the market depending on the kind of costs it faces it needs to make these choices so obviously that means that the cost is not exactly similar in the short run and in the long run and we are going to discuss what we mean by that short run cost long run cost and we are going to discuss about something called economies of scale.

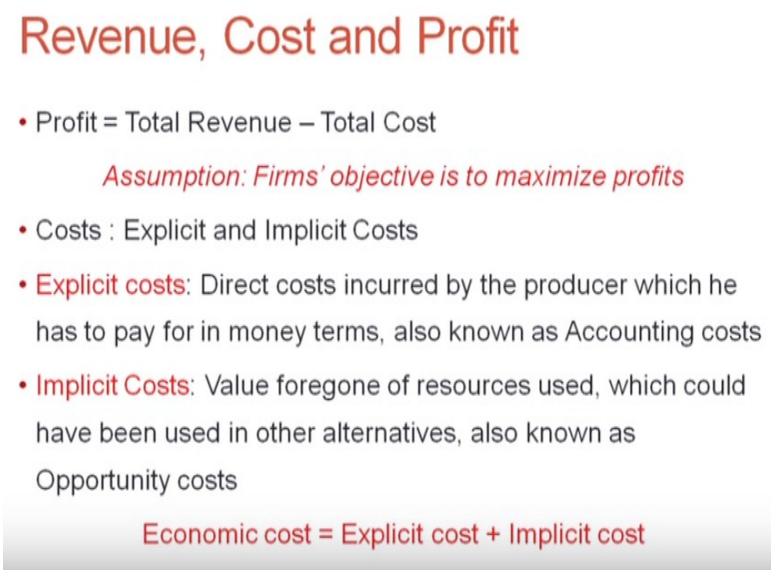
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So let us define revenue, cost and profit. So revenue everyone knows revenue is basically price multiplied by the number of units we are able to sell. So revenue is basically price multiplied by Q where Q is the number of units of output and the price that it fetches in the market.

So in the discussion in this week's discussion we are not going to talk about P or where P comes from P is exogenously given and we are going to discuss we are going to try to understand that where does Q come from what does Q look like. For a producer he needs to understand what is his production choices would look like and what his costs would look like so we would like to we are going to talk more about this Q in this week.

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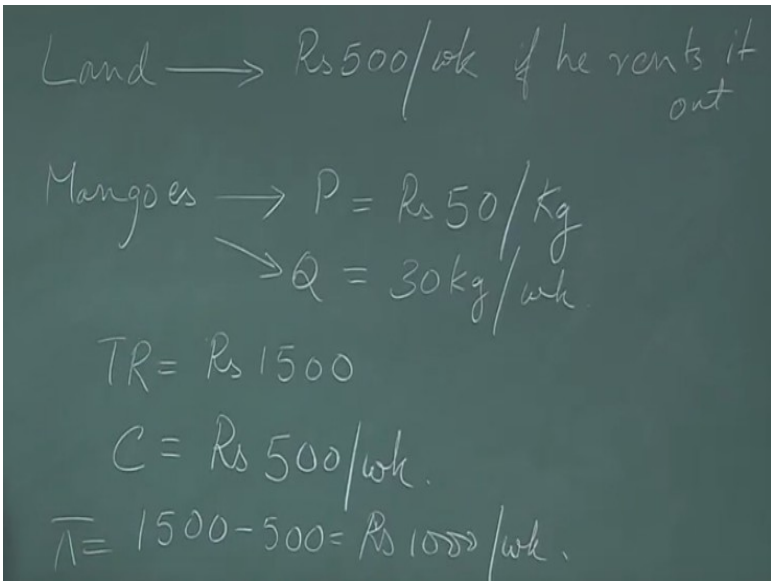
Coming back to the slide so profit is total revenue minus total cost and the assumption that we will be making throughout this lecture is firms objective is to maximize profit so not only in this lecture in the entire course we are assuming that the firm's objective is to maximize profit.

There is no harm in assuming that the that a firm's objective is maximizing say revenue or say minimizing cost but for simplicity let us assume that the firm's objective is maximizing profit but that does not mean that the firm that a firm always maximizes profit. It could have some other kinds of objectives also depending its depending on its long term strategy etc. but we are not going to going into those details in this course. It is beyond the scope of this course.

Next we are going to introduce something called explicit cost and implicit cost. Now explicit cost is basically defined as direct costs incurred by the producer which he has to pay for in money terms, also known as accounting cost and implicit cost is the value foregone of resources used, which could have been used in other alternatives, also known as opportunity cost.

So let me let me explain with the help of an example. Now say for example there is a person who owns a small piece of land and this land if he decides to rent this land out in the market probably he will get say Rs 500 per week

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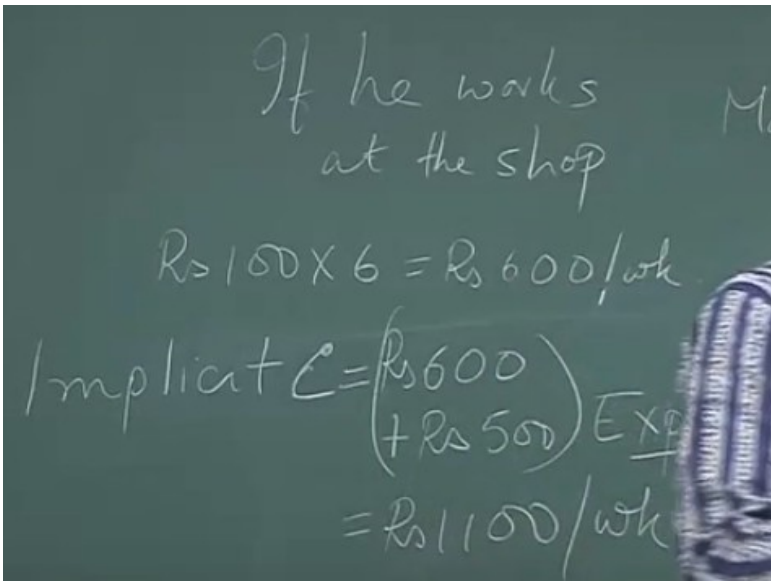


So there is a piece of land and this person has to decide he has a piece of land and he can get land can fetch him Rs 500 per week if he rents it out but instead say the person decides that I am going to use this land to grow mangoes and I am going to sell the mangoes in the market. So for example if he uses this piece of land, the amount of mangoes he is able to produce is say he is producing mangoes and he is which fetches the price in the mango price of mango in the market is say Rs 50 per kg. He produces say 30 kgs per week. He sells in the market.

So what is his revenue? He is getting a revenue of Rs 1500. What are his costs? What are his cost? The cost that he is incurring for producing the mangoes? Say for example he has to purchase seeds in the market, some fertilizers, some pesticide etc. in the market which costs him say so his cost is Rs 500 per week. This is the amount that he has to spend per week or he has to pay per week to produce this amount of mango or to get so much amount of revenue in the market.

So in that case his profit would be total revenue minus total cost, we would think that the profit is 1500 - 500 = Rs 1000 per week. Now here what we are missing out is he is could have rented out this land in the market and would have earned an amount of rupees 500 per week and say for example now given this situation say for example now the person knows that there is a nearby shop who are hiring people and this shop is going to pay him Rs 100 per day if he decides to work over there. Now assume that he is when he works over there then the amount say he works and the shop is open for 6 days a week.

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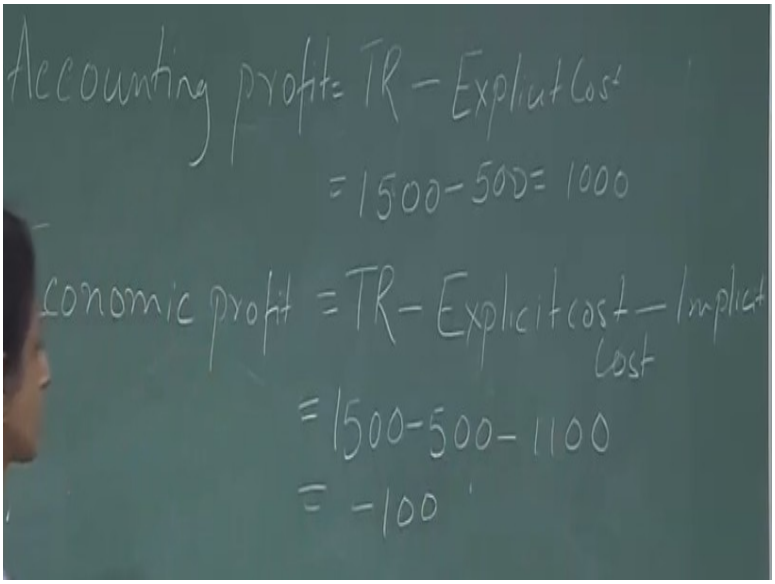
So in that case he is able to if he if he works at the shop he is 6 days a week he is able to earn Rs 100 into 6 per week so is equal to Rs 600 per week. So this is what he is able to earn per per week. Now what so in this calculation if we say that his profit is Rs 1000 per week and this is what is called the accounting profit I am just coming to that before let me explain about the implicit and explicit cost.

Now if we calculate this the profit as I have done here then what we are leaving out is what is the opportunity that he is missing? What is the opportunity that he is missing by growing mangoes? What he is missing is he could have rented out the land and earned an amount of Rs 500 per week. He could have worked at the local shop additionally like additionally he could have rented earned some rent out of here and he could have earned Rs 600 per week from here. These both these earnings he is having to forego because he has decided to grow mangoes and sell mangoes in the market.

So what we say is his cost is his explicit cost is Rs 500 per week but his implicit cost his implicit cost he is having to incur an implicit cost also. So his implicit cost is equal to Rs 600 + Rs 500 per week = Rs 1100 per week. So this is my definition of explicit cost or accounting cost which is basically the cost for which I am having to pay by cash.

This is something which I have to use some money to get the product to produce the output. But this is the cost for which I am not having to pay anything but this is something which I am foregoing. So this is the opportunity I am foregoing so this is called the implicit cost or the opportunity cost.

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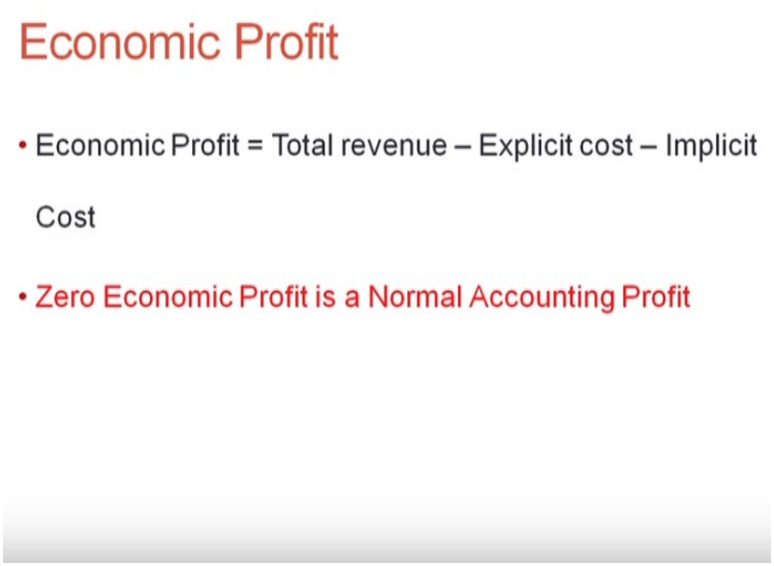
So this is my implicit cost or opportunity cost so accordingly I am going to define so accordingly I am going to define something called the accounting profit is equal to total revenue minus explicit cost which is equal to 1500 - 500 = 1000 in this case and I am going to define something called the economic profit which is equal to total revenue minus not only the explicit cost but also the implicit cost.

So this is equal to now 1500 - 500 - 1100 which makes my profit actually negative so this is -100. So basically this person is losing 100 rupees by choosing to grow mangoes instead of renting out his land and working in the nearby shop. So this is how we define cost in economics and this is how any producer to optimize his earning in the market that is maximize his profit this is how he should treat his costs and treat his profits.

So this is the this is the concept behind it and so explicit cost is coming back to the slide so the explicit cost is direct costs incurred by the producer which he has to pay for in money terms, also known as accounting cost and implicit cost is value foregone of resources used, which could have been used in other alternatives, also known as opportunity cost.

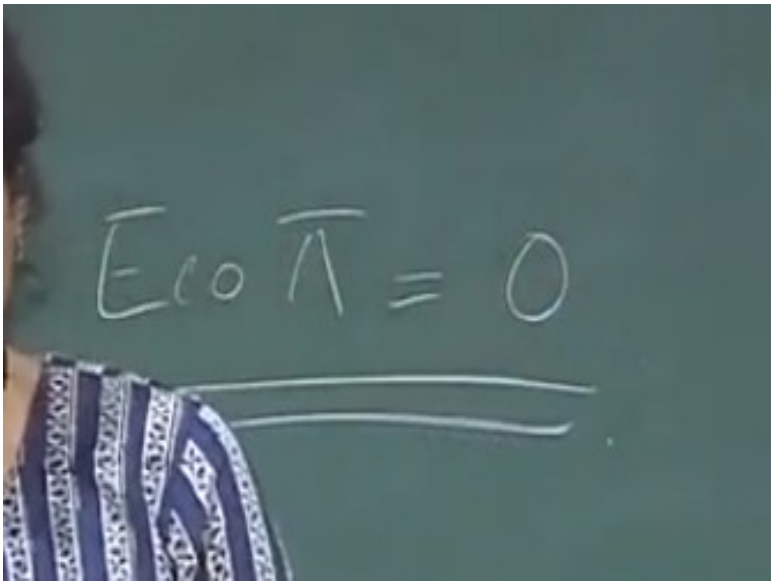
So implicit cost is also known as opportunity cost and opportunity cost is something which we are again going to use throughout the throughout economics. It is a very important concept in economics and I discussed about this even in the first lecture. So economic cost is equal to explicit cost plus implicit cost and accordingly we define economic profit.

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So economic profit is equal to total revenue minus explicit cost minus implicit cost. Now what happens when economic profit is 0? So we are going to see that later when we discuss about perfect competition etc. we are going to see that in the long run all the producers earn a profit which is equal to 0. Basically we are going to see that in all our calculation ends up in total revenue being equal to total cost.

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So what happens when economic profit now pi is a very common form of a common symbol for profit in economics and we are going to use pi for profit in most of our lectures. So economic profit equal to 0. What does this mean? Does this mean that the firm or the producers is running a loss? No. It does not mean it is running a loss because it is also including the implicit cost in its calculation.

Since it is including the implicit cost in the calculation that means that the firm or the person or the producer is at least earning the amount that he would have earned had he not pursued this production and had he done something else. So he is at least earning the amount that he would have earned from his next best alternative.

So zero economic profit is a normal accounting profit. So zero economic profit is a normal accounting profit and this is also something which we are going to use very frequently during our discussion on market structure so it would really help to remember that zero economic profit does not mean that the producer is not earning anything or he is running a loss.

So that ends our discussion of this module where we have introduced the concept of explicit cost, implicit cost. We have introduced the concept of economic profit, accounting profit and the reason for doing so is in the future lectures we when we discussed about cost curves always remember that we will be using both explicit we are using both explicit and implicit cost.

So when we are basically calculating the cost of any producer we are assuming it is the it is inclusive of both the explicit and implicit cost. So we will be talking about economic profit. We will be talking about both explicit and implicit cost of the producer and that is how we are going to try to understand what the production decision of the firm looks like what does the cost curves or what do the cost structures in the short run in the long run how do they look like. So I will end this module. Thank you.

Foundation Course in Managerial Economics

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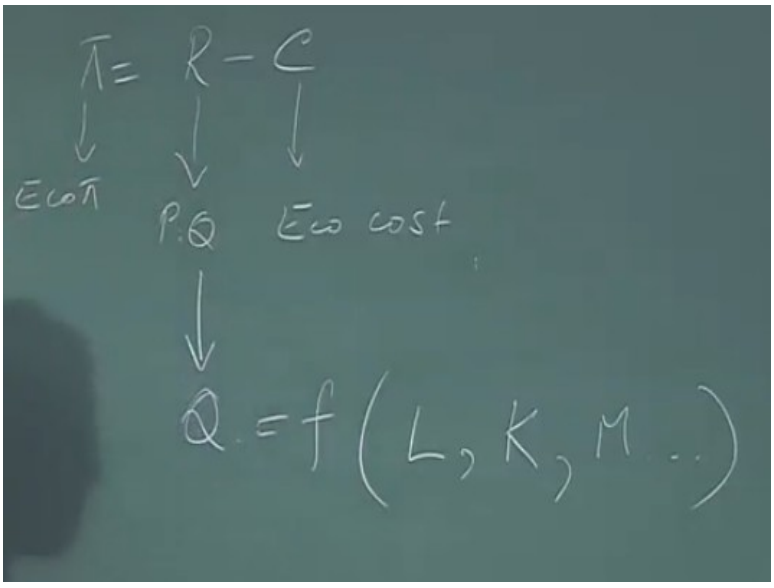
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Lecture - 16

Costs

Welcome back to our discussion on production and cost which we started with the previous lecture and we have already discussed about explicit cost, implicit cost, economic profit etc.

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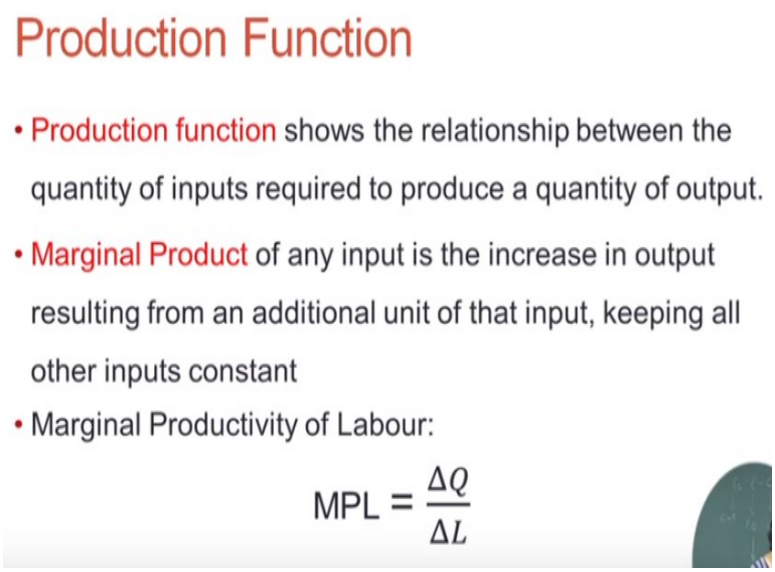
And we said that profit is equal to revenue minus cost and this is obviously economic profit and this is equal to P multiplied by Q and this is economic cost. So this is economic cost. So in this

module we will try to understand where does this Q come from. How do we determine Q? How much output will be produced how is that determined and basically production of any output like in the previous example we used the example of mangoes, mangoes getting produced.

So how much quantity of mango will be produced will be dependent on or will be a function of lot of inputs. It is a function of amount of say raw material used, number of seeds used, amount of pesticide used, fertilizer used etc. which is which will be probably the raw material, people working in the land people working on the land so labour then the land itself the amount of land that is there available to grow the mangoes. So we can call that the capital and if any machine is used to whatever machine equipment used in the that will be included in capital and raw materials etc.

So these are the kinds of inputs which are going to produce the output. So output is basically a function of lot of inputs. Now from here the producer has to decide how much output he would like to produce and before that he needs to know he should know that a certain amount of labour a certain amount of capital a certain amount of materials together is going to give me a certain amount of output. So this knowledge is essential for a person to start production in the first place. So this is called the production function. So this is not the how I explained it is not a definition of production function but this is how production how what we understand by production function.

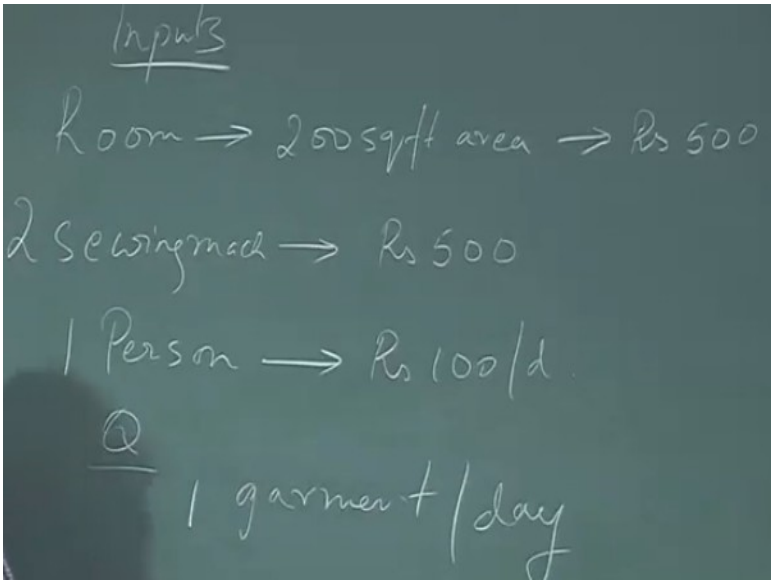
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So production function shows the relationship between the quantities of inputs required to produce a quantity of output and so this is our production function from here and before I go on to another technical term called the marginal product, marginal product is basically marginal products is about any input; say marginal product of labour.

So marginal product of any input is the increase in output resulting from an additional unit of that input keeping all the inputs constant. So that is the marginal product of labour. But before I go on to marginal product let me explain the concepts of production function, marginal product of labour etc. through a very simple example.

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Let us take the example of a tailoring shop. Let us take the example, a very simple example of a tailoring shop where basically there is a land say there is a land of rather than land say there is a room. There is a tailoring shop which is basically a room of 200 square feet area and there are 2 sewing machines. There are 2 sewing machines in that shop.

So the shop the what is the so what is the what so the room itself and the sewing machines they would be inputs in the production of garments in the tailoring shop. So there is a room of say 200 square feet area and say the so this is one input. So these are the inputs.

So there is a room and this input has a so every input has a cost. So when we wrote the production function in the first place Q is the function of all these inputs every input has a cost and that is the that cost is what is of concern right? So room is 200 square feet area and it has a rent of say Rs 500 okay so whatever like this is a flat amount of 500 that per week or month or whatever he is paying.

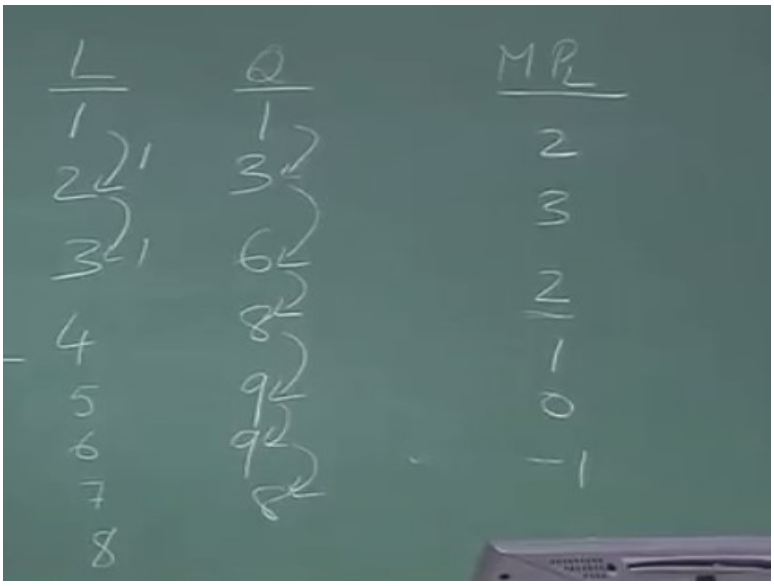
Secondly there are 2 sewing machines. So there are 2 sewing machines and sewing machines either they have been bought say for example someone has borrowed the money to buy the sewing machine or so basically you can imagine that the for the sewing machine he has to pay say Rs 500 say it is 250 per sewing machine that he has pay to whoever he has bought the machine from or borrowed the money to buy the machines from. So this these are the cost.

Now even before he buys purchases any cloth to produce garments a tailor or a person who decides to start the tailoring shop he is incurring these costs. He is incurring these costs. Now imagine he has engaged one person now he hires a tailor or he hires a person okay. He hires one person he hires one person and say he pays him Rs 100 per day.

So he pays him Rs 100 per day and he hires this person who is going to stitch clothes for him. He is going to stitch garments for him and so these are his inputs and when this person comes he starts so let us ignore about the let us not talk about the cloth etc. So imagine everything is included here or so this person is getting an amount of Rs 100 per day and he is able to stitch. So now we are talking about Q.

Next is we are talking about Q. So this person is able to stitch one garment per day. So when the person starts off with a shop he has to incur the cost of this entire 200 square feet area of room for Rs 500. He has to pay for the 2 sewing machines and then he hires a person and he gets to gets the output of one garment per day. So since there is only one person there is no way that he can increase the number of garments. So the only way he can increase his output is by hiring more people. So he can do this by hiring more people.

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So now in the same room with the same amount of sewing machines let us say, so now he has a fixed input of land plus sewing machine are his inputs which are fixed, he cannot change that. Even if he does not produce any garments he is having to pay for this these. So say for example he is paying 1000 Rs for these. So let us not talk about this part right now.

So number of people he is hiring is say first person he is able to stitch 1 garment. Then he hires one more person and as soon as there are 2 people there say due to whatever reason say the output increases to 3. Say for example this person really you know like sharing of knowledge two people are there, they are talking. So they he feels more motivated and they produce more clothes. It is not that 2 people are there 2 garments are getting stitched. So there is may be sharing of knowledge or whatever so little more number of garments are getting stitched here.

Now say for example he hires 3 people. So then we see that the number of garments that are getting stitched is 6. So although there are 2 sewing machines now maybe there is some division of labour so 2 people are stitching on the sewing machines and the third person is helping with the say buttons, some amount of thread work, this, that. So basically the productivity increases and there is more amount of output getting produced here.

Then he increases the number of people that he hires here to 4 and number of garments getting stitched is 8. Then say for example so similarly he increases 5, 6, 7, 8 and the for the fifth person that he hires his number of garments increases by only one unit. So there is only additional unit

that is getting stitched. When he hires the 6th person he sees that the number of garments that is getting produced is not increasing it is still the same.

He is able to produce still 9 units of garments and if he hires the 7th person he actually produces 8 garments. So why am I giving the example in such a way that what is the reason behind output getting stagnated after certain amount of input addition and later actually decreasing. What is the reason behind that or what is the intuition behind that?

The intuition is usually what happens is remember that the area of the room is still constant it is still 200 square feet and the number of sewing machines to work on that is also same that is only there are only 2 sewing machines. So now if you keep on increasing the amount of clothes or number of people who can work there they will be working with only 2 machines and in that land in that amount of space only.

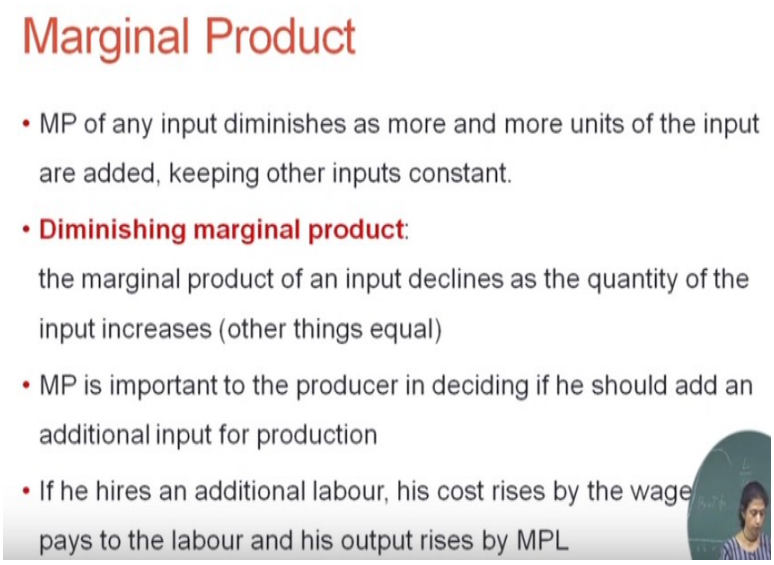
So they will probably not be able to increase the output any further and on the contrary it may quite happen that since the room is getting very crowded that there are too many people and as they say that too many cooks spoil the broth so similarly too many tailors are probably creating a mess over here and why the output is actually falling.

So this is very highly possible. So this is a very possible outcome which the producers are quite aware of and they know that the production just by increasing the amount of input there is only a limit to which you can increase the output and beyond that you have to increase the other inputs as well if you want to keep on increasing the output.

So in this situation now if at this stage if the person decides the producer decides that I am going to increase the space of the of my shop or better still I am going to add one more sewing machine then again probably his output is again going to grow much more than this. So but since he is not doing this so what we see is the additional output that the labour the input called labour is able to add or the that is able to contribute keeps on after a point of time it keeps on decreasing.

So initially it keeps on increasing as there is lot of fixed capital or fixed machineries to work with and gradually it gets saturated and later after a point of time the amount of output additional output that is produced it keeps on declining. So this is what happens.

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So then we are going to so then what is marginal product that we are talking about. So marginal product is MP of any, marginal product of any input diminishes as more and more units of the input are added keeping other inputs constant. So here coming back to this example then, what is the marginal product of any input. So what is the marginal product of the first input? So the first input that is added here 1 unit of labour is been added and what is the increase in output 2.

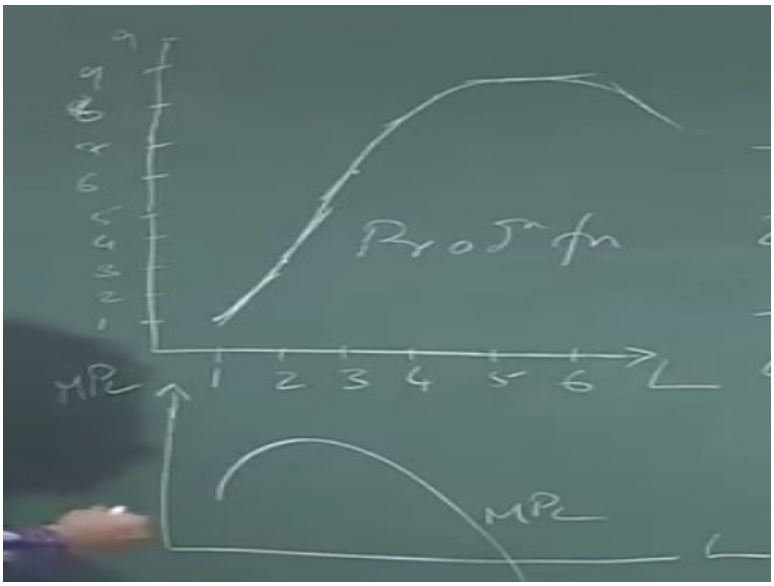
Now here again 1 unit of labour added, increased out is 3. So basically marginal product of labour is here marginal product of labour is 2. Here it is marginal product of sorry let me write it neatly. So from here to here marginal product of labour is 2. From here to here it is 3. From here to here it again falls to 2. From here to here it is 1. Here to here it is 0 and here to here it is actually - 1.

So this is how marginal product keeps on, initially it increases but then it keeps on diminishing. So this is called the diminishing marginal product. So this is called the diminishing marginal product and so diminishing marginal product is the marginal product of an input declines as the quantity of the input increases other things equal. So MP is very important to the producer in deciding if he should add an additional input for production.

So you can very well figure out from here that a person who is deciding how many units of labour to hire, to produce the amount of output, he is going to look at the marginal product. He is going to basically think at the margin.

He is a rational person and he is going to think at the margin and he is going to see that here if I produce initially if I hire 1 person here if I hire this person here I am getting 2 additional number of garments. If I hire 1 more person, I get 3. If I hire yet 1 more person I get a little less but still it is positive, still there is increase in the amount of output. So this is what he is going to consider. So before I move on to talking about costs or talking about his decision making process let me also draw the diagram, what it looks like.

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So what does the production function basically look like? So the production function basically looks like here is the number of units of labour he is basically deciding the producer is deciding whether to hire one more person or not. So say 1, 2. So here is the output that he is getting out of every additional hiring.

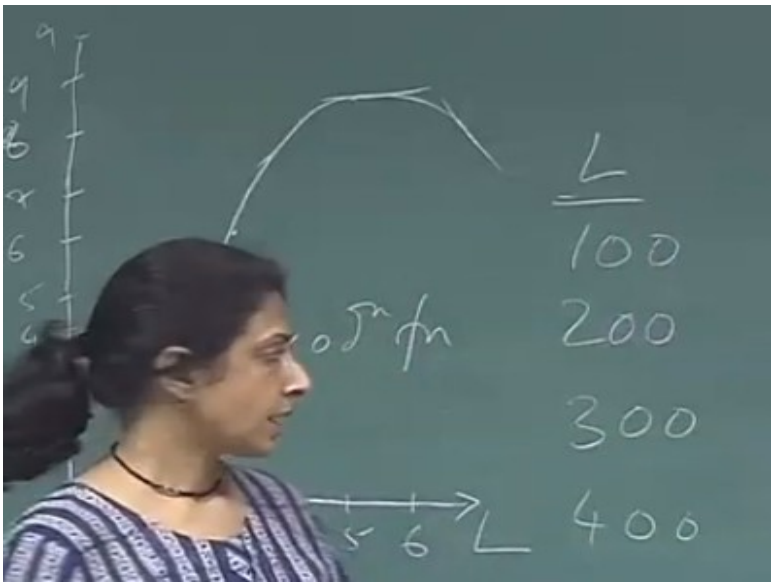
So for the first person he gets 1 unit. For the second person he gets 3 units. For the third person he gets sorry let me extend it. So third person he gets 6 units and then 8, 9, 9. So person for person 8 then 9 then again 9 and then declining. So basically his production function basically

looks like this. So production function looks like this and what does the marginal product look like.

Now so this is the production function and what is the marginal product. Firstly, the marginal product is the slope of this curve at every point. So this is the slope of the curve at different points and we are going to see that the slope actually initially increases and then becomes constant and then decreases. So this is the slope of the curve and if we plot the slope of the curve we are going to see that it basically looks like this. So this is marginal product of labour where you have labour on the x axis and marginal product of labour which is basically units of output in units of output. So marginal product of labour curve looks like this.

So coming back to the slides, coming back to the slides what we were discussing we said why is marginal product important. Why are we so concerned about trying to understand what marginal product of labour is and the reason that we are interested in this is because the producer while hiring the people what, he has to make a tradeoff here. So what is he doing here?

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So basically, now every person that he hires and say it is a fixed amount of he is paying an amount of 100 Rs per person. So every person that he hires he is having to incur a cost of first

one 100, second one 200, third person he has hired 300, 400. So he is having to incur these costs and for every person that he hires he is getting a margin additional output of these units.

So obviously while hiring he is going to say see if I hire the next person this is the amount of cost additional cost that I have to incur which is Rs 100 in this case and the additional revenue that I can hope to get by hiring this person is basically this additional amount that is getting produced multiplied by the price. So this is the tradeoff that he looks at.

He knows that if I hire one more person I am going to get some additional output out of him which is the marginal product of labour at that point but I have to incur an additional cost of hiring this person which is the wage that I am paying to this person. So if the amount of output gives me a revenue which is higher than the cost I am having to incur by hiring this person, I am going to hire this person.

But if my production or if my revenue falls below that I may not like to hire this person. The reason I said I may not like to hire because there are other costs involved, other decisions involved. We are going to look at the output choice of a producer later when we actually do the market structures but right now it suffices to know that marginal product is definitely important to understand because the producer makes his choices at the margin and he is going to look at his benefit and cost at the margin that is additional benefit he gets by producing an additional unit of output from hiring another person and the cost that he has to incur.

So this is why MP is important to the producer in deciding if he should add an additional input for production. Now if he hires an additional labour, his cost rises by the wage he pays to the labour and his output rises by marginal productivity of labour and this is why it is important to understand.

In the next module of this lecture we are going to look at cost curves and we are going to see, it will be basically a continuation of this discussion, and we will be able to see what the cost curves look like and we are also going to talk about the fixed cost which we have not spoken about in this module. We are going to talk about those also. Thank you.

Foundation Course in Managerial Economics

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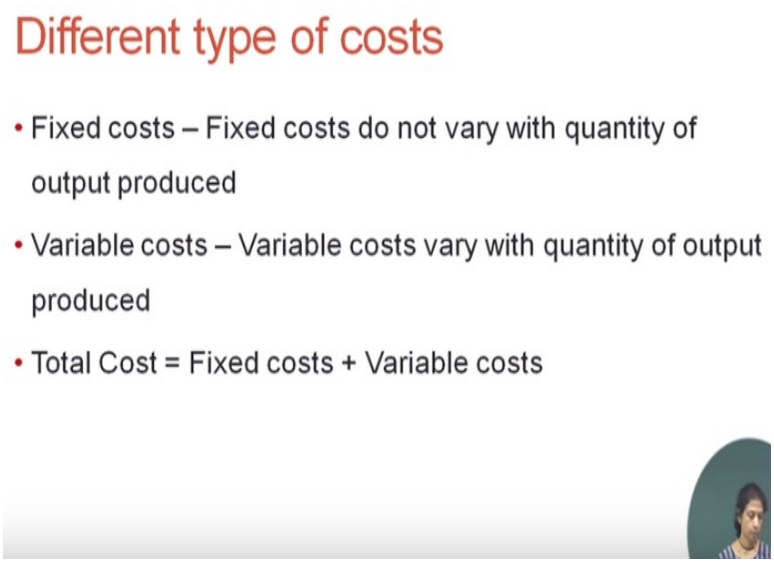
Indian Institute of Technology-Kharagpur

Lecture - 17

Cost Curves

Welcome back to the foundation course in managerial economics. In this module we are going to talk about cost curves. We are going to look at how the variable cost looks like with increasing units of output, how fixed costs look like. We are going to introduce various concepts of cost. So let me proceed.

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So there are different types of costs. Now one is called the fixed cost which is where fixed costs do not vary with quantity of output produced and we have variable costs where variable costs vary with the quantity of output produced. So probably you are being able to already relate to the example of the tailoring shop that I took in the previous module and there we said that there is a input of a room or the shop area of 200 square feet where basically the tailoring activity is happening and there are 2 sewing machines which are again fixed.

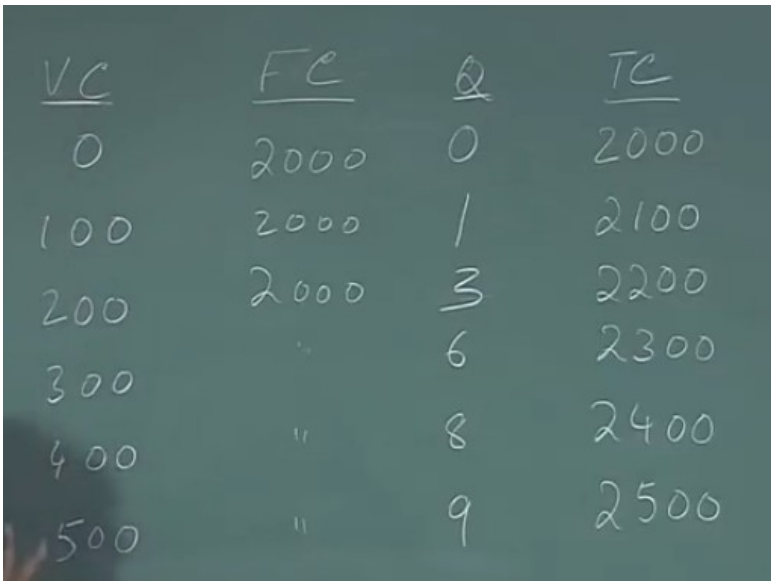
So even if the shop is not producing any garment then also the shopkeeper or the owner of the shop has to incur the costs of the room and the cost of the sewing machine and so these are the cost of the room and the sewing machine is basically the fixed cost that we are defining here and variable cost is basically if he wants to increase the number of garments that he wants to produce from his shop he has to keep on increasing the number of people he hires to produce the garment.

So however there is a diminishing marginal product of labour that we saw in the previous module that he cannot keep on increasing his output forever by keeping on increasing the number of units of labour. Say there is only a limit till which he gets some positive output of the out of the out of hiring more and more units of labour.

So anyway so these are the cost that he incurs by hiring the labour is called the variable cost because this cost depends on the number of units of output he would like to produce. So total cost for a producer is equal to fixed cost plus variable cost.

So next we are going to discuss about cost curves but before we talk about cost curves let me show with an example what we mean by the variable cost, the fixed cost etc. So let us continue with the example that we took earlier.

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So now he is trying to produce some output but first he incurs the fixed cost. He is incurring a fixed cost of the land and the sewing machines. So say he is incurring Rs 1000 here and for the 2 machines he is incurring an amount of Rs 1000. So basically his fixed cost is so let me write it as so basically his fixed cost or FC is 2000 and even when his output is 0. So even when his output is 0 he is having to incur this fixed cost of 2000.

Now he starts hiring labour and labour is his variable cost. His variable cost is labour and here if he has no person to work for him so his output is 0. As soon as he hires one labour his variable cost is not 1 but 100 Rs because one person gets a salary of 100 Rs. So this is variable cost is 100 and his fixed cost is 2000 and the output he produces is 1 garment. Then his variable cost is 200 as soon as he hires 2 people his fixed cost is still same.

He now is able to produce sorry he is able to produce 3 garments. So similarly so this is how his cost structure looks like 500 and these are the fixed cost stays same and 6, 8, 9, 9 and probably he is not going to hire this additional person because he is still able to produce 9 garments without hiring him. So we will talk about his decision till here.

So let us not talk about this portion because any producer, rational producer is not going to incur extra cost without increasing his amount of output. So this is his so what is his total cost then? So to produce every unit of output so this is his total cost is now fixed cost plus variable cost, for 0 units it is 2000, for 3 units it is 2200, for 6 units 2300, 2400, 2500. So this is his cost schedule.

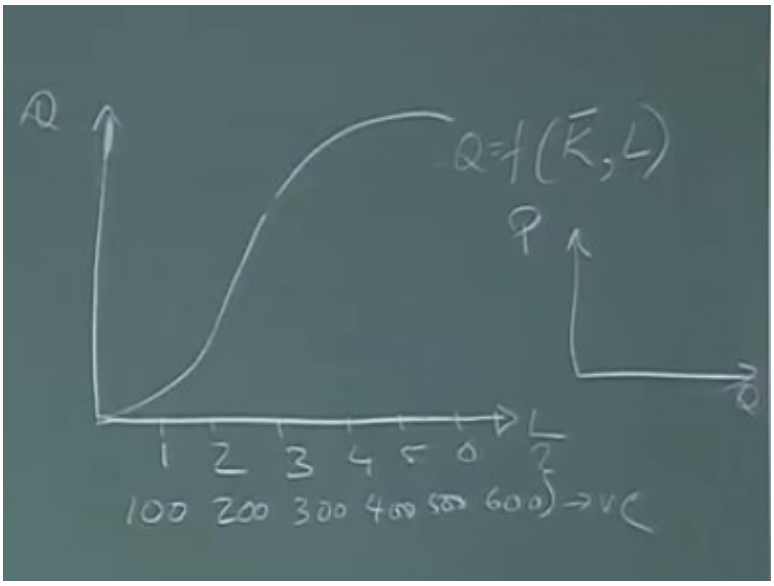
This is his total cost schedule and these are his and these are his output. So he knows it is like this.

Now let us try to draw the cost curve. Let us try to draw the cost curve because you already have seen the demand supply framework where we have a price on the vertical axis output in the horizontal axis and there was a demand curve the supply curve and similarly we would like to bring our cost curve or production decision everything in that framework of price and quantity and try to be able to see if we can figure out what is the amount of output optimum output the producer is producing, what is the cost he is incurring, what is the profit level that he is expecting, what is the profit that he is making etc. So let us try to draw the cost curve.

Now before I draw the cost curve let me start with the production function. See the production function basically gives me the okay before I talk about the cost curve one more thing is this is the variable cost and this is the fixed cost and when I say fixed cost it is as if right now the producer is not being able to change this like he is not at any point of time he is not adding any additional sewing machine or he is not renting another shop or renting some more amount of space to add to his existing shop.

So this is the reason that it is fixed cost. It is fixed. So he is not increasing the increasing or decreasing the capital that he already owns. All that he is changing is the variable cost. So basically all his decisions are about the variable cost. This decision he has already taken. So all his decisions are around the variable cost that is how many people to hire. Now so his so secondly his cost is about since this cost has already been taken into account like there is he knows that no matter how many units of output I produce this is going to stay the same. So his decision is about the cost incurred in hiring the labour, the variable cost. So let us start from the production function that we drew in the earlier module.

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So this is his production function and this was his labour, this was his output that he produced and the production function looked something like this. The production function looked like this. So this was the production function and where his capital was fixed and we were basically changing the units of labour. So this is for the fixed area of the land where he is operating and the fixed 2 inputs of machine, sewing machines that he has and he is basically just changing the units of labour here. So this is his production function.

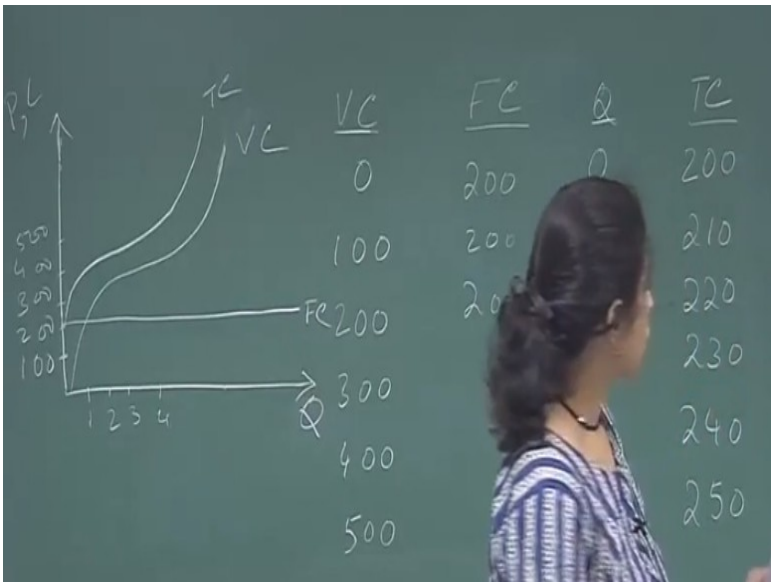
Now cost is about how much money he has to spend. So let us convert this let us look at instead of hiring every unit of labour so labour is basically every person that additional person he hires or additional number of hours that he makes them work. So if instead of writing labour here in physical units lets write it in terms of the wage that he is paying to the labour.

So if I convert it into rupees say first unit of labour he hires he is having to pay a wage of amount of money of 100 Rs. For 2 people he pays 200, 3 people 300, 4 people 400, 6 people 600. So this is in this is the cost that he is having to incur to produce every unit of output here. This is the cost that he is having to incur to produce or this is the variable cost. So this is the variable cost.

Now as you remember from our demand supply analysis that I said that my price is always going to be on the Y axis and my output is going to be the on the X axis. So I think it will be useful for us if we are able to look at cost also in this framework of having the price that I am having to pay on the vertical axis and the output getting produced on the X axis. So how can I get that from

here? So if I am looking at the cost curve how can I do that from this? So this is my production function what I basically do is what I basically do is let me draw on this side.

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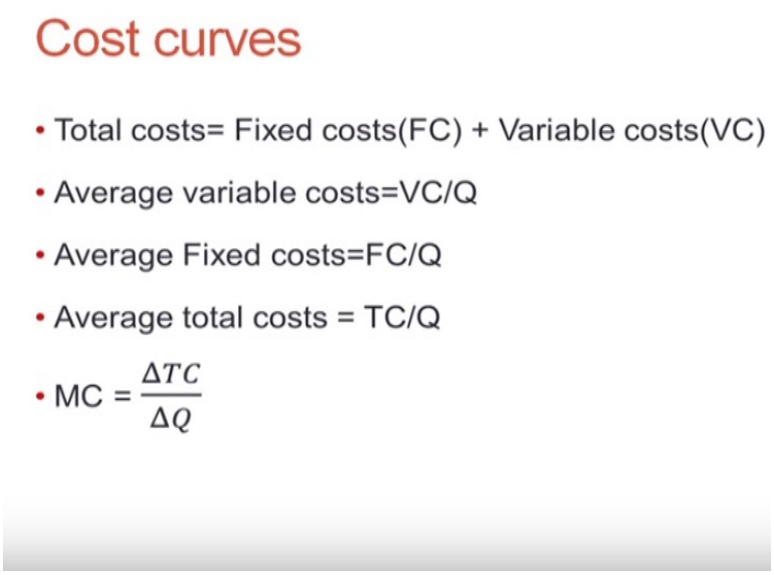
Now I am trying to change the axis of this graph or this diagram here. So I would like to have the price or cost figures on the X axis and sorry I would like to have the output in the X axis instead of Y axis and I would like to have the price or cost figures on the Y axis. To do that all that I have to do is flip this graph.

So basically I flip the curve so that I have the flipped curve in this space and what I now have is a curve which looks something like this, curve which looks like this. So when I flip the curve I have the output on the what output that I have on the Y axis output I have on the Y axis and I have the inputs in cost terms on the Y axis. Output I have on the X axis and I have the inputs in cost term on the Y axis.

So that basically gives me the cost curve. So is this what cost curve is this? This is basically the variable cost curve. So this is my variable cost curve because it shows me gives me the variable cost that is 100, 200, 300, 400, 500. So my variable cost is on the vertical axis, my output is on the X axis and this gives me my variable cost curve because this curve shows how much variable cost I have to incur to produce every extra unit of output.

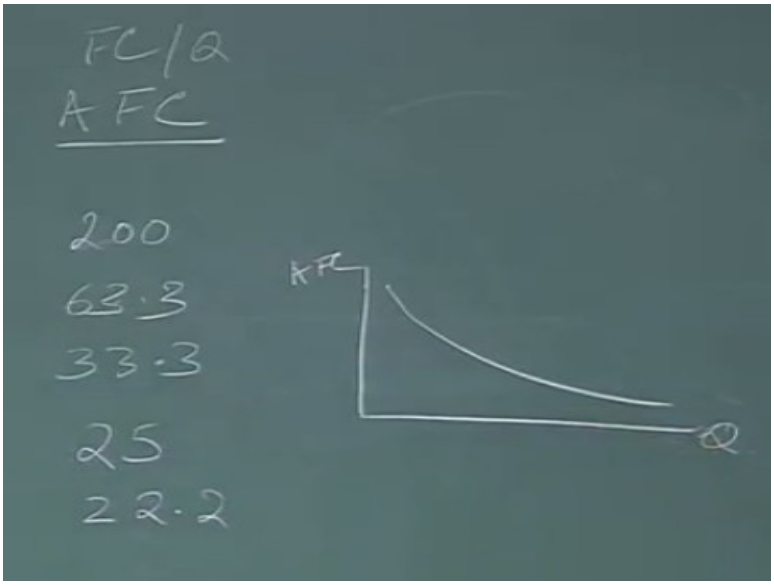
Now if this is my variable cost how should I draw my, what does my fixed cost look like? Now fixed cost is say fixed cost is say it is 2000 here let us bring it down to so that it is I am just making it 200 so that I can draw within this space. So say my fixed cost is 200. So my fixed cost looks like this. This is my fixed cost and so my total cost is basically this is my total cost. So this is my total cost. So this is 200 plus so this is my total cost curve and this is my variable cost curve.

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Now having drawn the cost curves a few more variations of the cost or different that concepts of the cost that we need to understand are the average variable cost that is variable cost produced per unit of output. The average fixed cost that is average fixed cost, that is fixed cost per unit of output. Average total cost which is total cost divided by output and marginal cost I will come to marginal cost later.

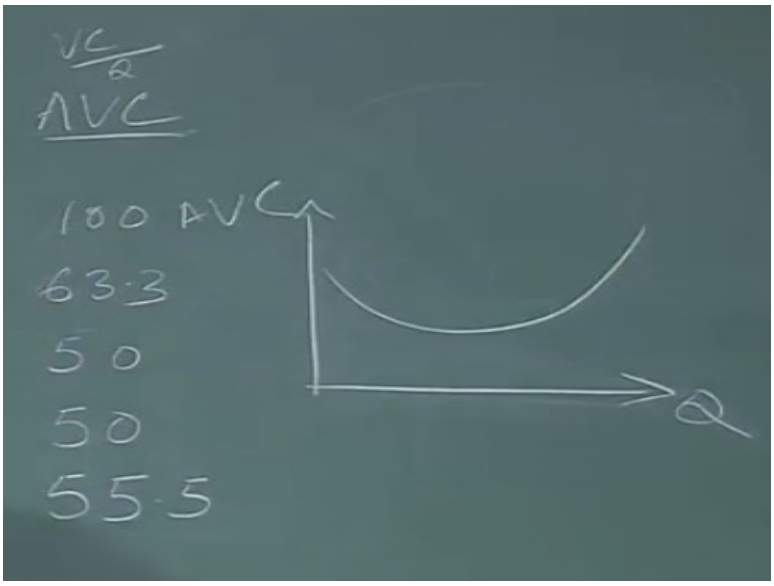
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Let me first show you what my let us first define the average costs. So first average fixed cost. So as you can see very well from here average fixed cost what would it look like. Average fixed cost what would it look like? Now average fixed cost would be in the first average fixed cost would be here it is not applicable only 0 units are produced. In the next it is 200, 200. Then it is 100, 2 units produced. Then it is 66, it is 200, 3 yes it is sorry 200, 3 okay I am sorry 200 then 200 divided by 3 it is 66.3 sorry 63. Then 300 Q fixed cost is 200, 33.3 and 200 then it is 25.

So average fixed cost is basically it keeps on decreasing. So what should it look like? So average fixed cost would look like this. So this is my cost again. So average fixed cost and output average fixed cost basically keeps on decreasing. So this is my average fixed cost. Next I am going to calculate my average variable cost.

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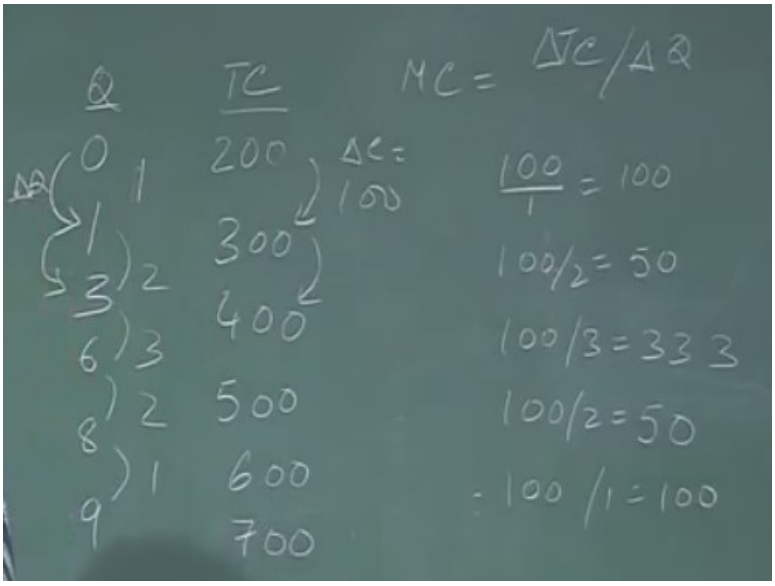


Now average variable cost is variable cost divided by output and that is variable cost it is 100 then 200 by 3 then 300 by 3, 30; 400 by sorry 300 by 6 and 400 divided by 8 then 500 divided by 9. So average variable cost what we see is average variable cost initially it declines and then it starts increasing again. So that means my average variable cost is a U shaped curve while my average fixed cost keeps on decreasing. So this is my average variable cost.

So this is how my average variable cost and average fixed cost look like and next we are going to discuss about marginal cost. Now as we said people think at, rational people think at margin. So we have already discussed about marginal product and we are going to see what happens to marginal cost. So basically the producer is trying to figure out what is the marginal cost he is incurring by hiring an extra unit of labour.

So marginal cost is basically marginal cost is not going to be influenced by the fixed cost, it is going to be influenced only by variable cost. So basically what we are looking at is let us look at the total cost and total output here.

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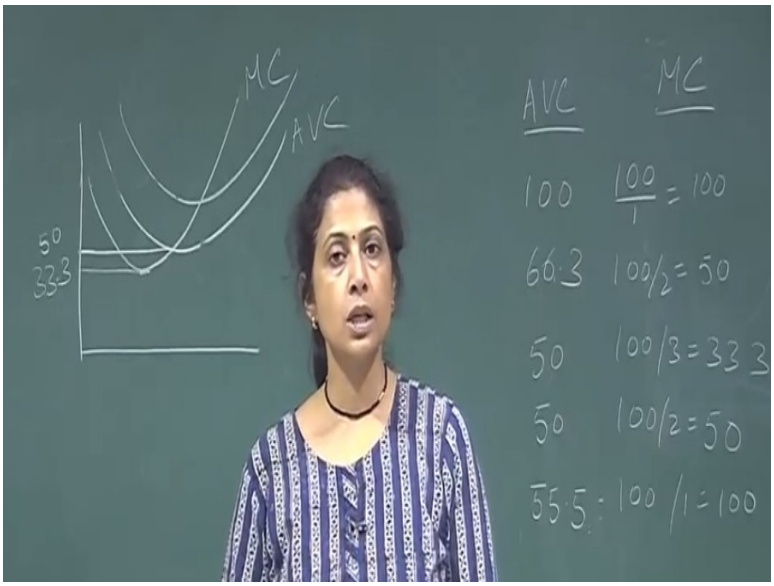
So marginal cost is equal to del total cost divided by del output. So additional output to be produced how much additional cost we incur. That is what we are trying to figure out here. So from 0 to 1 unit what is the del Q that is there. Now del Q is del Q is 1 here. Here del Q is 2. Here del Q is 3. Here del Q is 2.

Here del Q is 1 and here del cost del cost is equal to 10 here. It is again 10 here, 10 here. So it is my additional cost is 10 for every unit. (So this is my… oh sorry). I have done a major mistake. I deleted the zeros at the end. It is actually 200 now 200 plus this is a gross mistake that I have done here.

This is fixed cost plus the fixed cost I removed one zero from here so fixed cost plus variable cost is here 300. It is 400 here. It is 500 here. It is 600 here and it is 700 here. So apologies for the mistake this is my total cost here and additional cost that I am incurring for producing every unit is here del cost is basically 100. So del cost that I am incurring for producing the changes in output at every level the del cost is 100 and this is my here it is 1. This is 2, this is 3, this is 2, and this is 1.

So my marginal cost will be equal to marginal cost is equal to 100 divided by 1, 100. Second it is 100 divided by 2, 50. Then it is 100 divided by 3, let us put 33.3. Then it is again 100 divided by 2 is equal to 50 and 100 divided by 1 is equal to 100. So this is how my marginal cost curve looks like and let me now draw the marginal cost curve.

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So let me draw the marginal cost curve. So my marginal cost curve now looks like so it initially decreases, my marginal cost initially decreases and then it moves up again. So my marginal cost initially falls and then it moves up again and marginal cost is minimum at 33.3. Marginal cost is minimum at 33.3 and another thing if you may note from I have removed the average variable cost. You may remember that in the variable average variable cost the average variable cost kept on decreasing till it reached a level of 50 and then it started increasing again.

So average variable and when average variable cost was falling so let me write the average variable cost also here. So this is my marginal cost and this is my average variable cost. So my average variable cost looked like 100, then 66.3, 63.3 I think 66.3 and then 50 then 50 then 55.5. So I think my average variable cost looked like this and if I draw the average variable cost with the marginal cost I am going to see that initially my average variable cost is more than my marginal cost till it reaches a level of 50 and after that my average variable cost so this is my then my average variable cost is actually less than my marginal cost.

So my average variable cost is less than my marginal cost and this is the level 50 where my marginal cost intersects with my variable cost and after that my variable cost is actually less than

my marginal cost. Now why does that happen? So basically marginal cost when marginal cost is declining variable cost is marginal cost is less than the average variable cost and when marginal cost is increasing it is more than the average variable cost and the average variable cost and marginal cost they intersect at the point where average variable cost is minimum. They intersect at the point where average variable cost is minimum.

So what is the intuition behind it? So the intuition behind it is very simple that say for example we are imagining a CGP of a person, so the grade point average of a person, and say he has got the grades of all the subjects except economics and the average that he has got of all from all the subjects is say 5.0. So now if the grade that he receives from economics is the additional the additional amount that will be added to the average.

So that you can imagine as the marginal cost and that marginal amount when it is added it will reduce the average if it is less than the average. That is if the grade that he receives from economics is less than 5 that will bring down his average and if the grade that he receives is more than 5 that will raise his average.

So obviously it intersects, the marginal cost intersects the average variable cost at its minimum and that is the reason behind it and so that basically so this is how basically how our cost curves look like so this is my marginal cost, this is my average variable cost and another reason for the variable cost and the total variable cost is also looks like this.

Now the total variable cost also is U shaped because the average variable cost is initially decreasing and then it is increasing but average fixed cost basically brings down the total cost. So that is the reason that the average total cost is also U shaped and that brings us to the end of the discussion about cost curve.

So we have basically introduced the cost curves and in the next module we are going to see how the cost curves look in the short run and the long run. We are going to introduce the concept of long run and short run and how the producer basically makes his choices in the short run and in the long run. Thank you.

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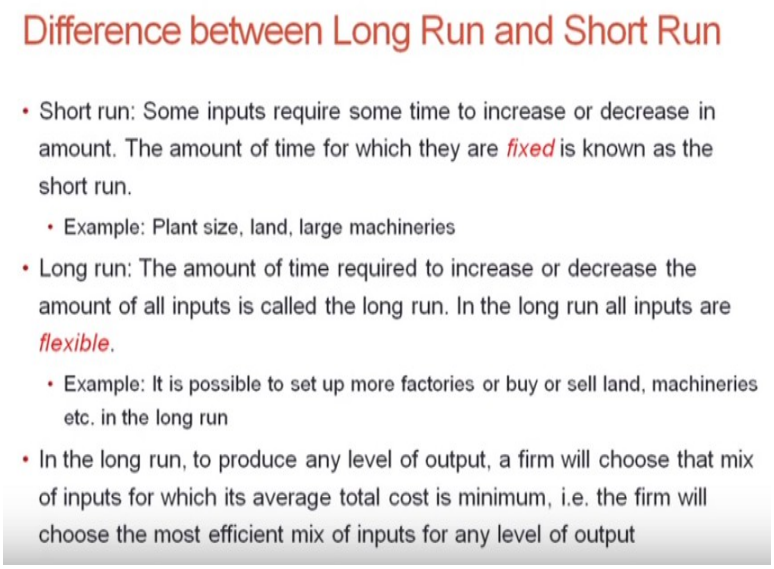
Indian Institute of Technology-Kharagpur

Lecture - 18

LR and SR Cost Curves

Hello and welcome back to our discussions on cost curves. We are going to in this module we are going to wind up our discussion on cost curves. We talked about the different the cost in different of its forms like the average cost curve the marginal cost curve etc. and we looked at their shapes and all. So now we are going to see if the cost curves change during the short run and the long run.

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So difference between long run and short run. Now how are we going to be keep on using this long run and short run all the time and we are going to use it a lot in the future also. So let me explain what is the difference between long run and short run. Now short run is something some inputs require some time to increase or decrease in an amount. The amount of time for which they are fixed is known as the short run.

For example, plant size, land, large machineries. So basically when we talked about costs and when we said that there is a fixed cost, there is variable cost. Now when we say fixed cost now this cost is not fixed forever but what we mean by fixed cost is basically that this these variables, these inputs require some time to change like we need some like the producer when the producer is using these inputs like land, machinery etc., huge machinery, land or a certain factory size it is using its already created a capacity of capacity and there is a certain factory size that it is that he is operating in.

Now these are all fixed inputs in the sense that they cannot be changed very easily or quickly. So they are called fixed variables and short run is basically the time period for which these all these inputs are fixed and the variable the variable cost that we discussed is basically the inputs which vary with the amount of output.

So if we if you if the producer wants to increase the output he needs to increase these variable inputs and this is called the variable cost that he incurs and this is flexible and this changes this can be changed in the short run also.

So in the short run some inputs require some time to increase or decrease in amount and the amount of time for which they are fixed is known as the short run for example plant size, land, large machineries etc. and long run is the amount of time required to increase or decrease the amount of all inputs and this is called the is called the long run. In the long run all inputs are flexible.

So basically the fixed inputs like plant size, land, machineries etc. they can be varied in the long run also. So a firm might like to increase its capacity, increase the plant size, increase the number of factories it has or increase the by a few more plots of land or large machineries but it requires some time to do that and that is the long run for a firm and the opposite can also happen.

A firm might decide to reduce its fixed cost and it may actually dispose of the land dispose of certain factories some of its factories or machineries. For doing that also it may require a little more time and that is called the long run.

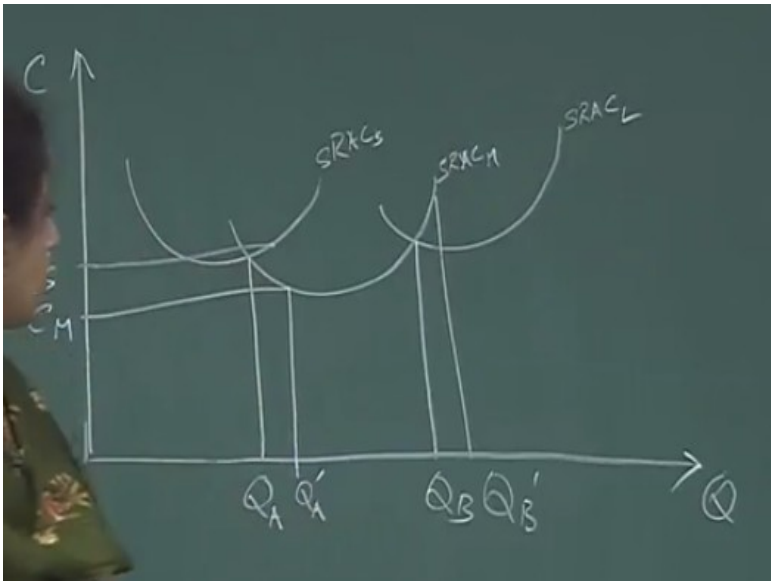
So example of long run it is possible to set up more factories or buy or sell land, machineries etc. in the long run. So in the long run to produce any level of output, a firm will choose that mix of inputs for which its average total cost is minimum. That is the firm will choose the most efficient mix of inputs for any level of output. So let me explain this with the help of a figure.

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Say for example we had ended our discussion the previous module with saying that the average total cost curve is U shaped, the average total cost curve is U shaped and that is because in the so when output is gradually increasing from a very low level average fixed cost is falling so that pulls down the average total cost and as output is increasing average variable cost goes up and that pulls up the average total cost. So the average total cost looks something like this. It is again a U shaped curve and this is the long run average cost curve.

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Now so this is just for recap and let me draw. Now say so this is the output and this is cost curve. Now say in the short run a firm is stuck with a certain factory size, a certain amount of land, a certain machinery etc. So in the short run it can choose various levels of factory size. Say for example when a firm decides that it is going to start production it can make a choice of how much fixed inputs to install or how much fixed input it will start production with.

So say it has 3 choices and this is one factory size, this is another factory size, and this is another factory size. So say he has 3 choices. There is a these are short run average cost curves. Say it the size small size factory size. This is short run average cost with a medium factory size. This is short run average cost with a large factory size. So these are the options that the firm has. These are the options that the firm has. So in the short run the firm can start producing at either this level of factory size or the medium or the large but once it has chosen its fixed inputs that is once it has chosen its fixed chosen its factory size it is stuck with that in the short run.

So say for example now the firm starts with the small factory. It starts with a small factory and then let me show 2 levels of output QA and QB. Say a firm is has started its production with SRAC S this is the factory size or small factory size that the firm is operating with and it is producing an output which is less than QA. So it is fine it is producing it here.

Now say for example the firm knows that given the demand situation in the market or whatever like the firm to maximize its profit it would like to increase the output. So given say there is huge demand for this product in the market and the firm would like to increase its production. So now if the firm wants to increase its production beyond QA it can do so with the existing factory size or it can decide to move on to a medium size plant in the long run.

So in the short run if it has to increase its output beyond QA it will operate on this small factory size and it incurs a cost say for example it produces QA prime. So to produce Q A prime since in the short run it is stuck with this cost curve it will incur a cost which is here. So this is CS. This is the cost it will incur but in the long run if the firm realizes that its intention of producing output is beyond QA in the long run it can decide to move on to the medium factory size and in that case its cost will reduce to CM. Cost will reduce to CM.

Similarly, but if the output the production requirement is less than QA it would not make sense for the firm to move on to a bigger factory size. Similar thing holds true for QB if it moves on to a level in the medium when it is operating on the medium factory size and its production requirement goes beyond QB then again it would make sense for it to change its factory size to the large one in the long run because in the short run in any case it is stuck with this factory size.

So in the long run it would make sense for it to change its factory size to the larger one and produce at a lower cost than the previous one. So depending on how much output it plans to produce in the long run the firm can change all its inputs. So when the firm does that it is basically producing at the minimum possible cost in the long run.

So in the long run since all the inputs are flexible the firm is going to choose that combination of all the inputs not only the variable cost inputs but the fixed cost inputs also because they are no longer fixed cost in the long run the firm is going to choose the that combination of input which makes its cost minimum in the long run. So in the long run the firm is going to choose that combination of inputs.

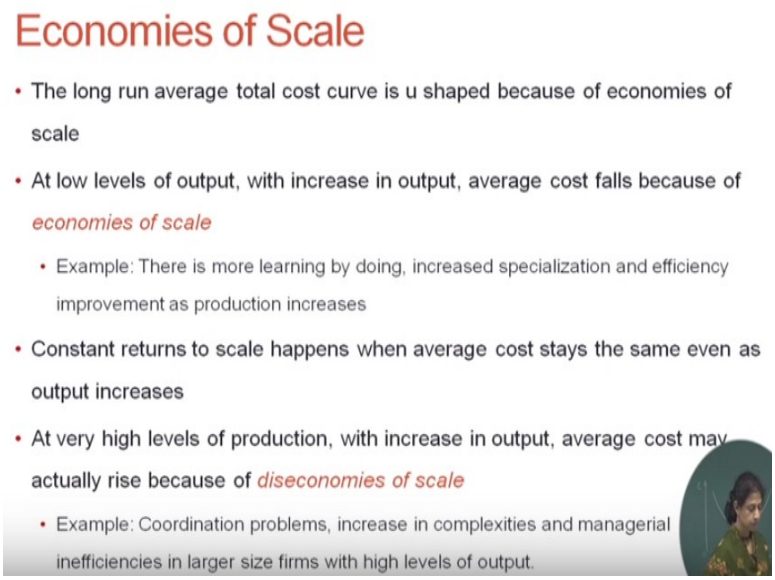
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So now here I have drawn 3 factory sizes but in the real world in reality it is possible to have it is possible to have numerous factory sizes. So for every unit of so for every unit of output it is possible to have various factory sizes and basically the firm will be choosing in the long run the firm will be adjusting its all its inputs and will be choosing the minimum of these cost curves which makes the firm's cost the minimum.

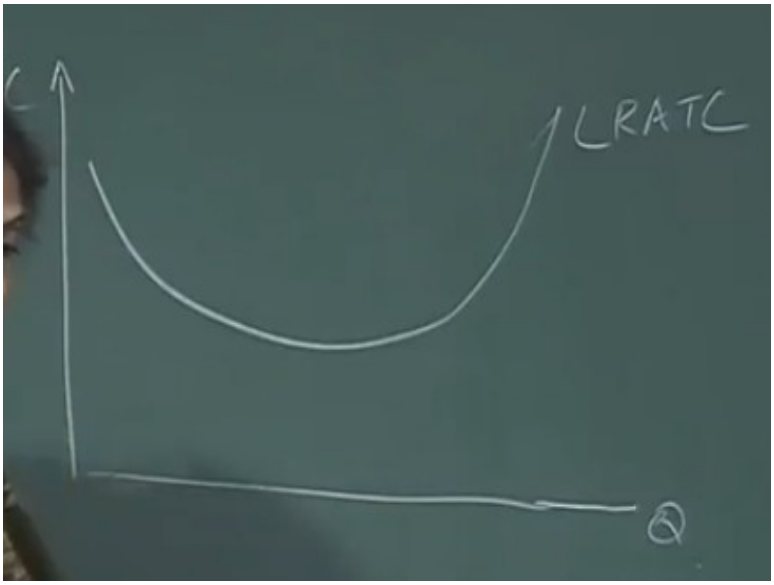
So basically the long run average cost is an envelope of all these short run cost curves. So the long run average total cost curve is an envelope of all these short run cost curves and in the long run to produce any level of output a firm will choose that mix of inputs for which its average total cost is minimum. That is a firm will choose the most efficient mix of inputs for any level of output.

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So now that brings us to another discussion about economies of scale. Now the long run average total cost curve is U shaped because of economies of scale.

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So as we have seen the long run average cost total cost curve that is again is U shaped because of economies of scale. Now at low levels of output with increase in output average cost falls because of economies of scale. So what we are saying here is a the so at low levels of output at

low levels of output as the firm keeps on increasing the output the with rise in production there is economy of scale which means basically the average cost falls.

The reason for this is as the firm keeps on increasing its production there is a lot of efficiency improvement it is something called learning by doing where the firm basically gathers knowledge about producing more and producing better and producing more efficiently and there is a lot of cost advantages as the firm increases its output level.

Then it reaches a level where there is it that the advantage actually goes on till a certain level where they after which there is no more increase in or decrease in cost because of increase in output and beyond that there is diseconomy of scale where basically when output increases lot of inefficiencies creep in say for example if there is a huge organization which is bursting to its brims and producing a huge level of output lot of managerial inefficiencies, lot of miscommunications these kinds of problems can creep in which can actually reduce the which can actually increase the cost per unit of output.

So at low levels of output with increase in output average cost falls because of economies of scale. For example, there is more learning by doing, increased specialization and efficiency improvement as production increases. Constant returns to scale happens when average cost stays the same even as output increases. At very high level of production with increase in output average cost may actually rise because of diseconomies of scale. For example, coordination problems, increase in complexities and managerial inefficiencies in larger size firms with high levels of output.

So that completes our discussion on cost and we will be moving on to discussing about market structure but one thing that we need to remember is the all the cost structures, the cost curve, the cost, the relationships, the long run, the short run, the definition, our understanding of cost everything stays the same no matter which market structure we are discussing.

So in the following weeks when we take up different market structures of whether perfect competition or monopoly, oligopoly or monopolistic competition we are going to see that the cost structures, the cost curves as we have seen them in this week they all remain the same in all the market structures. What will basically change is the demand situation in each of these

markets and what will change is how the price is determined in different markets and we move on to profit maximization objective of the firms and determine in each market structure how the firms decide how much to charge and how much to produce. Thank you.

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Lecture - 19

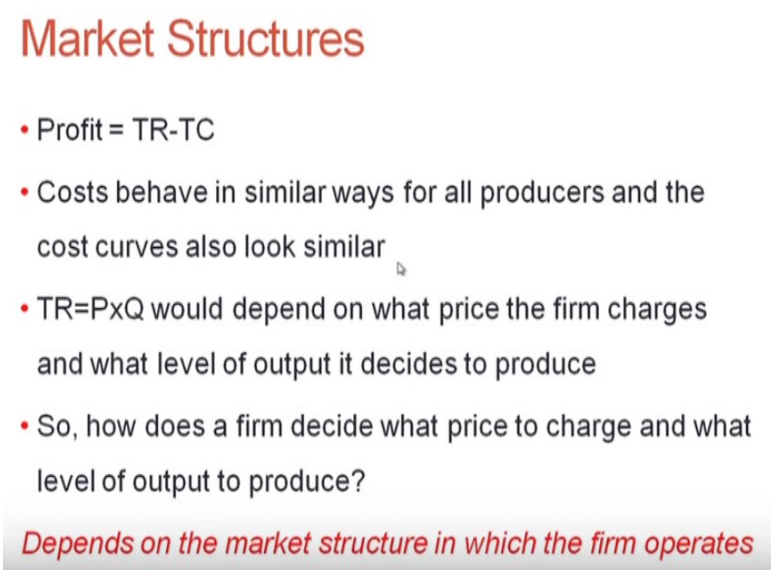
Market Structures

Welcome back to the foundation course in Managerial Economics. In the previous week we studied the production function, we saw the relationship between output and inputs and how that translates into cost. We looked at short run costs, the nature of costs in the short run, the nature of cost in the long run and how it changes how the cost decisions change from short run to long run. We discussed about fixed cost and variable cost and how that relates to long run and short run.

So in, in this week what we are going to do is we are going to see how the firm decides having known that the production function having known its own production function that is how much output can be produced through what combinations of inputs and costs. How does the firm basically decide how much output to produce and what price to charge because if you may remember we said that the objective of the firm is to maximize profit and profit is the difference between revenue and cost.

So we have developed our understanding about cost and this week we are going to look at how the revenue is determined by a firm. So revenue consists of 2 parts price and quantity. So the price and quantity gets determined by the firm in different kinds of market structures differently. So this is what we are going to study this week. So let us proceed.

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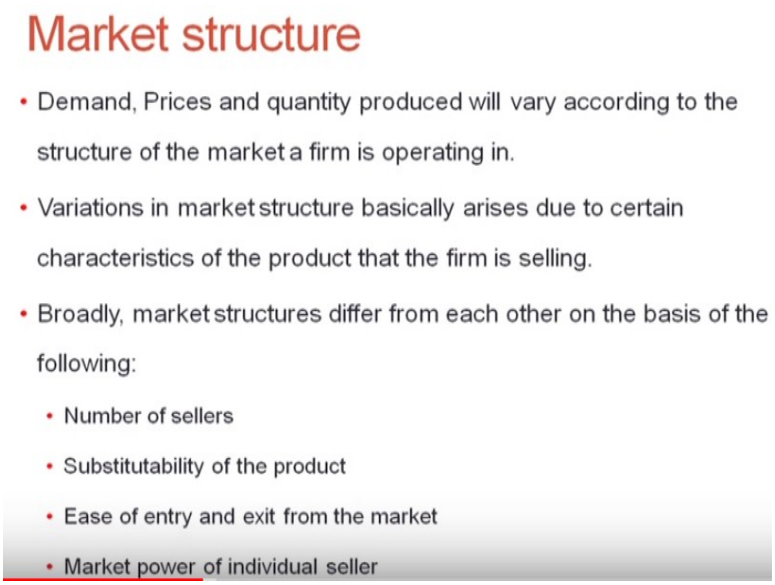


We are going to discuss about various market structures and we are going to assume that the objective of all firms is basically to maximize profit. So profit is total revenue minus total cost and costs behave in similar ways for all producers and the cost curves all also look similar. Basically, in the previous week the cost curves that we developed we looked at the marginal cost, the average cost, how they change overtime, how the average cost is a U shaped cost curve, how the marginal cost when it is decreasing it is below the average cost curve and when it is increasing it is above the average cost curve.

So all these the nature of the cost curves they do not change from firm to firm. So basically the for all the firms whether they are in a competitive market structure or they are in a monopolistic market structure they all have similar kinds of cost curves. So all that we need to understand now is how does price and quantity decision change for different kinds of market structures.

So coming back to the slides, so total revenue is price into quantity and this would depend on what price the firm charges and what level of output it decides to produce. So how does the firm decide what price to charge and what level of output to produce? So all this depends on the market structure in which the firm operates.

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So the demand for different goods the demand, the prices, the quantity produced all these will vary according to the structure of the market a firm is operating in. So variations in market structure basically arises due to certain characteristics of the product that the firm is selling. So we are going to see that the market structures are basically different because of some characteristic of nature of the product. It is inherent in the product itself which decides what kind of market structure the product will be sold in.

So broadly, very broadly we are going to see that market structures differ from each other on the basis of the following. The market structures will be different depending on the number of sellers. They will be different for number of sellers. The substitutability of the product; that is if the if there are alternatives which are available in the market that is going to determine what kind of market structure the product will be available in.

Ease of entry and exit, ease of entry into the market and ease of exit from the market. If it is very difficult to enter the market, then obviously the number of sellers will be low or if it is very difficult to exit the market. Say for example you are stuck with a very high cost. You incur a very high cost of setting up a business for that product in the market and you are stuck with for a stuck with it for a long time.

In that case we would say that there is a very high exit cost. So in that case also certain firms like the number of firms in the market may be very low. Market power of individual seller. Now

market power of individual seller this I am going to elaborate later when I discuss individually the different market structures but as is as you might already have kind of understood what market power of individual seller means.

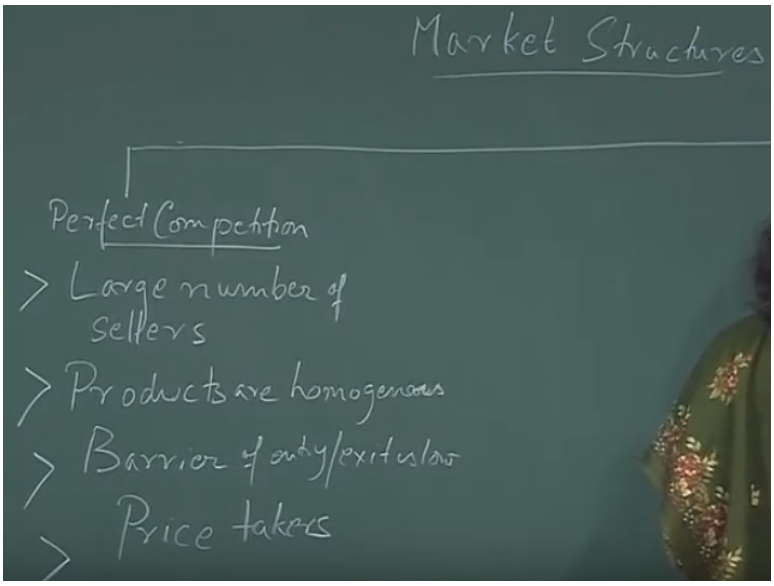
It basically means how much control the seller has on price. If the seller is able to determine what price to charge in the market so and that is indirectly that is kind of like whether the seller has a market power or not is highly related to the all the other 3 factors that we have listed above that is the number of sellers, substitutability of the product.

So if the number of sellers is low, obviously the market power of the individual seller will be high. If the substitutability of the product, if the product is easily substitutable like if it is possible to have it is possible to have alternatives of the, this product in the market then the individual seller cannot have too much power over the market or too much power to control prices in the market.

Ease of entry and exit from the market. Again if it is easy to enter or exit the market so the individual seller will always be faced with a large number of competitor. So in that case the seller will probably not have too much market too much market power in that kind of market structure.

So these are broadly the basis on which the market structures vary from each other and in this course we are going to focus primarily on 4 kinds of market structures which I am going to and how they differ from one another. Let me just elaborate on the board.

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So there can be 4 kinds of market structures. On one extreme, so on one extreme we have what we call perfect competition, we have perfect competition. So in perfect competition the all the 4 categories by which we say the market structures differ from each other so in under perfect competition firstly large number of sellers. So this is in case of perfect competition there are large number of sellers in the market and secondly substitutability of the product.

Basically products are homogeneous. When I say products are homogeneous that basically means the products are all very similar to each other. Not only very similar they are exactly similar to each other. When I say exactly similar that does not mean that you know like literally they are exactly similar. What that means is to the consumer they are deemed similar. So whether the product has been purchased from seller number 1 or seller number 2 the to the buyer it does not make any difference.

So it is very difficult to differentiate the products from each other and that is what I mean by mean when I say that the products are homogenous. A very the perfectly competitive market example is not we do not have an exactly perfectly competitive market for any product it is more theoretical concept but something which comes very close to perfectly competitive market is the agricultural products market.

Say for example the vegetables are sold or food grains are sold in the market, the wholesale market. So there all the products are very similar to each other and the price is determined in the

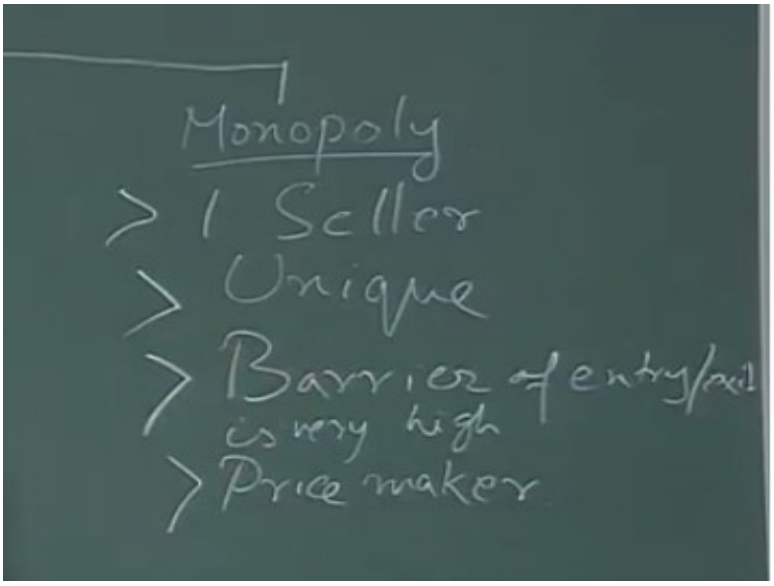
market depending on the amount that the sellers have bought the amount that the buyers are willing to purchase. So the demand and supply framework that we discussed in the earlier weeks that kind of setup determines what prices to is going to be charged in the market and basically all the buyers they accept that price and they just sell their products at that price.

So basically the all the products are very homogeneous or they are very similar to each other so the example is agricultural products. So products are homogenous to each other and thirdly ease of entry and exit from the market. So again the example of agricultural products.

So basically farm products. So it is not very difficult to enter this market. If someone wishes to grow vegetables or food grains or do farming it is not something very difficult to do. There is not too much high barrier of entry to it and similarly there is no too high cost of exit from the market. If someone wishes to stop producing agricultural product he may just stop and maybe sell off his land or use it to or put it to some other alternative use. So the barrier to of entry and exit from the market is very low. So basically this is the barrier of entry exit is low.

Fourth is the market power of individual seller. The market power of individual seller as we have already seen that there are large number of sellers, the products are homogenous, there is no barrier to of entry or exit in the market. So basically all the sellers in this market they are all price takers. So basically market power of individual seller is almost nil. So they are all price takers. So they are all price takers.

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So on the other extreme we have almost the opposite situation of perfect competition which is the monopoly, which is the monopoly which most of know that monopoly is basically there is one single seller in the market. So there is one single seller in the market so we have one seller. Okay one more thing is when I am describing all these market structures I am assuming that the number of buyers is many in all the market structures.

There could be situation where we have something like monopsony where there is one single buyer and many sellers in the market. So we are not going into the buyer side of the market yet. We will be discussing mostly about the seller's perspective if he is the sole supplier in the market or he is one of few or he is one of many or he is one of a huge number of sellers.

So this is what we are doing here substitutability of the product. The product is unique. The product is unique. The product is unique in the sense that there is one seller in the market. So he is the one who is selling this product and this itself that there is one seller in the market itself kind of implies that whatever he is producing no one else is producing that.

So to give an example this situation of monopoly could arise due to a lot of things. It could arise because the fixed cost of setting up this business could be very high. So we are going to go into the details of what causes natural monopoly, in what situation a monopoly arises and all that when we discuss about monopoly in the later weeks but it suffices to give some example here that one example of monopoly is basically patented medicines.

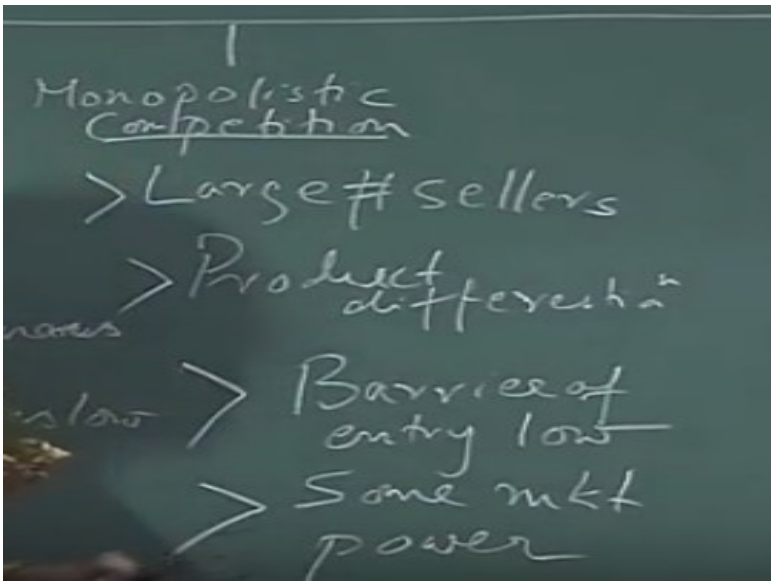
So pharmaceutical company which has patented a particular drug say a lifesaving drug that this particular company can produce and no one else can, is a that patent causes a high barrier to entry barrier of entry in this market and he is the, this pharmaceutical company is the single seller in the market and the product is unique.

Another example is say for example The Indian Railways. Indian Railways this is the one single organization which is catering to the railway requirement of the entire huge country of India. So barrier of entry is high because it is not easy to setup a railway network, a parallel railway network given that one is already existing. So that would be extremely costly and may not make sense actually.

So again we are going to discuss those in details later. So basically the examples are patents which create barrier to entry and there are natural barriers to entry in case of because of very huge fixed cost say in case of railways or water supply or setting up of power plants etc. So these are situations of monopoly.

So ease of entry. Barrier to entry barrier of entry or exit is very high okay and market power of individual seller. So all these 3 characteristics of the market itself determines that the market power of the individual seller is very high. Basically the individual seller gets to decide at what price he is going to sell the product. So he is basically a price maker. He is not a price taker but he is a price maker.

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So this is about monopoly. So we have perfect competition in one extreme and we have monopoly in the other extreme and we have 2 more market structures which we are going to discuss later. They fall within these extreme two categories. One is the called the monopolistic competition. One is the monopolistic competition and here like perfect competition there are large number of sellers.

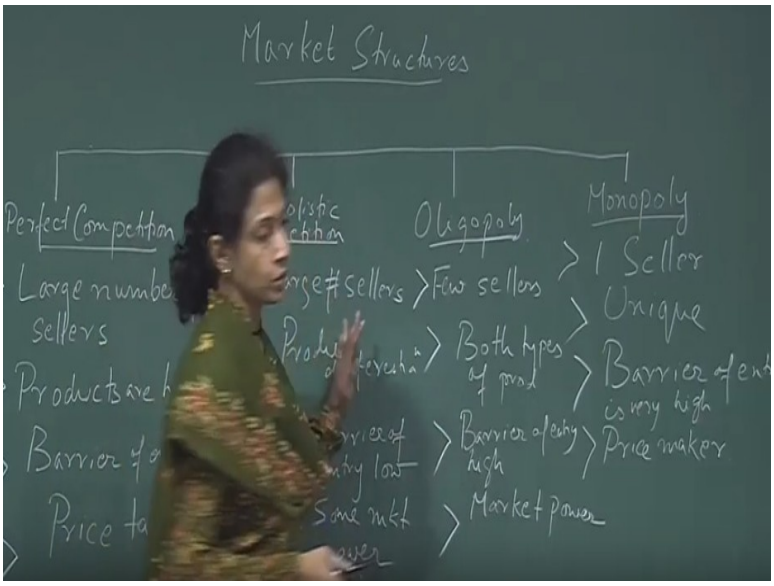
How is it different from competition? Products are not homogeneous. Products are not homogenous although there are large number of sellers who are selling this product, products are not homogenous but they are very similar to each other. So to give an example say for example FMCG products. Say for example toothpaste.

Now there are there can be whole lot of companies who are producing toothpaste. There can be number of brands, different types of products of toothpaste but they are at the end of the day they are all toothpaste. So they are substitutable but there could be minor differences between the toothpaste because of which the people can have preference for one variety over other. So there is some amount of product differentiation here and the products are not exactly homogenous.

So product differentiation happens here, product differentiation is there and the barrier of entry is not very high. It is not too difficult to set up business in this market structure so barrier of entry is not too high and the monopoly in monopolistic competition the firm has some control or some market power that is the firm has some control over setting up of price of the product.

Because the products are differentiated so the firm knows that there is some amount of loyal customers who would probably not shift to other products if the price change is a little bit so but even then since the products are at the end of the day they are substitutable so they cannot change the price too much. So this is the situation where they are they have some amount of market power. So some market power is there for the seller.

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Another and another the last type of market structure that we are going to discuss is the oligopoly. So in case of oligopoly there are few sellers. In case of oligopoly there are few sellers in the market products. Whether the products are homogenous or they are differentiated well in this in oligopoly the products could be homogenous perfectly substitutable or they could be differentiable.

Say for example the OPEC oil producing oil exporting countries. They form an oligopoly of oil producers or oil exporters and there are few of them so in the world market so that is an oligopoly and oil crude oil is perfect substitute if not like not going into technical details but crude oil is kind of homogenous and on the other hand we have the car companies or airlines. So there the airline services differentiated then car companies, different kinds of car manufactures different models, they are highly differentiated products.

So in oligopoly you can have both types of producer surplus possible. Barrier to entry is kind of high. Barrier to entry is kind of high and that is the reason we have few sellers in the market and the barrier of entry is high we have few sellers in the market. They are catering to the huge population of consumers who are looking for this product and the firms have substantial amount of market power but one thing interesting about oligopoly is in the rest of all kinds of market structures the firms in these market structures they decide how much output to produce what prices to charge and that is not dependent on what other firms are doing.

In none of the market structures the firm is concerned about what other firms or its rivals are doing but in case of oligopoly since there are few sellers in the market every sellers decision is dependent on the decision of its rivals.

So this is a special kind of situation and here different kinds of scenarios could emerge where the sellers could be competing on the basis of the output they produce or they could be competing on the basis of prices that they charge. So these are the different kinds of market structures that we are going to discuss during this week and one more to end the discussion one more interesting thing here is what is the role of advertisement in all of these market structures.

Now in case of perfect competition since products are homogenous since products are homogenous it does not make sense for a firm to advertise its products because they are all similar. So by advertising it is just going to increase its own cost. So it does not make sense. However, in case of perfect competition the industry level advertisement is possible which is going to shift the demand curve to the right.

So to take an example, say for example in we have advertisement for eggs or milk. So you generally want people to increase their consumption of eggs or increase their consumption of milk and it does not matter you do not care who they purchase it from, which of the dairies they purchase it from or which of the poultries they purchase it from. You just want the industry demand curve to shift to the right.

So the role of advertisement is at industry level in this case of perfect competition. Same is the case in monopoly. In monopoly the seller is can advertise to basically increase the consumption

to basically to shift the demand curve to the right but obviously it does not have any requirement or any need to attract the consumers to itself because it is the only seller in the market.

In case of monopolistic competition advertisement pays the maximum role. Because in case of monopolistic competition, the products are highly substitutable. The product differentiation that can be created can be created in the minds of the consumers and that is the reason that in case of monopolistic competition the firms are going to spend a lot on advertisement because they can attract the consumers to itself and in case of oligopoly also depending on whether it is a differentiated product if it is a differentiated product they may go for advertisement. If it is a undifferentiated product homogenous product they are not going to go for advertisement.

So this was our broad discussion about market structure and every week we are going to take up each of these market structures and discuss about them. Thank you.

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Lecture - 20

Perfect Competition

Welcome back. We discussed in the previous module, we discussed about the different kinds of market structures. Now we are going to this week, we are going to spend most of the time discussing about the first kind of market structure that we mentioned which is the perfect competition. So we are going to discuss about perfect competition this week.

When we say competition, the use of the term competition in here the use of the term competition in economics, kind of is very different from the way we usually use it in fields of say like sports. In sports when we say that 2 parties are 2 people are competing and wherever we say that there is competition it basically means that the parties are trying to the rivals are trying to establish their supremacy over each other. They are trying to they beat each other in a game they are trying to up their game beat each other and emerge as the winner.

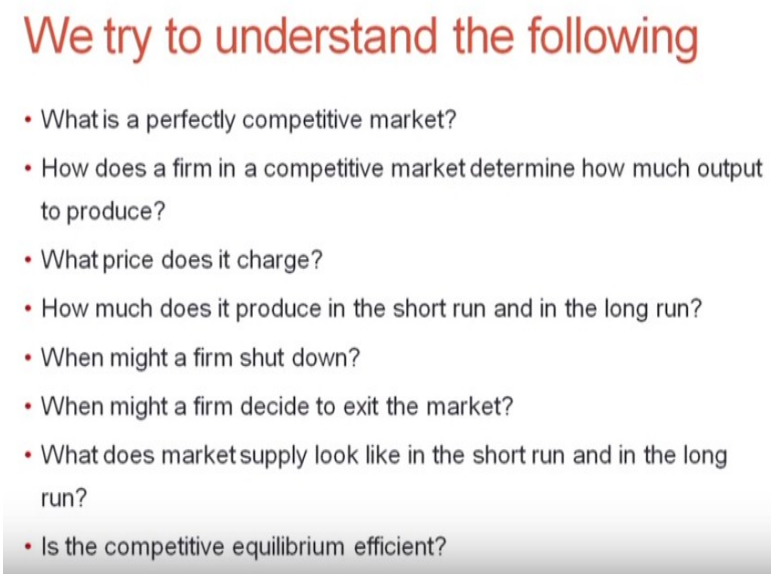
So that is what we mean by competition but here in case of the perfect competition we are going to see that competition here does not mean that the products of the different firms the different competitors in the market they are different from each other. They are not competing on the basis of products.

Basically competition the use of the competition here in this context is more to establish the establish the fact that there is level playing field for everyone in this kind of market structure.

Basically what we mean is there is level playing field for all the firms who enter this market and compete with each other. So they cannot compete on the basis of the products because the buyers know that the products are exactly the same. There is not much difference between the products and say so there is no special loyalty to any particular seller.

So they cannot compete on the basis of the products they are selling and basically there is level playing field. So let us proceed and see what are the typical characteristics of a perfectly competitive market.

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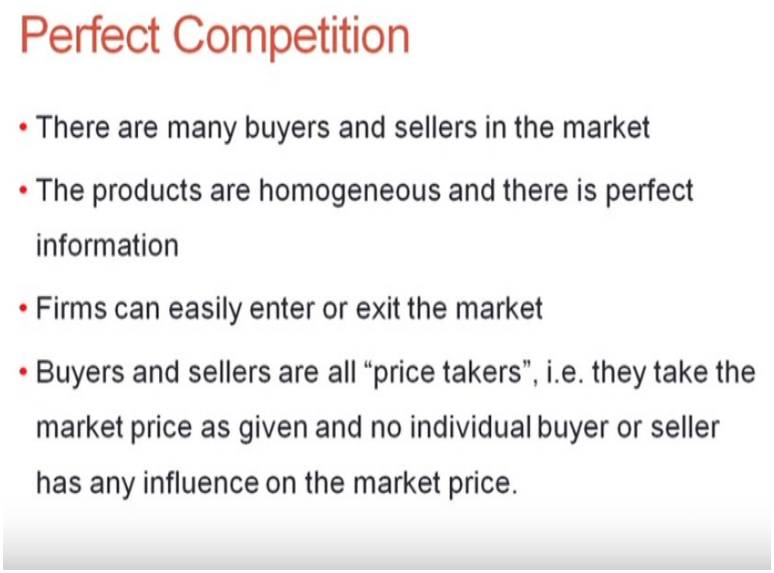


So in case of when we study perfectly competitive market we try to understand the following. What is the perfectly competitive market? How does a firm in a competitive market determine how much output to produce? What price does it charge? How much does it produce in the short run and in the long run? When might a firm shut down? That means in what situation would it be better for the firm to shut down production than keep on producing. When might a firm decide to exit the market altogether? What does market supply look like in the short run and in the long run in a perfectly competitive market?

So first we are going to see the look at the decision making process of a competitive firm and finally we are going to look at what the decision making process of individual firms leads to and what happens to the market supply in case of a competitive market. Is the competitive

equilibrium efficient? This is the question we will be asking at the end of discussing each of the market structures. We are going to look at the market equilibrium, of the market equilibrium, price charged, and the market equilibrium quantity produced and we are going to ask ourselves is this a efficient outcome or not?

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So these are the assumptions in case of a perfect competition which we discussed in the previous module also when we talked about market structures. So there are many buyers and sellers in the market. The products are homogenous and there is perfect information. Why do I use the word perfect information? So the products can be homogenous but that would not mean anything unless and until the buyer is aware of the fact that the products are homogenous.

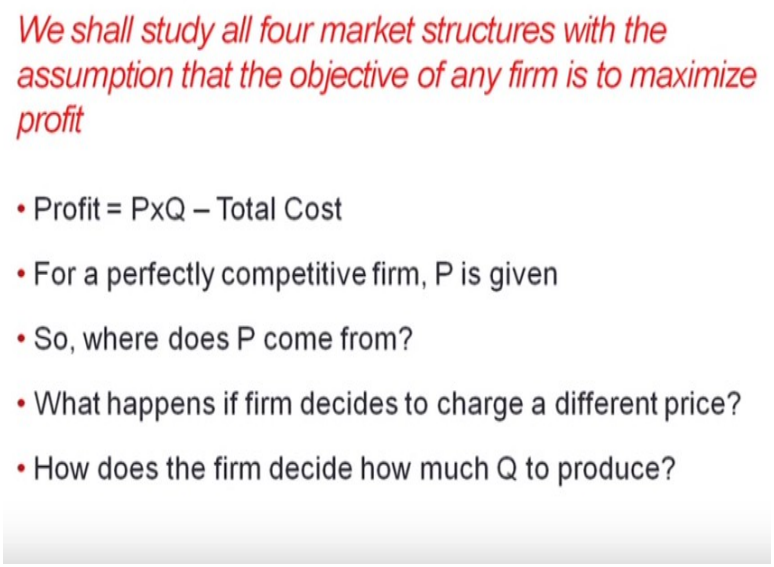
Say for example 2 sellers are selling milk in the market and the milk are exactly the same. They are probably procuring it from the same dairies or the same farmers or local farmers who produce the milk and sell it to the dairies. So but they are all that they are doing is packaging the milk and selling it in the market. So the buyer needs to understand that the content of the milk cartons are exactly the same no matter what kind of celebrity endorsement has been done, what kind of claims are there on the packaging and how attractive the packaging is.

In case of perfect competition, we are assuming that the buyer sees through all these kinds of what we say marketing gimmicks and they understand that the products are exactly the same. So it does not make sense for the seller to put so much of effort in trying to differentiate the products when the products are not different.

So the firms can easily enter or exit the market. It is not difficult for the firms to start production and if it is not profitable for the firm if he wishes to discontinue if it wishes to exit the market then also it is not stuck. So that is another assumption of a perfectly competitive market and finally buyers and sellers are all price takers. That is, they take the market price as given and no individual buyer or seller has any influence on the market price. So this basically happens because we have assumed there are huge number of buyers and sellers in the market so if say for example one seller drops out of the market he withdraws his produce from the market that does not reduce the supply so much in the market that it affects the price in the market.

Similarly, the number of buyers in the market is very high. So if one buyer or a few buyers leave the market they stop buying the product the price it that does not their action of discontinuing the consumption of the product does not reduce the price in the market. So these are the certain assumptions of perfect competition and to before we discuss about what happens in case of the questions that we asked in the beginning of this session, we said that how does the producer decide how much output to produce and what price to charge.

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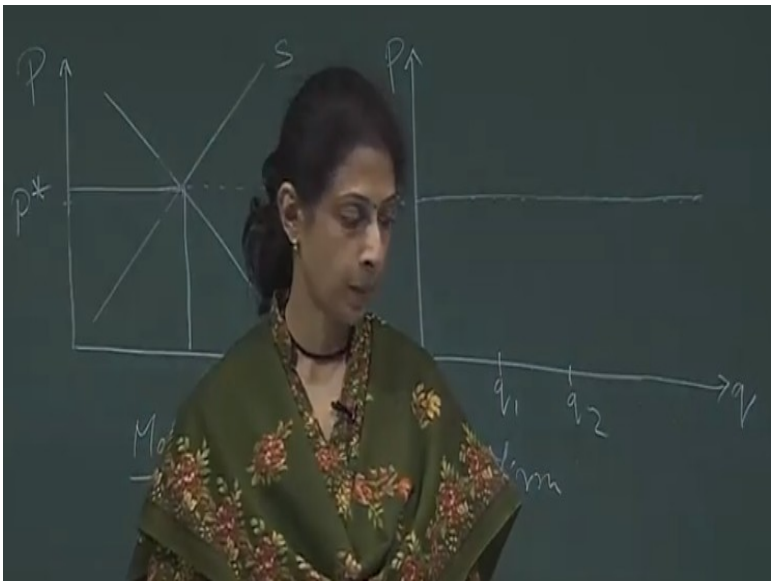


So before I am that is the question that we will be asking in each of the market structures when we discuss about them and so before that we are making this assumption that we shall study all 4 market structures with the assumption that the objective of any firm is to maximize profit. So profit maximization is the objective of the firm that is something which we are going to assume all throughout the course.

So as we know profit is total revenue minus total cost. Total revenue is price multiplied by quantity and for a perfectly competitive firm P is given. This was our assumption that price is given but where does this price then come from. So that is the question and what happens if the firm decides to charge a different price and then we are once we understand about the pricing then we are going to ask how does the firm decide how much output to produce.

So let me show so let us try to answer the question where does this P come from? So when I am saying that the for all the buyers and sellers in the market price is given and they are all price takers basically where is this price coming from.

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So remember that when we discussed about market demand and market supply we had shown that in a competitive market this is price, this is quantity, this is the demand curve, this is the market supply curve and this is the market this is not the firm but this is the market this is the aggregate of all the sellers and buyers who are there in the market.

The buyers come to the market with their set of demand and the market demand curve is a negatively sloping demand curve which follows the law of demand and all the sellers they come to the market with their supply curve and they are willing to sell the products at certain price they are willing to sell at certain amount and which gives us the aggregate market supply curve and this is the positively sloping supply curve and the intersection gives us the market price. The intersection gives us the equilibrium price in the market.

So now this price we shall be using the small q for showing the firms output level and we are going to use the capital Q to show the market output level. So the price is basically same. So this is for the firm. This is the decision making for the firm and the firm basically so this is the price that the firm has to accept. So he is a price taker so he has to he knows this is the price line. He knows this is the price line and he can supply and he can sell as much quantity he wants at this price. So he knows that this is the price that is prevailing in the market and he knows that at this price he can sell all the products will be all his output will be sold at this price.

So to the firm the demand curve that the firm is facing the demand curve that the firm is facing is basically horizontal. The demand curve that the firm is facing is horizontal. The reason that I am repeating this is because this is important. Look at the market demand curve. The market demand curve is negatively sloping market demand curve but a individual seller in the market who is a very small part of this supply curve he is one of the numerous sellers that they are in the that there are in the market. So to him this demand curve does not matter.

What matter to him is he knows that at this price he can sell whatever product he has. So basically he knows that at this price this is the demand curve of the market. So to the seller the demand curve is horizontal. What amount he is going to sell that is going to depend on his cost situations. That we are going to come later.

So what whether he is going to sell this much or he is going to sell this much that depends on his cost situation and that we are going to discuss in details later. The all that I would like to emphasize here is to the individual firm the demand curve that the individual firm faces in a perfectly competitive market is basically horizontal at the given market price.

Now next question is what happens if the firm decides that I am going to charge a price which is higher than P star or he decides I am going charge a price which is lower than P star? Can that happen? Now say the firm decides and before that before we talk about that let us take an example.

Say for example there is the local wholesale vegetable market where there are farmers coming from lot of places of nearby places and they are bringing their produce there. There are small farmers who are probably bringing you know like a few kilos of the product and say they are bringing spinach and there are farmers who are who are bringing say truckloads. So all of them are bringing their produce to this wholesale market to sell it over there and they do not have any influence on the price.

They just reach the market and they will ask what is the what is the going price? So they will be told that the price is P star and they know depending on how much produce they have they can sell q 1 or q 2 or q n or anywhere. So for each of these sellers this is the prevailing price and they all face a horizontal demand curve.

Now what happens if one of the sellers decide I am going to charge a price which is higher than P star. What will happen? Now if this firm or if this seller decides to charge a price which is higher than P star since all the buyers have perfect information they know that the prevailing price in the market is P star and they would not like to pay a price which is higher than P star. So this seller will see that all his customers are leaving him and his shop is empty. So everyone will go away from him and buy from other sellers.

What happens if he charges a price which is lower than P star? That is good for the buyers right? No. Because if he charges a price a price which is lower than P stars there are numerous buyers in the market and everyone wants to buy from him. So everyone will rush to his shop to buy his produce but he cannot cater to all, he cannot meet all the demand and it will be extremely costly for him to supply to all the people who are asking for his product and then he has to increase its price. So in a perfectly competitive setup there is no way but for the sellers to follow the market price, the market determined price. So they are all price takers in this perfectly competitive market setup.

So this is in the next module we are going to see we are going to discuss about what how the firms decide how much output to produce. Thank you.

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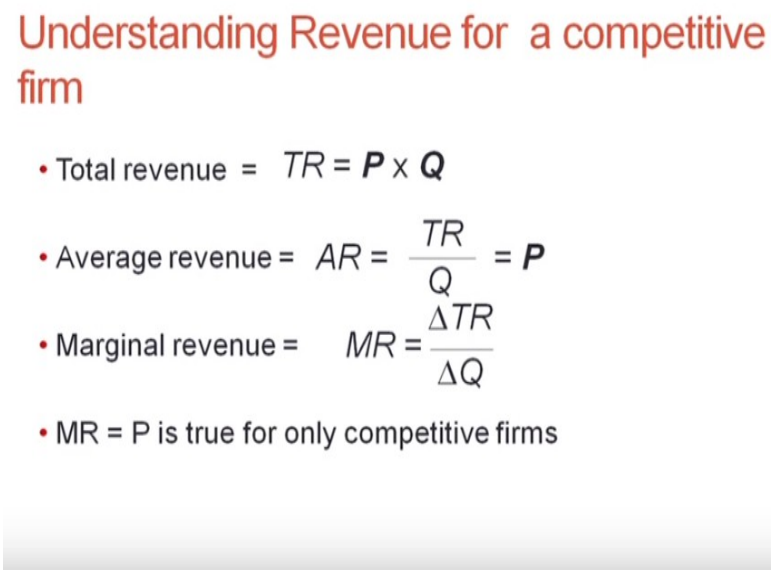
Indian Institute of Technology-Kharagpur

Lecture - 21

Perfect Competition and Profit Maximization

Welcome back to our next part of our discussion on perfect competition. We introduced the market structure in the previous module and we discussed about where the price comes from in case of perfect competition, where does the price taker take his price from and we discussed that basically for every individual firm the individual firm faces a horizontal demand curve at the level of price that is prevalent in the market. So we started of with that and our assumption is the firm is trying to maximize its profit. It has taken its price. Now the last part of the entire puzzle is the firm has to decide what level of output to produce.

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Now before we discuss about the level of output etc. and talk about profit maximization, let us understand how does the revenue look for revenue look like for a competitive firm. So to understanding revenue for a competitive firm now total revenue is equal to price multiplied by the quantity. Now here the P is given. P is a P star. It is given from the market intersection of demand and supply in the market.

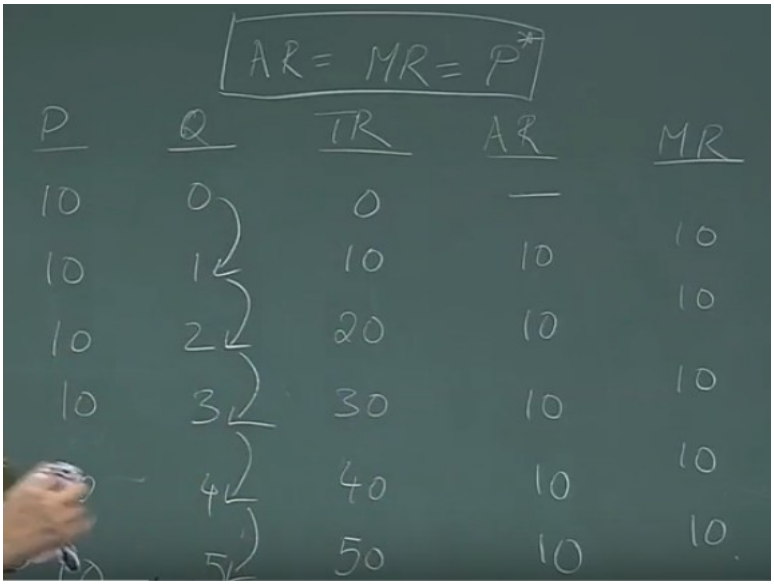
So the average revenue is total revenue divided by the amount of output produced. So that is PQ divided by Q and that is equal to P. So average revenue is equal to the price that is given in the market.

How much is marginal revenue? Marginal revenue is basically it is differentiation of the total revenue divided by the marginal amount of outputs. Basically marginal revenue means that when the firm is selling an additional unit of output how much additional amount of revenue the firm is receiving. That is the marginal revenue.

But in case of a perfectly competitive firm since the price for every individual unit it does not change it is given by the market and the price remains the same so for the competitive firm marginal revenue that is del TR divided by del Q is equal to the price in the market. So this is true for only competitive firms.

This is true for only competitive firms because in other kinds of market structures we are going to see that the price changes with different levels of output. The price changes with different levels of output and we are going to discuss about them in details later. Let me not get into those market structures right away.

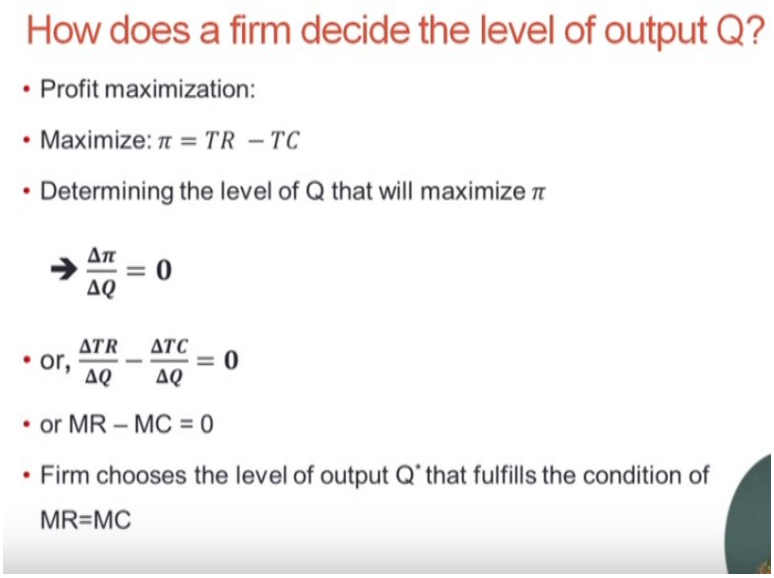
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So this is our understanding of revenue for a competitive firm and to elaborate this more with an example let me take a very simple example where we have the price is given. So the price is always say 10 Rs. So the price is always say 10 Rs because the firm is facing a horizontal demand curve. So no matter how many units of output it sells or decides to sell the price is always going to be 10 Rs. So Q is 0, 1, 2, 3, 4, or 5 price is always 10. How much is total revenue? Total revenue is price multiplied by quantity. So it is 0, 10, 20, 30, 40, and 50. How much is average revenue? Average revenue is always 10. Average revenue is always 10. What about marginal revenue. Marginal revenue is basically every single so additional amount of output that I am selling the next marginal unit of output here I am selling 1 unit I am getting 10 Rs. Another additional unit I sell I get again 10 Rs. Another unit I sell again I get 10 Rs. So my marginal revenue in all these situations is 10.

So we can see that in case of a perfect competitive firm average revenue is equal to marginal revenue is equal to price. So this holds true for only a competitive firm. So this holds true for only a competitive firm. So let me clean this up. Okay so moving on.

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Now, now that we know what the revenues the different forms of the revenue looks like now that we know that price is given we know about the cost curves of the firms now how does the firm decide the level of output. How does it decide how much output to produce? So for this the firm is going to our assumption is the firm is maximizing profit.

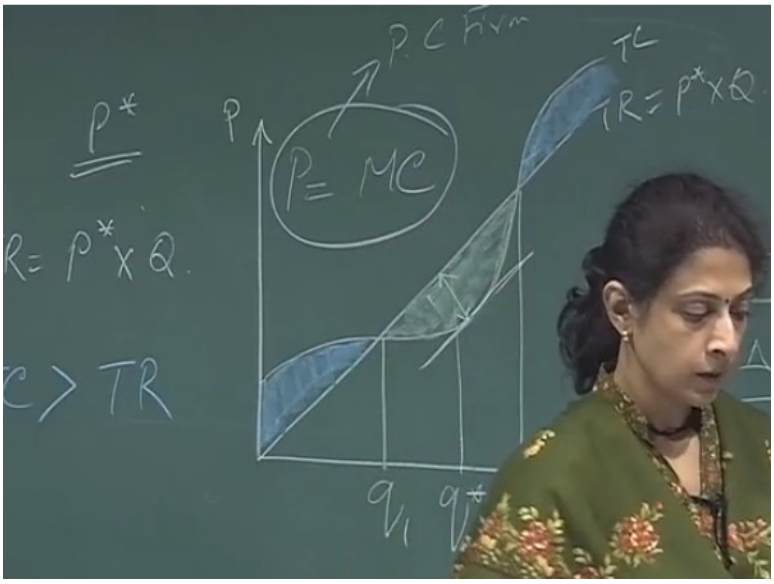
So as is shown in the slide so pi is our typical symbol for profit which we are going to use during the entire course and pi is equal to total revenue - total cost. So determining the level of output that will maximize pi. So we would like to determine that level of output which is going to maximize profit and so that is by the first order condition for maximization. It implies that del pi del Q is equal to 0 that is del TR del Q minus del TC del Q is equal to 0 or marginal revenue - marginal cost equals 0.

So the firm chooses the level of output Q star that fulfills the condition of marginal revenue equals marginal cost. Now marginal revenue equal marginal cost equals marginal cost is the profit maximizes profit maximizing condition for any firm. Profit maximization maximizing

condition for any firm is that, that the additional revenue that the firm expects to receive from selling the last unit of output that should be equal to the cost of producing that additional unit of output.

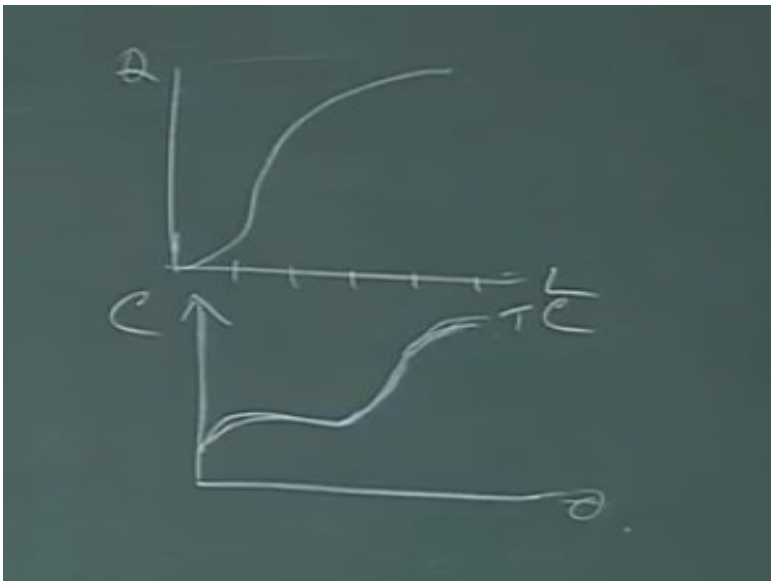
Now for a perfectly competitive firm let us look at the intuition. Let us look at the profit maximizing condition graphically and see how the output is determined.

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Now we know that price is fixed. So P star is given. So since P star is given, what is my total revenue? My total revenue is equal to P star into Q. So total revenue is P star into Q. So the total revenue curve will be a straight line through origin where slope is P star. So this is the total revenue curve.

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Now what does my total cost curve look like. So you may remember that when we were discussing about costs we developed the cost curve from the production function. If you remember this is the production function that we drew. Then we flipped the production function and this is the reason that we flipped the production function to translate it into the cost curve and see and put the cost curve in this kind of framework to understand the difference between revenue and cost so this is what we did and our cost curve after including the fixed cost it looked something like this. So this was my cost curve.

This is how my total cost curve looked like. So if you may remember we had labour here the variable cost and we had the output here Q. So we translated the labour into the monetary units of the wages that is the cost and basically we flipped the curve so that we had cost here and we had output here.

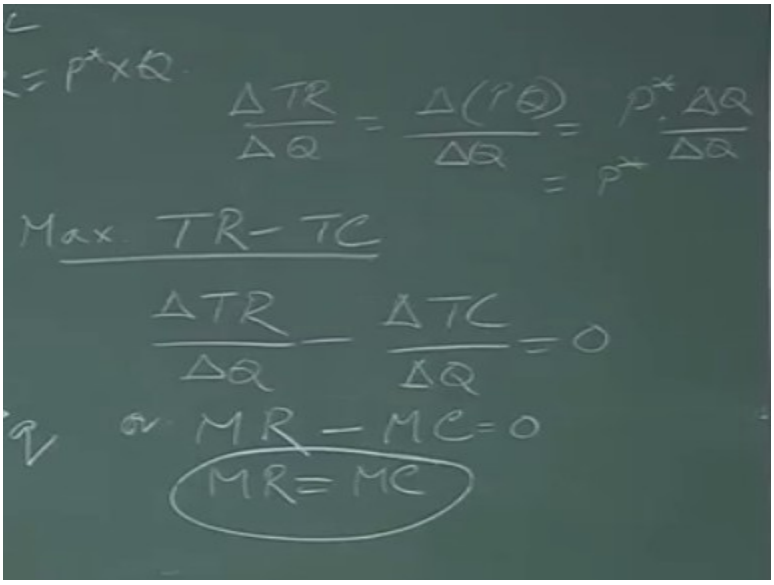
So if you can recollect this cost curve then basically for this firm we are going to draw the cost curve here. So this is assuming it is it fixed cost so the cost curve looks something like this. So the cost curve looks something like this total cost curve and so this is the cost curve. What do we have over here, in the blue region? What do we have in the blue region here and over here?

What we have in this blue region is that total cost is more than total revenue in these regions. So total cost being more than total revenue implies that the firm is incurring losses because its cost is more than its revenue. So this is the region where it is incurring losses and where is it incurring

profit? It is incurring profit in this region the green region. So it is incurring profit in the green region. So that basically means that in the range from q 1 to q 2 the firm is incurring profit and the firm can very well operate or produce any amount in this range. So if the firm produces any amount in this range it is going to incur a profit.

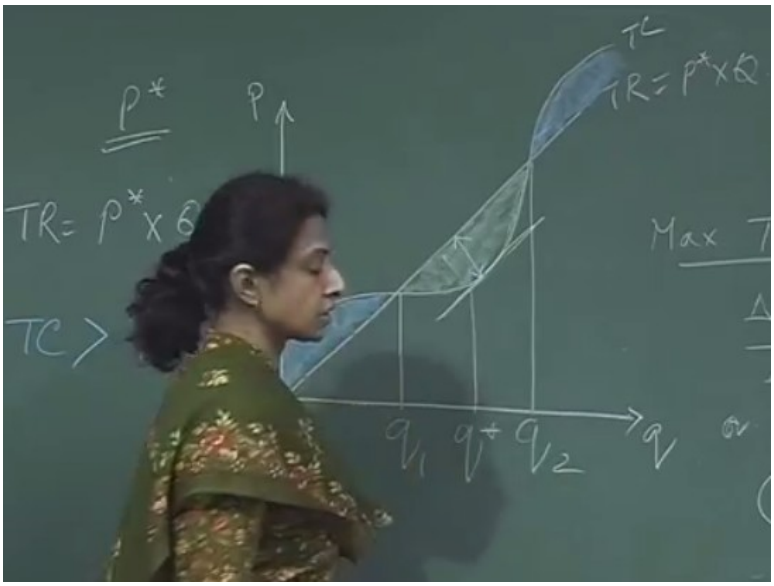
But what amount is going to maximize its profit. What amount is going to maximize its profit? The amount that is going to maximize its profit is the region where the distance between the revenue and the cost curve is maximum. So this is the so where total revenue minus total cost is maximum that is where that level of output should be the profit maximizing output level of the firm. So this is so this gap is maximum where the slope of the cost curve the slope of the cost curve the slope of the cost curve equals the slope of the revenue curve and that is nothing but the first order condition of maximization.

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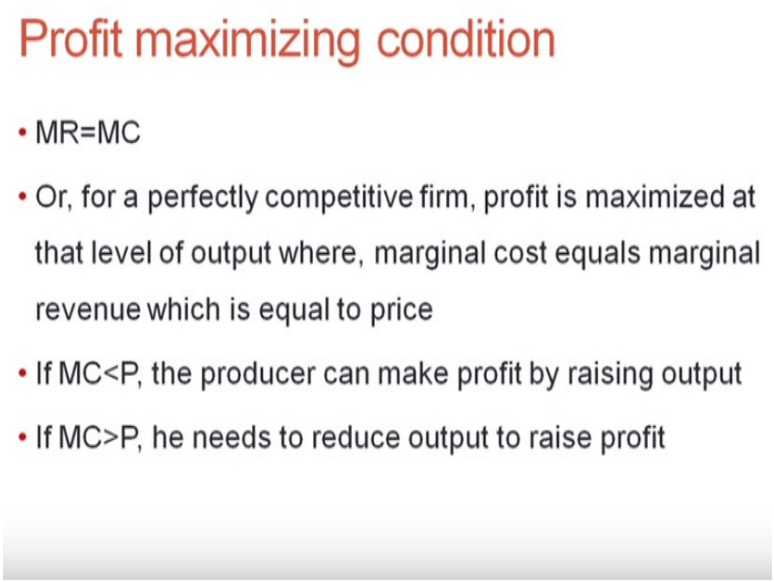
So when we maximize this our del TR del Q minus del TC del Q is equal to 0 or so del TR del Q is nothing but the slope of the revenue curve and del TC del Q is nothing but the slope of the cost curve and the slope is called the marginal revenue minus marginal cost. This is equal to 0. So marginal revenue equals marginal cost is my profit maximizing condition. So this is my profit maximizing condition and this happens at this region. So at this region my the difference between the 2 is maximum and marginal revenue equals marginal cost.

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Marginal revenue as we saw from the previous slide that this is the definition of marginal revenue and so this is the level at which the firm should basically produce. So it is going to the firm is going to produce the output Q star which is going to maximize marginal revenue and the difference between marginal revenue and marginal cost.

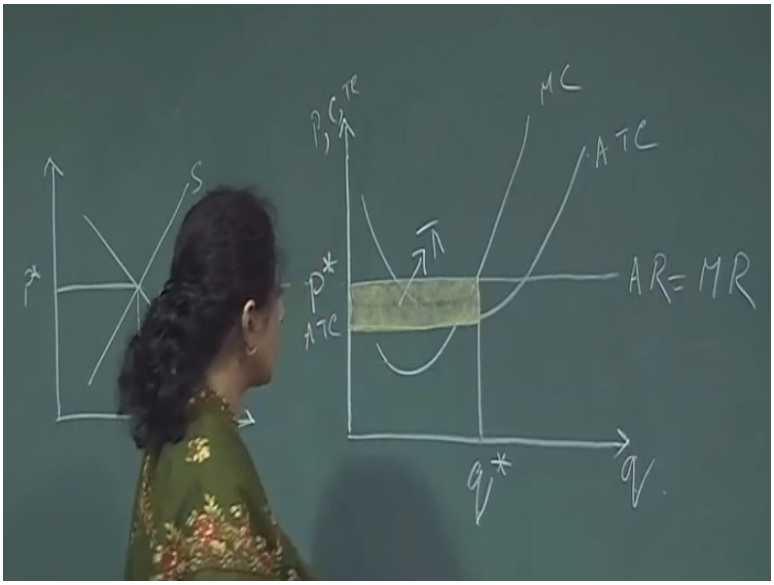
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So coming back to the profit maximizing condition, so profit maximizing condition is marginal revenue equals marginal cost or for a perfectly competitive firm profit is maximized at that level of output where marginal cost equals marginal revenue which is equal to price. Now coming back to this graph now what is the slope of the total revenue or what is this marginal revenue. This marginal revenue or del TR del Q is nothing but del TR del Q is equal to del PQ del Q since P is constant at P star so this is P star into del Q is equal to P star.

So for a perfectly competitive firm the profit maximizing condition is the price should be equal to the marginal cost. This is for only the perfectly competitive firm. So price equals marginal cost. Now what happens if price is more than the marginal cost? Let me clean this clean the board and show you a more familiar diagram that you are going to see in all your text books how profit is determined and how can we quantify the profit, the revenue and the cost. I am going to draw that diagram.

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So coming back to the this is the market demand and supply. This determines the price P star and the firm is basically so in this diagram we are going to draw all the revenue, price, cost, all the curves. So I am writing here P, C, R etc. So on the X axis you always have the quantity. It is rather the small q. This if for the firm and since the P is given so we draw the price line first. So

we draw the price line first. So this is P star and as we have seen from our understanding of the revenue so P star is equal to average revenue curve this is also your marginal revenue curve.

This is not only the price line this is the demand line that the firm is facing. This is the average revenue line of the firm and this is also the marginal revenue line of the firm. Now our profit maximizing condition we have already understood it is clear that profit maximizing condition is marginal revenue equals marginal cost.

So let us draw our very familiar marginal cost curve here. Now this is the marginal cost curve. This is our marginal cost curve. So profit maximizing condition says that the intersection between where the point where MR is equal to MC that level of output should be profit maximizing. So basically the firm is producing the firm should produce q star. Q star is going to maximize its profit. So does our problem end there? Does our problem end there?

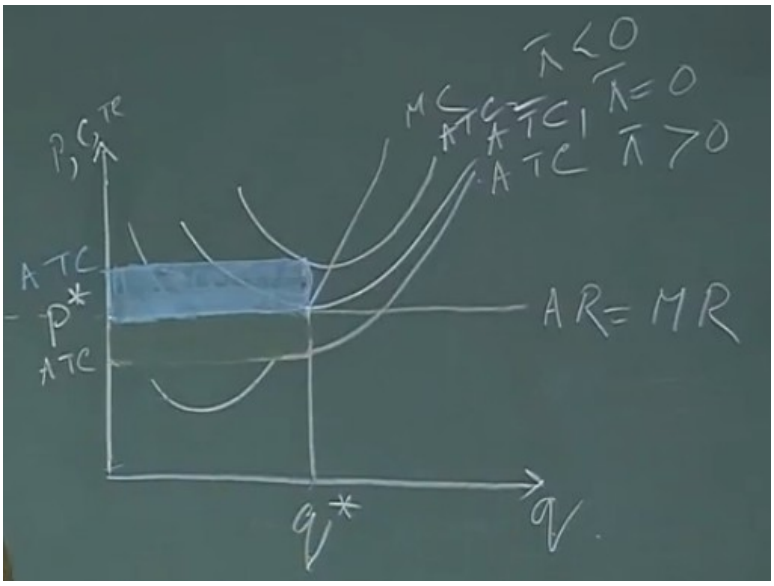
So we have determined price. We have determined quantity and we know about the cost and this is our profit maximizing condition. So is it that our problem ends here? No. because we need to understand this could be very well our profit maximizing level of output but is it actually giving me any profit or not? That is important to be understood.

So let us draw the average cost curve. So let us draw the unless and until I draw the average cost curve here I will not be able to understand how much cost I am incurring in producing this amount of output. So here my revenue is P into q. My total revenue is this rectangle but what is my cost. For the cost for understanding the total cost I need my average cost curve here. So in the situation that my average cost curve is somewhere here.

If my average cost curve is somewhere here so this is average total cost. So if my average cost curve is somewhere here then at q level of output my total cost is this is my sorry so this is my average cost. So average total cost so average cost multiplied by output this is my actual cost. So the entire rectangle is my total revenue minus the cost gives me my profit.

So we can very well see that since we can see that since my price line is above my average total cost so this firm is basically managing to get some profit here. So it is basically getting a profit here and this is the profit. So it maximizes its profit and it has an amount of profit, has a positive amount of profit.

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Now what happens if my average cost curve is what happens instead here instead say my average cost curve is here. Say this is my average total cost. If this is my average total cost, then also you may have noticed that I am trying to draw the average cost in such a way that the marginal cost intersects the average cost at its minimum because this is what we found out when we discussed about the cost curves that the marginal cost intersects the average cost at its minimum.

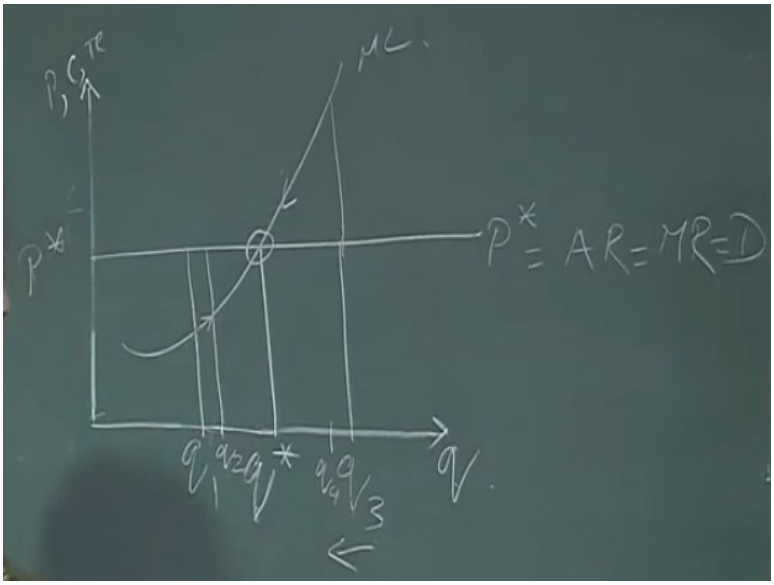
So that is how I am trying to draw and so when the average cost looks something like this. So this is another average cost. Say this is my average cost. So here my price is equal to my average cost which means my revenue is equal to my cost and here my economic profit is equal to 0. So here my profit is equal to 0. So when in case of ATC 1 so in case of ATC my profit is more than 0. In case of ATC 1 my profit is equal to 0.

Now what happens if my average total cost is somewhere here? It is high up. My average cost is higher than my price. So my average cost is higher than my price say it is ATC 2 so then what is my cost like? So here again this is my output. So at this output level this is my cost, at this output this is the cost that I am incurring, average total cost. So average total cost multiplied by my output this is my total cost and this is my revenue the lower box is my sorry this is not the revenue this part is the revenue.

So this is my price line marginal cost and this is my revenue and this is my cost. So my so this part is the, this part is the loss that I am incurring so out of this is the so this is the loss that I am incurring because this part is more than my price. The cost is more than the price that I can charge in the market so I am incurring a loss over here because the lower part is my this is my revenue. So in case of ATC 2 my profit is sorry this is not, ATC profit is more than 0, ATC 1 profit is equal to 0, ATC 2 profit is less than 0.

So this is how I determine how much profit I am going to so profit maximizing condition does not mean if you are able to determine the profit maximizing level of output the problem does not end there because the profit maximizing level of output may very well give you a loss as we are showing here. So we have to find out by using the average cost curve that actually how much profit or loss the firm is incurring in such a situation. Now what happens let me clean the board again.

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So this is my P star equals revenue is equal to MR is equal to D so everything and this is my marginal cost curve. Now what happens if marginal cost is less than price? So the produce can make profit by raising output. So what happens if the producer decides that let me produce at a level over here q 1. So what happens at q 1? Now at q 1 marginal cost although it is increasing but still it is much less than the price.

So marginal cost is less than the price. So additional unit of output if the firm is to produce it is going to incur a marginal cost which is much less than the price or the marginal revenue that the firm can expect by selling that unit of good. So it makes sense for the firm to produce that additional unit of output. So here q1 say this is q 1 to q 2.

So these are additional units of output it has to make a decision whether to go to q 2 or not. So obviously although its marginal cost is increasing still it is much less than P so it is going to earn some additional revenue, P is the additional revenue that it is going to get by selling this extra unit of output. So it keeps increasing the output till it reaches q star where its marginal revenue is equal to its marginal cost.

Now beyond q star what happens again beyond q star. Say for example by chance it is operating at this level where it is say q 3. Now here clearly it is incurring a high cost. So if it reduces its output by another unit say here q 4 if it moves this way if it moves this way it is its marginal cost is falling but it is still above the price that it is going to get.

So it keeps on reducing the output till it reaches this point. So till it reaches this point and here it is the marginal cost that it is going to incur by selling another unit of output is going to be the marginal revenue that it is going to get by selling that additional unit of output. So this is the optimum point where the firm it makes sense for the firm to operate. So the so if marginal cost is more than P he needs to reduce output to raise profit and if marginal profit is less than P the producer can make profit by raising output.

So that so this was our discussion about profit maximizing condition about how the firm decides what level of output to produce and will continue this discussion in the next module. Thank you.

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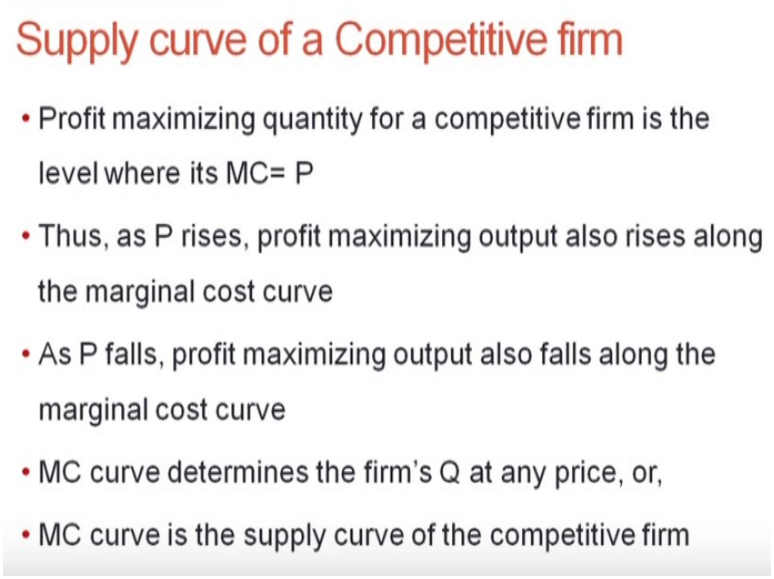
Lecture - 22

Supply Curve of Firm in Perfect Competition

Hello and welcome back to our discussion on perfect competition. We in the till the last module we discussed about profit maximization behaviour of a perfectly competitive firm and we saw that a perfectly competitive firm is a typical case where, since price is given by the market average revenue is equal to marginal revenue is equal to the price and that is also the demand line a horizontal demand line that the perfectly competitive firm faces in the market.

So what is the supply curve of a perfectly competitive firm? This is what we are going to find out because we need a supply curve of the firm and eventually we are going to develop the supply curve of the entire market and see what it looks like in a perfectly competitive market. So we are going to discuss about perfect competition and supply decision of a perfectly competitive firm.

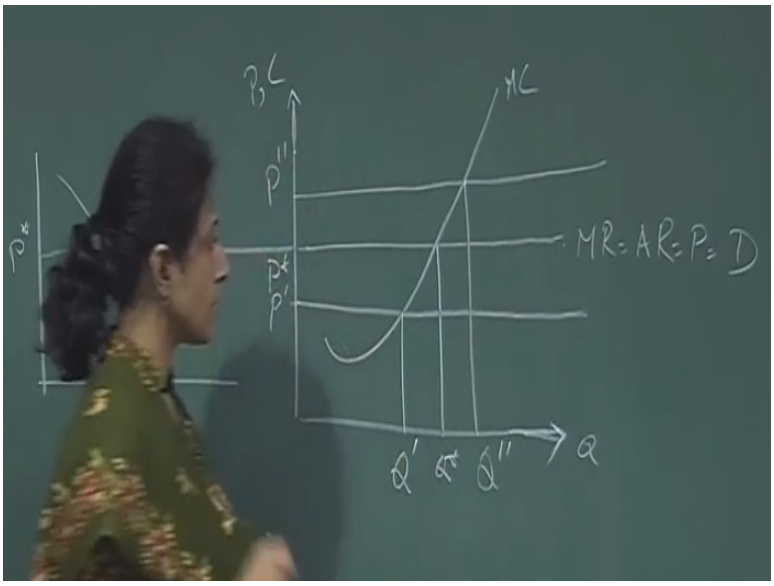
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Now the supply curve of a competitive firm let me just walk you through the slide then I am going to show on the board. The profit maximizing quantity for a competitive firm is the level where its marginal cost is equal to the price. So we saw in the earlier module that the intersection of marginal cost with price is going to determine the output that the firm would like to supply.

So thus as P rises you can imagine that as the price line shifts up or down profit maximizing output also rises along the marginal cost curve. As price falls profit maximizing output also falls along the marginal cost curve. So that kind of determines that the marginal cost curve determines the firm's output at any price or marginal cost curve is the supply curve of the competitive firm. So let me just show what we just discussed in the slide. Let me show through diagram.

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So we have price and cost here and as we saw earlier the market demand and supply they determine how much price is to be charged. So this is the price line which is given to the firm from the market and what it basically does is it tries to see where is this price line which is also its marginal revenue and average revenue and price and demand.

It tries to see the firm tries to see where does it intersect with its marginal cost curve. So this is the output level that the firm is going to supply. So the firm is going to its perfect maximizing level is marginal cost equals marginal revenue and so Q star is the optimum amount of output that the firm is going to supply when price is P star.

Now what happens when price changes? Say price falls to P prime. So when price falls to P prime again the firm is going to see where does its marginal cost intersect with its price line and so now this is the output level Q prime that the firm is going supply to the market. What happens when price goes up? When price goes up again it is going to see the intersection between MC and P and it is going to determine the new level of output say Q double prime.

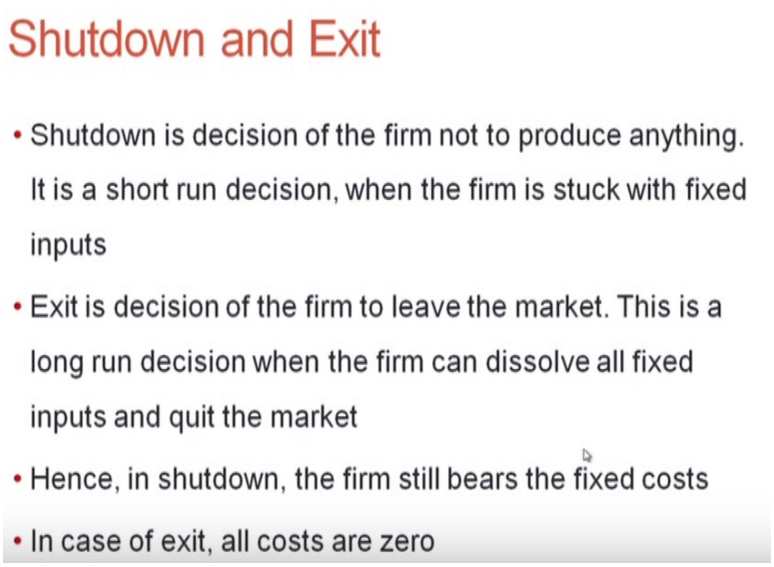
So basically what we are seeing is marginal cost curve is basically tracing out the output level at various price lines. So every point on the marginal cost curve shows the supply that the firm is willing to do at that price. So at that price, any price, the quantity that the firm is going to supply is given by the marginal cost curve. So effectively the marginal cost curve itself should provide the supply curve of a firm. But there is more to it which we will discuss later but so far it suffices

to say that the supply curve of a perfectly competitive firm is traced out by the by its marginal cost curve.

Now we move on to some like we when we started our discussion on perfect competition we also asked this question is the firm because it is a price taker is it always going to just keep on producing? Is it going to stay in business? What determines if the firm is going to stay in business? If it should produce any output at all or not. Whether it should think of quitting the market? How are these decisions taken in a perfectly competitive setup.

So we are going to talk about 2 situations. One is shut down which means that the firm is temporarily shutting down its production. It is not entirely quitting the market, it is just shutting down its production. It is just stopping its production and exit is when it completely winds up. It sells off everything and it just quits the market. That is the exit. What is the difference between the 2? Let us see.

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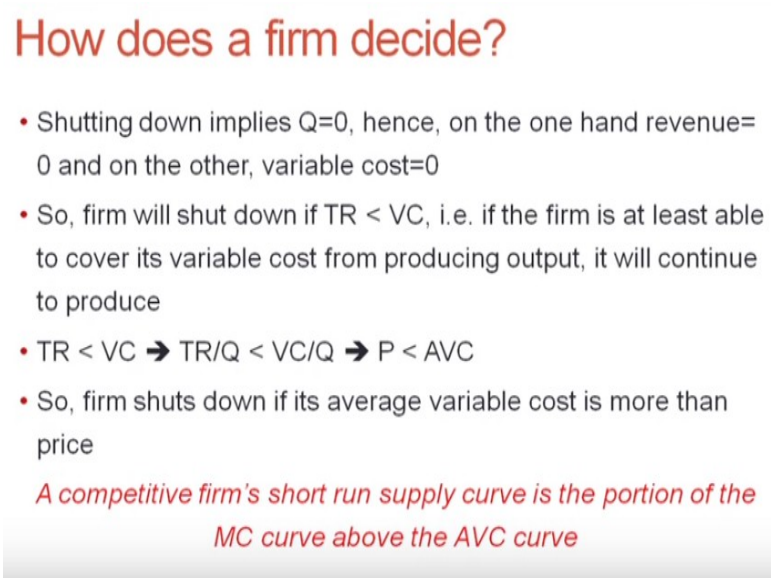
Shutdown is decision of the firm not to produce anything. It is a short run decision when the firm is stuck with fixed inputs and exit is a decision of the firm to leave the market. This is a long run decision when the firm can dissolve all fixed inputs and quit the market. So shutdown is something which the firm can do in the short run because in the short run it is not easy for the firm to come out of its fixed costs. It is it has already incurred certain fixed cost, it has probably

setup a plant, it has bought some land where it is operating or it has a huge machine which it has either borrowed.

Probably the firm is paying some mortgage some interest to which for borrowing money from the bank to buy these machineries. So all these are short run costs of the firm which the firm even if it stops production even if it does not produce anything it has to incur these costs. So the firm all that the firm can do in the short run is continue to incur these fixed cost but at least stop incurring the variable cost by stopping its production and that is called shutdown. Exit is a decision of the firm to leave the market.

So in the long run as we already discussed when we discussed about long run and short run cost curves we said that in the long run all variables are flexible. In the sense the firm can either decide to acquire more of its fixed inputs or it can decide to dispose of its fixed inputs. So everything is possible in the long run. So in the long run the firm if it is incurring losses the firm can decide to dissolve all its assets or all its fixed inputs and quit the market. That is called exit. Hence in shutdown the firm still bears the fixed cost. In case of exit all costs are zero but this cannot happen in the short run. This happens in the long run.

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So how does a firm decide? How does a firm decide whether it should shut down or not or whether it should exit or not? Now when a firm is shutting down, when a firm is shutting down

in the short run that implies that its output is 0. So when output is 0 on the one hand revenue is equal to 0 because it is not selling anything but on the other hand its variable cost is also 0 because variable cost depends it is directly proportional to the amount of output it is producing.

So this is like imagine there is a pizza joint a person a producer has come up with. He has opened a pizza joint and he has incurred certain fixed cost. He has rented that piece of he has rented some area some space to set up his shop and he has hired some people and he has bought some ovens and paying some rent for the paying some paying the electricity bills etc.

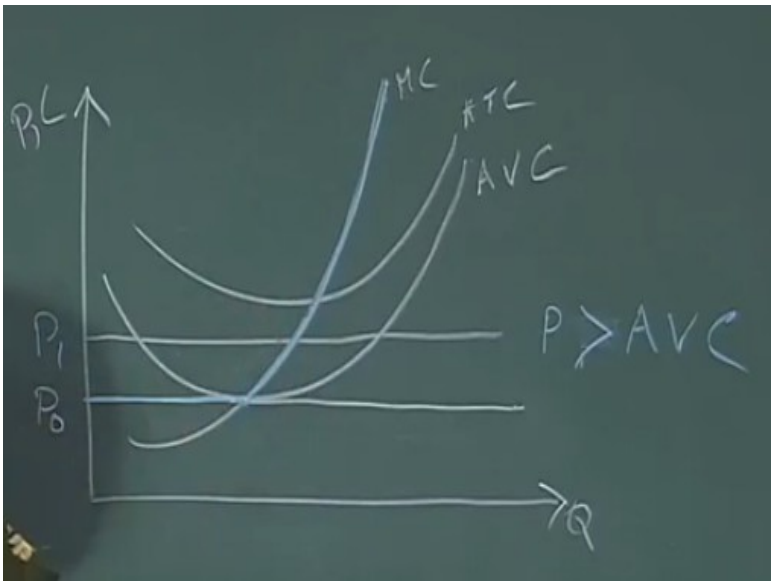
So it has already set up an infrastructure to produce pizza there and he has incurred some fixed cost and then he is running at a loss so his variable cost is basically the flour, the fuel that is required to run the ovens, the cheese, the even some of his labour force could be dependent on how much pizza he is producing. So these are his variable cost which he can cut down in the short run if he is running a loss and he may decide that fixed cost he has in any case he has to pay. Variable cost he can reduce by not producing any pizza.

So shutting down implies Q is equal to 0. So firm will shut down if total revenue is less than variable cost. That is the firm is at least able to cover its variable cost from producing output, it will continue to produce but the firm will shut down if the revenue that it is incurring by producing is actually less than the variable cost that he is incurring because he is producing those output the firm is going to shut down.

So this condition is total revenue is less than variable cost which implies that when we divide both sides by output and this is total revenue divided by Q is less than variable cost divided by Q and TR by Q is nothing but average revenue or which is equal to price P into Q by Q and VC by Q is average variable cost. So when price is less than average variable cost, the firm shuts down if its average variable cost is more than price.

A competitive firm's short run supply curve is the portion of the marginal cost curve above the average variable cost curve.

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So basically the same diagram that we drew earlier. So this is the marginal cost curve and this is the average variable cost curve and this could be average total cost curve. So basically what we are seeing here is that the firm will continue to produce in the short run if it's if the price is at least able to cover the average variable cost.

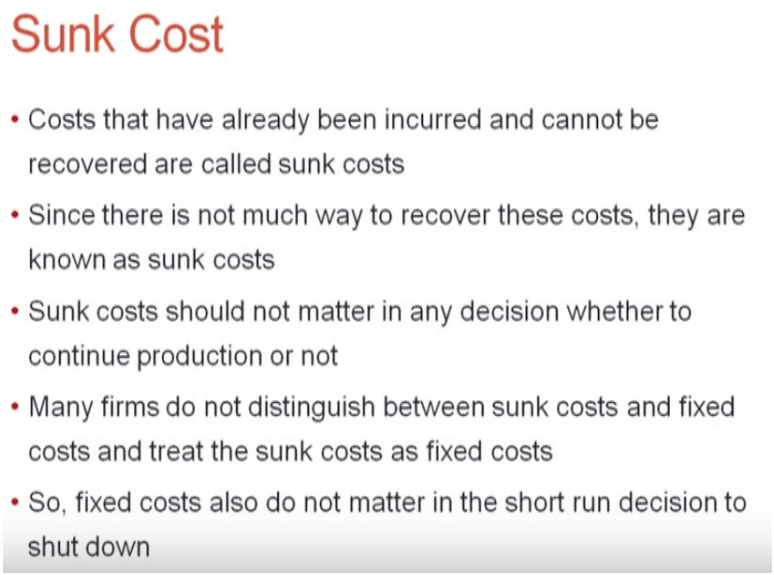
So this is the average variable cost and if the price falls here. Say for example the price is falling here. Say this is the price P 0. If the price falls below P 0 then the average variable cost is above the price line so the firm is going to produce only till here. So below this level if the price falls even below if the price falls below P 0 the firm is not going to produce. It is going to shut down.

So basically as we know that the marginal cost curve traces the supply curve but in this case basically the supply is till here. So the average the minimum of the average variable cost and the so this blue line is the supply decision of the firm.

In the short run the firm is going to supply till the average minimum average variable cost and below it will refuse to supply because if the price falls below the minimum of the average variable cost then the firm is going to incur a loss and it does not make sense for the firm to continue production so it is going to produce as long as the price is above the average variable cost and note notice here that average total cost is much above the price line although the average variable cost is below the price line.

Say if the price is P 1 the price is P 1, the firm will continue to produce because here the price is covering the price is covering the average variable cost. Price or price here price is at more than the average variable cost. So in the short run the firm will continue production because here the price is above the average variable cost. But the firm is not going to stop production below P 0 because here it this these price levels will not be able to cover the minimum average variable cost curve. So this is the average this is the supply so the blue line is the supply curve of the in the short run supply curve of the perfectly competitive firm.

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Next we are going to discuss about something called sunk cost. What is sunk cost? The costs that have already been incurred and cannot be recovered are called sunk costs. So since there is not much way to recover these costs they are known as sunk cost and sunk cost should not matter in any decision whether to continue production or not and many firms do not distinguish between sunk cost and fixed cost and treat the sunk cost as fixed cost. So fixed costs also do not matter in the short run decisions to shut down.

So in the short run decisions fixed cost does not matter because fixed cost you have to in any case pay. Fixed cost say for example the producer has rented a building to start his production he has leased a property for 1 year.

Then he is stuck with that property for 1 year and where he had decided to manufacture something and maybe his short run variable cost is not getting covered by the price that is prevalent in the market but still he has to pay the rent for the property that he leased for that period of 1 year of time. So 1 year is the short run for which he is stuck with this fixed cost. But after 1 year he can dispose of this fixed cost and come out of this situation.

So fixed cost is not exactly the sunk cost as we are defining here. Sunk cost is like let me give an example. Say for example someone has spent 2 years getting a law degree. He has spent a lot of money to get a law degree and then he gets the degree and after that he decides that I would really do not I would not like to pursue a legal profession. Instead I am going to be a pizza joint owner.

So I am going to just sell pizza so where he is not using his law degree. Now the cost that he has incurred in getting the law degree that is a sunk cost. That has already been done and that should not influence any of his future decisions. That should not influence any of its future decisions because that cost has already been incurred and that is not going to affect any of his production decisions in the future.

So similarly another example is say for example the pharmaceutical industry. They spend a lot of money on R&D and to come up with new drugs etc. and the huge amount of money spent on that. But whether the company is able to come up with new drugs or not that is not guaranteed. So the amount of money that the company spends in R&D is basically a sunk cost for the company and that should not influence any of its production decisions for the future. On the other hand fixed cost is not exactly sunk cost.

Fixed cost is basically the infrastructure it has created to produce and which has a short in the short run it is stuck with that but in the long run it is possible to salvage it. It is possible to dissolve it. It is possible to dispose it off. So that is fixed cost which is not exactly the sunk cost that we are defining here but nevertheless some producers some producers treat the sunk cost as fixed cost and they basically spread the sunk cost they have that they have incurred.

Say for example they have the pharmaceutical company can do that, that they have incurred certain amount of cost in the laboratories and they basically spread this cost over future years of production and show it as fixed cost. So that is possible technically but here the way we are

defining sunk cost is the sunk cost should not influence any decision of the firm whether to stay in business or quit.

Secondly so to give another example so it is the distinction between fixed cost, sunk cost, variable cost is important for a firm or an organization to know it should understand which of the 3 costs is more predominant in its line of business. Say for example the hardware industry. Say for example a industry where personal computers are manufactured.

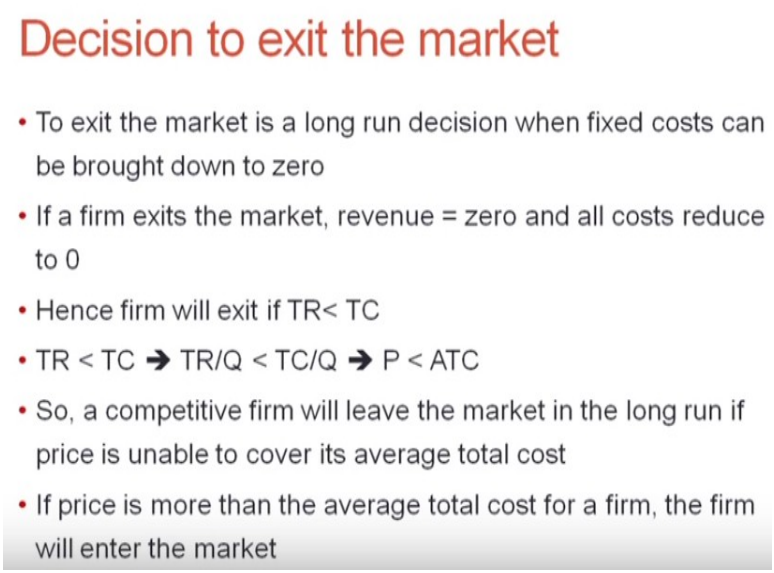
So in this kind of production the variable cost is predominant because the micro processing units, the hard disks, the sound and video cards, these are all variable costs because they depend or they will be dependent on the number of units of production. So they will be variable cost and that will be more predominant in the entire production process.

On the other hand a software industry or software firm is going to incur a lot of sunk cost. It may spend a lot of money in developing certain applications which there is no guarantee whether they will be bought or not where there will be any demand for them or not. So a software industry has a lot of sunk cost. A sunk cost is a major part of its cost.

Again say for example the or coming back to the example of the pizza joint in case of a pizza joint probably the proportionately the predominant cost is the fixed cost where to start the pizza joint one has to incur a lot of fixed cost for setting up the place, for decorating the place, to have the kitchen and equipments in place the chair table setup ambience everything.

That is the fixed cost and variable cost is comparatively lower. Variable cost is comparatively lower in case of a pizza joint. So these are the distinctions which are more important which are important for a business to understand when it is getting into the business to understand more of its cost, the nature of the cost that it is going to incur.

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Next we are coming back to our shut down and exit decisions. Now we are going to talk about decision to exit the market. To exit the market is a long run decision when fixed costs can be brought down to 0. If a firm exits the market revenue is equal to 0 and all costs reduce to 0. Hence firm will exit if total revenue is less than total cost. So total revenue is less than total cost implies that price is less than average total cost. So a competitive firm will leave the market in the long run if price is unable to cover its average total cost. If price is more than the average total cost for a firm the firm will enter the market. So this is not only about exiting the market it is also about entering the market. So if price is more than average total cost, if the price is more than the average total cost for a firm the firm will enter the market.

So basically the firm looks at the price and the relationship between the price and the difference between price and average total cost. So it can be a decision making variable for the firm whether to enter the market or to exit the market. So if the price is more than the average total cost of the firm and it has not entered the market yet it may it sees the potential to make some profit in this market and it is going to enter the market.

On the contrary in the long run if the firm is not able to cover its average total cost so in the long run the firm may decide to quit the market and basically dissolve all its fixed as well stop as soon as it stops producing its variable cost goes down to 0 and in the long run it has the choice of disposing of its all fixed assets or fixed inputs and come out of the market so that its cost come down to 0.

So that brings us to the end of our discussion on firms, how does the how does the firm, a perfectly competitive firm decides decide its level of output. How does it decide its supply decisions in the short run and how does it decide whether to stay in business whether to shut down in the short run whether to exit the market in the long run. So these are the discussions that we have done so far.

In the following module we are going to talk about the market supply. So far we have talked about the firm but in the following module we are going to talk about the perfectly competitive market like in totality how does the market supply curve look like and how do the firms behave in the long run as a part of the market in a perfectly competitive environment. Thank you.

Foundation Course in Managerial Economics

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Lecture - 23

Supply Curve of Market in Perfect Competition

Welcome back to our discussion on perfectly competitive market where we have been discussing about the supply curve of the firm. So far we have discussed how the supply curve is determined in the short run, in the long run by the perfectly competitive firm and now we are going to move on to look at how does the supply curve look like in a perfectly competitive market. This is what we are going to do but before that before we proceed to determine what the market supply curve looks like we shall make some very simplistic assumptions.

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So the assumptions that we make are all existing firms and potential entrants have identical costs. So what we are assuming is basically the all the firms are exactly the same. They look exactly the same. They have exactly same kinds of cost structures and any potential entrant who might enter the market later also they also have very identical costs.

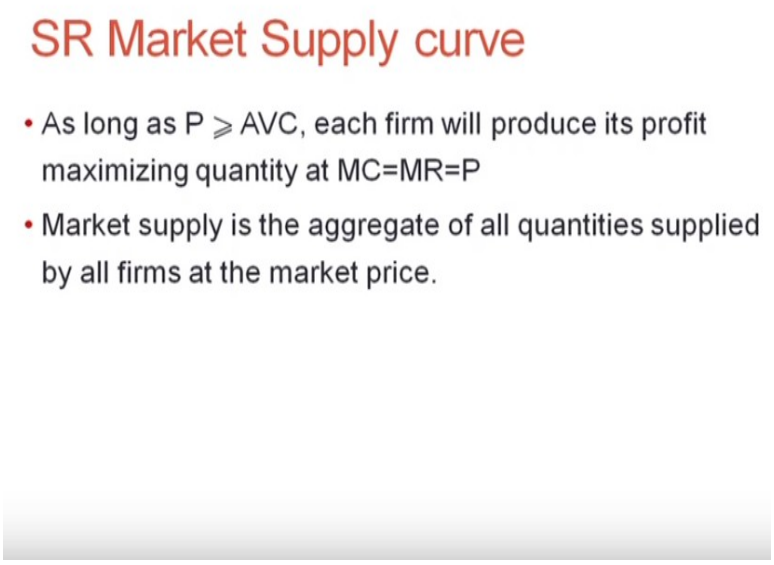
So this is not a very realistic assumption because we know that even in a perfectly competitive setup also there are various firms of various sizes and they have different kinds of cost structures. So it is not a very realistic assumption and we are going to see what happens if we relax this assumption later but for the time being let us assume these and see what the market supply curve would look like.

So the first assumption is all existing firms and potential entrants have identical costs. Each firm's costs do not change as new firms enter or exit the market. So we are assuming that if new firms enter the market or exit the market that does not change the cost situation for the existing firms in the market.

So that also is not a very realistic assumption because we may think that if there are huge number of entrants in a market there will be a lot or pressure on the existing inputs. That can happen right. So in those situations the cost of inputs may go up and costs may not remain the same as new firms enter or exit the market as we are assuming here. This assumption also we are going to relax later and see how that changes our long run market supply curve.

Third we are assuming that the number of firms in the market is fixed in the short run because of the fixed cost they can shut down but they do not exit the market and the number of firms in the market is variable in the long run due to free entry and exit because it is easy to enter or exit the market so in the long run the firms which are running at a loss they are going to easily exit the market and firms or new entrants who are hoping for some potential profit in this market they can enter the market. So this is a realistic assumption.

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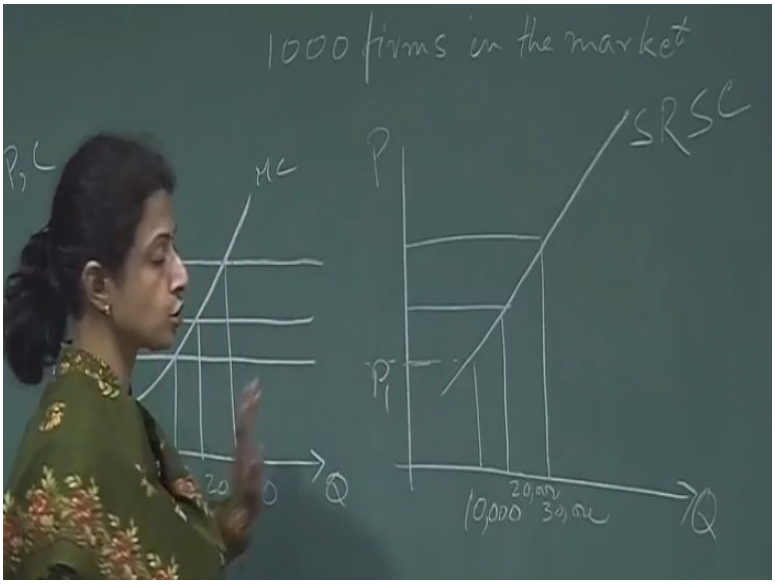


Now first the short run market supply curve. What does the short run market supply curve look like? In the previous discussion we already said that as long as price is above the average variable cost in the short run each firm will produce its profit maximizing quantity at marginal cost equals marginal revenue equals price.

So this is what we did in the previous module and we showed that the short run supply curve of a firm is basically the minimum of the average variable cost and from that point onwards it is the marginal cost curve of the firm. This is how we drew the short run supply function of a firm. Now market supply is nothing but it is the aggregate of all quantities supplied by all firms at the market price.

So at any market price all the firms whatever output they are supplying the aggregate of all that is going to give us the market supply curve but we know from our assumption from the previous slide we have already assumed that all the firms are identical. So let me draw the market supply curve what would it look like.

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So if this is a firm this is a typically a typical firm and this is the price line and this is the marginal cost and at P 1 say there are it produces 10 units of output, it produces 10 units of output. At P 2 it produces 20 units of output. At P 3 it produces 30 units of output and so on. So this is for the firm this is for a typical firm but by our assumption all firms look like this. All firms look like this and if we are assuming there are 1000 firms in this market so what does the supply curve look like for the market.

So for the supply curve also this market so for the supply curve if there are we are assuming there are 1000 firms in the market and these firms are all similar by our definition by our assumption so at P 1 10,000 units of output is produced. At P 2, 20,000. At P 3, 30,000 and so on.

So this basically gives us the market supply curve. So this gives us the market supply curve. Here we have P, here we have Q and this is the short run supply curve of the market. This is the short run supply curve of the market and which we arrive at from the individual firm's supply curve.

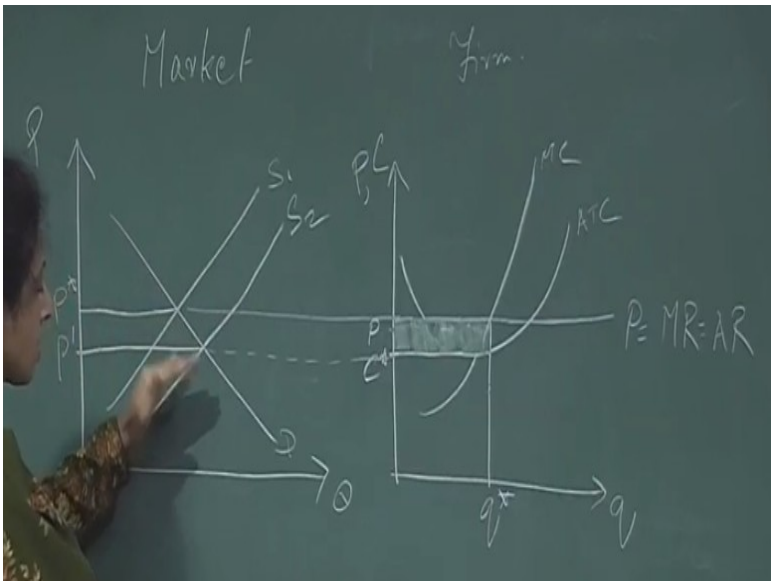
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Okay now moving on in the long run what happens? In the long run there is entry and exit in the market. So in the long run firms enter and exit the market and if existing firms earn positive economic profit new firms enter the market. So if there is positive economic profit new firms are going to enter the market.

As the number of firms in the market goes up supply curve shifts to the right and hence prices fall in the market slowing entry. This process continues till all existing firms earn zero economic profit in the long run. The firms which incur losses will exit the market in the long run thus shifting the supply curve to the left and raising prices thus reducing losses of the remaining firms. Now what happens here?

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So what happens is basically say let me show what is being said in this slide. Say this is the this is the market. This is the market, price, quantity. This is demand and this is supply. Now there is this firm which is like any other firm and this is the price, this is the price and this is small q, this is P and costs and this is the this is its price line which is equal to MR, AR and this is the marginal cost of this firm and say this is the average cost of the firm.

So say this is the average cost of the firm in the short run, this is the market, this is the firm. So in the short run the firm is incurring some profit why because this is the price line and at the intersection of marginal cost and marginal revenue this is the q star is the optimum amount of output that the firm is going to produce. At q star the cost of the firm is C star and the profit that the firm makes is this green region.

So in the short run the firm is making the firm is basically making a profit of the amount of this green rectangle and looking at the profit in the long run what will happen is more firms are going to enter the market. Since there is positive economic profit to be made in this market and since we have assumed that all the firms are similar in their in terms of cost they are they all are making these profits. So new firms enter the market.

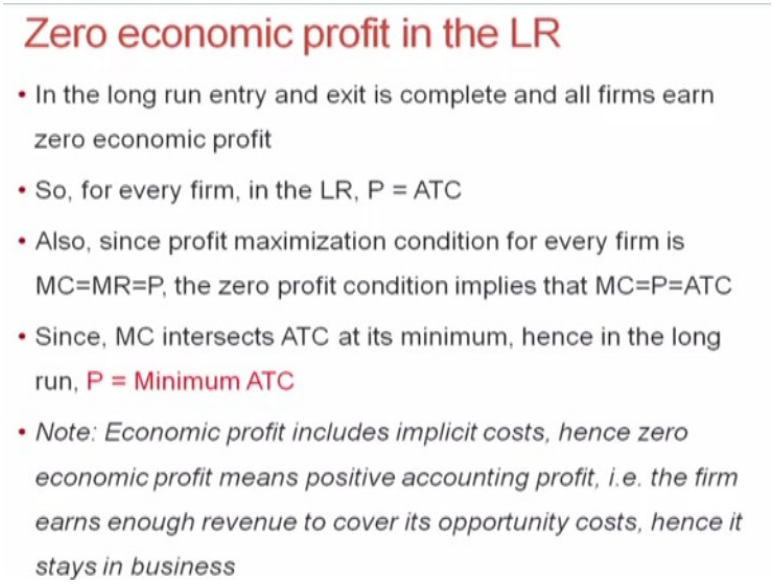
As new firms enter the market the supply curve in the market shifts to the right. So as supply shifts to the right so supply shifts to the right as new entrants, new firms enter the market supply shifts to the right and then the equilibrium price falls from P star to P prime. As it falls from P

star to P prime we will see that gradually this process is going to continue. This process is going to continue till all this profit is wiped out.

So basically firms will keep entering the market and as firms keep entering the market the supply curve will gradually shift to the right and as the supply curve keeps shifting to the right price falls in the market till price reaches the minimum of the average total cost of the firm. So when the price reaches this minimum there is zero economic profit to be made in the market and this is where the firm's entry will stop.

After this no more firms are going to enter in the market because already the existing firms are making zero economic profit. So in the long run the price is going to fall here and in the long run price will come down to the minimum of the average total cost curve and all firms will be operating at the level where price is equal to marginal cost is equal to the minimum of the average total cost so that all firms incurring zero economic profit. So this is what has been said in the slide.

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Now let us discuss a little more about zero economic profit. So in the long run entry and exit is complete and all firms earn zero economic profit. So for every firm in the long run price is equal to average total cost. Also since profit maximizing condition for every firm is marginal cost

equals marginal revenue equals price in case of perfectly competitive market marginal revenue is equal to price as we have discussed earlier.

So the zero profit condition implies that marginal cost is now equal to price is equal to average total cost because if the average total cost is less than price then there will be positive economic profit. But since zero profit condition has been reached so it is basically marginal cost is equal to price is equal to average total cost.

But at what point is marginal cost equal to average total cost? At what point of the average total cost curve is marginal cost equal to average total cost? You may remember we had discussed this when we had talked about the cost curves and we said that the marginal cost intersects the average total cost at its minimum. The marginal cost intersects the average total cost at its minimum. Hence in the long run price is equal to minimum average cost.

So basically price is falling till where the price is equal to the minimum average cost of average total cost of all the firms in the industry because we have assumed all the firms have very similar cost. All the firms have exactly similar cost is what is our assumption. So in the long run prices will fall as firms keep entering the market prices will fall till the prices reach the minimum of the average total cost of all the firms.

And just a quick recap, why does the marginal cost intersect the average total cost at its minimum? Because as we discussed earlier we took the example of grade point average and this is another example I am going to give. Say for example the average cost of producing pizza is say say 100 Rs and the cost of producing an additional unit of pizza is going to be 105 Rs. So is the average if I produce that additional unit of pizza is my average cost going to go up or down?

Now since the average cost, the marginal cost is higher than the average cost so my average cost is going to go up if I produce that additional unit of pizza. On the other hand if the cost of producing an additional unit of pizza is actually 98 Rs which is less than the average cost right now so if I produce that additional unit of pizza my average cost is going to go down.

So in that case my marginal cost when my marginal cost is below my average cost the average cost is falling. When my marginal cost is above my average cost my average cost is rising. So my marginal cost will be equal to my average cost at the point where average cost is minimum.

So this was the logic and coming back to our discussion so it basically means that if prices fall keep on falling and price is equal to marginal cost is the profit maximizing condition so basically price is for zero economic profit the price has to be equal to the average total cost of the minimum average total cost of the firm. Okay so zero economic profit does that mean that the firm is running at a loss or the so where is the income coming from?

So the firm is basically not getting anything because it is zero economic profit? No. If you remember that we discussed about economy, difference between economic profit and accounting profit and we said that economic profit includes implicit cost or opportunity cost. So when you have zero economic profit that basically means that this profit includes implicit cost hence zero economic profit means positive accounting profit. That is the firm earns enough revenue to cover its opportunity cost, hence it stays in business.

So economic zero economic so in the long run all perfectly competitive firms are going to earn zero economic profit which means they have positive accounting profit and which means it is they have they earn enough revenue to stay in business yet they operate at a level where price is equal to their minimum total cost.

In the next module we are going to extend this discussion a little more and look at what happens when there are demand changes in the market and what happens to supply and we are going to look at the long run supply curve of the firm in the next module. Thank you.

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Lecture - 24

Market Supply (Contd.)

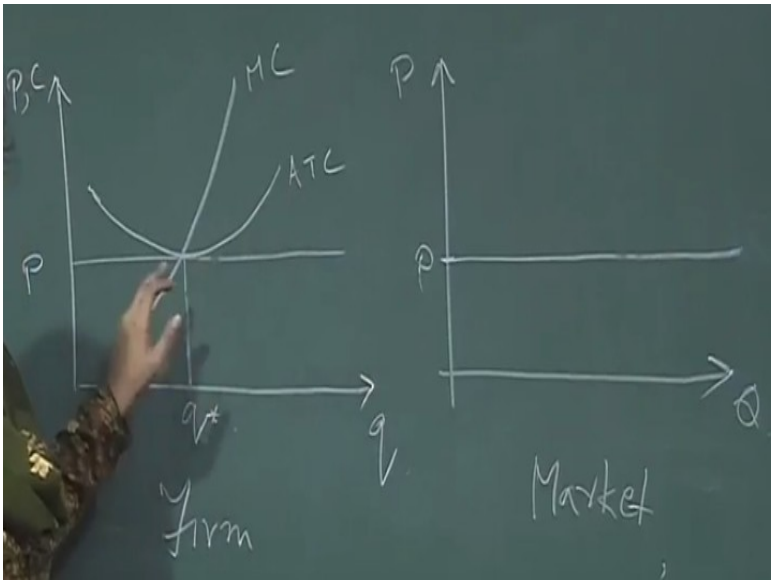
Welcome back to our discussion on perfect completion and market supply curve we left off the earlier discussion with talking about how the market supply curve is arrived at and we made certain assumptions and to move ahead.

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So basically the long run market supply curve so the long run market supply curve as we have assumed that all firms are identical and as all firms produce at price equals minimum average total cost the long run market supply curve is horizontal at price is equal to minimum average total cost. So let me explain what this means.

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This basically means that here we have a typical firm, here we have a typical firm and they are all identical and so this is the average total cost, this is the marginal cost of the firm and this is the price. So this is for the firm and this is for the market. This is for the firm and this is for the market but we know that all the firms are exactly similar. In the long run they all are operating at the minimum of the average total cost where marginal cost equals average total cost equals price and this is the situation for all the firms in the market.

So basically so basically at this price at this price the supply in the market is basically a horizontal supply curve for the market. So the long run supply curve in the market is horizontal because all the firms are exactly similar. In the long run they operate at minimum of their average cost and minimum of the average cost is equal to the price.

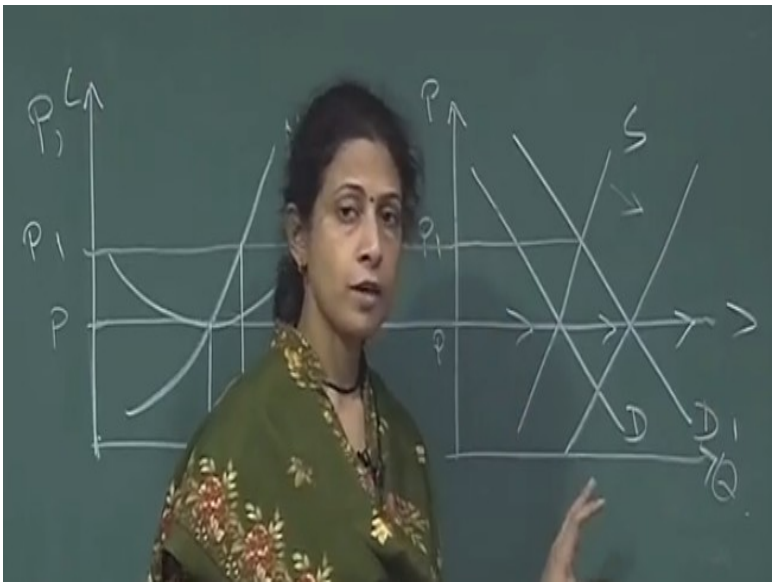
So depending on how many firms are there or how much output is can be produced like depending on how many firms enter the market and they produce. So at this price the amount of

quantity supplied is unlimited because the price is not at the, the price gets fixed at the minimum of the average total cost and the supply curve of the market is horizontal.

So basically this quantity gets the depending on how many firms are there in the market the market supply will be determined but the price is not going to change. So for that because this price is the price at which this price is the level at which average total cost of all the firms is minimum. So the firms are operating at their minimum cost and that is the price that they are charging in the market and at that price unlimited supply is possible in the market. So long run market supply curve is horizontal.

So now to discuss a small variation in what we have been discussing what happens if there is a impact of an increase in demand in a market. So far we have talked about supply and demand is something exogenous in the market. It is so far we have not discussed about impact of demand in the market.

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Now say so basically again I will draw a typical firm and I will draw a market. So this is a typical firm. Again they are all similar and I will draw a draw the aggregate the market. This is price and say this was the price that was being charged, this is the price. Okay, so this is the price

and this was the market demand and supply curve. So this was the market demand and supply curve which determine the price and so and this is the marginal cost.

Now in the short run what happens is say the demand shifts to the right. All of a sudden there is more demand for this good and the demand curve shifts to the right so because of which price increases from P to P 1. So demand curve shifts to the right price increases from P to P 1 and when that happens so initially it was say here. So this is P 1 and now the firm is producing a higher level of output so it was earlier producing here q and this is q 1.

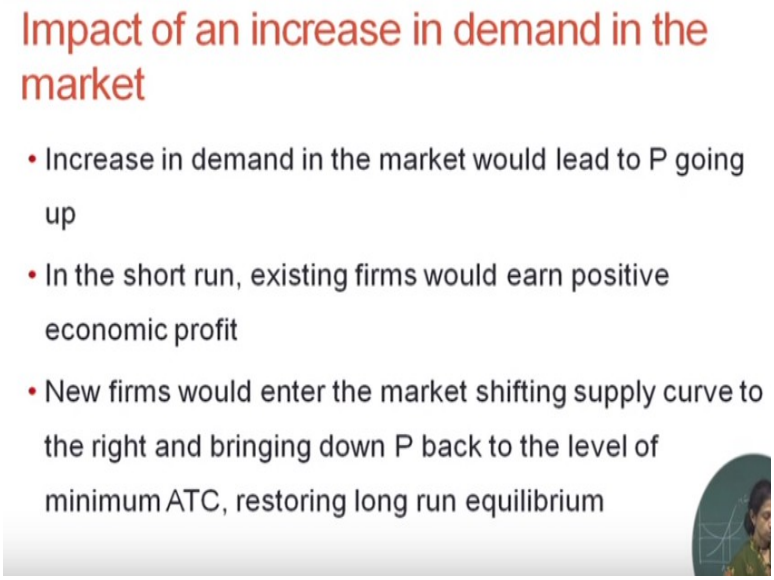
So it is producing a high level of output and it is getting a profit. So since the price has gone up so the average cost is now lower than price and this firm is basically earning a profit. So this firm is basically earning a profit. Now what happens? When this firm is earning a profit, looking at the profit other firms enter the market.

So other firms are going to enter the market which shifts the supply curve to the right and the process is going to continue as we discussed in the previous module P so that all the firms are now again operating at the minimum of their average total cost. So the so due to demand increase initially there was potential for some positive economic profit.

Looking at the economic profit new firms enter the market which shifts the supply curve to the right and when supply curve shifts to the right price starts falling in the market and price will keep on falling in the market till price hits the minimum of the average total cost so the situation is again back to where it started from that all firms are operating at zero economic profit and all firms are producing an output at the level where their marginal cost is equal to their minimum average cost is equal to the price in the market.

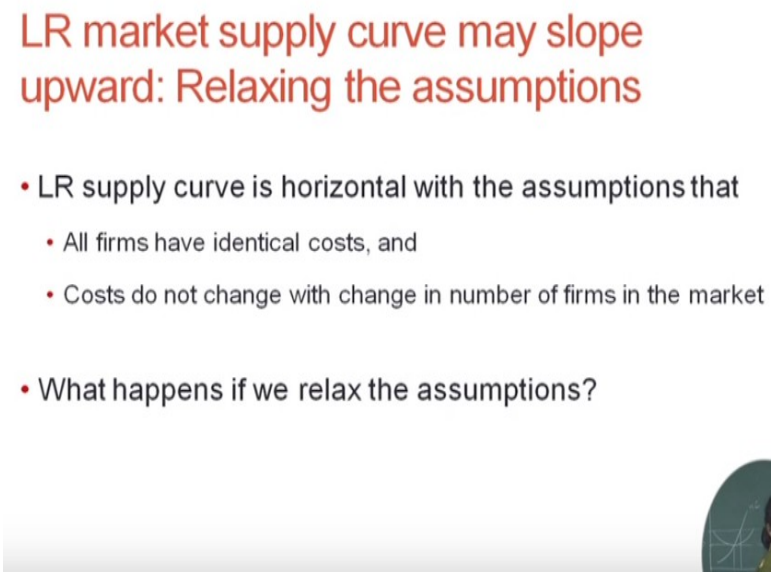
So this situation happens and the price finally goes back to P and the output has just increased. So in the long run so this is how it increases. So in the long run the supply curve in the perfectly competitive market it is a horizontal supply curve at the price level which is equal to the minimum of the average cost of all the existing firms in the market.

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So to summarize increase in demand in the market would lead to P going up. In the short run existing firms would earn positive economic profit. New firms would enter the market shifting supply curve to the right and bringing down P back to the level of minimum average total cost restoring long run equilibrium.

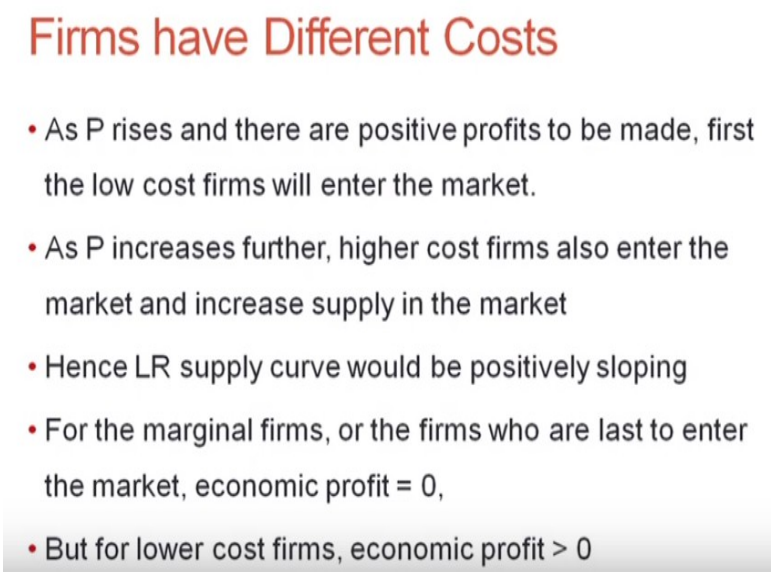
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Now what happens if we relax the assumptions that the cost are that the cost structure for all the firms are exactly same and costs do not change when firms enter or exit the market. What if we relax these assumptions? What happens to the long run market supply curve and we will see that the long run market supply curve can actually slope upward if we relax this assumption.

So long run supply curve is horizontal with the assumptions that all firms have identical costs and costs do not change with change in number of firms in the market. What happens if we relax the assumptions?

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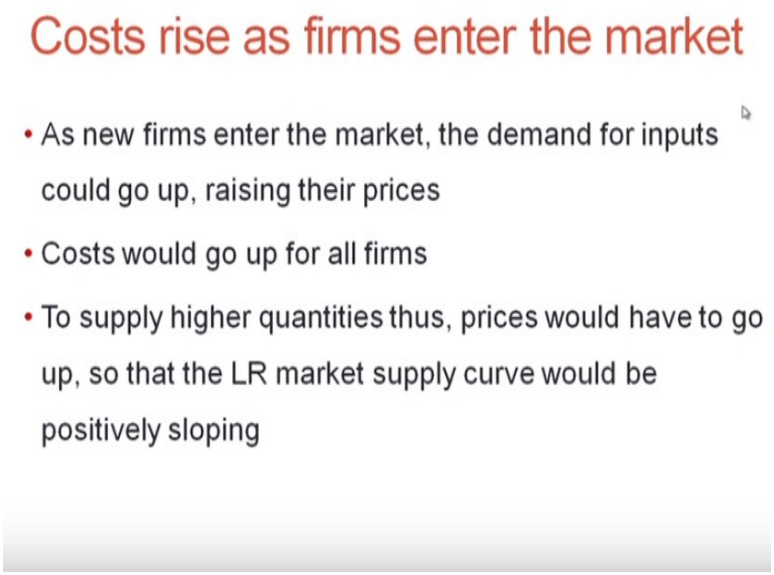
So first let us assume that the firms have different costs. So it is a very realistic assumption. Firms may have different costs and as price rises and there are positive profits to be made first the low cost firms will enter the market. So in this situation the firms or the entrants are not exactly same as all the entrants all the firms they are not similar to each other. So what happens when price rises initially it will be the low cost firms who will enter the market.

So as price increases further and further even higher cost firms find it lucrative or find the potential of making profit in the market and they also enter the market and as they enter the market that increases the supply in the market even more.

Hence long run supply curve would be positively sloping. So since costs are also increasing with higher cost firms entering the market so the long run supply curve would as high cost firms enter the market on the one hand the costs are increasing but on the other hand the quantity supplied is also increasing. So we have a positively sloping supply curve in this situation.

For the marginal firms or the firms who are last to enter the market. Obviously the firms which have higher costs are the ones who are last to enter the market. So for the marginal firms the economic profit is equal to zero but for more efficient firms or firms with lower cost they can manage to continue to have positive economic profit in this situation. So lower cost firms economic profit may be greater than 0.

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Now what about the second assumption? What happens if cost rise as firms enter the market? What happens when cost rise as firms enter the market? As new firms enter the market the demand for inputs could go up raising their prices. So this is a very realistic assumption or this is a very realistic situation that when there is more and more firms entering the market for production of the same good the pressure on input can go up. The inputs demand could go up.

Say for example there is say land for producing certain cash crop the amount of land is limited so the cost of land is going to go up or say for example beach front property. So if the cost of if

more and more firms try to build beach front property then where is the land going to come from? So the land cost is going to go up for all the firms.

So as new firms enter the market the demand for inputs could go up raising their prices. So cost would go up for all firms. So to supply higher quantity thus prices would have to go up so that the long run market supply curve would be positively sloping. So in both the situations we see that if we relax both the assumptions it is possible to have a positively sloping long run supply curve in a perfectly competitive market.

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Coming to the last question that we asked in the first slide when we initiated our discussion on perfectly competitive market that, is the equilibrium efficient or not? Is the equilibrium efficient or not? When we try to discuss this what do we mean by efficiency? So there could be 2 levels of efficiency. One is called the productive efficiency another is called the allocative efficiency.

Now what is productive efficiency? Productive efficiency is basically production at minimum cost. Now productive efficiency is achieved in a competitive market as the most efficient firms enter the market first and firms operate at their minimum average cost. So if we go by the first set of assumptions that all the costs are similar there we saw that all the firms are operating at the minimum of the average cost. So which means basically all firms are operating at the most efficient combination of inputs possible.

In the second set of assumptions if we assume that different firms have different costs there also we are seeing that the most efficient firms are entering the market first and as price goes up the less efficient ones are entering the market but nevertheless to make any amount and the less efficient firms are basically not earning any positive economic profit. It is the more efficient firms which are earning the positive economic profit which basically means that production productive efficiency is achieved in a perfectly competitive market.

What about allocative efficiency? Now allocative efficiency what it means is basically that the allocative efficiency is who gets the goods? Who gets the good at what price and who supplies? So is there efficiency there in the allocation of the resources and allocation of the output. Basically in perfectly competitive market allocative efficiency is achieved in a competitive equilibrium because it ensures that the willingness to pay that is the price by the buyers where does this P come from. Where is where does this price come from.

This price basically is coming from the demand curve of the market and demand curve basically represents the willingness to pay by the consumers for every level of output. So if Q is the amount that is getting sold in the market then for Q level of output P is the willingness to pay of the buyers and in case of perfect competition we are seeing that the willingness to pay the competitive equilibrium is ensuring that the willingness to pay that is P by the buyers for the last unit produced is equal to the marginal cost of producing it.

So basically the supply the cost of supply of the product is equal to the willingness to pay for the product. So the cost of supply of the last unit which is the marginal cost of supply is equal to the willingness to pay for that last unit which is equal to price. So that ensures that allocative efficiency is achieved in this perfectly competitive market.

So this happens because in perfectly competitive market price is equal to marginal revenue and hence the profit maximizing condition of the firm implies that at equilibrium marginal cost is equal to marginal revenue is equal to price. So basically the cost and benefit of that single output the last unit of output is equal in a perfectly competitive market and that ensures that the equilibrium of perfectly competitive market is efficient.

Now we will be talking about this efficiency will coming will be coming back to this efficiency aspect again when we discuss about monopoly and when we look at the equilibrium situation in a monopoly market and we will discuss more of efficiency in that context and it will be interesting to contrast the equilibrium in case of monopoly to the equilibrium in case of a perfectly competitive market.

So that ends our discussion on perfectly competitive market. In the following week we are going to take up the other extreme of the market structure which is monopoly and see how the producer decides how much to since the producer is a price maker in case of a monopoly unlike a perfectly competitive firm. So we are going to see how he decides what price to charge and how he decides how much output to produce. Thank you.

Foundation Course in Managerial Economics

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Lecture - 25

Monopoly

Hello and welcome back to the foundation course in managerial economics. Till the last week we discussed about we started our discussion on different kinds of market structures and we discussed in details about the perfect competition. We understand that perfect competition has certain assumptions.

We went through the assumptions of perfect competition that it has large number of buyers, the products are homogenous so basically they so the products are highly substitutable across different sellers and that leads to almost zero market power of all the sellers as a result of which all the sellers and buyers in the market are price takers.

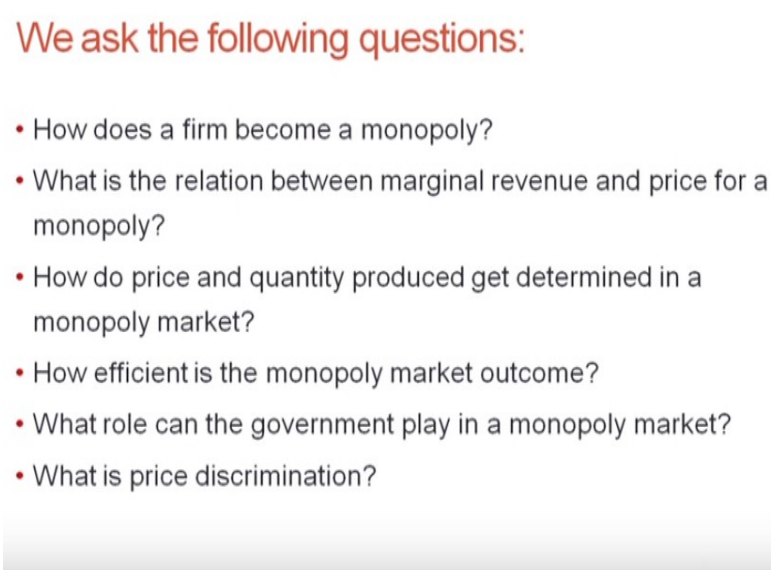
So and then we went on to when we looked at the profit maximizing condition and where the perfectly competitive firm chooses its output. We saw that the final market outcome or the market equilibrium that is reached in a competitive market that is highly efficient. We saw that there was productive efficiency that is the products were getting produced at the minimum cost across all firms; the most efficient of the firms produces first and we also saw that there is allocative efficiency that is the willingness to pay for the marginal unit is equal to the cost of producing that marginal unit.

So that was our discussion of perfect competition and since the market outcome is efficient it is the most desirable outcome for the economy but does perfectly competitive market perfect competition does it exist in reality although it is desirable but do we find enough perfectly competitive markets in the real world and what we are going to when we look around we see that the assumptions of perfect competition are far from fulfilled in any of the markets that we see around.

So we are going to see that in most of the markets most of the markets for different kinds of products that we see around us the markets have few number of sellers or very few number of sellers. The products are highly differentiated and because of and so they are not perfectly substitutable products as a result of which the sellers have some amount of market power. That is they have some control over the price that they can charge and the most extreme case of this situation is the monopoly.

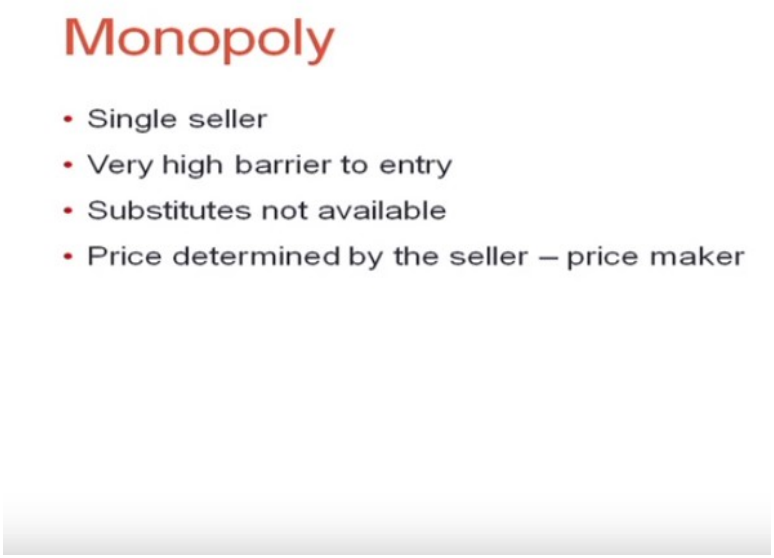
As we discussed when we discussed about the different market structures we already discussed this that the other extreme of perfect competition is monopoly where we have one single seller. So in this week we are going to discuss about what happens in a how does a firm become a monopoly and what happens what does the market equilibrium look like. What is the final market outcome and is it efficient. These are the kinds of questions that we are going to ask this week and let us see.

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So we ask the following questions. How does a firm become a monopoly? What is the relation between marginal revenue and price for a monopoly? How do price and quantity produced get determined in a monopoly market? How efficient is the monopoly market outcome? What role can the government play in a monopoly market? So since perfectly competitive market outcome we found out that it was efficient so if we see some kind of inefficiency in a monopoly market outcome can the government have some role to play there to correct it. That is the question that we are going to ask and finally we are going to ask what is price discrimination. Since we say that the monopoly seller has is a price maker, has some control over price can he price discriminate that is can he charge different prices to different of its buyers. So these are the questions that we are going to ask this week.

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To quickly a quick recap of the what a monopoly basically what we mean by monopoly? So monopoly we have a single seller. Very high barrier to entry. Very high barrier to entry and that is the reason that we have a single seller because others are not being able to enter the market and the monopoly seller knows that. Substitutes are not available and so price is determined by the seller and hence the seller is kind of a price maker. He is not a price taker but he is a price maker.

So the question that next comes to mind is how does a firm become a monopoly. How does it happen that one firm emerges as a monopoly player. How does it happen that if the barrier to entry is so high one firm is there in the market others are not being able to enter. So what how does a firm basically become a monopoly and we are going to see that there are very broadly there are 3 reasons because of which a firm can end up becoming a monopoly.

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The first is the firm is major holder of a key resource. The firm is a major holder of key resource. So it often happens that let us take the example of the biggest diamond raw diamond company which is De Beers Company. They have owned most of the diamond mines all over the world mostly in Africa since the 19th century and they have been the single largest seller of raw diamond during this entire period from late 19th century to almost the beginning of 21st century.

So and they have held their position as the largest owner of most of the raw diamonds across the world through various of their business strategies and that is how they ended up being the single largest seller or the kind of they were able to attain a monopoly status in the market for raw diamonds. So that is how they became they emerged as the as a monopoly player in the market and another example is although this is not a natural resource that I am talking about but I would also take the example of Microsoft.

So when Microsoft came up with a extremely user friendly operating system during a time when there were not too many user friendly operating system existing in the market. So as a result they were the major holder of this key resource although this is not a natural resource like diamonds but this is a key holder of a resource which everyone was looking for.

There were it is not that operating system did not have major substitutes in the market, obviously Linux and all were there, but not everyone could use these operating system as easily as the

Windows. So basically having come up with something like the Windows was something which made the Microsoft a monopoly player in the market for operating systems.

Now the second way by which a firm can emerge as a monopoly player is monopoly by law. That is the law gives the firm a status to be the single seller in the market. So how can that happen? That can happen through say patents through copyright laws and the example of public monopoly. So let me give examples.

I discussed in the previous when in the earlier weeks when I was discussing about sunk cost. I had taken the example of pharmaceutical companies. I had said that pharmaceutical companies have immense sunk cost in the sense that they are they are investing a lot of money on research to come up with come up with new drugs new formulas new compositions. So they a spending a whole lot of money for it and the end result may or may not bring any returns to them.

So there is a huge sunk cost for the pharmaceutical companies. So what happens if fearing this kind of sunk cost, the pharmaceutical companies stop spending on research and development. So that is not something very desirable and hence we have the patent laws which kind of gives a protection against protection to the pharmaceutical companies that if they come up with something really useful they will have the sole right to sell that composition for some period of time.

So that is by law the law is giving the power to that firm to be the single seller in the market or giving the monopoly status to the firm. So that is the patent and copyright laws and on the other hand we have public sector companies public sector monopolies like the Indian Railways in India.

So Indian Railways although it although the railway network as I discussed again in the during when I was discussing about cost I said that certain businesses have very high fixed costs and railway setting up railway networks etc. obviously it has huge fixed cost and which has been basically borne by the government and we have the single Indian Railways but it is not that many countries actually have private railways also running in the in the country.

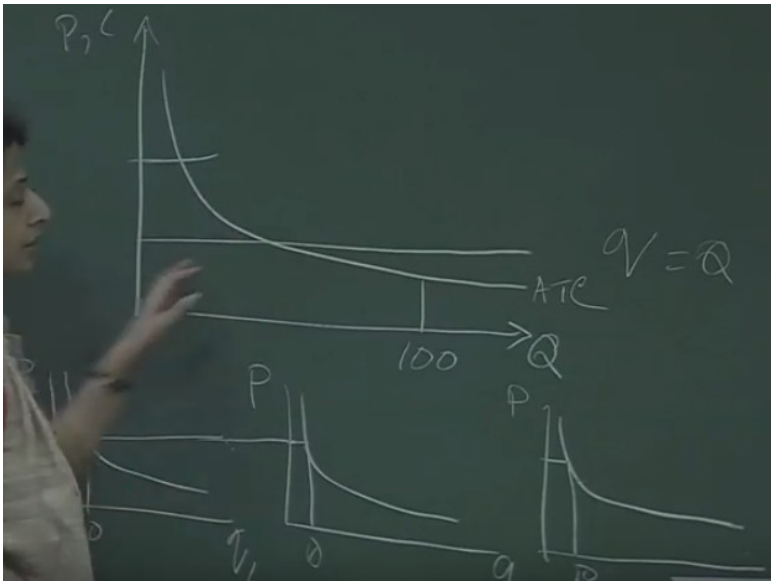
So basically the status of the monopoly status to the railways or to the postal service has been given the monopoly status has been given by the law by the government by the law of the land that it says that the postal service is completely it is a monopoly player.

Another kind of monopoly is the natural monopoly. Natural monopoly is why do we call it natural monopoly because it makes sense for a single firm to produce for the entire market to utilize economies of scale and bring down price for the buyer. So natural monopoly basically means that the market is such or the nature of the product is such and the production process and costs are such that it makes sense for one single large firm to produce the output for the entire economy for all the consumers because in that case when the amount of output is very high the firm is able to get economies of scale as a result of which prices will also come down for all the consumers. So this is called a natural monopoly. An example is an example of natural monopoly is say setting up a power generation plant.

So in case of power generation plant the initial cost of setting up the plant the fixed cost is enormously high say for example it is a coal fired thermal power plant. So the initial fixed cost of setting up the power generation unit is extremely high and not only the generation unit the distribution, the network, the towers, the lines the electric lines that have to be set up to distribute the powers to reach the power to the final consumers the entire setup is extremely costly.

So the initial fixed cost of such a unit is extremely high and in comparison the variable cost is low. So the variable cost for generating power is what it is primarily the say the fuel cost and some other incidental cost to produce the electricity. So the so the initial fixed cost is extremely high and the variable cost is extremely low so that gives a typical shape to the average cost which I am going to show here on the board which will may clear that why it makes sense why such kind of setup actually makes this kind of output or this kind of market a natural monopoly.

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So one more thing it is a very which I would like to point out is that I said in the previous week that we are going to distinguish between the firm and the market the the quantity produced by the firm we are going to denote it as small q and the output produced in the market as big Q but since in the monopoly the firm is the single sole producer or seller in the market so small q is equal to big Q for a monopoly.

So I am going to use it synonymously, so I am going to use Q here. So what does the average total cost then look like? So in a typical any typical firm we said that average total cost is a U shaped curve but in case of the natural monopoly that we have described here and we said that the fixed cost is extremely high so the average total cost takes the shape of the average fixed cost and if you may remember the average fixed cost keeps on falling with rise in output.

So the average fixed cost so the average total cost looks very similar to the average fixed cost. So this is the average total cost and so if the price if the total so if this is the single large power generating unit in the economy and it is catering to all the households in the economy then say it is being able to sell over here. Say it is able to sell around 100 units of power.

So when it is selling such a huge amount of units its average cost comes down. Its average cost is as low as here and in such a situation it is possible to charge a price it is possible to charge a price somewhere here and even then make a profit because average cost is here lower than the price here or even bring down the price even lower.

Now what happens if we say that okay we do not want a monopoly and we want lot of firms to produce electricity then we are going to have lot of small firms we are going to have lot of small firms who again have to incur the fixed cost because they cannot avoid the fixed cost of before they want to start producing power they have to incur the fixed cost and for them also the average cost will look like this. but now since they have split up the market amongst themselves so they will be each of them will be selling to only a part of the market.

So earlier 100 units were getting sold. So now maybe every so there are 10 such small power plants. Each of them is say selling around 10 units. So each of them if they are selling around 10 units which is maybe here so the cost becomes very high for all the small power plants. So they are able to so they are able to sell only a small amount of power that is around 10 units per power plant and so their cost becomes very high and they will not be able to sustain at a low price.

So they will have to charge a very high price which will be somewhere here and the buyers will also have to pay for this very high price because there are too many sellers in the market who are all operating with very huge fixed costs. So this is called a natural monopoly where it makes sense to not have so many small power plants and have a single big power plant which caters to the entire economy and thus produce a huge amount of output and having economies of scale which brings down the cost for this firm and hence the price for the entire economy and price for the all the buyers in the economy.

So this is called a natural monopoly where it makes sense to have a single firm but we will however this is a very atypical case where you have very huge cost and this is but it is not the only case where you have monopoly in the economy. There can be various others reasons of having a monopoly in the economy. So we are going to stick to the usual cost diagram where we have the fixed cost, the average cost and the marginal cost.

So we are going to stick with these typical cost curves and see what is the monopoly outcome, what is the monopoly combination of price and output that happens in the market when the monopolist is trying to maximize its profit. So that is what we are going to do in the following modules of this week. Thank you.

Foundation Course in Managerial Economics

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Lecture - 26

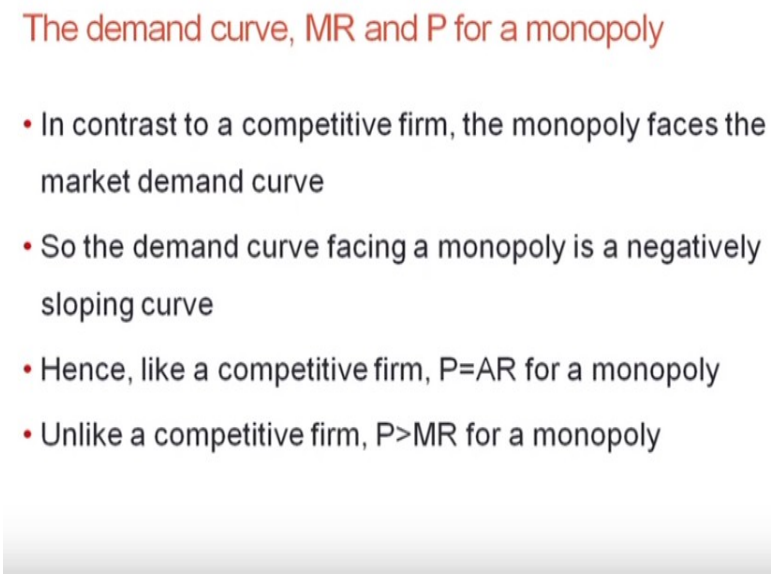
Monopoly - Determination of Price and Quality

Welcome back to our discussion on monopoly. We started with introducing monopoly. We said how does monopoly happen? How does a firm gain the status of a monopolist and we saw that there could be 3 reasons for very broadly there could be 3 reasons because of which a firm can end up being a monopolist.

And so we basically looked at and the 3 reasons that we discussed were it could be the owner of a single it could be the single largest owner of a certain important resource or it could be by law it is a monopolist because it has some patent or copyright or by public directive it is a monopolist or it is possible that it is a natural monopolist where it natural monopoly where it makes sense for the market to have a single large producer who is catering to needs of all the consumers to keep the prices low using economies of scale.

This is what we discussed in the previous class and now today we are going to discuss about determining the price and output. So now that we have a single seller in the market and we know that he is he has some control over price how does he determine the price? How does he determine how much output to produce? How does he determine what price to charge? So these are the questions that we are going to ask in this section.

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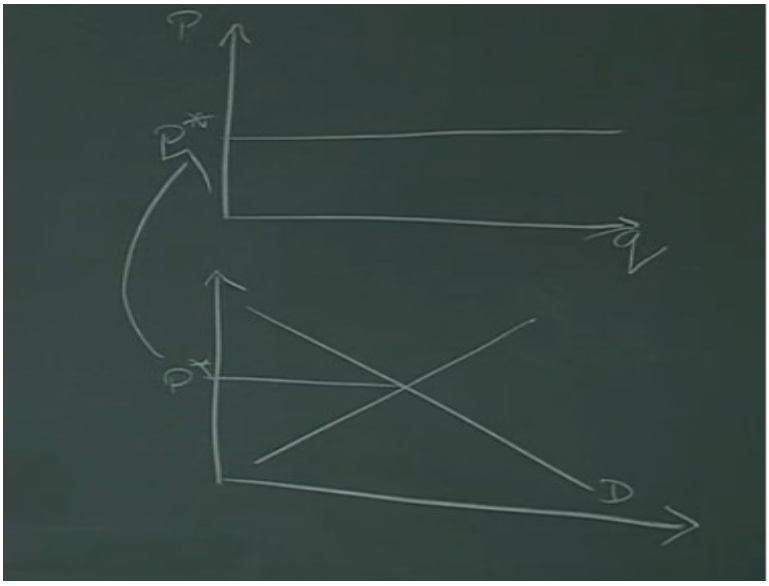


So before we go on to profit maximization we have to familiarize ourselves with the demand curve that the monopoly faces, the marginal revenue, and the price that finally the monopoly charges in the market. So this is the very familiar track that we followed in the case of determining the equilibrium price and output for perfect competition also and that is what we are going to do again for monopoly also. So let us see.

So what is the difference between a competitive firm and a monopoly firm? So in contrast to a competitive firm the monopoly faces the market demand curve. So the monopoly faces the market demand curve. So the demand curve facing a monopoly is a negatively sloping curve. So we all know that from our initial discussion of demand and supply and we have always said that the market demand curve is negatively sloping.

Hence like a competitive firm price is equal to average revenue for a monopoly. However unlike a competitive firm price is actually more than marginal revenue for a monopoly. So let me explain. So let me explain.

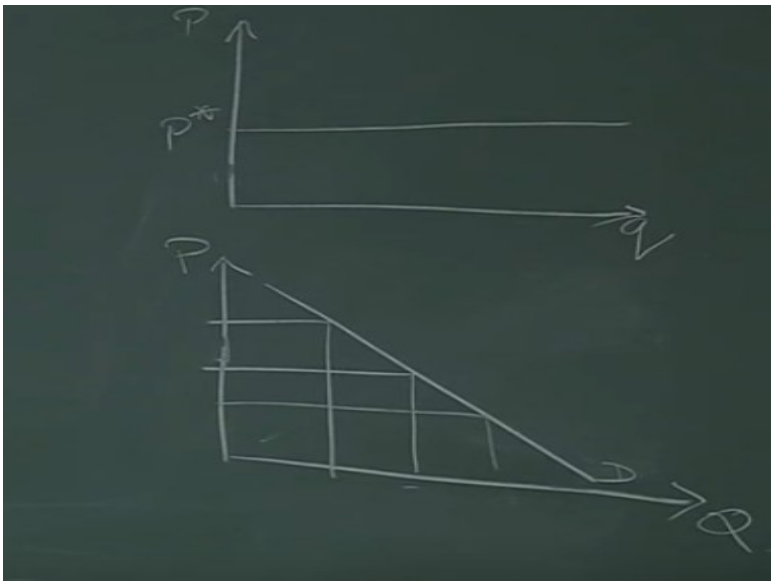
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So what we are saying here is if you may remember from our discussion on perfect competition we said that the perfectly competitive firm faces a demand curve which is horizontal. So the for the perfectly competitive firm when it enters the market it knows that it cannot charge price anything other than P star and it knows that at this price any amount could be sold in the market.

So this is the demand curve facing the perfectly competitive firm. But that does not mean that the market demand curve is not negatively sloping. So the market demand curve is still negatively sloping and this equilibrium P star is basically this P star in case of a perfectly competitive market.

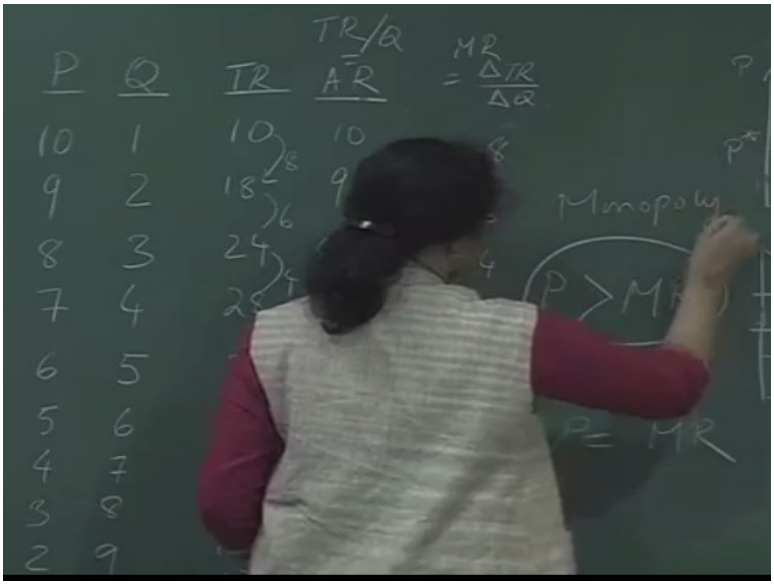
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But in case of a monopolist, but in case of a monopolist since he is the single supplier in the market, he is the single seller in the market he faces the market demand curve which is negatively sloping. So he faces this demand curve and for him he has to make a decision along this demand curve where to produce.

That is he knows that if he produces any output here this is the price that he has to charge. If he produces here this is the price that he has to charge. If he produces here this is the price that he has to charge. So this is this is the knowledge to the monopolist that the demand curve is negatively sloping and he just basically has to choose what to produce and how much to produce and what price to charge. So this is the demand curve, is a negatively sloping demand curve for the monopolist.

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So let us take an example of a very simple demand curve. Say for example the monopolist faces a demand curve where this is the price, the series of prices, quantity. So this is the demand curve. Say price is 10, 9, 8, 7, 6, 5, 4, 3, 2 and 1 and corresponding quantity is say 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10. So as price is falling the quantity demanded is increasing. So this could be a demand curve. So what is the total revenue?

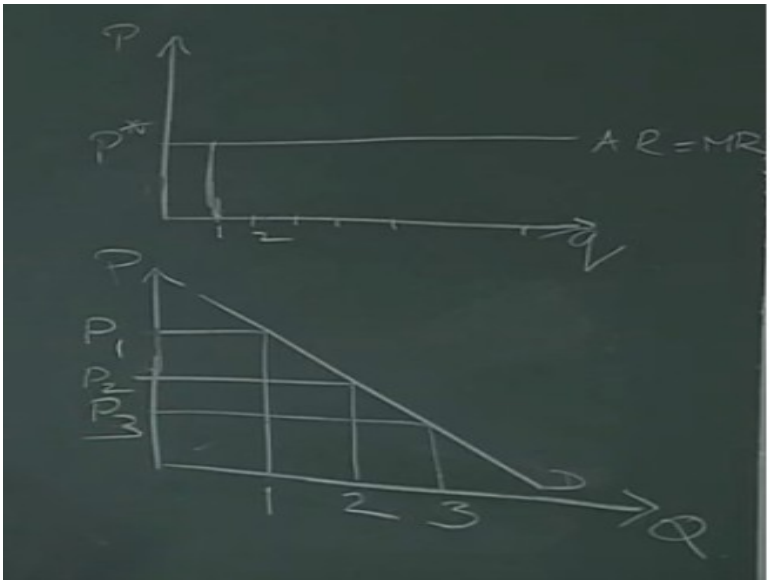
So total revenue is 10, 18, 24, 28, 30, 30, 28, 24, 18 and 10. So this is the total revenue. So we would like to know what is the so you can all see from here that average revenue is price because total revenue divided by quantity is average revenue. So average revenue is always price. So like perfect competition average revenue is 10, 9, 8, 7, 6, 5, 4, 3, 2 and 1. So average revenue is total revenue by Q is equal to average revenue.

So this is so P is equal to average revenue like in the case of perfect competition. Now what is marginal revenue? So marginal revenue is del TR del Q. So here the change in quantity is 8. Here it is 6. Here it is 4. It is 2, 0, -2, -4, -4, -6, -2, -4, -6 and -8. So marginal revenue is basically 8 let me write it here 8, 6, 4, 2, 0, -2, -4, -6, -8.

So marginal revenue is this. So price so what we see from here is price is actually greater than marginal revenue. So at all the points we are going to see that price is greater than marginal revenue. So price is more than marginal revenue. So this is what we are seeing here in case of monopoly, price is more than marginal revenue while in the case of competition price is perfect

competition price is equal to marginal revenue and for a monopoly price is more than marginal revenue. So this is what we see from this example. But what is the intuition? How do we explain this over here? So let me clean the board.

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Okay, so the marginal revenue in case of a perfect competition is the marginal revenue in case of perfect competition is this is demand this is demand is AR is equal to MR and why is the marginal revenue equal to price? So imagine that this is 1, this is 2 and so basically when that the next unit of output that the seller sells if the producer decides that I am going to sell or produce additional one more unit of output he sells that additional unit for price P for price P star because the demand is a horizontal line.

So no matter what he sells till what amount he sells every unit that he sells he sells that additional unit for price P star and the revenue that he gets out of selling one additional unit which is actually the marginal revenue so the amount that he gets out of selling one additional unit or marginal revenue is equal to P star or the price.

So that is the reason that price is equal to marginal revenue in case of a perfect competition. But what happens in the case of the monopolist. Now in the case of monopoly what happens is say this is the first unit, this is the second unit, and this is the third unit. Now when the monopolist is charging price P 1 for the first unit, for the second unit if now he decides that I am going to

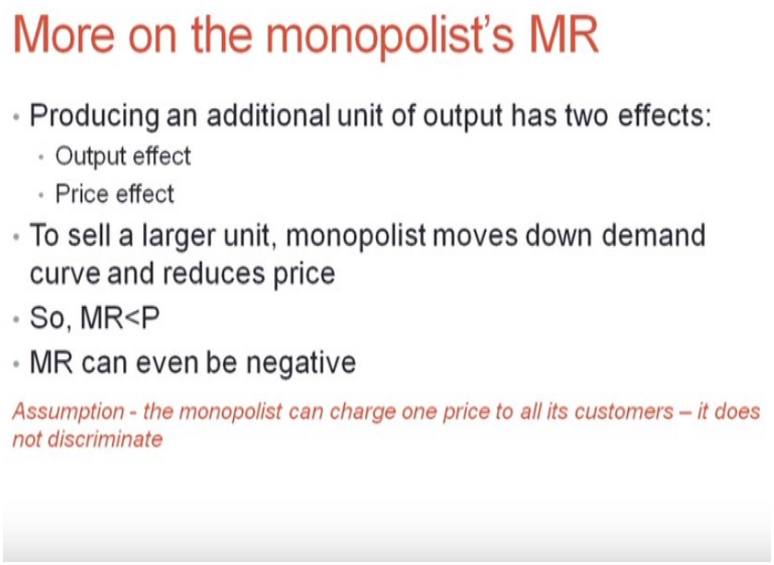
produce one more unit and sell it in the market how much does he get for each of his units. He does not get P 1 because as soon as he increases the units that he is selling in the market as soon as he sells 2 units his price falls to P 2.

Similarly, when he reduces his when he sells the next unit or the third unit his price falls to P 3 and when the price falls when he is selling the third unit and the price falls to P 3 he is not selling the third unit only for P 3. He is selling all the 3 units at price P 3. So there his marginal revenue goes down when he sells an additional unit his marginal revenue is going to may go down actually.

So that is the difference between the perfect competition and monopoly is that for the monopolist he has a decision to make. For the perfectly competitive firm he does not worry he does not have to worry about marginal revenue. He knows that marginal revenue is going to be P no matter what like the no matter how many units he sells in the market his marginal revenue is always going to be P.

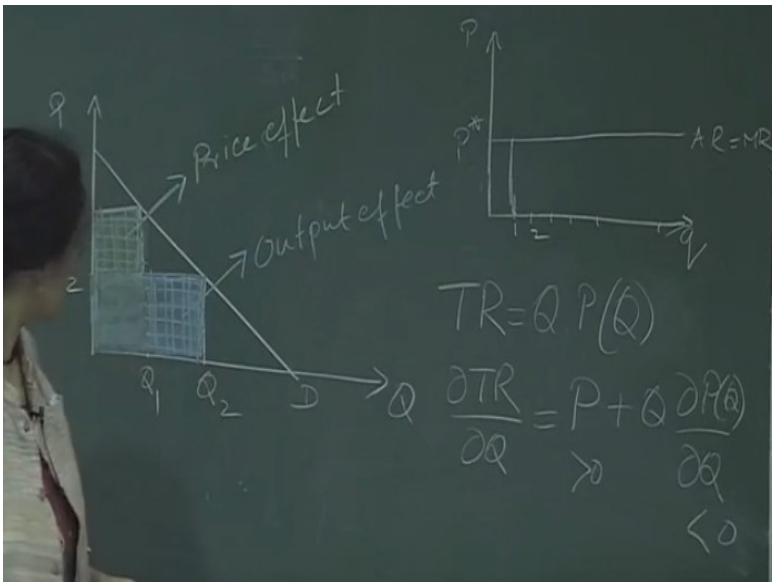
But in case of the monopolist he knows that if he keeps on increasing the units of output his price is going to go down and down and his marginal revenue is also going to change. So the marginal revenue is not same for him at all the for all the units that he sells. So there is a decision to be made for the monopolist. So let us explain a little more. So more on the monopolist's marginal revenue.

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So producing an additional unit of output has 2 effects that is the output effect and the price effect. Let me explain. Let me explain.

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So here is the monopolist's demand curve and say he is charging a price P 1 to produce Q 1. Then he decides to increase his output to Q 2. So if he decides that I am going to sell Q 2 instead of Q 1 immediately his price falls to P 2 for not only Q 2 but for all the units before Q 2. So the entire amount everything he has to now sell at price P 2.

So the price, so when we say output effect and price effect so as soon as the unit of output is increased so there is a the new revenue is this let me, so the earlier what was the revenue that he was getting? He was getting this revenue. So this was the revenue that he was getting from this was the revenue that he was getting earlier P 1 Q 1 and now that the revenue that he earns by increasing his output to Q 2 is this. So he gets this is the revenue that he earns now. So this is the revenue that he earns now. So what is happening here? He is losing this revenue.

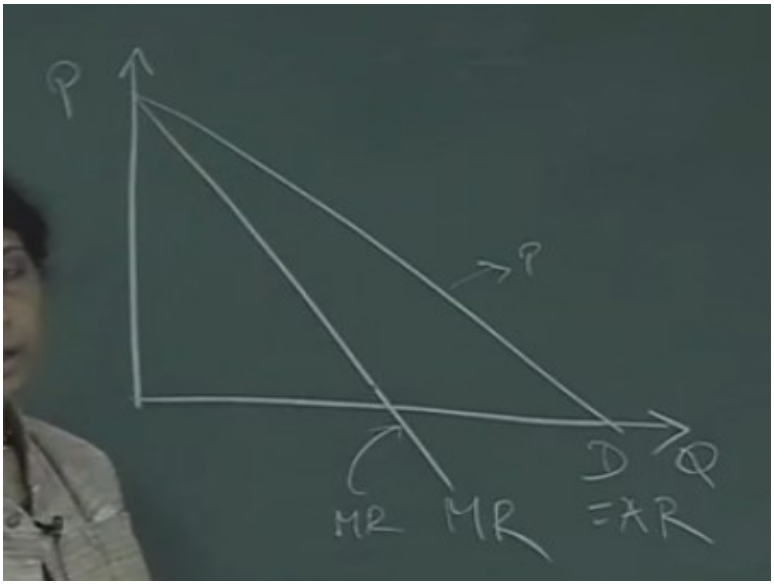
He is losing this rectangle here and he is gaining this revenue the blue rectangle here. So he is gaining this rectangle and he is losing this rectangle. So this part this part is his price effect that is price has fallen and this is what he is losing and this is the output effect that is output has increased and hence his revenue has increased. So this is what happens.

This is the dilemma that the monopolist faces because he has a negatively sloping demand curve and to show it so let me show it mathematically also. So let me show it mathematically also. Say total revenue is equal to P into Q. Total revenue is P into Q and so del TR del Q del TR del Q because which is marginal revenue is equal to P + Q into del P del Q. So why are we having del P del Q here because is now a function of output.

So we have let me write it this way. So P is a function of Q. Since we have a negatively sloping demand curve so price is a function of Q and marginal revenue is equal to P + Q del P del Q. Now in this entire expression P is always positive. Price cannot be negative. So P is always positive. So this is greater than 0. What about this term?

Now this term del P del Q is always negative. Why is it negative because the demand curve is negatively sloping? The slope is negative so del P del Q term is negative. So marginal revenue has 2 parts to it. One is the positive part, other is the negative part and the net is going to determine what is the marginal revenue. So and since this is negative so since this is negative so P is always greater than marginal revenue. So let me draw it.

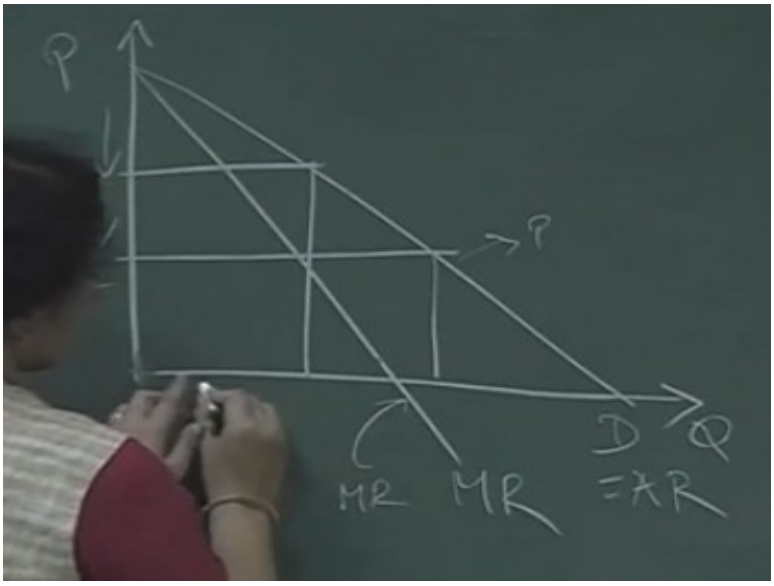
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So what we have basically is P and this is the demand line which is also the average revenue line and we have the marginal revenue line here which is always less than the P line. So this demand line is the price line and this is the marginal revenue line and marginal revenue is always less than P except the first unit where P is equal to marginal revenue.

So coming back to the slides, to sell a larger unit monopolist moves down demand curve and reduces price. So marginal revenue is less than price. So marginal revenue can even be negative as we move down marginal revenue it is possible to have marginal revenue less than less than zero but in this entire discussion the assumption is that the monopolist can charge one price to all its customers. It does not discriminate.

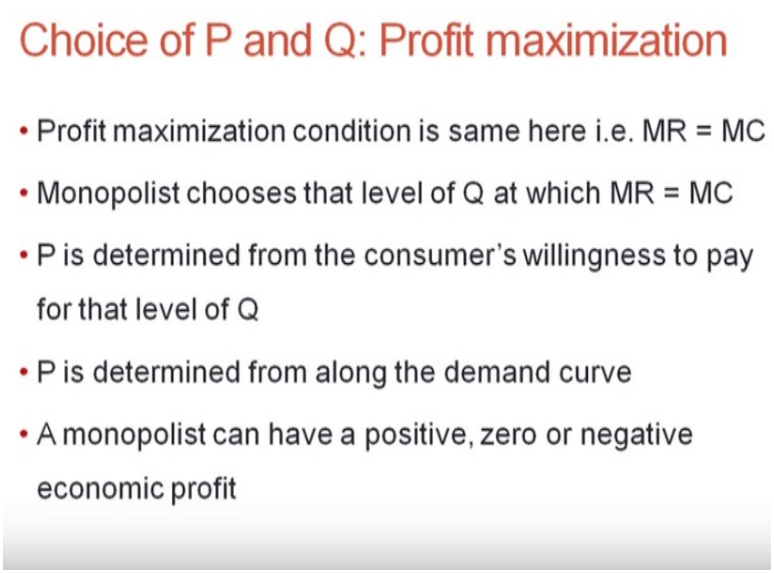
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That is we are always assuming that when he reduces his price so when he reduces his price when his price goes down from P 1 to P 2 so he has to reduce the price for all the previous units. It is not that he gets to charge different price to different customers. So this is one assumption that we are making now and this we are going to relax when we discuss about price discrimination but and then it will make more sense why this assumption was important to understand about monopolist when we are discussing all this.

So what is this, so having discussed, so now that we have a understanding about the marginal revenue curve, about the average revenue, about the demand line for monopolist and the choice that he has to make and we have an understanding about the output effect and the price effect so what does the monopolist do. How does he determine what quantity to produce and what price to charge.

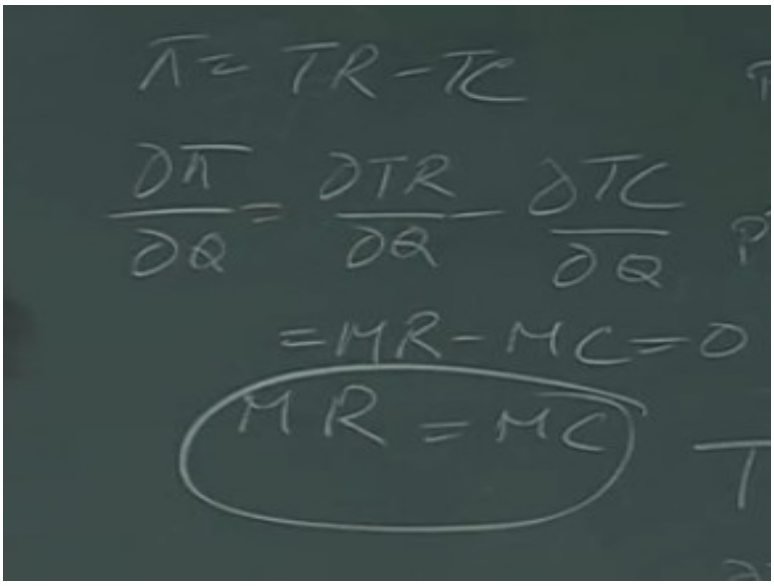
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So the profit maximizing condition maximization condition stays the same. The profit maximization condition is the same here as we had in case of perfect competition and that is marginal revenue equals marginal cost. So the monopolist chooses that level of output at which marginal revenue equals marginal cost. So price is determined from the consumer's willingness to pay for that level of output.

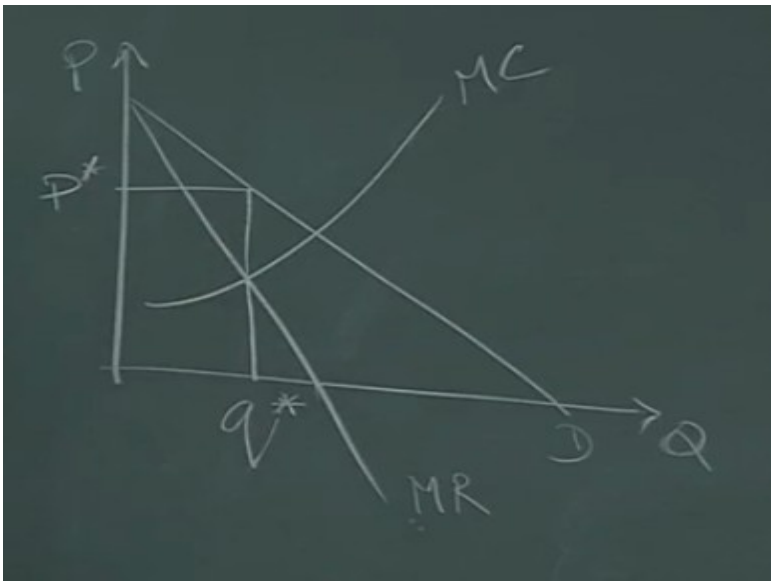
And price is determined from along the demand curve. So a monopolist can have a positive, zero or negative economic profit. So let me explain all these.

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So when we say profit maximization condition is the same here as in case of (monopoly) perfect competition that is because the profit maximization condition that P is equal to TR - TC. This is a quick recap and del pi del Q is equal to del TR del Q - del TC del Q is equal to MR - MC zero as per first order condition and which MR is equal to MC. So MR is equal to MC is always are profit maximization condition no matter what market structure we are discussing. So it is the same. So when we know that it is the MR is equal to MC is our profit maximization condition so let us see what the equilibrium looks like in case of a monopolist.

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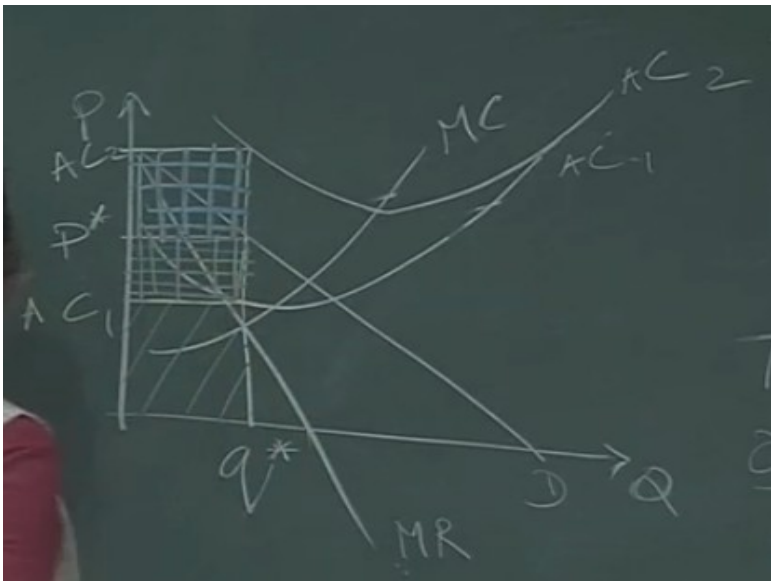
So this is the demand line for the monopolist and this is the marginal revenue and once we have the marginal revenue in place we have we just impose the very familiar marginal cost curve. This is our marginal cost curve. Intersection of marginal revenue marginal cost gives us the profit maximizing level of output. So the MR is equal to MC. This is the, the intersection of MR and MC is where the monopolist is going to choose his output level.

So what price is it going to choose now? Does it choose the price along the MR curve? No. He does not why? Because he is facing the demand curve in the market. He knows if this is the amount that he is willing to sell in the market or if he sells this amount in the market he is going to charge the maximum price that the buyers are willing to pay for this q star and where is he is going to find this price from?

He finds this price on the demand curve. He finds this price on the market demand curve. So basically at this output level he finds what price the consumers are willing to pay and he charges that price P star. So this is his price that he charges in the market. So q star is his MR MC intersection gives him gives us the monopoly output and once we have the monopoly output that output what price the buyers are willing to pay we find from the demand line and that is the price that the monopolist is going to charge.

So what from this we understand that although the monopolist is a single seller in the market yet he has to respect the demand curve. He cannot just charge any price in the market. He respects the demand curve and he charges a price that the buyers are willing to pay for that amount in the market and that he is doing along the demand curve. So now so this is the price quantity combination that the monopolist can charge in the market but is it always going to give him profit? No. It is it does not give us any guarantee that the monopolist is always going to get a profit because profit will depend on what is the average cost of producing this output for the monopolist.

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So if the average cost curve looks something like this, if the average cost curve looks something like this then at this level of output q star this is the average cost for the monopolist and he is his total revenue is Pq and this is his total cost. So this is his, the horizontal line the box with the horizontal line this is his or square this is his profit.

But if the average cost is say somewhere here sorry it will it cannot look like this. If his average cost looks something like this, then his cost of producing q star is very high at AC 2. So he is incurring a loss. He is incurring a loss equal to this blue rectangle. So he is incurring a loss in this case.

So basically whether the monopolist is going to get a profit or not will depend on his average cost of producing the output and we can see from here that it is possible for the monopolist to get a profit and it is also possible for the monopolist to get a loss in the market. So that ends our discussion on profit maximization in the case of monopolist, how does he decide what price to charge and what output to how much amount of output to produce.

In the following sections we are going to look at other issues related to monopolist and we are going to discuss those and we are also going to see what happens when the demand curve what is the, we discussed about elasticity earlier; the elasticity of the demand curve that has some implications on the monopolist's decision. We are going to discuss about price discrimination. We are going to see what happens in case of monopolist, is it an efficient market output or not

and eventually we are going to talk about the role of government in case of monopolist. Thank you.

Foundation Course in Managerial Economics

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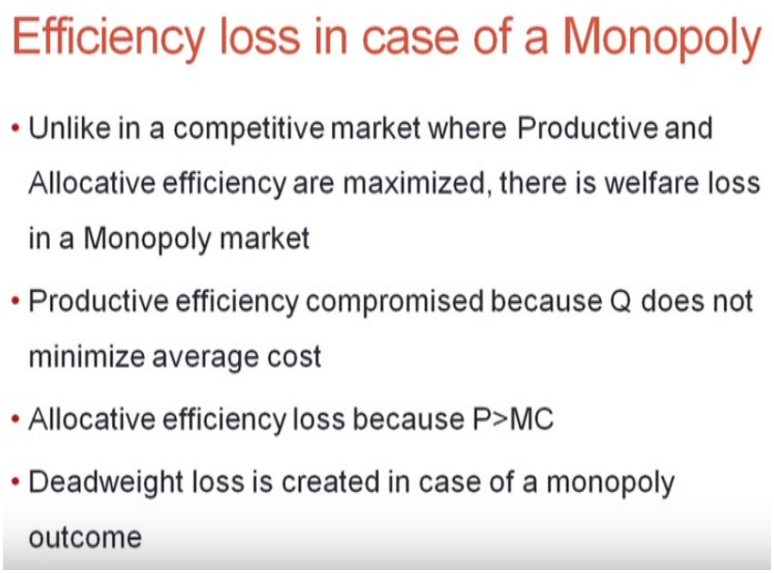
Indian Institute of Technology-Kharagpur

Lecture - 27

Monopoly - Supply and Efficiency

Hello and welcome back to our discussion on monopoly. Till the last class we discussed how does a firm attain the monopoly status and we went on to see what is the equilibrium level of price and output that the monopolist decides and today we are going to look at monopoly supply. Does the monopolist have a supply curve like the perfectly competitive firm and what is the efficiency level when monopolist decides the level of price and output? Is it as efficient as the perfectly competitive equilibrium outcome? So these are the questions that we are going to ask in today's class.

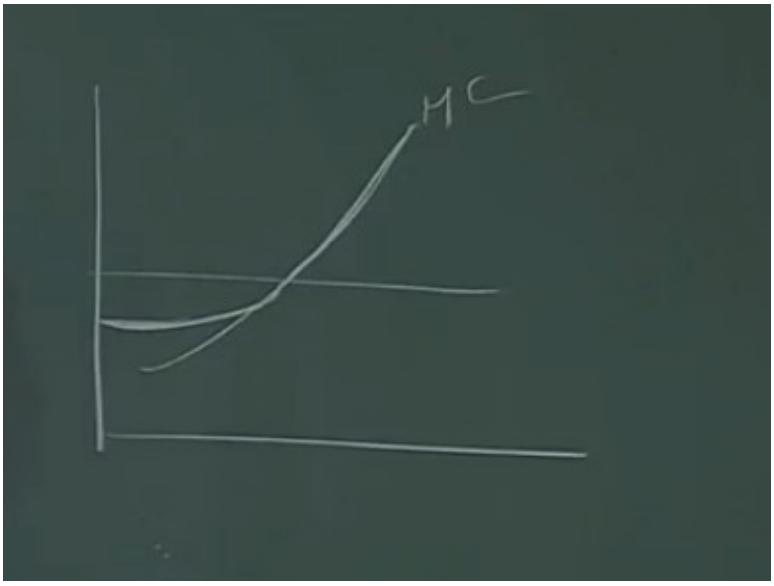
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So supply curve of monopoly. Now unlike a perfectly competitive firm there is no supply curve for monopoly because output for a monopoly is not determined by price as in the case of perfectly competitive firm but jointly by marginal revenue, marginal cost, and the demand curve. This is what we saw when we looked at profit maximization for a monopoly firm and we saw that the monopoly firm basically does not have a supply curve because it decides on the basis of intersection of marginal cost with marginal revenue.

It decides what level of output to produce and then basically he traces back to the demand curve and sees what is the willingness to pay in the market by the buyers for this amount of output.

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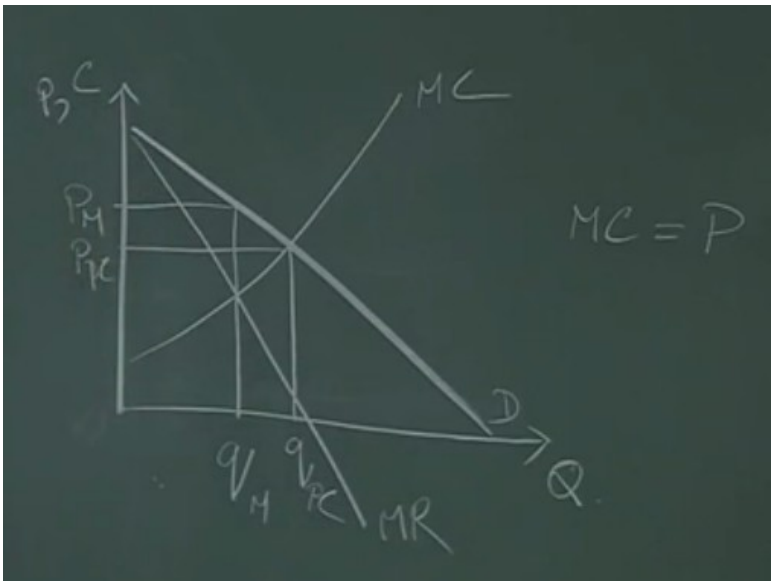
So basically the monopolist does not have a supply curve unlike the perfectly competitive firm who has a so in the case of a perfectly competitive firm this is the price line and this is the marginal cost and we saw that the marginal cost till the average variable cost is the, the minimum average variable cost is the supply curve of a perfectly competitive firm but that is not the case in case of a monopoly.

So next we also discussed in the case of a perfectly competitive firm that the long run equilibrium the final market equilibrium that is achieved in the case of a perfectly competitive market the productive efficiency is reached and allocative efficiency is reached but in the case of a monopolist unlike in a competitive market where productive and allocative efficiency are maximized there is welfare loss in a monopoly market.

Productive efficiency is compromised because output does not minimize average cost or monopolist does not decide on the level of output does not produce that level of output which minimizes its average cost and allocative efficiency is lost because price is more than marginal cost in the case of a monopoly firm.

So deadweight loss is created in case of a monopoly output. So let us see what this means like all the points that I have shown in the slide let us see what we mean by this.

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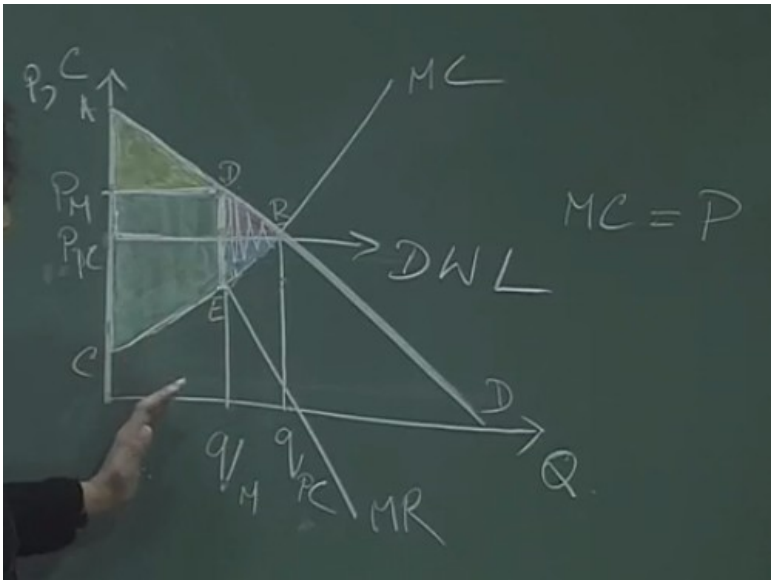
Now in case of a monopoly firm it has a negatively sloping demand curve and it has a marginal revenue curve which is not the same as the demand curve. So demand curve is basically the price line in the market and we have a marginal revenue curve in case of a monopolist because the monopolist because the monopolist knows that if it reduces the amount of output price is going to increase or if it increases the next unit of output price is going to fall. So there is a output effect and there is a price effect which we discussed in the previous class.

So this is the so this is the demand curve that is facing the monopolist and this is the marginal revenue curve and the monopolist basically this is the marginal cost curve and let us draw the equilibrium output in case of a monopolist. This is the output that the monopolist produces and he charges a price along the demand curve, he charges a price along the demand curve. So he charges a price of P M and he produces amount q M.

Now this is our monopoly outcome. What happens in the case, if we imagine this monopolist be a to be a perfectly competitive firm? In that case what is the output that is going to be produced? In the case of a competitive firm the marginal cost is the supply curve of the firm. So if marginal cost is the supply curve and the perfectly competitive firm basically produces at the level where marginal cost is equal to price so the competitive output is here, competitive output is here and price that is charged is here.

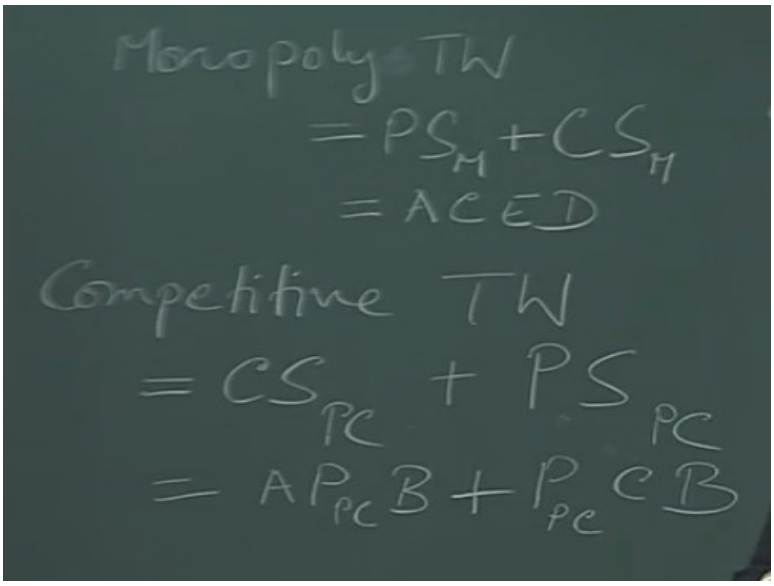
So P PC is the price that is being charged in a competitive market and q PC is the amount that is produced in a competitive setup. So now what is happening in the case of the monopolist? Let us for the time being ignore the competitive firm and look at the monopolist. Now what is the monopolist doing here. If you remember we had discussed about producer surplus. We had discussed about consumer surplus and we had showed what is the consumer surplus and what is the producer surplus. So looking at the outcome for the monopolist what is the producer surplus that the monopolist is basically getting. So he is charging a price P M so he is getting a revenue of P M multiplied by q M which means he is getting this rectangle is his revenue. Out of this here is the marginal cost.

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So everything that he gets above the marginal cost that is his producer surplus. So his producer surplus is basically this green region. So the producer surplus that he is getting is basically this so this green region is his producer surplus and what is the consumer surplus? Consumer surplus is this yellow region. So this yellow region is the consumer surplus so in case of the let me call it region A, B, C. So C, A, B, C, D, and E. So the region between the ABEC is the total consumer surplus and producer surplus.

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So in case of the in case of monopoly total welfare TW in the economy is basically equal to PS producer surplus of the monopolist plus the consumer surplus of the monopolist is equal to the region APMD plus PMCED so which is basically ACED is the region called ACED. Now what so having said that now had it been a competitive equilibrium, had it been a competitive equilibrium we know that production would have happened here. There would be intersection between the demand and the supply curve and total output would be PC and total price charge would be P total quantity would be q PC and total price charge will be would be P PC.

So in that case producer surplus would increase by this blue amount but this green rectangle would then go to the consumer surplus. So this rectangle of so let me put say I call it O. So this rectangle would now then go to the consumers because so watch carefully now in case of had it been a competitive output then the price that the firm would be charging would be P PC and in that case everything above the line BPPC that is this line, everything above this line would go to the consumers.

So the consumers would get this yellow triangle plus this green rectangle plus the consumers would get this pink region as well. So the consumers surplus in case of perfectly competitive output would be this entire triangle, would be this entire triangle.

So writing the, so let me write the competitive total welfare in the economy would be equal to consumer surplus in case of perfect competition plus producer surplus in case of perfect

competition. Now consumer surplus in case of perfect competition would be the triangle APCB, so would be AP PC B plus now the producers in that case the producer would lose this small rectangle which has gone to the consumers so now the producer gets continues to get this green region plus the blue triangle plus the blue triangle. So now the producer surplus in case of a perfect competition is the triangle P PC CB.

So now the producer surplus is this region. So clearly the monopoly total welfare is only this region monopoly total welfare is this region while perfectly competitive output welfare has a additional triangle of EDB. So this region of EDB is lost when the market is a monopoly market and this region is basically that the pink triangle and the blue triangle together this region is the deadweight loss that the economy is suffering when the market has a single seller when the market has a single seller or market has a monopoly this is the deadweight loss that is happening in that case.

So basically welfare loss happens when the economy shifts from a perfectly competitive setup to a monopoly setup. So this is what we were trying to explain in this slide and if you may remember we have defined productive efficiency and productive efficiency is compromised because output does not so productive efficiency is compromised why because the monopolist does not end up producing at the minimum level of the of its average minimum cost average total cost.

So because of that since the monopolist is not operating at its minimum cost hence it is possible to still reduce cost further and keep on producing so productive efficiency is not achieved. Secondly what happens here in this case, this is monopoly price. So P M is clearly more than the marginal cost. So P M is clearly more than the marginal cost.

So since this is P M and this is the marginal cost since price is more than the marginal cost it is possible so what does that mean when price is more than marginal cost that means it is possible for the monopolist to keep on producing and it there will be people in the market who would be willing to pay for this keep on the price in that case would keep on falling. Yet one would have customers or consumers in the market to purchase that good.

So this amount of output q M q PC this amount of output is not getting produced in this market because the monopolist is trying to only maximize its profit and it is charging a price which is more than the competitive price. As a result, the monopolist is eating away from the consumer surplus. So the monopolist, although the monopolist is getting a high producer surplus, if you can see from here what is it like one might think then what is the what is the interest of the monopolist to curtail output?

So clearly if we see from here for the monopolist the monopolist output is better, in a monopolistic output the seller is better off than the buyer. The seller is even better off in the absolute sense because this triangle is much, the triangle that he is losing by producing less amount of output is much smaller than the rectangle that he is gaining by charging a high price in the market.

So since this is happening so the for the monopolist it makes sense to produce a smaller amount of output, charge a higher price, eat away from some of the consumer surplus and still be able to stay in the still be able to stay in the market.

So that is the reason that it is an inefficient outcome because for the entire economy as a whole so many buyers are getting left out, the seller is not producing the optimum quantity although he would be able to so that is the reason that the monopoly outcome is considered as inefficient and there is a deadweight loss happening here.

So we are going to discuss more on this later and see what can what happens when the monopolist tries to we are going to discuss something called price discrimination and we are going to see what happens when the government tries to intervene in a monopoly market because the monopoly outcome is inefficient. Thank you.

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Lecture - 28

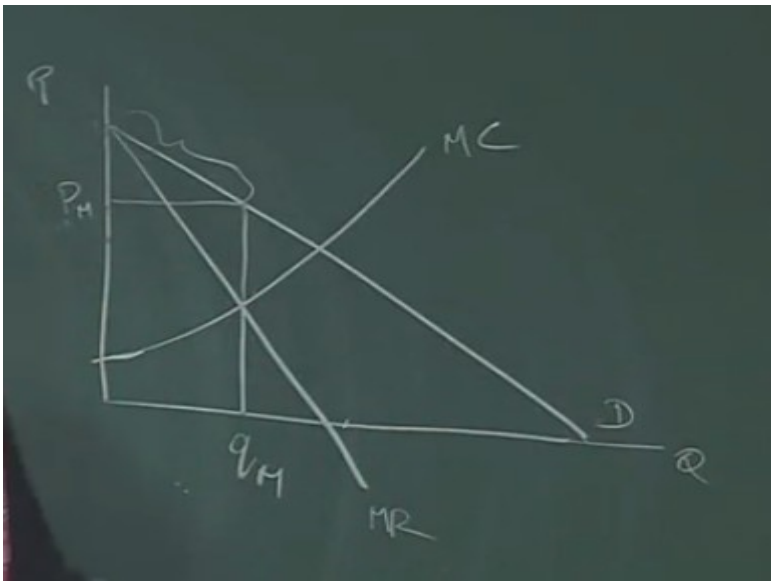
Price Discrimination

Hello and welcome back to our discussion on monopoly outcomes. We discussed in the last class that monopoly outcome is inefficient for the market. It does not maximize welfare in the market and we discussed that perfectly competitive outcome is always much better than a monopoly outcome because in the case of a competitive outcome, the output is much more than the monopoly output and the price is much less than the monopoly price and both productive efficiency and allocative efficiency are achieved in the case of a perfectly competitive outcome.

So today but throughout this discussion if you remember when we discussed in the first couple of lectures in monopoly when we discussed about profit maximization and when we discussed about the marginal revenue we made a very important assumption if you may remember and I did not go into too much details of the assumption so let me recapitulate that assumption right now.

I said that when we are discussing about marginal revenue and when we are talking about the output effect and the price effect there is a very important assumption we are making and that assumption is when the monopolist has to reduce the price because he is increasing the amount of output in the market he has to reduce the price for all the consumers. It does not happen that he gets to charge a higher price to the consumers who are high up on the demand curve. So let me explain.

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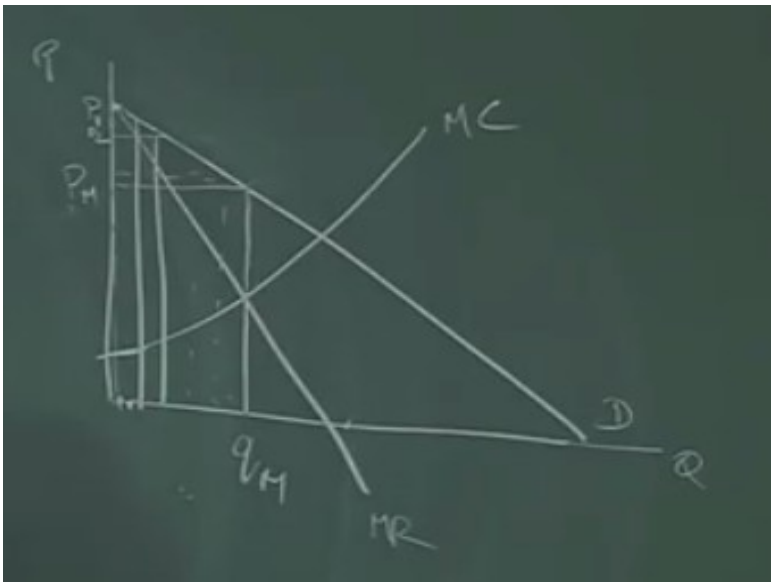
So if this is the demand curve and this is the marginal revenue curve and this is the demand curve. This is the marginal revenue curve and this is the marginal cost. This is the monopoly output and monopoly price. So here we discussed so this is the equilibrium, so this is the equilibrium for the monopolist but here the assumption is when he is producing output of q M he has to charge this price P M to all the consumers who are here.

So he so we know on the demand curve there are some consumers who are high up on the demand curve who are willing to pay a very high price for the product. They are willing to pay very high price for the same product. So when the monopolist is charging a price of P M these consumers over here who are getting to stay in the who are getting the product from the market they are having to pay a price which is less than their willingness to pay because P M is less than every consumer's willingness to pay except the last consumer whose willingness to pay is exactly equal to P M.

So that is the reason that this consumer surplus stays in the market and these buyers actually get a consumer surplus out of buying the product from the monopolist. So now what happens if we relax this assumption? What happens if we say that the since monopolist is the only single seller in the market, also we know that the monopolist knows what the demand curve is.

So now assume that the monopolist knows what is the what is there in every point of the demand curve he knows who all are the buyers who are willing to pay these prices on this demand curve. If that is possible, if that situation is possible and say the monopolist is able to charge a price that each of them is willing to pay. Since he is the only seller in the market so it is possible for him to actually charge different prices to different buyers. So say he knows by the look on the face of the buyer that this person is willing to pay a higher price for the product that I am selling than his than the next person.

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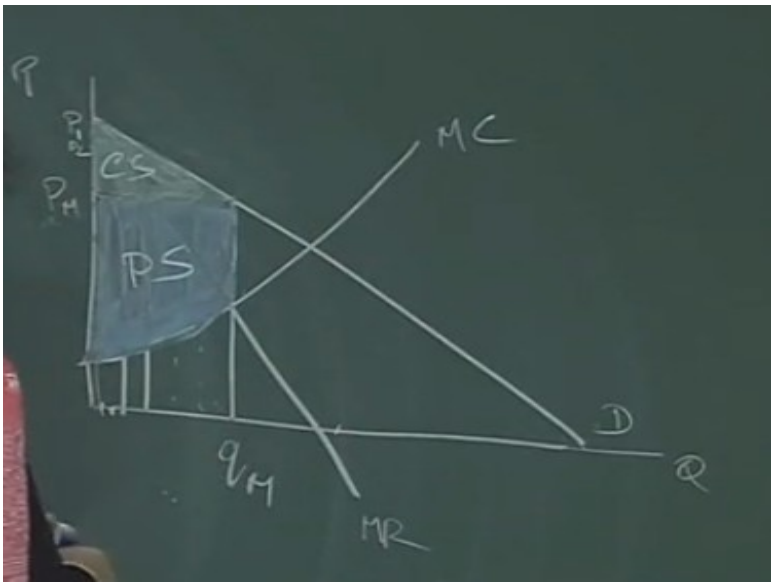


So if such a situation is possible then he gets to charge prices, so if that is possible, for the first unit he is going to charge this person very high P 1 say. The next unit the next person is or say the next few people they are willing to pay P 2. So the next person is for next few people are willing to pay say P 3. So he gets to he say, for example he is a single seller in the market. He knows the market and his consumers through and through and he knows how much each one is willing to pay and he charges those prices to each of these buyers. In that case what is his revenue? In that case what he gets is he gets P 1.

So basically what he gets his revenue in that case is this entire amount he gets. Then the P 2 q 2, then say P 3 q 3. So this way he is going to get the entire revenue is his. So he gets to charge each and every person so the entire amount that is entire revenue in the market is this entire area. Now

the next question is, so the last person so nothing changes for the last person because he was willing to pay P M and he is having to pay P M so 2 things are happening here.

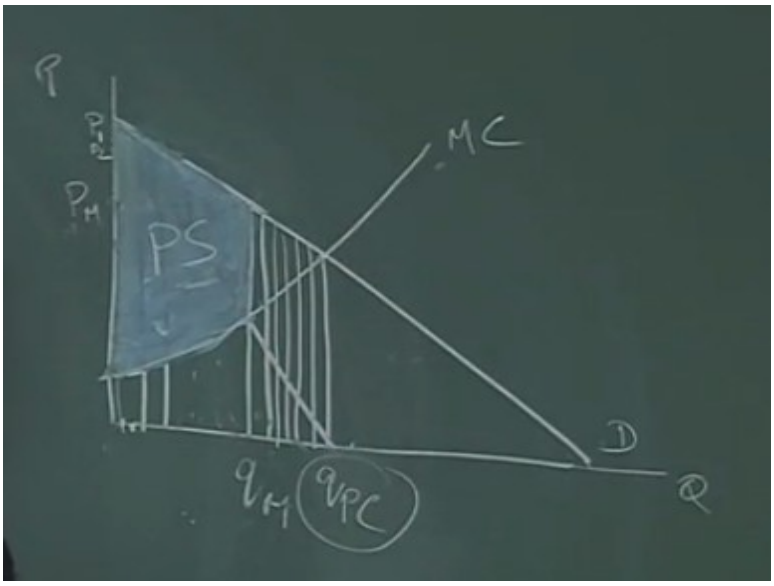
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So the first question is what is happening to the consumer surplus that we drew in the previous class, this was the consumer surplus. In the previous class we drew that this is the consumer surplus in case of a monopoly outcome and we said that, I do not remember what colours I used but let me draw a new one, and we said that this area is the was the producer surplus. So we said this was the producer surplus and this was the consumer surplus in the case where the monopolist has to charge the same price to all the consumers.

But in case of a perfectly price discriminating monopolist or where the monopolist actually charges different price to different consumers that is the monopolist charges different prices to different consumers so in that case this entire amount he is being able to get where that will be his revenue and what happens to consumer surplus.

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Now consumer surplus then gets wiped off. So the consumer surplus now gets wiped off because no one gets anything. So no consumer gets any surplus because every consumer is having to pay what he is willing to pay so hence this entire area becomes the producer surplus. So this entire area becomes the producer surplus and in case you are wondering what is this area. This is basically his cost, marginal cost. This is his marginal cost. So this part is what he is spending to produce the output. This is his expense. So the producer surplus is the blue area.

Now my next question is so in case of a price discriminating monopolist we understand that the monopolist is basically eating away the entire consumer surplus but my question here is, is he then going to produce q M? Is he going to stop his production at q M like we saw in the case of profit maximization MR is equal to MC? So does he keep does he stop at q M or is there any chance that he is going to produce more?

So what goes on in the mind of the monopolist? What is the logic there? The logic is since he is able to charge what the firm what the buyer is willing to pay it makes sense for him to keep on producing because if he produces the next unit of output and price falls he does not have to reduce the price for all the buyers preceding this output. He does not have to reduce the price for all the buyers.

So it makes sense he knows that he has already sold q M to all these buyers who paid him a much higher price but to sell it to the next buyer he has to reduce the price and the next buyer is

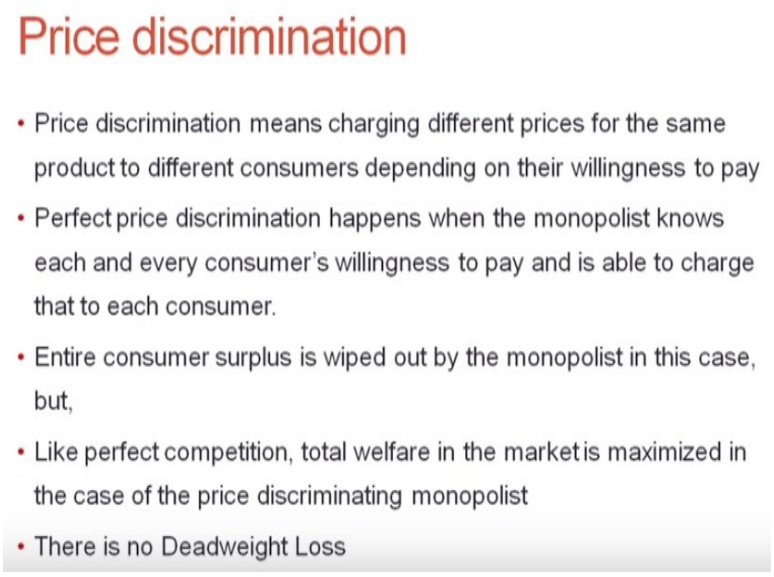
going to pay a smaller price and buy some additional output but that does not change anything for the preceding buyers.

So he is going to sell the next unit of output and he keeps on selling. It makes sense for to him to keep on selling till he reaches the point where marginal cost intersects the price line. So what is this equilibrium? So what is this point? This point is the same as the perfectly competitive equilibrium and the output that he produces here which is the perfectly competitive output.

So the output that he produces here is the perfectly competitive output and so this is the competitive perfectly competitive output and he charges the so price is prices are different for different consumers but the output that is getting produced here is the same as the q PC.

Now what happens? Is he going to sell the next unit of output as well? No. That is that he is not going to do because beyond this q PC his marginal cost of producing the output is more than the price that he is going to get out of it. So he is not going to produce beyond q PC amount of output. So here monopoly output is also equal to q PC. So let us look at the slides now before I move on with the diagram.

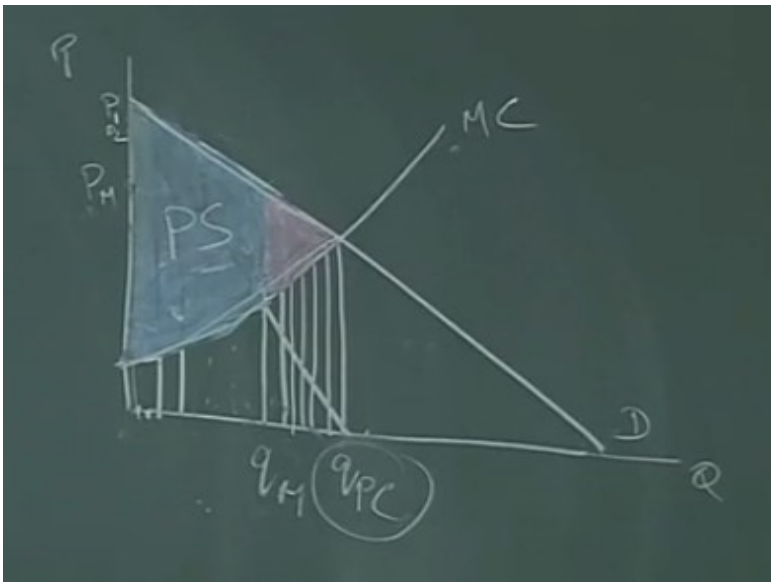
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Now, so price discrimination, price discrimination means charging different prices for the same product to different consumers depending on their willingness to pay. Perfect price

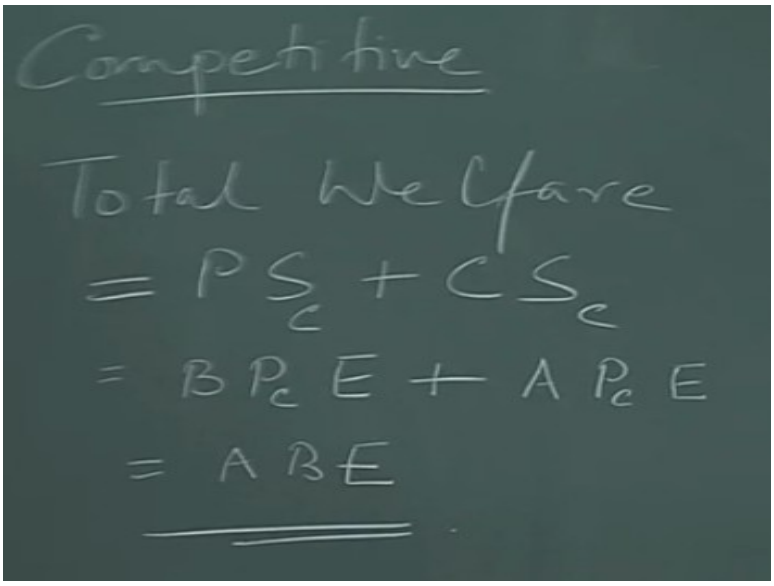
discrimination happens when the monopolist knows each and every consumer's willingness to pay and is able to charge that to the consumer. Now entire consumer surplus is wiped out by the monopolist in this case but like perfect competition total welfare in the market is maximized on the case of the price discriminating monopolist and there is no deadweight loss.

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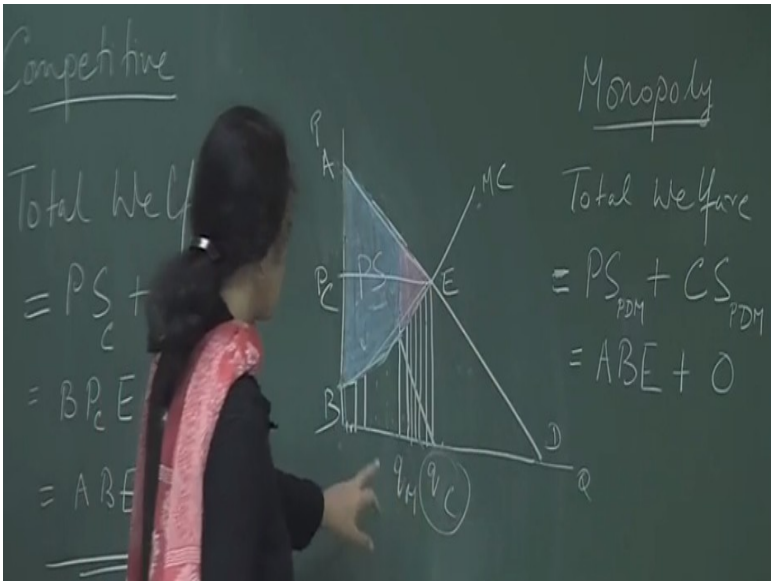
Now let me explain a little bit about the last 2 points that we have discussed here. If you remember the diagram that we drew in the earlier class our deadweight loss was this portion, our deadweight loss was this triangle but in case of a perfectly price discriminating monopolist there is no deadweight loss because he keeps on producing till q PC and the total welfare in the market is this entire triangle. So what is the difference in welfare between the competitive output and perfectly price discriminating monopolist output? So let us see.

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So in case of competitive output competitive in case of competitive market you have total welfare is equal to producer surplus plus consumer surplus, let me write here C for competition, and this is price in case of competition and q let me call it q C. So in case of in a case of competitive output let me now name the triangles. Say this is E, this is A, B. So in case of total welfare in case of competitive output producer surplus is this triangle BPCE and consumer surplus is this triangle that is APCE. So total welfare is equal to ABE.

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Now what happens in case of the monopolist? In case of monopoly total welfare is equal to producer surplus plus let me write producer surplus in case of monopolist. This is perfectly price discriminating monopolist. So let me write PDM and this is consumer surplus in case of price discriminating monopolist.

So what is the producer surplus in case of a price discriminating monopolist? It is basically this entire area. He is able to wipe out the entire consumer surplus so this entire area of ABE is his producer surplus and consumer surplus is zero. Consumer surplus is zero so in both the situations total welfare is exactly the same.

Total welfare is this triangle ABE. So total welfare is equal to ABE but consumer surplus is equal to zero in case of a perfectly price discriminating monopolist. So should that make any difference. So the next question is should that make any difference because total welfare in the economy is still the same in both the cases because the allocative efficiency is achieved because this is a this is the output is exactly the same as the competitive output.

So both our productive and allocative efficiencies are achieved because every buyer is paying what he is willing to pay and for that unit of output he is the and finally the producer is producing till the point where price is equal to the marginal cost. So productive and allocative efficiency is achieved in this situation. The only difference is if in the eyes of the economy there is difference between the producer and the consumer then probably there is difference between the two outcomes. But if we do not differentiate between a producer and a consumer but in that case total welfare is the same in the case of price discriminating monopolist as in the case of the perfectly competitive output.

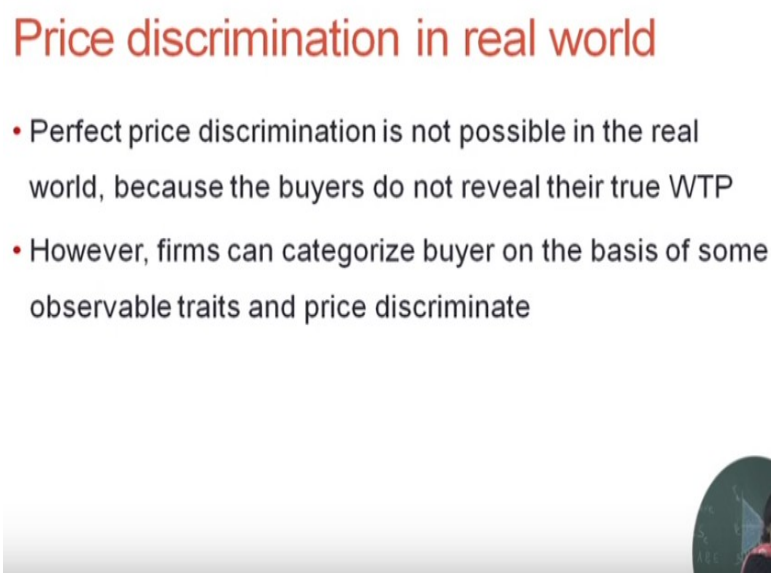
So now this is a price discriminating monopoly so at least in the case of price discriminating monopolies there is no deadweight loss. So we know that monopoly output is not as good as the competitive outcome but at least price discriminating monopolist outcome is better than the monopoly outcome and it is close to the competitive outcome in the sense that the total surplus for the economy still remains the same.

Now does price discrimination happen in the real world? So we have said here that the monopolist knows each and every buyer's willingness to pay but is that possible? No. That is

perfect price discrimination is not possible in the real world because no buyer is going to reveal his real preference to the buyer to the seller. He is not going to say that yes I really want your product and I am going to pay as high a price as this.

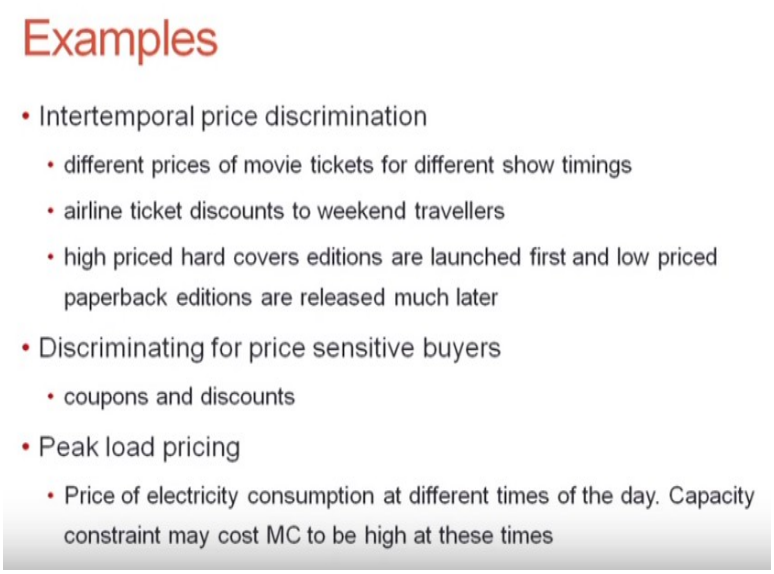
So it is not possible for a seller to actually know the exact willingness to pay of each and every buyer but there are ways in which the sellers basically figure out the willingness to pay of different categories of consumers.

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So say price perfect price discrimination is not possible in the real world because the buyers do not reveal their true willingness to pay. However, firms can categorize buyers on the basis of some observable traits and price discriminate.

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So let us take some examples. Say for example the intertemporal price discrimination. So price discrimination, intertemporal price discrimination can happen when the intertemporal price discrimination basically means that the seller is discriminating across different time of selling the products. Say different prices of movie tickets for different show timings. Now the assumption of the seller is the busier people are richer people and who have better willingness to pay for say movie tickets or entertainments but they cannot come in the morning, during the weekdays, during the morning hours when they are working.

So it is better to charge higher prices in the evening shows when these busy people who are basically earning people will turn up for the movies and the morning timings, morning show timings, afternoon show timings are for people who are usually either students or retired people or homemakers who do not have such high level of income and probably lesser willingness to pay or lesser ability to pay. So these are the people who the sellers are trying to categorize by selling the movies movie tickets at a lower price in the morning hours.

Similarly airline ticket discounts to weekend travellers. So weekend travellers are usually people who are travelling for personal reasons maybe holiday or vacation travellers or who are paying from their own pockets in contrast business travellers who usually travel during the weekdays. So the weakened travellers weekend airline tickets are usually cheaper.

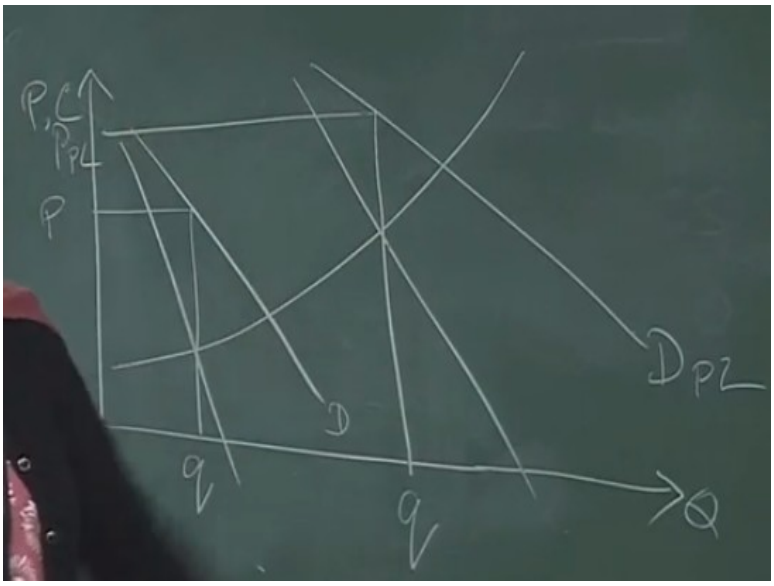
High priced hard cover editions are launched first and low price paperback editions are released much later. So often we see that if there is bestselling author's books is coming up, it is going to be launched. Initially the people who are really hardcore fans of the author they are really waiting eagerly for the book and they would like to buy it as soon as it is launched. So to charge a higher price to these groups of readers often the book is launched in hard covers with some fancy jacket etc. and the cost is not so much higher than the ordinary paperback version but the price is considerably higher.

So the purpose is to target or categorize or differentiate between 2 categories of buyers, one is more price sensitive who are patiently waiting for the prices to come down and then they can buy the paperback version and the other people who are willing to pay a much higher price for the to read the book to be one of the first people to read the book in the market.

Second is discriminating for price sensitive buyers. So typical example is coupons and discounts. We often see that there are fliers or coupons which are mailed or distributed at departmental stores and it is said that if you really want this discount or coupon please mail it back to the seller or mail it back to the company and you get the discount you get a cash back or something like that. The purpose is people who are really busy are not so much price sensitive. They would not bother to respond to these coupons. They will probably throw it away as opposed to people who are really price sensitive and who would take the time to fill up these coupons mail them to get that additional discount.

Another example is the peak load pricing as happens in case of specially electricity in the case of pricing of electricity; so price of electricity consumption at different times of the day. So capacity constraint may cause the marginal cost to be high at these times. So this is slightly different from the examples that we have taken in the earlier, we have taken earlier in the other kinds of situations where the assumption is cost is not so much different but basically the seller is categorizing people on the basis of their willingness to pay but our last example is where let me quickly draw it.

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The last example is kind of interesting that so it is possible that so at normal times this is the demand for electricity which is probably inelastic that there is there the people are sensitive about the price of electricity and they are going to use it so this is where it is like a necessary good. So it is inelastic and we have the marginal cost curve.

Now at low amount of generation, so this is the quantity that is sold here and this is the price that is being charged. But what happens when there is too much too high demand for electricity. So our demand curve would be somewhere here. So in that case our marginal cost has gone up. You will see that with higher levels of output the cost is going up and so here for here the price is different.

So you have a higher amount of output that is produced. So this is D and this is peak load D. So this is peak load D and peak load price. So basically there are what is being said is there are 2 categories of consumers who are consuming the electricity and this is in peak load the demand is too high and hence the cost that is being incurred by the seller is also high so he charges a very high price for this for peak load electricity generation and supply.

So this was our example. We are going to discuss lot of more different ways in which the not only a monopolist but also any producer who has some amount of market power what are the

various innovative ways that the sellers with at least some amount of market power they come up with to price discriminate across different categories of consumers.

So these examples we are going to take up later once we have discussed the different kinds of market structures and look at the different amount of market power different levels of market power that a seller could have because the monopolist is a very extreme case where he has complete market power but at least but in situations where there are a few sellers in the market there also the sellers have some amount of market power and they come up with lot of ideas to discriminate across different categories of consumers and try to make a as much profit as possible.

In the following section we are going to talk about what the government does to see to it that the monopolist does not charge too high a price and monopolist can be regulated so that consumer welfare is at least improved if not the competitive output cannot be achieved but at least some amount of welfare improvement is possible, is attempted by the government through public policies and regulation. Thank you.

Foundation Course in Managerial Economics

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Indian Institute of Technology-Kharagpur

Lecture - 29

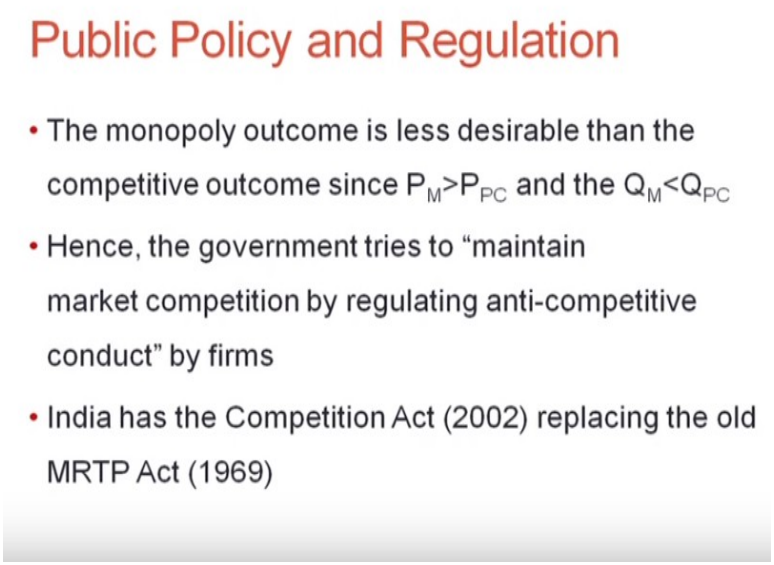
Public Policy and Regulation

Hello and welcome back to our last module in monopoly. We discussed in the last class that there is considerable efficiency difference between a monopoly outcome and a competitive outcome. And we saw that the monopolist if the monopolist is able to price discriminate it can increase its output and it can charge prices differently to different consumers.

So a price discriminating monopolist behaviour is at least better than a monopolist who charges the uniformly same price to all consumers and hence ends up producing less amount of output in the economy.

So today we are going to discuss what happens when there is regulation of public policy. That is the government intervenes. So basically the government knows that a monopoly outcome is not desirable because the full potential amount of output is not getting produced in the market so the government may like to control a monopoly outcome.

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So what does the government do? The government can go for some public policies or it can decide to regulate the monopoly. So the monopoly outcome is less desirable than the competitive outcome since the price charged by a monopolist is more than price that gets charged in a perfectly competitive market and a monopoly output is less than the amount of output that gets produced in a perfectly competitive market.

So the first point however should indicate that when we say P M and P PC they are not they may not be strictly monopoly and strictly price perfect competition outcome. What we mean by the first point is whenever there is some amount of market power with the seller the price is going to be more than competitive outcome and quantity is going to be less than competitive outcome.

So that is what that is the difference or that is the difference that market power brings to the market outcome. So coming back to the slides. Hence the government tries to maintain market competition by regulating anti-competitive conduct by firms. So hence the government strives to keep competitive environment in the market so that competitive outcome is achieved in the market and consumers get lower prices and higher amount of output in the market.

So when we discussed perfect competition you may remember that I had said that although perfect competition is a is not realistic we do not have absolute perfect competition in the real world. Yet the competitive outcome is important because it increases output it improves efficiency in the market it implies lower cost and lower prices to the consumers. It implies

production at minimum cost or implies highly efficient level of production and hence a competitive outcome is highly desirable for the consumers in a economy.

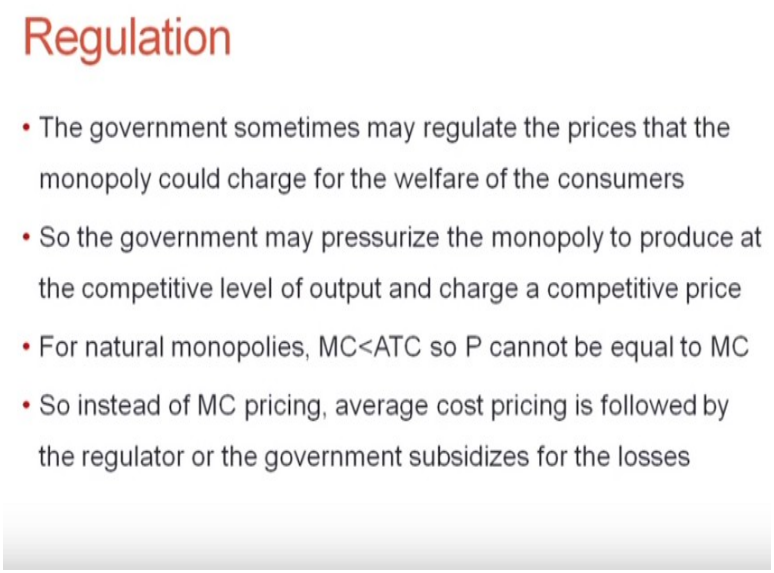
So hence the government tries to maintain market competition by regulating anti-competitive conduct by firms and India has the Competition Act of 2002 replacing the old monopoly and restrictive trade practices act of 1969. So quickly although we are not going to go into the details of the competition law or why it is there in place etc.

But let me quickly just tell you this that the competition act has replaced the MRTP Act because the MRTP Act according to the ARTP Act monopoly per se was bad but now at least it has been recognized that monopoly per se is not bad. There are situations where a monopoly outcome or a monopoly monopolistic production actually brings about better efficiency and lower prices to the consumers.

How is that possible? If you remember in the last class we discussed about natural monopoly and in case of a natural monopoly the market outcome is better than too many firms existing in the market in which case the firms basically operate at very high cost and they cannot sell their products at lower prices to the buyers.

So in cases of natural monopoly it makes sense to have a single seller in the market. So hence MRTP Act has been abolished and we have the competition act which basically promotes competition in the economy than trying to remove monopoly in the economy.

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So coming to regulation, regulation is basically where the government exercises its power on the market and government basically says that any monopoly which is operating in the market it should not exploit the consumers in the sense that it should not charge too high prices especially say for say imagine a situation where the monopolist is either a natural monopolist where the monopoly is selling something as necessary good like say electricity.

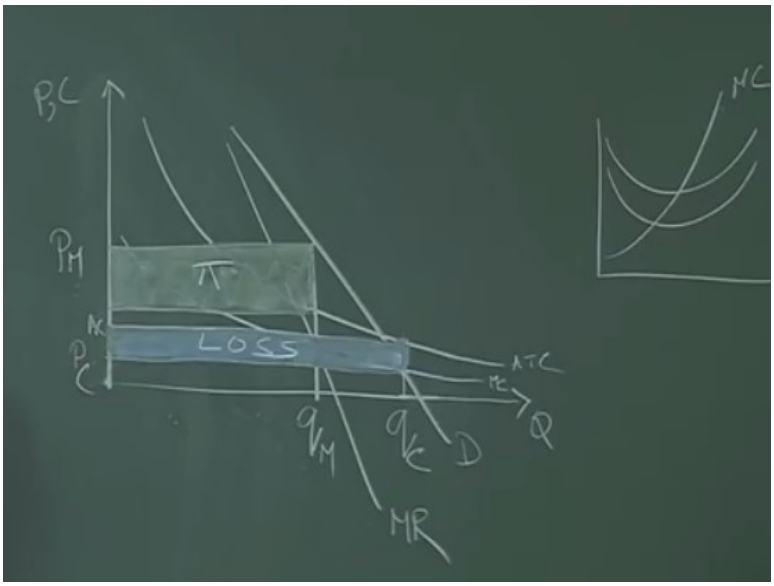
So that is a natural monopoly and there if the monopoly outcome implies very high prices and low amount of output is produced in the market compared to the competitive outcome then it is not very desirable for the economy.

Secondly say for example the economy is a monopolist is selling something like medicines. It is a lifesaving drug that the monopolist is able to sell because it has a patent but at the same time it charges too high a price and so the drug is not affordable to the consumers. So in such kinds of situations it is possible for the government to step in and regulate the prices that the monopolist can actually charge in the market.

So coming back to the slide so the government may pressurize the monopoly to produce at the competitive level of output and charge a competitive price. For natural monopolist marginal cost is less than average total cost so price cannot be equal to marginal cost. So instead of marginal cost pricing average cost pricing is followed by the regulator or the government subsidizes for

the losses. So let me explain this through a diagram on the board. So as I said the first two points are understandable but what happens in case of a natural monopoly?

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So say in case of a natural monopoly if you may remember from the earlier lectures so if you may remember from the earlier lectures that in we said what is a natural monopoly? A natural monopoly has very high fixed cost. Hence for the natural monopolist the average total cost looks like the average fixed cost and for a very huge amount till a very huge amount of output is reached, marginal cost is lower than the average total cost.

So let us so it looks something like this and marginal cost also goes on like this. So this does not mean to say that marginal cost is always lower than the average cost or it does not mean that this monopolist does not have the typical cost curves that we are familiar with that this is the average variable cost, the average fixed cost and average total cost and this is marginal cost. So this figure does not mean to say that we do not have a U shaped curve.

What this means to say is this goes on for a very huge amount of output till it is produced and then the marginal cost is going to intersect the average cost at the minimum point. So for till a very huge level of output is produced this is what the cost curve looks like to the producer and this is the very reason that it does not make sense to have too many firms in the economy

because the amount that will be produced will be so small that the cost will be extremely high for the seller.

So this is a natural monopoly case and we have the say for example we have the demand curve, this is the demand curve and we have the marginal revenue curve. So we have the demand curve and this is the marginal revenue curve. So marginal cost intersecting the marginal revenue this gives us the equilibrium output q M is produced by the monopolist and it charges a price of P M. It charges a price of P M and what is the cost to the firm?

The cost is average cost at q. This is the cost to the monopoly, natural monopoly. So this is the cost to the natural monopoly. So green area is the huge amount of profit that the monopolist is incurring. So the monopolist is getting a huge amount of profit equal to the green rectangle and so the and the output is q M and price is P M.

Now what would have been the competitive output? The competitive output would be where marginal cost intersects the price line. But before that say the government is not happy with this amount of output. Say it is something a necessary good. Say it is electricity and electricity production more the better in the economy for the benefit of the economy and lower the price is it is better for the economy because it is a necessary input in most production processes.

So if you leave the power generation in the hands of a monopolist and it controls output to q M charges are high price P M this is not at all an ideal situation which the government would like to continue. Or say for example it is not electricity say for example it is something like a essential drug and this monopolist has a patent over the drug because of which it is a monopolist and it exercises and its cost is, why is the cost like this because he has say for example he has amortized the what we call sunk cost in the previous classes instead say he is treating the cost of inventing this drug as fixed cost and he spreads it over the output for all the periods to come.

So his fixed cost is very high and he decides I am going to produce only this many amount of medicines in the market and charge a very high price but say it is a lifesaving drug and the government says that no you have to produce a higher amount of output. So then what happens? So if the government would like to regulate this regulate the production of this drug or say electricity the government might say that you have to produce a competitive level of output.

What if the government says that you have to produce a competitive level of output in that case the production has to be where the marginal cost equals price. So competitive outcome will be let me call it q C is where marginal cost equals the demand line or the price line. Now what happens here? So lot of output is produced which is q C is much higher than q M. What is the price that is that the monopoly gets to charge? It charges a price over here P C.

But is it feasible for the monopolist to continue in this market if it charges a price P C? No because at q C the average cost is much higher than the price. So the basically the monopolist is facing a loss. So the monopolist is facing a loss. So this is his cost. So the monopolist faces a loss equal to this blue rectangle. So this is the loss that the monopolist will face if he is if it is directed by the government to produce at the competitive level of output.

So this is loss and this was the profit in the earlier case. So this is the loss that the monopolist has to incur and how does so if it is a loss then the monopolist is going to just shut down his production and say that I am leaving the market. I cannot produce at this produce this amount of output and incur such a huge loss.

So what are the options available to the government? What the government can do is government might say that okay I will subsidize you for these losses. So one option that is available to the government is government may pay the monopoly this blue rectangle or cover for the losses and force the monopolist to produce at q C and charge a price of P C.

But that is not again a very desirable output because where is this subsidy going to come from. It is going to come from taxes to be paid by the people. So they may not like it and it is not a very desirable output. So the next thing the next option that the government usually exercises is the government says that let it not be marginal cost pricing that is you need not keep your price equal to your marginal cost, keep your price equal to your average cost.

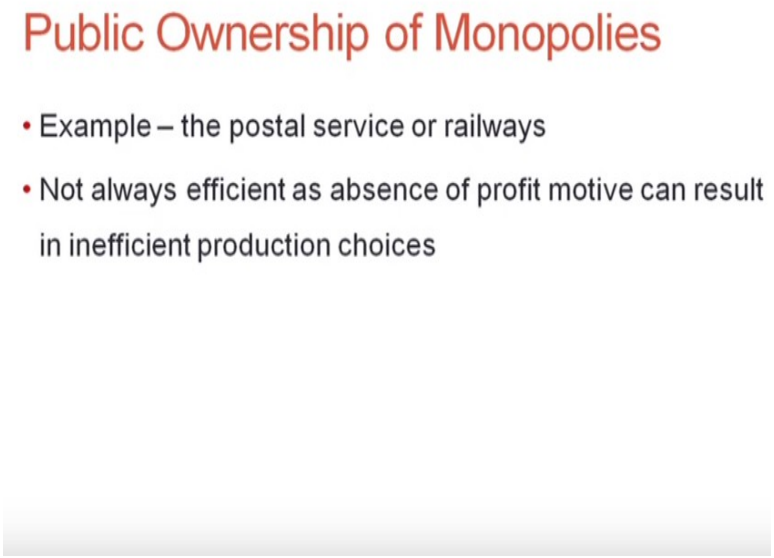
So that is called average cost pricing and the government says that okay fine you are allowed to charge a price along your average cost where average cost is equal to the price that you charge. So you can charge a price slightly higher so this is the price that you get to charge and you produce an output which is slightly less say q C1.

You produce an output slightly less but it is still much higher than the monopoly output. So this is what the government can do and say that so instead of marginal cost pricing average cost pricing is followed by the regulator or the government would subsidize for the losses. So this is the second option that the government has and what is the third option?

The third option is the government might say that please go for price discrimination. Please go for price discrimination and charge you may charge higher prices say for example peak load pricing in case of electricity. So you get to charge higher prices to say industries who consume a lot of power and during the peak time of the day and so you can charge high prices to them but you have to charge low prices to other consumers say households. So this is the price discrimination is one option that the government may give to the monopolist.

So these this was our discussion about regulation and thirdly what else is possible to control so if the government is not government is wary about any market being in the hands of a monopolist because it knows that the quantity is going to be less and price is going to be more. In those situations, the government may own the monopoly. The government may say that let this be run by the government so that public welfare is the topmost priority for this producer and it becomes a public ownership of monopolist.

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So the examples are the postal services or railways. As I said in the earlier class, in India we have the postal service is a monopoly which is by the government. The railway is a monopolist. So however this is public ownership is not always efficient as absence of profit motive can result in inefficient production choices.

So what happens is in case of postal service or railways the government may being a extremely benevolent government may try to produce a competitive level of output here and incur a lot of losses which are covered from the government revenues but since it is already known that the loses will be covered so the efficient production choices may not happen.

Efficient production choices may not happen and since there is no profit motive probably the efficiency is compromised in case of a public ownership of monopoly which may not happen if the monopoly is monopolies has a profit motive and is under private ownership where there the profit motive basically leads to the incentive to reduce cost and hence earn higher and higher profits.

So this was about this entire week we have discussed about monopoly and we have seen that the monopolist has the output and price combination that the monopolist can decide on is slightly tricky for the monopolist because he has to not only decide the level of output he has to also decide the level of prices. However, the monopolist knows the demand curve. It has an understanding of the demand curve.

So it respects the demand curve and after deciding on the profit maximizing level of output it decides its price on the basis of the negatively sloping demand curve and hence he decides on the price and output and he has to always be concerned about a negatively sloping marginal revenue curve because when he produces an extra unit of output his price basically goes down and there is a tradeoff between the output effect and the price effect.

And we also saw that from the unlike the perfectly competitive outcome in case of monopoly outcome there is some amount of deadweight loss or there is some efficiency loss and welfare loss in the market and which can be corrected through price discriminating price discrimination where efficiency level or total welfare in the economy is exactly same as the perfectly

competitive outcome but consumer surplus is completely wiped out in favour of producer surplus.

So if the economy is indifferent between a producer and a consumer in that case the price distributing monopoly outcome is exactly same as the competitive outcome and we also discussed different examples of price discrimination where basically the supplier looks at price sensitivity of different categories of buyers looks at their demand curve of different categories of buyers at different time periods of buying the product.

And this is how any supplier any producer with some amount of market power he could price discriminate across different consumers and since monopoly outcome is not as desirable as competitive outcome we also discussed that the government may like to step in to correct the inefficiencies of monopoly outcome or basically to improve the welfare of consumers so that consumers get more supply of the output at a lower price so the government can have different kinds of policies.

One is the government by law tries to create a level playing field for all kinds of producers in the market so that a competitive environment is created in the market to encourage more and more producers to produce and sell the same commodities through the competition act.

And secondly the government may like to regulate the monopolies in case of specially, natural monopolies the government may direct the monopolist to produce at produce higher level of output than a monopoly output and charge lower level of prices. This the government can achieve through either subsidizing the monopolist or going from for average cost pricing or asking the or allowing the monopolist to go for price discrimination.

And the another option that the government has is it basically controls the monopoly as a public ownership where instead of profit maximization output maximization could be the objective of the monopoly firm where it is the objective is not profit maximization but output maximization so that the necessary output is provided to all kinds of buyers in the economy.

However, this is criticized because since profit maximization is not an objective it could lead to a lot of inefficiency in producer in the in this kind of monopoly and the cost could be too high

which finally at the end of the day would have to be borne by the government from the tax payers money.

So in the following week we are going to discuss about monopolistic competition where it will be somewhere the market will look somewhere in between perfect competition and monopoly where there will be lot of buyers but at the same time each of the lot of sellers at the same time each of the seller has some amount of market power because the product that he is selling is slightly different from the products of his rivals in the market. So this monopolistic competition we are going to take up in the following week. Thank you.

Foundation Course in Managerial Economics

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Lecture - 30

Monopolistic Competition

Hello and welcome back to the foundation course in Managerial Economics. We had we have been discussing about different kinds of market structures and we have completed 2 extreme kinds of market structure with very opposite almost opposite kinds of assumptions. One is the perfect competition we discussed in the last to last week and last week we discussed about monopoly.

So in case of perfect competition we saw that there are whole lot of buyers and sellers in the market who really do not have much control over price. So basically the producer decides how much to produce. He does and just generally accept accepts the price that is given in the market. So he is basically a price taker.

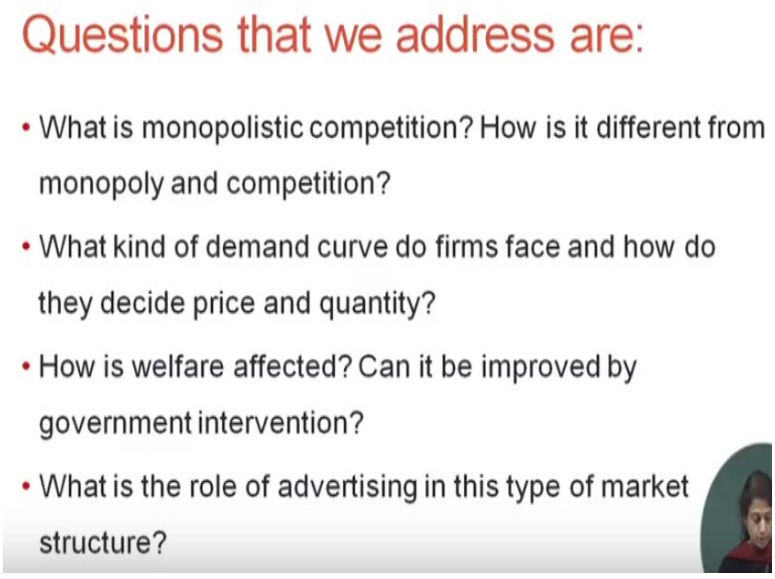
And on the other extreme when we discussed about monopoly we saw that in case of monopoly there is only one single seller in the market and he is the single seller so he has a lot of control over the market. He faces the negatively sloping demand curve and he has a lot of choices to make what amount of output to produce what price to charge. So everything he decides and depending on the demand curve he decides the price to be charged in the market and he earns a it is possible for him to earn a greater than zero economy profit.

So this is what we saw and the very central difference between perfect competition and monopoly basically comes from the nature of the product that is sold in this market because we saw that in case of perfect competition the products are highly substitutable or they are homogenous. So the buyer basically does not differentiate between the different sellers.

It is fine for him to buy from whoever is there to sell the product. On the other hand, in case of monopoly the product is unique. It is not available. If it is if this single monopoly seller does not sell the product, then there is no other seller available in the market to sell this product. So basically it is a unique product and its substitute is not available in the market and this is the kind of market that we studied when we talked about monopoly.

So today we are going to discuss about monopolistic competition. So as the name suggests this kind of market structure has some qualities of a perfectly competitive market but on the other hand it has some qualities of a monopoly market. So let us see what monopolistic competition looks like.

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So the questions that we are going to address in this week are what is monopolistic competition and how is it different from monopoly and competition? What kind of demand curve do firms face and how do they decide price and quantity? How is welfare affected? So we have seen in

most of the in all the market structures that we have discussed so far we have always tried to not only see what the market output is, what is the market clearing price, what is the market clearing quantity but we have also studied what are the welfare implications of this market outcome. The same thing we are going to do in this in the case of monopolistic competition as well.

So we are going to study how is welfare affected and can it be improved by government intervention. What is the role of advertising in this type of market structure?

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So coming to monopolistic competition. Now monopolistic competition, in monopolistic market structure lies somewhere, monopolistic competition market structure lies somewhere between the 2 extremes of perfect competition and monopoly. There is a mistake in the first point in this slide; it should be monopolistic competition market structure. So I am going to correct it when the slides are shared with the students.

Next is like competition there are many sellers and buyers and entry and exit are easy. However, products are differentiated but substitutable to some extent and like monopoly firms have some amount of market power over its buyers, though not absolute.

So let me explain a little bit. So in case of monopolistic competition market structure as I already said that it lies somewhere between perfect competition and monopoly. Like in competition in

monopolistic competition there are whole lot of buyers and sellers in the market. Since we are focusing on sellers primarily so since there are large number of sellers in the market no single seller can actually influence the price too much.

So no one can influence the price or rather the market too much and the entry and exit is easy as is already indicated by the fact that there are so many buyers and so many sellers in the market. That basically means that entry and exit are possible. They are easy. It is not difficult to enter this market and it is not difficult to exit this market.

So this point is very similar to perfectly competitive market structure. So there are whole lot of buyers and sellers in the market. So any single seller cannot influence the market too much. Yet the products are slightly differentiated. What does that mean? That means that products are such that there it is they are not perfectly substitutable like in the case of say perfect competition where products are like to give example the products are like say agricultural products say rice, wheat, or say milk, eggs which are fundamentally they are all exactly same products.

So a perfect competitive market perfectly competitive market one can imagine to sell such kinds of products which are not at all differentiated. They are basically fundamentally the same products. On the other hand, and since we are taking example so let me also complete it by giving examples of monopoly and in case of monopoly the products are very unique say patented drugs, lifesaving drug, or say the railways the water supply.

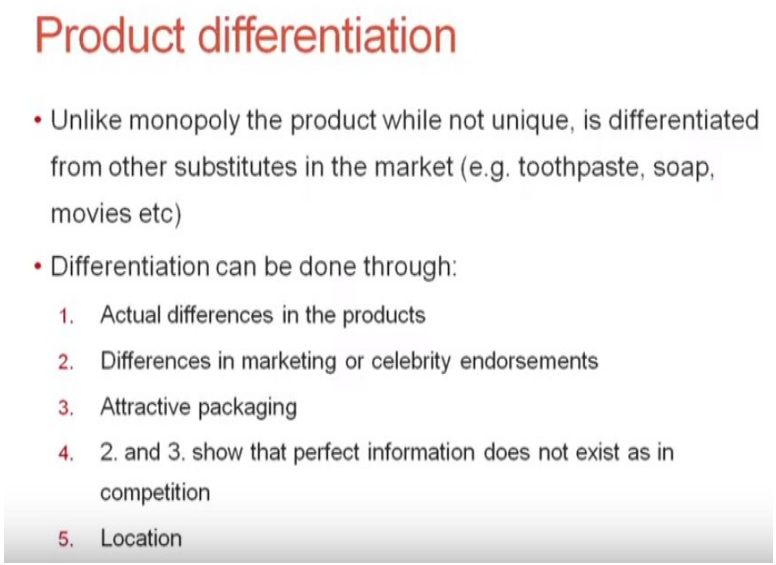
So there is one single producer or supplier of these products in the market and monopolistic competition lies somewhere in between and examples of monopolistic competition are actually far and wide. Most of the products that you are going to see around you are belong to monopolistic competition market structure. Say for example the fast moving consumer goods they are monopolistic; they belong to monopolistic competition. Say movies, novels.

So all these are monopolistic competition, products which are sold in monopolistic competition market structure. So what is how are these products different from the examples that we took for perfect competition or say for monopoly. Now the products are different in the sense that say fast moving consumer goods. So there can be whole lot of examples where different kinds of the firms are selling goods like say toothpaste.

Now toothpaste can be of various types and categories and one could have some inclination towards repeatedly using a particular brand of toothpaste and so they are slightly differentiated. The products are differentiated. So basically the consumer has some amount of loyalty towards consuming a particular type of product although fundamentally the products are all same. At the end of the day they are all toothpaste.

So similarly so it is said that the products are differentiated but substitutable to some extend and since the products are differentiated and since the products have their own consumers every kind of variety of products has its own group of consumers who would not like to shift from this variety of the product to any other variety so some amount of market power is possible over these consumers. So let us move on the next slide and as we discuss the difference will be even more clearer.

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So what do we mean by product differentiation? What does it what does it mean product difference when I say product differentiation and they are substitutable but they are differentiated what does that mean? Now unlike monopoly the product while not unique is differentiated from other substitutes in the market. For example, toothpaste, soap, movies, etc.

Now differentiation can be done through a lot of ways. Now the seller has a has, the seller will come up with whole lot of ways to differentiate his product from the other products in the market; so one is the actual difference in the product. Say for example again coming back to the example of toothpaste you will get whole lot of varieties of toothpaste in the market. Some will be having will be claiming to have clove, some will claim to have mint, some will claim to have salt, some are white, some are red, some are striped. So the products are different from each other. Although they are all toothpaste but still they are different from each other and one can imagine that different consumers will have different preferences. Some may like to stick to a toothpaste which has mint. Some may like to stick to like the colour red for his toothpaste. So these are ways in which the seller creates differences in the products and this is how a seller makes his product different from the other products in the market.

Second is differences in marketing or celebrity endorsements. So apart from having the product exactly, products different actually, the making the products different, one can imagine that the seller will try to create notions in the minds of consumers that the products are different. Say for example we took the example of milk, eggs when we discussed about perfect competition.

Now one can imagine different companies they are selling milk in tetra packs and they might create difference through marketing saying that our packaging is good or we the milk is fresh, it contains these vitamins which it is not easy for the consumer to check or even find out when he consumes the milk what is the difference between 2 different cartons but it is basically the marketing or the endorsement say in the case of other product say soap, shampoo, etc. it is being endorsed by a physician or doctor.

So these are the perceptions that are created in the minds of the consumers that the product is better than any other product in the market. So one carton of milk is better than another brand of milk that is available in the supermarket. So this is differentiation of perception that is created in the market.

Another possibility is attractive packaging. So one might like a soap because it is it has a better shape or it has a better packaging so attractive packaging. So 2 and 3 show that perfect information does not exist as in competition. So the points 2 and 3 that one or the seller is able to

create differences in opinion about the product through just market advertising or attractive packaging in the minds of the consumers that shows that perfect information does not exist.

So it is possible to sell exactly the same product through different packaging and charging different prices for them. So that shows that perfect information does not exist in this kind of market structure and perfect information is something which we had assumed in the case of perfect competition. So in this kind of market structure actual difference in product does not matter as long as there is difference in the mind of the consumer that the products are different.

Another interesting differentiation that can happen is through location. Now when I say location say for example there is say for example there are 2 petrol pumps on 2 sides of a road and during rush hour during office hour when there is very high traffic on the road you are driving on the left side of the road and there is a petrol pump both on the left and the right and on the right side petrol pump you know that they charge a price which is 50 paisa less per litre of petrol than the petrol pump on the left.

So during rush hour you may not like to so if you are planning to buy 10 litres of petrol that means only 10 Rs that you are going to or 5 Rs that you are going to save by making a U turn and buying the petrol from the right side of the highway or right side of the road. So you may not like to do that during the rush hour because you know the traffic is too heavy and it will be it will take you a lot of time to actually turn the car and fill up your tank and then again come back into this road.

So you are going to prefer paying a little extra and filling up your tank from the left side of the road. So basically that means that although on the left side of the road the petrol pump is charging 50 paisa more than the one charging than the petrol pump which is on the right so that shows that because of location there is a locational advantage which the petrol pump on the left has and it is basically leveraging that advantage and charging a little extra from its consumers.

So it is possible to do that if you have a preferable location and we see that happening all the time. Say for example in a very in a residential complex in a residential area there is only one stationery shop. So probably they are going to sell you stationeries say pencils, notebooks etc. at a at MRP or a high price than shops which are there in the office area where there are whole lot

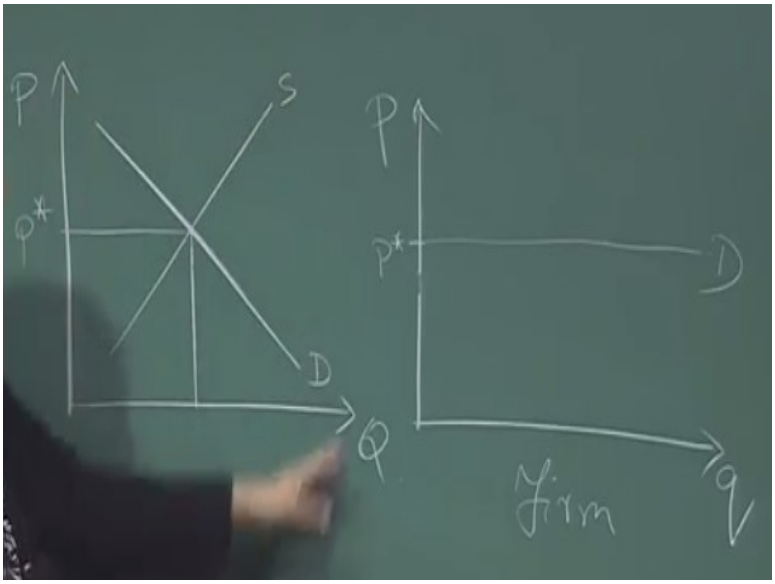
of stationery shops and who are probably going to give you a discount on the maximum retail price that is printed on the products.

So these are locational advantages that these kinds of sellers' leverage and this is how they have differentiated their product from the rest of the products available in the market. So now that we have understood what is the difference what does the product in case of monopolistic competition look like and what is the fundamental difference between monopolistic competition and perfect competition, let us look at what the demand curve looks like.

Now a quick recap in case of perfect competition you may remember that the demand curve facing each of the buyers, there are whole lot of buyers in the perfect competition. Similarly, in case of monopolistic competition also there are numerous buyers. It is very easy to enter or exit the market so there are lot of buyers in this market.

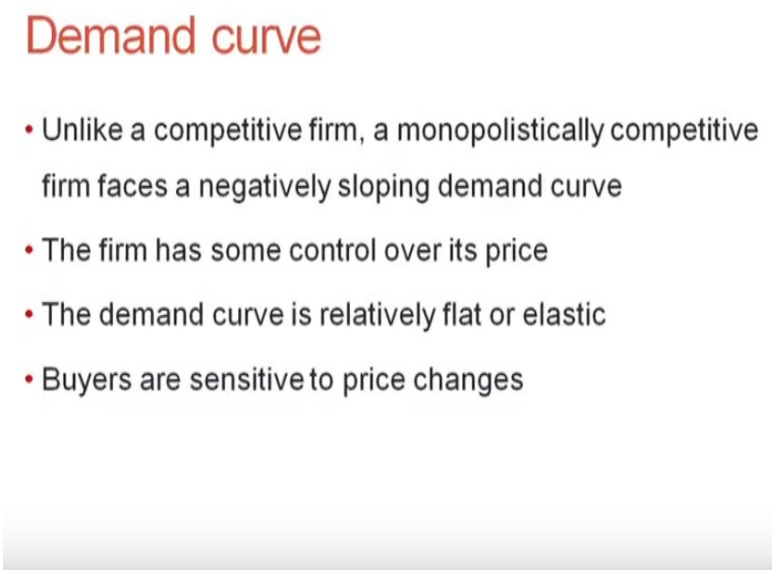
Similarly, in case of competition also there were whole lot of buyers in the market and if you may remember in case of competition each of the buyers was facing a horizontal demand curve. That means each of our each of the firms I am sorry not each of the buyers each of the sellers. So each of the firms was facing a horizontal demand curve and they were all price takers and that price was determined from the market demand and market supply curves.

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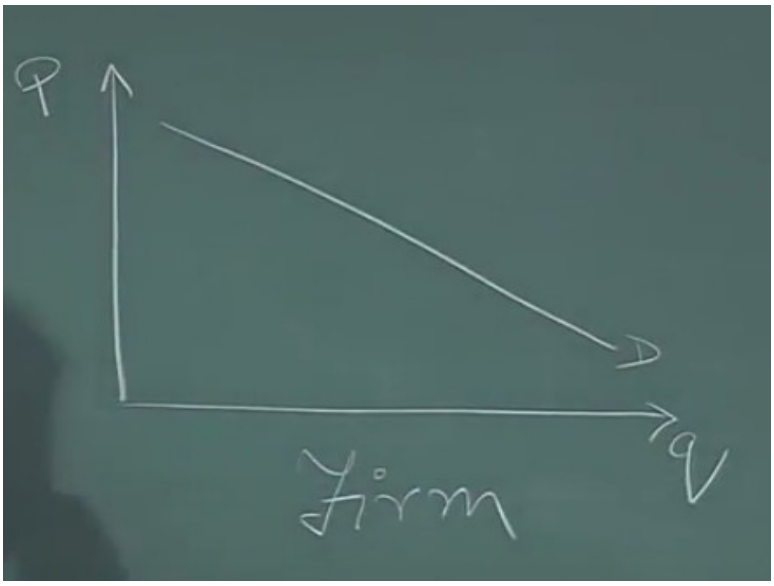
So basically so this was the P star was determined in the market and for every firm any firm where small q was the output that each firm produced and P was the price that it charged so this was for the firm. So firm basically took this price as given. So the firm knew that at this price this is the demand curve that is he knows that that the buyers will buy as much of the product as possible at his price. So this is the demand curve that the firm was facing in case of a perfectly competitive market. But what happens in the case of the monopolistic competition, let us see.

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So unlike a competitive firm a monopolistically competitive firm faces a negatively sloping demand curve. The firm has some control over its price. The demand curve is relatively flat or elastic. Buyers are sensitive to price changes. So let us see what the demand curve looks like to the monopolistically competitive firm. So there are whole lot of firms in a monopolistic competition market structure also but in case of a monopolistic competition market structure there is no single price. The firm there is no single price determined in this way and to the firm they the demand curve does not look like this. So let us see what the demand curve looks like.

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So again see we have used the small q here because we are showing it for one typical firm in a monopolistic competition market setup. So this is a firm we are talking about. So this is a firm, this is the price on the y axis. Now this firm sells a product which is nothing unique but the product is slightly differentiated from the other products in the market. So there are other products in the market of similar nature but not exactly same.

So this firm, say for example this is a firm which sells minty toothpaste. So he has a whole lot of he knows that there are whole lot of consumers in the market who are fond of minty toothpaste. So they are loyal to the products of this firm; so this firm faces a negatively sloping demand curve similar to a monopolist. So for this firm his product is unique in certain way and he knows that there are some people who would really not who would really like to stick to this particular type of product, say minty toothpaste.

So he has a negatively sloping, so he has some amount of market power over these people who are fond of minty toothpaste and so he has a negatively sloping demand curve but the demand curve is flat. The demand curve is it is not exactly flat but the demand curve has a flat slope or it is a highly elastic demand curve. Why is that so?

So we are saying that it has a demand curve which looks like this. It is very flat. Why is that so? Now flat demand curve means that the consumers are highly price sensitive, why? They are price sensitive because if there is some amount of price change here since this firm has some amount

of market power over these consumers who like minty toothpaste so this firm might like to change the price a little bit.

The firm might decide that I am going to increase the price a little bit. Now these people who buy this type of toothpaste they may not mind to pay a little bit of extra to continue to consume the same amount of same variety of product. But if the firm decides I am going to increase the price a little higher, it decides that I am going to make a substantial increase in price, then he is going to lose his consumers**.**

Then he is going to lose his consumers because the consumers will then shift to other products in the market because other products are available in the market. They are not perfect substitutes yet they are substitutes in the market. So the firm, the buyers are going to shift to another seller in the market and shift probably shift their preferences from minty toothpaste to some other variety of toothpaste.

Hence the demand curve is flat or it is highly elastic which means that in monopolistic competition the firms have the buyers are highly price sensitive and they respond to any reduction in price. There will be a whole lot of increase in demand and little increase in price lot of people are going to shift away from the consumption of the product. So buyers are sensitive to price changes hence demand curve is relatively flat or elastic.

So this was our preliminary discussion about how the monopolistic competition market structure looks like and we have seen that there are many sellers in the market as in perfect competition but yet unlike perfect competition where we have a horizontal demand curve in case of monopolistic competition the firm faces a negatively sloping demand curve which implies that the firm has some amount of market power like in case of monopoly yet the demand curve is quite flat or it is highly elastic because the buyers although they have preferences for this differentiated product yet they know that substitutes are available in the market although they may not be exact substitutes.

So if the price changes too much, if the price increases too much, people are going to reduce their consumption. So basically the monopolistic competition competitive firm faces a negatively sloping demand curve like in case of monopoly and yet the products yet it is very easy for other

firms to enter or exit the market. In the following section we are going to see how the market outcome is arrived at by a monopolistic competitive firm. Thank you.

Foundation Course in Managerial Economics

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Lecture - 31

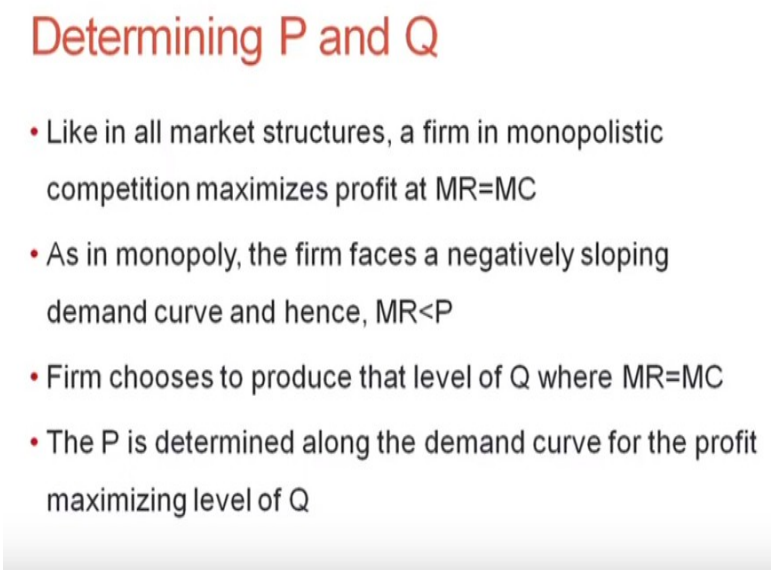
Monopolistic Competition - Determining P and K

Welcome back to our discussion on monopolistic competition. We in the previous lecture we developed the concept of monopolistic competition. We discussed how the monopolistic competition lies between the competitive market structure with lot of buyers and sellers in the market with easy entry and exit conditions and on the other hand like monopoly in case of monopolistic competition the products are differentiated.

In case of monopolistic competition products are differentiated. So although the products are substitutable to some extend there are differences in the minds of the consumers about the product because of which the consumers have some amount of loyalty or some amount of preferences towards one type of one sellers one type of product against another and because of which the monopolistic competition seller has a negatively sloping demand curve where he has some control on the market and some control over the price.

So that is what we are going to see now how does the monopolistic competition in case of monopolistic competition how does the firm choose what amount of output to produce and what price to charge.

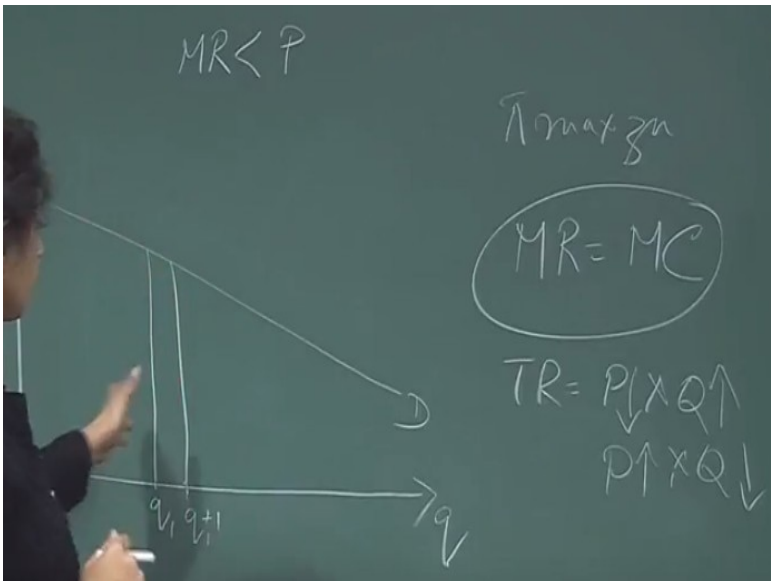
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So determining P and Q, again there is a typo here it is it should be P and Q. Like in all market structures, a firm in monopolistic competition maximizes profit at MR is equal to MC. Let me correct this. So determining P and Q like in all market structures the firm in monopolistic competition maximizes profit at MR is equal to MC. As in monopoly, the firm faces a negatively sloping demand curve and hence marginal revenue is less than P. the firm chooses to produce that level of output where MR is equal to MC. The P is determined along the demand curve for the profit maximizing level of Q.

So let me explain. So like in all market structures, the firm in monopolistic competition also maximizes profit at MR is equal to MC. So we already discussed this that the profit maximizing condition for any firm in any market structure profit maximization happens at MR is equal to MC.

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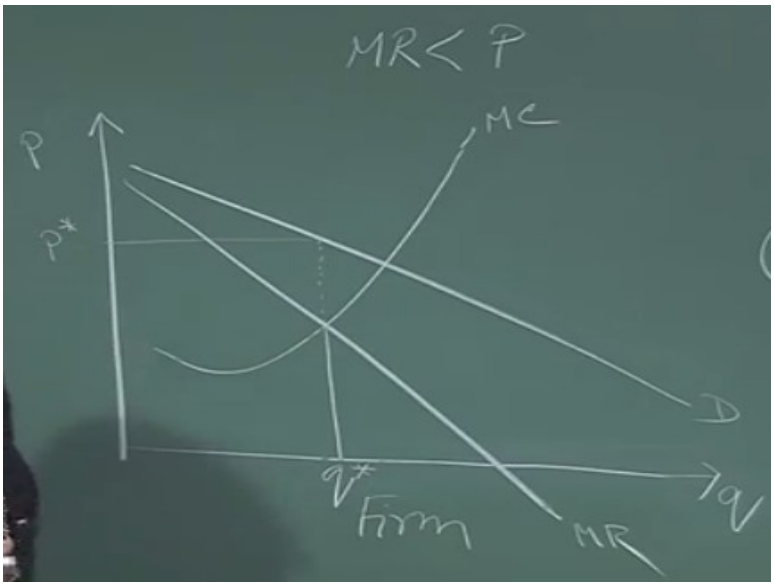
This is always going to hold true for profit maximization no matter what kind of market structure the firm is operating in but since in case of a monopolistic competition it faces a negatively sloping demand curve, it faces a negatively sloping demand curve, so at every point again as in the case of monopoly in this case also the firm faces a dilemma because it knows that if it changes the price, say for example it reduces the price there is going to be a price effect and there is going to be a output effect.

So basically if it reduces the price there is going to be a increase in output. So in total revenue so as soon as he reduces price this part falls but output increases. So they move in opposite direction. So when P falls output increases or when P increases output falls. So there is one price effect and then output effect always and as in case of monopoly in case of a monopolistic competition also the firm has is always in a dilemma whether to change price because it knows that if it reduces the price output is going to increase so there is going to be a positive impact on revenue through output but there is going to be a negative impact on revenue through price.

So this he knows and accordingly his decision becomes difficult and that is what we saw in case of monopoly that MR is always less than P. So MR is always less than P because what is MR? Marginal revenue is basically equal to the additional revenue that the firm is going to get out of producing another unit of output. So another unit of output, production of another unit of output, is possible in case of a negatively sloping demand curve only by reducing the price.

So if the firm is producing here q 1 and it decides to produce q 1 + 1 an additional amount of output immediately it has to reduce its price from here to here. Immediately it has to reduce the price from here to here and so immediately there is an impact on the total revenue. So marginal revenue is always going to be less than price for a negatively sloping demand curve unlike in the case of competition where price is always the same. No matter how many units of output the firm decides to sell it never has to worry about price because price is fixed by the market. So here in case of a monopolistic competition also, let me draw it little more nicely.

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So this is for the firm and this is the price and here is the flat demand curve and since so this is the flat demand curve and this is the marginal revenue curve. So demand curve is basically the price line and marginal revenue is always less than the price, is always less than the price line. So this is the marginal revenue curve and which is always less than P and so the firm chooses to produce that level of Q where MR is equal to MC.

Now what do we do here? You are probably quite familiar with this now and you probably have got a hand of the exercise that once we draw the demand curve we draw the marginal revenue curve, and we draw the J shaped marginal cost curve. So we draw the J shaped marginal cost curve and the profit maximization condition is marginal revenue equals marginal cost.

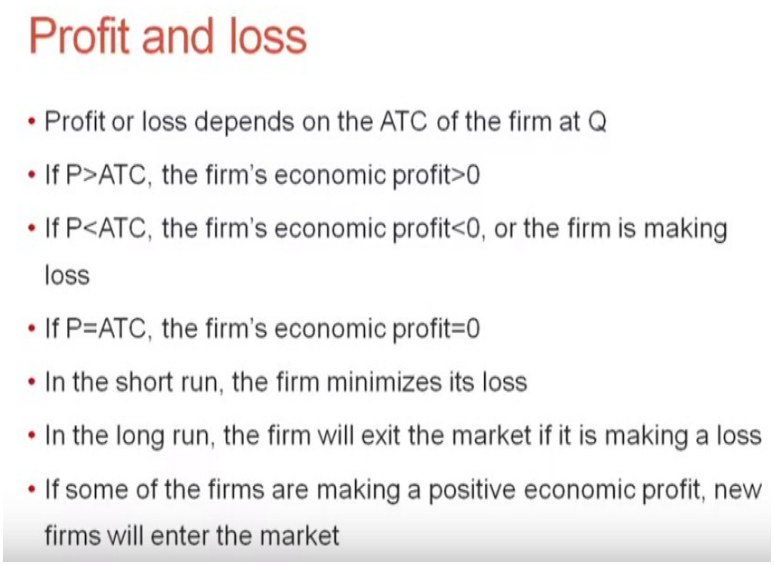
So marginal revenue equals marginal cost at this point. So this intersection is going to give me the equilibrium output or is going to give me the profit maximizing output. So q star is the amount that this firm is going to produce. How does it decide what price to charge? How does it decide what price to charge? Again the exercise is exactly similar to the monopoly situation so what does he do?

He knows that he has to produce q star and how much are the people willing to pay for q star? How much are the people willing to pay for q star so for that he traces back to the price line or the demand curve that he faces in the market so this is the demand curve that he faces in the market and this is the price P star that he is going to charge in the market.

So this is a differentiated product in the say toothpaste market so this is the this is the firm which sells a particular variety of toothpaste and this is the output that he decides, he is going to produce and this is the price that he is going to charge and this price he determines from the demand curve of the differentiated product. So this is the demand curve for the mint toothpaste that he is selling in the market and he charges a price P star from here.

So this is so the P is determined along the demand curve for the profit maximizing level of output. Now coming to the next slide to discuss about profit and loss. So once so now we know what the what how the equilibrium condition for the how the firm perfectly monopolistic competition monopolistically competitive firm decides what output to produce and charge what price. So next question is what is the profit and loss that the firm is going to make?

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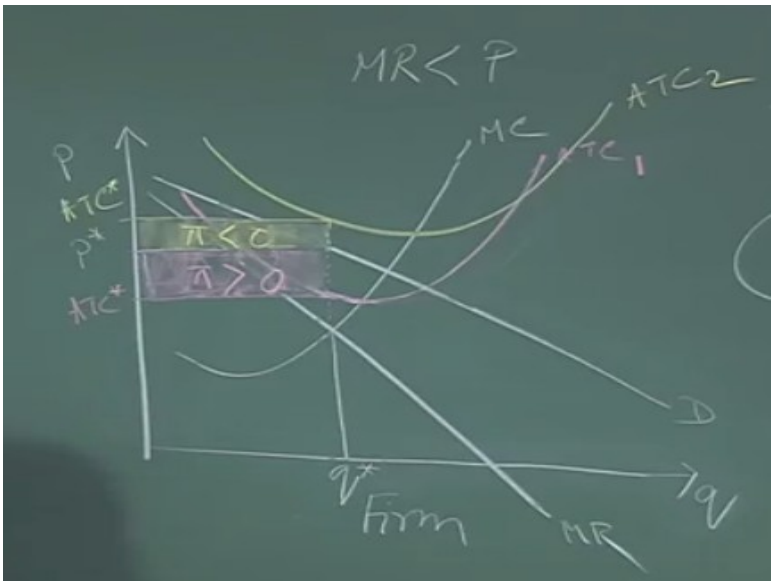


So profit or loss depends on the average total cost of the firm at Q. If price is more than average total cost the firm's economic profit is more than 0. If price is less than average total cost the firm's economic profit is less than 0 or the firm is making a loss. If price is equal to average total cost the firm's economic profit is equal to 0.

So in the short run the firm minimizes its loss. In the long run the firm will exit the market if it is making a loss. If some of the firms are making a positive economic profit, new firms will enter the market. So let me explain here. So what do we mean by this? We know that this is the profit maximizing level of output but profit maximizing itself does not mean that there is actually a profit being generated. It does not mean that.

It just means that q star maximizes the profit that is possible there. So that profit could be negative also. There could be a loss also. So profit maximization is basically it is minimizing the loss. So what does profit actually depend on. Profit depends on, what does the average total cost curve look like? Say for example the average cost curve say looks like this.

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Say the average cost curve looks like this. So let me call, use a different, sorry I. Okay, this is supposed to be the minimum point of the average total cost curve. So I hope you have been able to figure out why I changed the average total cost curve because I wanted the marginal cost to intersect the average total cost at its minimum point.

So this is just to draw it as correctly as possible. The marginal cost always intersects the average total cost at its minimum. So this is the average total cost for example. So it is average total cost for example. So when the firm is producing an output of q star what is the cost that it is incurring? So along q star I find out here the q star that q star it is incurring a cost of this much, ATC star. So this is the cost, this is the price. So obviously for each unit of output it is getting a profit of P star minus C. So this is the amount of profit, the pink portion is the amount of profit that the firm is getting. So this is economic profit is more than 0.

Now what would happen if the cost curve looked like this? If the cost curve was here ATC, had the cost curve looked like this had the cost curve looked like this then what would be the cost like at q star? At q star then cost would be ATC star in yellow. So this would be the cost and in that case there would be a loss of the amount equal to the yellow rectangle.

So there here profit is greater than 0. Here profit is less than 0. So here profit is greater than 0, here profit is less than 0. So again to repeat this profit maximizing condition is MR is equal to MC. So profit maximizing condition tells me that the firm should produce q star if at all it

produces. Now when the firm produces q star it is possible that the firm is going to earn a positive economic profit and it is also possible that the firm is going to earn a negative economic profit and that would solely depend on what is the average total cost of the firm at that level of output and we took 2 examples. One, the example of a pink average total cost which we drew like this and showed that if it looks something like this then there is a positive economic profit. Again we took another example of a ATC 2 and we said that if the cost looks something like this then the firm is having a negative economic profit. So this is what we said.

Now and in the typical case of where ATC is exactly equal to price where basically ATC at q the ATC basically passes from here at this point, if the ATC passes through this point so here basically cost is equal to price for every unit of output. So there economic profit will be equal to 0. When economic profit is equal to 0 means accounting profit is more than 0 and if that also happens the firm is happily going to continue to produce in the market.

Now what happens, so this is the case of a typical firm. So there are whole lot of firms, similar firms in the market and they are facing different demand curves depending on the kind of differentiated products they are selling in the market. Now all the firms like there are whole lot of firms in a market some have cost curves which are similar to the pink ones, some have cost curves which are similar to the yellow ones.

Now what happens in the long run? Now in the short run if the firm is incurring loss here, what is the condition? If you may remember when we discussed about producer theory, we said in the short run what is the firm going to do? It cannot exit in the short run. So in the short run it is going to see if it is getting a price which covers at least the average variable cost.

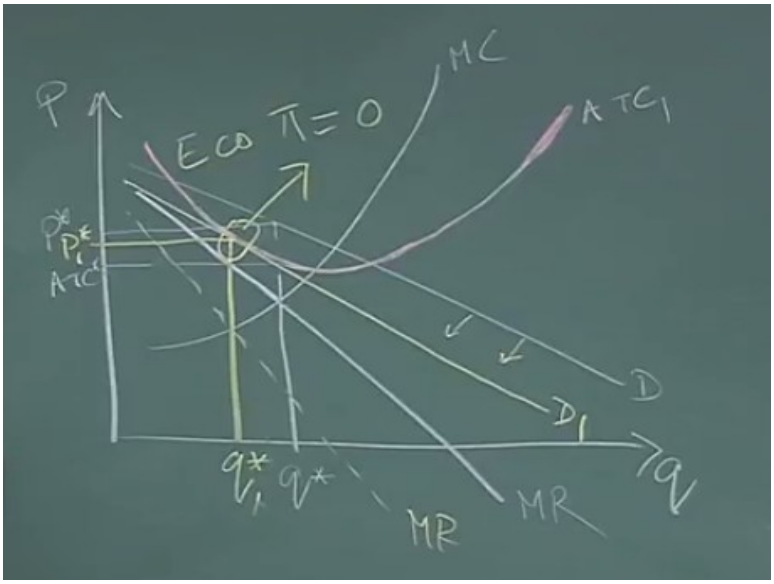
So for firms where the cost curve looks like the yellow ones, the firm is going to see that if the average variable cost lies below the price, so in that case price is able to cover at least the variable cost. So the firm is going to stay in business. Now and if it is above the price line, even the variable cost, then the firm is going to shut down. What happens in the long run? In the long run, in the long run the first let us talk about the yellow ones.

In the long run these firms if they see that they are not able to cover the, the price is not able to cover their cost, they are going to exit the market. They are going to exit the market. What

happens to the firms which have positive economic profit? Now there are firms which have positive economic profit in the market and looking at this there are whole lot of producers outside the market who are keeping a watch on this market and who are potential producers in this market or potential sellers, they will see that there is profit to be made in this market.

There is there are firms which are producing which are getting positive economic profit and that attracts these potential sellers into this market. So these producers are going to enter this market. So these producers are going to enter this market and what will happen is, I will so at the producers, so more producers enter this market and let me remove this and draw another figure to show you what happens when there are new entrants in the market. What happens when there are new entrants in the market?

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So again this is a typical firm, it is operating in this market and this was his demand curve, this was his MR, this was his marginal cost curve and he was producing q star, charging a price of P star okay and he was incurring a profit. So basically I have to so this is the average cost curve of this firm. This is the average cost curve of this firm, and so this firm was, so the firm was incurring a profit which is equal to this rectangle. Now, no let me keep it clean. So this is average total cost and this is the profit that this is the profit that the firm is making in the short run.

Now what happens if you remember from just the earlier diagram we saw that this the profit that the firm is making and looking at this profit new entrants will enter the market. So new sellers will enter the market and they will also start offering new differentiated kinds of products so there will be some so this was a minty toothpaste so there will be new products. We will say not only ours have a mint flavor it also has this colour. It has these properties. See this is endorsed by this film star. So all these are possible.

So once that happens when there is new product in the market some of the people who were loyal consumers of this product they are going to move to the new products that are being offered in the market. So they are going to move away and because of which the demand curve is going to shift inwards. So this demand curve is going to shift inwards and this continues to happen in the market till there are so it is going to shift inwards but still there may be positive economic profit.

So as long as there is positive economic profit being made by some of the firms in the market new entrants will keep on entering the market so till these consumers they disperse and these consumers they some of them at least move away to other sellers in the market move away to other differentiated products as a result of which this demand curve shifts inwards.

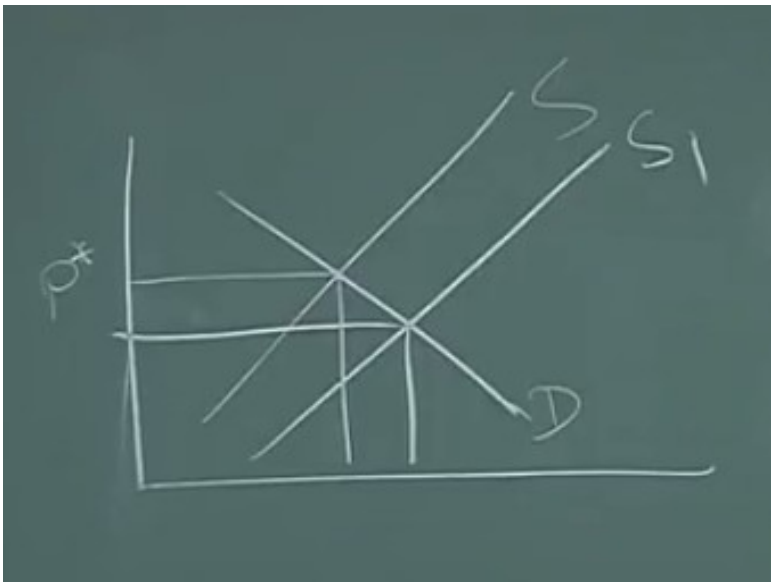
So this demand curve is going to shift inwards till the time till the demand curve so this is the new demand curve till the demand curve shifts so much inwards that it so the it is tangent to the average cost curve of the firm. So this is the new demand curve and this will be the new marginal revenue curve and this will be the new output that the firm is going to produce and this is going to be the new price of P 1 star.

So this is going to be the new price and you can clearly see from here that the demand curve has shifted so much inwards that the demand curve is now equal to the, the demand curve is tangent to the cost curve as a result of which at this level of output the price that the firm is charging is exactly equal to the cost that the firm is incurring for producing that unit of output.

So here at this point, economic profit is equal to 0. Economic profit is equal to 0. So in the long run the firms what so it is very similar to the perfect competition case here also in case of

perfectly competitive market what happened? As more and more buyers, sellers entered the market the demand curve the supply curve would shift.

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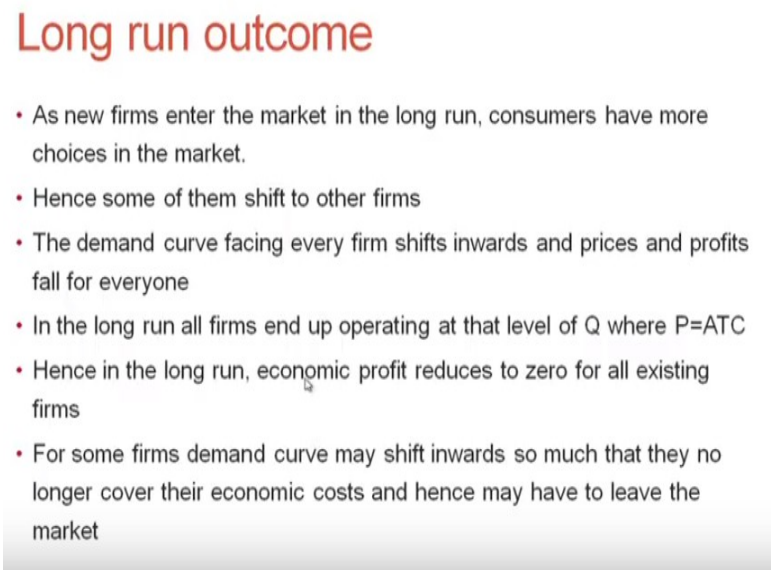


If you may remember, in case of perfectly competitive market as more and more sellers enter the market, so this is the demand curve, this is the supply curve and this was the equilibrium price. So as more and more sellers entered the market the supply curve would shift to the right and price would come down and bring down the economic profit of the existing sellers in the market.

But in case of the perfect in the case of monopolistic competition nothing happens to the supply curve in the market or the supply curve in the market obviously the supply is going to increase but that is not changing the economic profit. What is changing is for each of the firms the number of consumers or the demand in the for each of these firms that goes down.

So each of the firm has its loyal set of consumers whose demand curve it is facing but that demand curve shifts inwards as more and more suppliers enter this market, as more and more suppliers enter this market, so their economic profit is wiped out till the time when economic profit of all existing suppliers in the market reduces to 0 and in the process if it happens that some of the suppliers demand curve shifts inwards so much that economic profit reduces to less than 0 these firms are going to exit the market. So these firms are going to exit the market.

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So what is the long run outcome? As new firms enter the market in the long run consumers have more choices in the market. Hence some of them shift to other firms. The demand curve facing every firm shifts inwards and prices and profits fall for everyone. In the long run all firms end up operating at that level of output where P is equal to average total cost.

Hence in the long run economic profit reduces to zero for all existing firms. For some firms' demand curve may shift inwards so much that they no longer cover their economic costs and hence may have to leave the market. But one thing is of interest here which you may have noticed is that how is this outcome different from the perfectly competitive outcome.

In case of perfect competition in the long run so in case of perfect competition in the long run all the firms have zero economic profit. Similarly, in case of monopolistic competition also in the long run all the firms have zero economic profit. However, the difference is in case of perfect competition the firms charge a price equal to the minimum average total cost of every firm. So the firm is going to charge a price which is here.

So the competitive outcome will be price will be here where the marginal cost intersects the average total cost and that price will be charged in the market. So that will be the single price that is going to be charged in the market and price will be equal to the minimum of the average total cost. However, in case of monopolistic competition each of the firm charges a price which

is equal to the average total cost at that level of output yet that price is not equal to the minimum of the average total cost for that firm.

So basically the firm is charging, it is slightly short of the average total cost. It is charging a price over here which is although along the average total cost yet here average total cost is falling and it is not the minimum. So this is a fundamental difference between a competitive outcome and a monopolistic competitive outcome. So in the following section, in the following module we are going to see what is the welfare implication of a monopolistic competition as compared to say perfect competition. Thank you.

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Lecture - 32

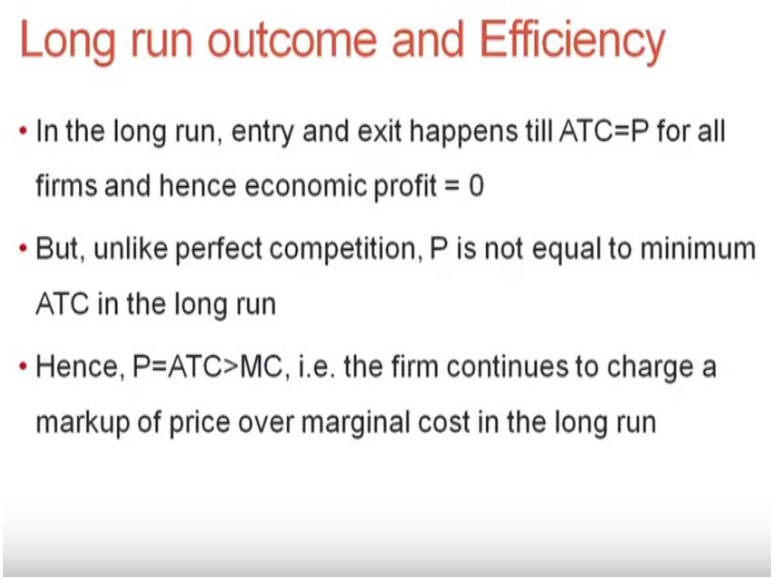
Monopolistic Competition - Efficiency and Welfare

Welcome back to our discussion on monopolistic competition. In the earlier 2 modules we have developed the assumptions about monopolistic competition. We have understood what the products look like in a monopolistic competition, how are they differentiated, how are they substitutable and what does the demand curve look like and eventually we developed the profit maximizing condition for monopolistic competition and we saw that saw the profit maximizing output level that the firm chooses to produce and then it basically traces back the output along the demand curve and decides what price to charge in the market.

Then we went on to discuss that the average total cost that the firm has that is going to determine if the firm is making any loss or profit or zero economic profit and if the firm is making profit we saw that lured by that profit other firms are going to enter the market and the firms which bear a loss even in the long run they are going to exit the market.

So this is this was the standard theory about monopolistic competition basically discussing on about how profit maximizing output and price is determined in a monopolistic competitive setup and so like all our previous discussions on perfect competition monopoly etc. where we have looked at the welfare implications of the market outcome so in this module also we are going to discuss about welfare or efficiency implications of the monopolistic competition market outcome.

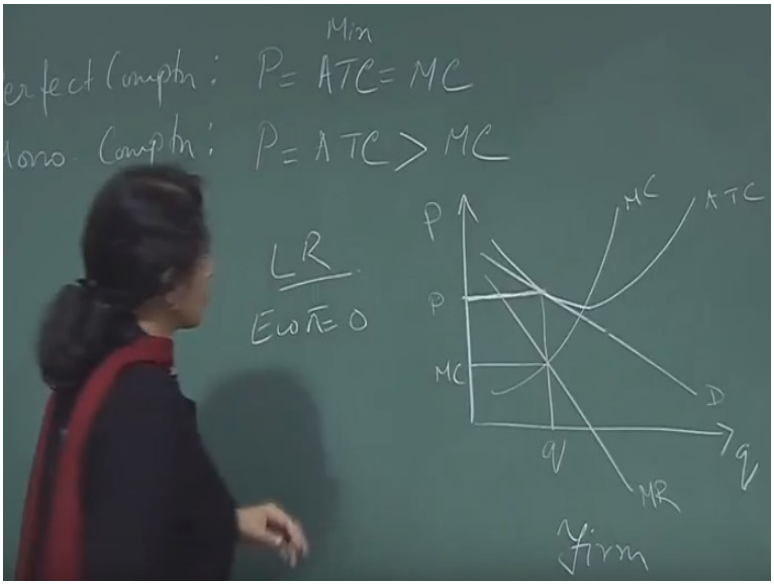
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So talking about long run outcome and efficiency, in the long run entry and exit happens till average total cost is equal to price for all firms and hence economic profit is equal to 0. But unlike perfect competition price is not equal to minimum average total cost in the long run. Hence price is equal to average total cost but it is more than the marginal cost that is the firm continues to charge a markup of price over marginal cost in the long run.

So in the long run although price is equal to average total cost and the firm is basically having a economic profit equal to 0 but still the price has a is a markup over the marginal cost of the firm at that level of output. So let me draw the figure again.

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So this is output, this is price. So in the long run this is the q and this is the price. So in the long run the average so in the long run the cost is equal to price for this firm. So economic profit is equal to 0. This is a typical firm that we are talking about and economic profit is equal to 0 in the long run, yet the price is more than the marginal cost, this is the marginal cost.

So price is more than the marginal cost and hence this is a difference that is there between perfect competition and monopolistic competition. In case of perfect competition, in case of perfect competition in the equilibrium, price is equal to average total cost is equal to marginal cost. So average total cost is equal to marginal cost at the minimum average total cost.

So this was the equilibrium, this was the long run equilibrium in case of perfect competition but in case of monopolistic competition price is equal to average total cost but is more than marginal cost. So this is the equilibrium in case of monopolistic competition. So this is not minimum of average total cost but average total cost is falling. Yet it is more than marginal cost.

Hence there is a markup between price and marginal cost in the long run for a monopolistic competitive outcome. Now monopolistic competition is less efficient than perfect competition. So this is kind of evident from this from the comparison of the 2 outcomes it is the difference in

efficiency or difference in welfare is quite apparent from these 2 because in case of perfect competition the price that is charged is minimum of average total cost.

So basically in the market the price that is getting charged is the minimum price that is possible. It is the minimum average cost that the firms are incurring and they are charging that price. But in case of monopolistic competition although the firms are incurring zero economic profit yet the price that they are charging is not the minimum cost at which they can produce.

The average total cost is still higher than the marginal cost and average total cost and marginal cost are the average total cost is basically falling here. So basically that means that the firm is operating at excess capacity. It is possible for the firm to increase output and lower cost yet the firm is producing at a level where average cost is higher than the marginal cost and average cost is falling.

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So the monopolistic competitor operates at excess capacity. It operates at downward sloping path of the cost curve and hence produces less than cost minimizing output. Production does not

happen at minimum cost as in perfect competition. Since price is more than marginal cost market quantity is less than socially efficient quantity. However, government cannot intervene to reduce prices or increase quantity because the firms are already operating at zero economic profit.

So basically this means coming back to this diagram it basically means that production does not happen here but production happens here where the average cost is falling. So average cost is falling. In case of perfect competition, the production happens here, in case of perfect competition but in case of monopolistic competition there is excess capacity.

So it is possible for the firms to increase the output yet its cost is going to keep on decreasing yet the firm does not do that and it produces at a level where the average cost is falling. Hence there is excess capacity at which the firm is operating. Yet the firm is producing zero economic profit since here it is tangent to, the price line is tangent to the average total cost hence there is inefficiency in the market in the sense that the price that is getting charged in the market is more than the marginal cost.

That is the benefit from the product the benefit that the or the willingness to pay for the product by the buyers is higher than the cost at which the additional unit of output can be produced by the seller. So there is clearly an inefficiency existing there because here in this situation it is possible for the firm to keep on increasing output and the buyer will keep on demanding higher output at a lower price and it is possible for the firm to keep on increasing output yet getting a yet incurring an average cost which is yet it is in this region basically in this region it is see this is the perfectly competitive output.

So it is possible for the firm to keep on or it is possible to imagine that the firm will continue to increase its output because till the point where its average cost is minimum. So it is it is possible to imagine the price also to fall to that level where average cost equals marginal cost equals the price charged in the economy. In that case there would be perfect efficiency would be reached but that is not happening here because the firm is charging a price which is higher than the marginal cost.

So however the problem is in case of monopoly also the same thing was happening. In case of monopoly also the although in case of monopoly there was not zero economic profit but in case

of monopoly also this inefficiency was there. There was this deadweight loss and in case of monopoly also the firm was charging a price higher than the marginal cost because it has some amount of market power but in that case the government could step in and say that you have to increase your output.

So it could it was possible for the government to actually direct the monopoly to increase its output, lower its price etc. but in case of monopolistic competition that is not possible because in the long run the monopolistic competitive market where there is free entry and exit that ensures that the firm is producing at a level where it earns zero economic profit.

So in such a situation if the government intervenes and tries to reduce the price even further in that case the firms are going to incur negative economic profit or loss. In that case so the firm is going to exit the market. So because of that it is not possible for the government to intervene in such a case because the firms are already earning zero economic profit. So although there is inefficiency in the market yet the government cannot do anything here. So what are the welfare implications?

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Now there are broadly 2 welfare implications of monopolistic competitive market output. The first is the product variety externality of consumers. That is introduction of new products and new entrants in the market raises surplus for consumers. What does this mean? This basically

means as I said that whenever the firm is incurring any positive economic profit other firms are going to enter the market.

Now these new firms are going to bring in more differentiated products. Remember that every firm is striving to make its product different in the eyes of the consumers from the existing products. So as that keeps on happening, the consumers are benefited because they have more and more choice and the product could be differentiated on the basis of quality also remember.

So one can imagine that the product differentiation actually makes the consumer better off. The consumer gets a whole variety of choice to choose from and the consumer also gets better quality, better variety of products and because of that the consumer surplus actually increases. The benefit, its willingness to pay for a product that keeps on increasing because it is getting better and better product and so its consumer surplus increases.

Also because as more entrants are there, more entrants are entering the market this demand curve gets split between lot of across different other firms and because of which the individual firms get to charge lower and lower prices. So as they keep on charging lower and lower prices the consumer is better off because on the one hand he is getting better choices, more variety of products, better quality products but also lower prices for products as there are lot of sellers in the market.

But there is a flip side to it because welfare is not only about consumers. A economy consists of both consumers and producers. So welfare is dependent on the welfare of both consumers as well as producers. So although the consumers' welfare is increasing what happens to producer's welfare? Now as the second point says the business stealing externality faced by the producers is another welfare implication. When new firms enter the market, existing firms may lose customers, face losses and sometimes may have to leave the market.

So one can imagine that in a monopolistic competitive market setup since it is easy to enter the market it is possible for new firms to enter and bring in new varieties of products and it is possible for the firms to actually steal ideas, steal products, product ideas from the existing firms and make them even better and in the process they steal business from the existing firms who may have to leave the market. So that is also possible.

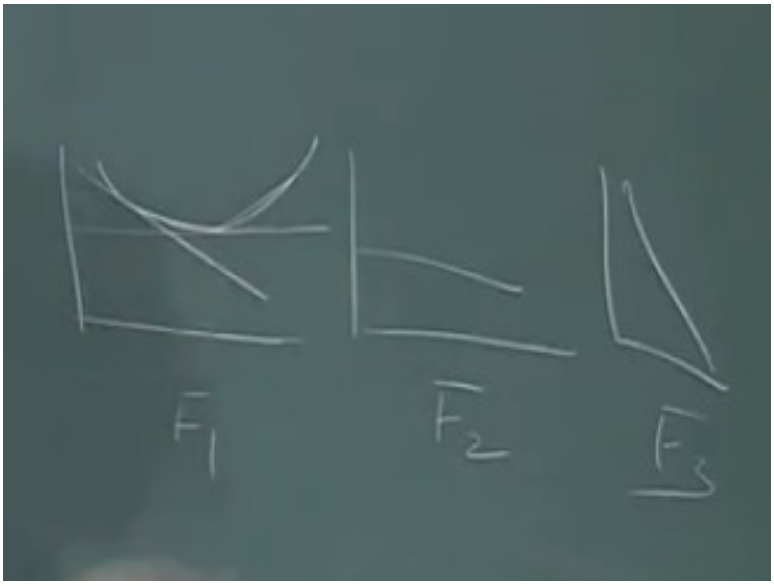
So such amount of competition in the market because it is free entry and exit. It is possible to imagine that in such kind of market setup there are continuously churning of new producers in the market, new sellers are entering the market and as a result the old sellers are getting wiped out from the market as new and new products are developed and circulated in the market so that is a negative welfare implication on the producers in the market.

The net result of the above are hard to measure and hence there is not much that the policy makers can do. So on the one hand while consumers are better off on the other hand one can imagine a whole lot of insecurity on the parts on the amongst the producers because as new and new entrants enter the market and wipe off the existing profits, wipe away businesses altogether, wipe away products altogether from the market.

So however the it is a natural phenomenon that keeps on happening and there is little that the government or any policy maker can do in such kind of market. So coming back to another so we already saw from the diagram that the firm is operating at excess capacity. It is operating at a level where average cost curve is declining and it is tangent to the price line or the demand curve and hence the firm is operating at zero economic profit.

But one may wonder what is the logic behind it. Why does the firm operate at excess capacity? Why does the firm operate at excess capacity and not at the minimum point? What is the logic or intuition behind it? Now there could be 3 or 4 reasons behind it one could imagine. First is the very obvious which very obvious reason is these are all differentiated products.

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So these are all differentiated products so there are whole lot of products which have a negatively sloping demand curve. So they have negatively sloping demand curve. So these are very highly differentiated products and they have different negatively sloping demand curve and hence there is no way they can be tangent to the minimum point of the average total cost curve for the firm.

So each of these firms, so this is say firm 1, firm 2, firm 3. So each of these firms have a negatively sloping demand curve and negatively sloping demand curve can be tangent only to the declining portion of the average cost curve and hence this is the outcome where it is operating at a level where there is excess capacity because there is there is a negatively sloping demand curve. Had this been a homogenous product then all the firms would be facing a flat demand curve. All the firms will be facing a horizontal demand curve and in that case they would be operating at a minimum of the average cost.

Now the second reason is that if you imagine such a market where there are whole lot of product differentiation and all the firms are existing coexisting in this market and each is trying to behave like a not a monopoly but each firm has some control over its consumers and they are focusing on product differentiation and catering to this demand curve.

What they are not focusing on is that they are not focusing on minimizing cost. They are not focusing on minimizing cost but they are focusing on retaining these consumers and charging them a price which is minimum in the long run that they can charge. So intuitively one could

imagine that the firms are coexisting in a live and let live kind of environment where they each of them is focusing on product differentiation and not really focusing on price competition. They are not competing with each other in terms of price but they are competing or they are trying to create their own niche clientele by product differentiation and that is how we have developed this model and hence there is excess capacity at which they have to operate because they each of them is facing a negatively sloping demand curve and another intuition behind excess capacity is when we discussed about the long run when we said that there is a profit to be made and there are firms existing in the periphery who are sitting and watching the market and as soon as there is profit to be made these firms enter the market.

Now what happens in such a scenario? In such a scenario what happens is this demand curve it gets split up into too many firms. So imagine this is a group of consumers who used to buy from this single seller minty toothpaste and now they know that there are whole lot of other flavours available in the market, different kinds of products, packaging, etc. So now they get as new and new entrants are there in the market this demand curve gets split up between 3, 4 more firms or even more firms.

So in that case the existing firms or the new firms each of these firms are now going to cater to even lesser and lesser number of consumers. So obviously they will be operating at excess capacity because capacity has already been created. It is like having an if you imagine that there is a residential new residential area coming up and initially there was only one, initially there was only one store one grocery store and gradually more and more grocery stores coming up in that area because they know that profit is to made.

This one single store is making a profit. So more and more grocery stores enter the market or enter that in that residential area more and more grocery stores come up. So now instead of the one grocery store now 10 more grocery stores are all going to cater to the existing number of residents who are residing in that area. So basically each of them is now going to operate at excess capacity.

So it is not possible for each of them to operate at marginal average cost instead they are all going to operate at excess capacity and since each of them is product differentiating since the firms each of them is focusing on product differentiation so that also is kind of reason for the

firms to operate at excess capacity and come up with more and more ideas for better product differentiation and maybe have different this basically is reason enough for the firms to operate at excess capacity so that they can keep on improving on the product differentiation that they are already doing in the market and because of this for better differentiation of product they probably would like to keep on operating at a excess capacity.

So this these were the welfare implications of a monopolistic competitive market and we have seen that although unlike monopoly the monopolistic competitive firm does not have more than zero economic profit or positive economic profit like perfectly competitive firms the monopolistic competitive firm has zero economic profit in the long run because of entry and exit. But like a monopolist the monopolistic competitive firm has its own unique product in the sense that they keep on product differentiation to make keep their products different from the other products although all these products are kind of substitutable.

So in generally speaking these products are substitutable but yet the firms strive to make their products different or create the perception that the products are better than the other products in the market in the eyes of the consumer. This they do through different either making their products themselves slightly different or through marketing, through advertising, through locational advantages, through packaging.

So this is what the firms strive to they strive to keep on doing and it makes sense for them to operate at excess capacity to keep on differentiating their product from the rest of the market. So in the following module we are going to see the role of advertisement in monopolistic competition and we are going to see that if what are the opinions of people whether advertising is good or bad. This is what we are going to do in the following module and complete our discussion on monopolistic competition. Thank you.

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Lecture - 33

Monopolistic Competition - Advertising

Welcome to the final module of our discussion on monopolistic competition. We have looked at how the monopolistic competitive firm decides its level of output and prices and we have looked at the welfare implications or efficiency implication of a monopolistic competitive market output and we have discussed a lot about product differentiation and how does the firm differentiate between its products.

We said that the firm can create slightly different products. We took the example of the toothpaste or one can imagine say candy, ball pens etc. so the products are different but at the end of the day they are basically the same products. So the firms basically keep on striving to make their products different from other products available in the market.

And we said apart from creating different products, marketing, advertising, celebrity endorsement these are other tools by which monopolistic competitive firm creates perceptions in the minds of the consumer. They try to differentiate their product on the basis of these also.

So in this module we are going to discuss about advertising. What happens in case of advertising or brand names are they really helpful? What are their welfare implications on the society? These are some of the interesting things that we are going to discuss in this module. Now before we go

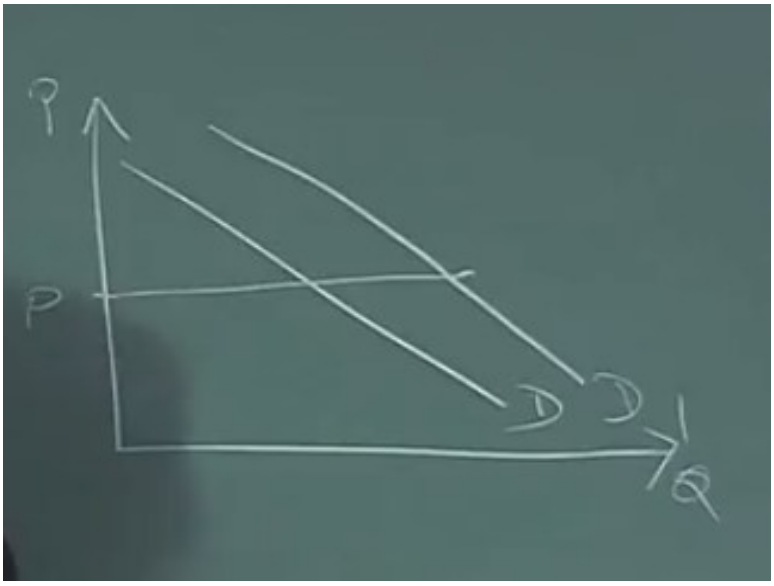
on to talking about monopolistic competition let us discuss about the role of advertising in different market structures.

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We already have discussed this once when we introduced market structure. Let us just do a quick recap over here. Advertising helps in shifting the demand curve to the right. So but advertising also increases costs of firms. In competition and monopoly, industry level advertising may be done to shift the overall demand curve of the market to the right.

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So as we know that so there is a demand curve in the market. So any amount of any advertisement in any kind of market structure what is that going to do? In any kind of market structure any amount of advertisement that is going to increase the demand for the product; so more and more people are going to know about the product. So they will enter the market. They will start wanting the product and hence at the same level of price at the same price more of the product will be demanded and hence the demand curve shifts to the right.

So this is what advertisement does. So advertisement basically shifts the demand curve to the right. However, the advertisement is done by a firm. When a firm does a advertisement it has to incur the cost of the advertisement. So when it incurs the cost of advertisement the cost for the firm goes up.

So there is both a positive and a negative aspect to advertisement for the firm. So the if the firm decides to advertise, its demand for its product is going to increase but if the firm decides to advertise it also has to incur a higher cost. So of the different kinds of market structure in which kind of market structure does it make sense for the firm to advertise?

Now in case of perfect competition where the firms are all selling more or less the same kind of product that is the products are homogenous in the eyes of the consumers. So it does not matter who is selling the milk or who is selling the eggs. To the consumers they are all same. So if one firm decides to advertise so obviously the demand for eggs is going to go up and the demand is

going to shift to the right but there is no guarantee that these people are going to buy from this particular firm which has advertised; so obviously no firm in a perfectly competitive market setup is going to advertise.

However, in case of competition in perfect competition one can imagine industry level advertisement. So if there is an industry level advertisement where all the dairies together advertise about milk consumption or all the poultries together they advertise there is an industry level advertisement for eggs. So if you consume eggs it is good for health. So what happens?

The market demand curve shifts to the right and because of which price increases and more and more output product is sold in the market and for some time some period of time the existing firm actually have positive economic profit because firms have gone up because the demand for eggs has gone up. So similar is the case for monopoly.

Now a monopolist need not advertise because it is the single seller in the market and there is the existing buyers are not going to go anywhere else whoever needs this product are going to come to this seller. So it does not really the producer to the producer it is not too much necessary to advertise yet the monopolist may decide to advertise again to increase the or shift the demand curve to the right so that it can charge a higher price and get a higher amount of profit.

Now what is the role of advertisement for the market structures which are in between perfect competition and monopoly and right now we are discussing about monopolistic competition. So in case of monopolistic competition advertisement is of utmost importance. Why is it so important because in case of monopolistic competition the firms are always trying to differentiate their products in the minds of the consumers. Hence advertising is a very powerful tool to create perceptions in the mind of the consumers. So in case of monopolistic competition the firms are going to advertise.

So coming back to the slides, the competitive firm will not engage in advertising since it is a price taker and would not like to increase its cost. In monopolistic competition firm level advertising will be done to differentiate the products. So also in the case of oligopoly with differentiated products. In case of oligopoly say for firms which are selling cars these are differentiated products and it would make sense for the oligopolist to advertise. Now we know so

now we know what advertising does to the demand curve in the economy. Now what is the what are the social values of advertisement.

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Now there are various criticism against advertising like the following that advertising increases cost for everyone. So if the firm is advertising to differentiate its product, its cost is going to go up and when its cost goes up it has to charge a higher price to cover that cost. So basically the cost is going up not only for the firm but the consumers also end up paying a higher price.

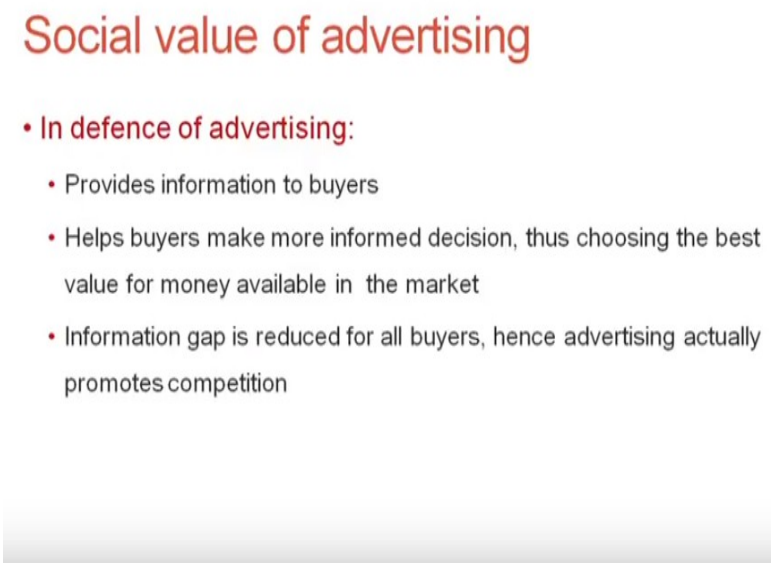
Advertising have the potential to misrepresent facts and influence people's taste and choices. So this is another criticism against advertisement that like when we said that product differentiation is important we said that whether the products are different or not is not important. What is important is in the minds of the consumers the products should be different and that is what the monopolistic competitive firm is concerned about only.

It is only concerned about creating a perception in the minds of the people that the product is different. So if the same milk is packaged in different ways they can create different kinds of perceptions in the minds of the consumers and that is what is the fundamental criticism against advertisement that basically you are selling the same product but because it has been endorsed by a celebrity or you have a very flashy ad people are willing to pay more for the same product and

advertising is anti-competitive as it creates perception of product differentials thus increasing markup over cost.

So the same product which could have been sold at a lower cost now because of advertisement the cost has gone up and the consumers end up paying a higher price for it and people end up paying higher prices for differences which are just notional. So these are the criticisms against advertisement and however there are arguments in defense of advertising.

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First is that it provides information to buyers. So the same way that the criticism is advertisement creates distortions in information in the minds of the buyers or it misrepresents facts or it creates information asymmetry in the economics through advertisement. In defense of advertisement, advertisement is a very powerful medium through which information can be disseminated amongst the buyers.

It helps buyers make more informed decision thus choosing the best value for money available in the market. So basically buyers get to know what all are available in the market, at what prices and they get to choose and basically they are able to decide what which of the products gives them the best value for money.

Information gap is reduced for all buyers, hence advertising actually promotes competition. So it is the opposite argument that is provided that since advertisement reduces information gap so it increases competitive environment amongst the sellers in the market or amongst the products in the market so advertisement actually promotes competition.

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Now let us look at a couple of examples about the impact of advertising as has been studied in literature. First we take the example of a case which was this was published in the journal of law and economics and authored by Benham in 1972.

What Benham basically did was he studied the market price for eye glasses in different states of the United States of America. In the USA he studied the price of eye glasses in different markets across different states and price advertising in some of these states was prohibited and in some of these states price advertisement was allowed.

And when Benham studied the prices of eye glasses across these different states he found that price of eye glasses were actually lower in states where there was advertisement and the states which had prohibited advertisement the price of eye glasses were actually higher. So this gives a this is this gives clear evidence that advertisement thus provide information about prices amongst

consumers and there it creates a kind of competitive environment for all the sellers in the economy because of which prices go down in the market.

On the other hand, if there is in states where there was no advertisement there was probably less information and prices settled at a higher level in these markets. Another example that is given is advertising is a signal of quality. So it is firm's willingness to spend on ads is a signal to the consumers that it is confident about repeat buyers. So expensive ads indicate that the quality of the product must be good.

So let me take an example of say breakfast cereal. Say there are 2 firms who are coming up with a new variety of breakfast cereal which they are introducing into the market. Now one of the firms is very confident about the quality of the breakfast cereal that it is introducing and it knows that once people start consuming it they there will be repeat buyers of this breakfast cereal because it is so good.

On the other hand, there is another seller in the market who have come up with another variety of breakfast cereal and they know that it is it may not be so good and people may not like it and yet since they have already produced it they have to sell it in the market. So what are they going to do? Which of these 2 types of producers is going to spend more on advertisement. So you could probably spend 2 minutes thinking about it which of the 2 sellers is probably going to spend more on advertising their products.

Now the first firm which has developed a very good quantity breakfast cereal who knows that once people start consuming it they are going to buy it repeatedly. It makes sense for this firm to spend more on advertisement because they know that what they spend on advertisement as they rope in or bring if they have a very high quality advertisement or very expensive advertisement it is probably going to bring in more and more consumers for this breakfast cereal and the demand curve is really going to shift to the right and is going to continue. These buyers are going to repeatedly buy the breakfast cereal once they are hooked on to it. So they are going to spend lot of money on advertisement to rope in as much as many consumers as possible.

On the other hand, the other firm knows that I have to sell this breakfast cereal first and probably people are not going to buy it again. So it does not make sense for me to spend too much on

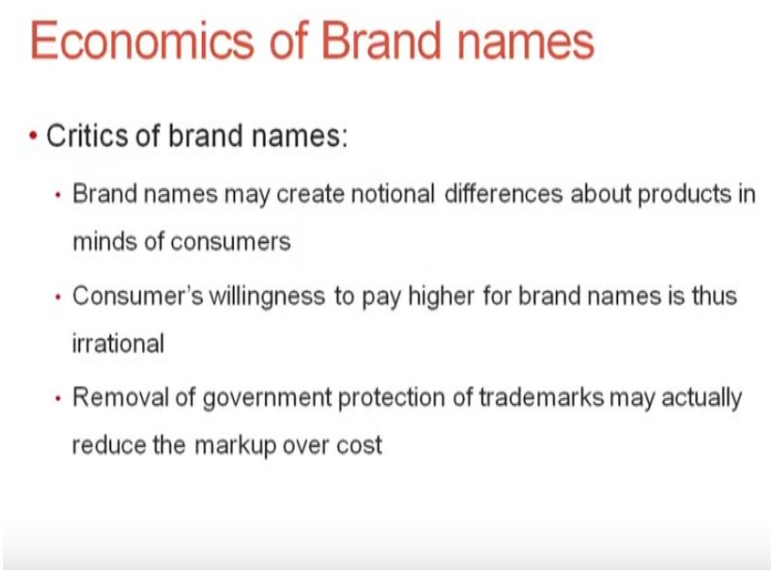
advertisement or spend too much money on advertisement because in any case there are not going to be probably repeat buyers. So he is not going to put so much money into advertisement. So that means indirectly that means that advertising is a signal of quality. When a consumer sees a high quality advertisement a high expensive, a grand expensive advertisement the it is a signal to the buyer that the firm is confident about its product is confident about the quality of its product and hence the buyer is going to buy it.

On the other hand, if the buyer sees a poor quality advertisement or not a very a very budget budgeted advertisement which is not so grand is not so expensive so that kinds of kind of hints the consumer that probably the firm is not too confident about its product hence it is cutting down on advertisement cost.

So expensive ads indicate that the quality of the product must be good. Now this was our discussion on advertisement and very closely following behind the advertisement is another signal in the market which is brand name. Brand name is another signal in the market and and you will understand that brand name itself differentiates the products.

You can have the same jeans probably but having a brand name with the jeans creates a whole lot of difference and sometimes the buyer is willing to pay a much higher price for a branded pair of jeans than a jeans which is not no one knows the name of the company. So brand names in themselves are product differentiators.

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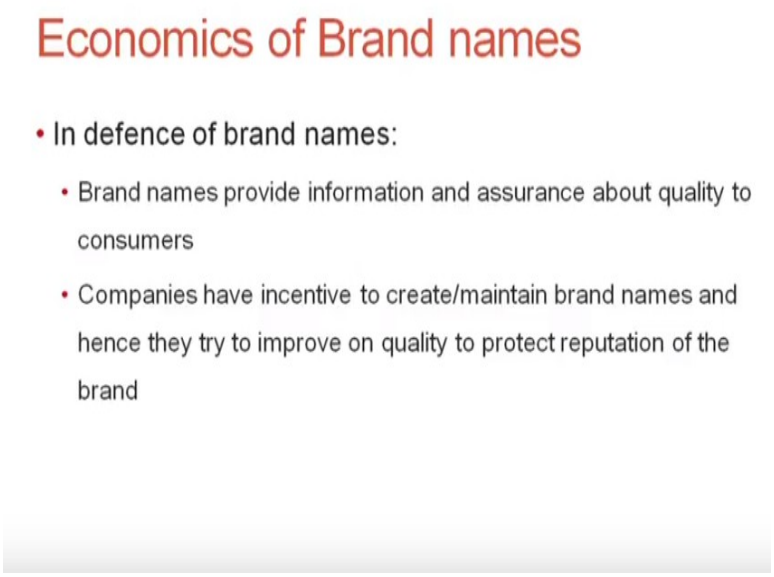
So what is the economics of brand names? Now similar to advertisement you will have the almost similar kinds of arguments in favour of and in defense of brand names so look at looking at the critics of brand names, brand names may create notional differences about products in the minds of consumers. Consumer's willingness to pay higher for brand names is thus irrational. Removal of government protection of trademarks may actually reduce the markup over costs.

So brand names, critic of brand name is as I already gave the example of the jeans it could be probably the almost the same similar quality of jeans but because one has a brand name a tag which the consumers are going to they are almost paying for the tag.

Sometimes few years back I remember there was that there was a case against a very reputed brand of t-shirts. A consumer had actually sued the sued this particular company because the tag or the label of the brand had got ripped off and the consumer basically the customer said that I had paid so much money for the brand name itself and if it is not showing or it is ripped off from the t-shirt then this t-shirt is of no value to me.

So this is the kind of impact that the brand name creates and basically it creates a notional difference about products in the minds of the consumer. So basically the cost is the same but the producer the buyer ends up paying a much higher markup over cost because of the brand name. So removal of any protection of trademarks etc. may actually reduce the markup over costs.

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Now in defense of brand names, brand names provide information and assurance about quality to consumers. Companies have incentive to create or maintain brand names and hence they try to improve on quality to protect reputation of the brand. So similar to advertisement brand names provide information and assurance about quality to consumers because the brand name is established. So it is kind of assurance to the consumers that whatever the brand is claiming or the company is claiming that is true and it kinds of kind of assures the consumer and this assurance is what the consumer is paying the money for.

And since the brand name has its impact of product differentiation so all the companies have incentive to create or maintain the brand names and to try to create or maintain brand names they try to improve on the quality and to protect the reputation of the brand they will keep on improving the quality or the product that they are offering in the market and that is the kind of assurance that brand name provides to the consumers.

So to retain the reputation of a brand the quality is assured by the company and that assurance is the markup over the cost that the buyer is paying is basically a cost of that assurance or that is the willingness to pay for that assurance if one may one could imagine it that way. So basically we have discussed how the firms can create different kinds of products by making the products

different themselves but also they can create notional differences about the products in the minds of consumers through tools like advertisement and brand names.

And we have discussed the pros and cons of advertisement and brand names and there are people there are arguments in favour of advertising and brand names that they uphold the quality of the product the brand name etc. and they provide information and on the other hand there are criticisms against advertisement and brand name in the sense that they create unnecessary differences in the minds of consumers because of which the consumers end up paying a higher price over cost.

So this ends our discussion on monopolistic competition and this discussion on advertisement and brand name is not completely confined to monopolistic competition. It holds true for any market structure but since in monopolistic competition advertisement and brand name plays a huge role so I included it in the discussion on monopolistic competition and basically monopolistic competition is the example of monopolistic competition is the maximum that we are going to find around us.

Most of the market most of the products operate in monopolistic competitive market framework or market structure and we see lot of product differentiation that happens all around us. Different firms are trying to create different kinds of products so different ways of product differentiation and because of which at the end of the day each of these small firms which are catering to a clientele of their differentiated product.

So each of these firms end up producing at excess capacity and hence it is not a perfectly efficient outcome as in the case of perfectly competitive market but nevertheless it is a better situation than monopoly outcome where the firm gets, firm basically ends up making supernormal profit. It makes a more than a positive economic profit while in the case of monopolistic competition the firms at least earn only zero economic profit in the long run.

They charge a price which is equal to their average cost and not more than their average cost although the markup is still there because of the excess capacity that is existing in the case of monopolistic competition and however monopolistic competition is better in an in the sense that in monopolistic competition the consumers get a whole variety of choices to choose from.

They get a whole variety of differentiated products which make them better off. So even if the difference is notional in the minds of the consumers but still it is providing some utility to the consumer when he is buying a soap which is better packaged than any other soap no matter not mattering that the soap may be same in the 2 packages.

Yet the utility or satisfaction or the pleasure that the consumer is getting from buying a better packaged soap or the utility that the consumer is getting from buying a product which has been advertised by his favourite celebrity that also nevertheless improves the consumer surplus of the different consumers in the market and creates more and more product differentiation; creates more and more varieties of products to choose from for the consumers. Hence monopolistic competition is probably provides a more interesting market scenario to the consumers than the perfect competition. Thank you.

Foundation Course in Managerial Economics

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Lecture - 34

Oligopoly

Hello and welcome back to the course on foundation Course in Managerial Economics. We have completed our discussion on different forms of market structure like perfect competition, monopoly, and monopolistic competition and now we are going to take up the model for oligopoly.

So far we started with the most simplest case of perfect competition where basically the market determines the price and none of the stakeholders have any influence on price. That is they cannot increase or decrease the price and they have to take the price for granted.

They have to take the price as exogenously given by the market and decide only on how much output to sell in the market which again depends on their cost structure and on the other extreme we have discussed about monopoly where there is only firm and this firm basically caters to the entire market and hence has a lot of control over price and quantity and the monopoly the monopolist basically faces the market demand curve and hence it maximizes its profit and decides the price accordingly.

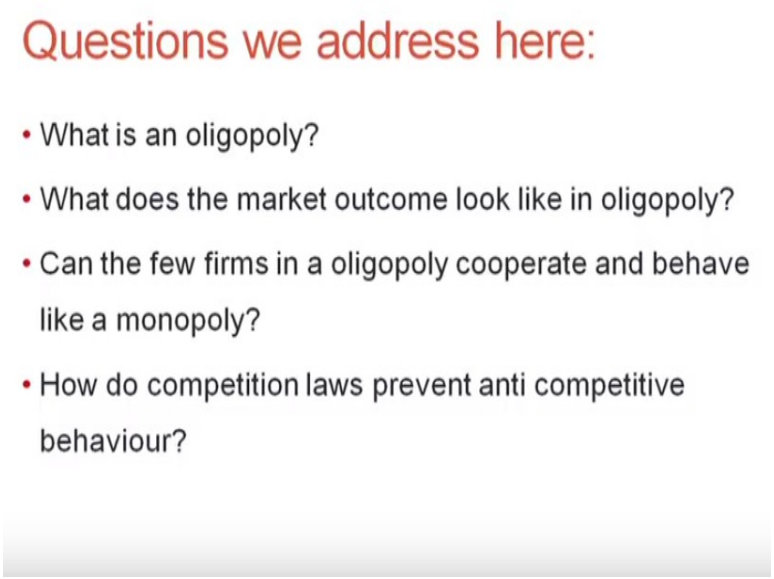
And in between the 2 cases we have talked about monopolistic competition which has a lot of buyers because of which the buyers do not have too much influence of price yet the buyers try to differentiate their product and hence they face a product specific negatively sloping demand

curve that is they face consumers who consume their differentiated products which the producer is supplying in the market and they know that the products are not exactly unique.

They have some differentiation from other products available in the market yet if prices are too high consumers can shift or change their preferences or their choices .or tastes and move to other firms or other producers in the market. So hence the monopolistic competition also has very limited control over prices and they have to be they have to constantly differentiate they try to differentiate their product in the eyes of the consumers to stay in business.

So these were the 3 different kinds of market structures which we have discussed so far and today we are going to start our lessons on oligopoly. So let us move on and see what are the questions that we are going to address in this when we cover this chapter on oligopoly.

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So the questions we address here are what is an oligopoly. What does the market outcome look like in oligopoly? Can the firms in oligopoly cooperate and behave like a monopoly and how do competition laws prevent anti-competitive behaviour in a oligopolistic market structure. So once we have defined oligopoly we are going to see we are going to try to find out if there is any unique price or price and output combination which brings the market into equilibrium because

we have been able to determine price and quantity on the basis of profit maximization objective of the firm in the other cases of market structures that we have discussed so far.

In case of perfect competition, we know the P and Q, the definite P and Q produced by each firm and eventually what is supplied in the industry we have seen it is a horizontal supply curve.

And in the case of monopoly we have been able to arrive at a specific combination of price and Q which maximizes the profit of the monopolist and in case of monopolistic competition also there are combinations of P and Q for each of the firms which brings the market into equilibrium so that in the market in the long run none of the firms is actually making any positive economic profit.

They make a economic profit equal to zero, not negative because if there is a loss they are going to leave the market but nevertheless the economic profit is equal to zero in the long run in case of monopolistic competition and we know the price and quantity combination for each of the firms in case of monopolistic competition.

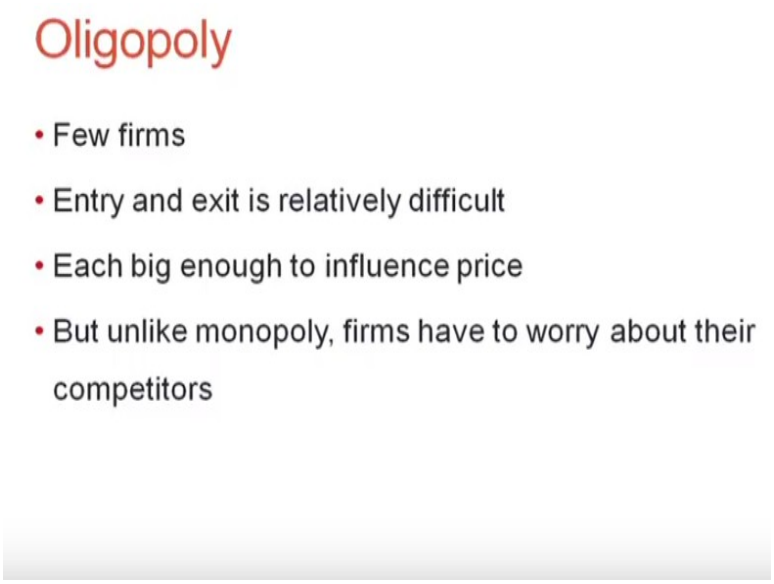
So the question that we are going to ask here in case of oligopoly after we understand what are the assumptions or how does the oligopolistic market structure look like, we are going to try to we are going to tread the similar path and see that if profit maximization is a goal of each of the firms do we have a unique combination of P and Q or not?

Then we are going to see that oligopoly in case of oligopoly there are few firms in the market unlike perfect competition and monopolistic competition there are few firms in the market so since there are only few firms and the producers are likely to know who are the rivals who are or competitors who are their competitors who are operating in the market, is there any way that these few firms can tie up together?

They can come together and decide how much to produce, what price to charge, and I am sure when I asked this question some of you might have already understood that I am hinting towards a cartel formation. So if not then please be a little patient and we are going to discuss about cartel formation.

We are going to discuss about cooperation in case of oligopoly later and as many of us already probably known that cartel formation is illegal, that firms cannot firms are not allowed to collude in a market and decide prices to charge and so there are various competition laws which again we are going to discuss in details later and they prevent any anti-competitive behaviour so what happens when there is some law and what happens to what happens when there is a law to prevent collusion and the oligopolistic firms what do they do in such situations. So we are going to discuss all these in details. So let us go to the next slide.

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So this slide basically gives us the characteristics of an oligopoly market structure what happens in case of oligopoly market structures. In an oligopoly typically there are few firms. Entry and exit is relatively difficult which can be kind of inferred from the fact that few firms or few firms can be inferred from the fact that entry and exit is relatively difficult so that is the reason probably there are only a few firms in the market and each firm is big enough to influence price.

Now since there are only few firms in the market and they are basically catering to the entire market or they are catering to the entire demand that is existing in the market so each of the firms is big enough to influence price. So in real life example a very good example would be the automobile sector where there are few car manufacturers.

They come up with whole lot of models and they have a quite substantial control over the price that they are going to charge to the customer. So one might ask the question that in case of oligopoly are the products then differentiated because I had taken the example of differentiated, of cars when I try to explain about differentiated products in the earlier lectures.

So yes cars can be highly differentiated, even within a section of car like say SUVs. There can be whole lot of different models of SUVs with different kinds of features and accessories etc. So huge amount of product differentiation is possible yet the car manufacturing sector is such or the automobile sector is such that it is not very easy to enter the market and it is not very easy to exit the market also.

Lot of fixed cost is involved and lot of R&D is involved and to some extent reputation, brand name, advertising, these are also involved so once a company, a car company has been able to establish itself in the market then only it can compete with other rivals in the market. So it is not so easy to do that in case of say cars.

So car is a very good example of oligopoly and on the other hand we have example in case of oligopoly we have the example of OPEC or Oil Producing and Exporting Countries who basically collude or cooperate or decide together how much what price to charge or how much amount or what quantity to produce. Each of the countries they decide what quantity to produce and together they decide what price to charge in the market.

So this is also an example of oligopoly where there are only few producers, these are producing countries rather than companies but they are producing a product which is which can be considered as highly homogenous. They are all oil, crude oil.

So it can be considered as homogenous. Yet it is not possible to enter this market so easily because you have to have the oil. You, unless you have the resource, if unless and until the country has the resource endowment it cannot enter this market. It is as simple as that. So there is a barrier to entry, entry in this market.

So unlike monopoly, firms have to worry about their competitors. This was the last point in the slide which basically means that in case of monopoly there is, entry and exit is difficult. There are there is only 1 firm and since there is only 1 firm and the product is unique or the entry and

exit is really difficult so the monopoly can need not worry about any competitors and it can charge a profit maximizing price and it can eventually get a high, huge amount of profit in the long run.

But does that hold true for oligopoly or not? Because, one might think that if there are only a few firms in the market, if it is only the oil the countries which have oil, they are only catering to the entire demand for oil all over the world then they basically have lot of control over the market so they can exercise so they have a lot of control over the price as well and they can get away with earning a huge amount of profit.

Similarly, in the case of car manufacturers one might think that there are only a few companies which are producing cars and they are basically supplying to the entire market which has a huge demand for cars. Say imagine in India the demand for cars is exploding and there are only a few manufacturers who are basically coming up with whole lot of models to cater to different kinds of needs and tastes of customers in the country.

So in that case one might think that since each of the firms is catering to a whole lot of consumers it is possible for the firms to have a lot of control over the price or have a lot of market power and can stay at a very high level of price like a monopolist and continue to earn, enjoy profit in the long run. But does it happen that way in case of oligopoly? That is what we are going to study here.

And what we see the very preliminary discussion in this for this slide I am just going to say that also remember that it is although there are few firms but it is not a single firm. So since there are few firms in the market and they all know each other they know who all who else is producing cars or who else is producing oil.

So they also know they have pretty much good idea about the cost conditions of the rivals. They know what quality what quantity what they are producing, the rivals are producing and what kind of prices the rivals are charging. So unlike monopolistic competition where the firms do not bother to look at the what the rivals are doing because there are so many of them.

It is not possible to compete with all the other rivals in terms of so it is not possible in case of monopolistic competition to gather information about all your rivals and try to decide a price and

quantity. So what every firm does in case of monopolistic competition is strive to retain its demand curve. It strives to retain its consumers by constantly product differentiation by constant advertisement etc. so that it can hold on to the demand curve or shift the demand curve to the right as much as possible and earn a decent profit and zero economic profit in the long run and decide on a price and quantity.

But in case of oligopoly it is very easy to find a have full information about your rival and know what your rival what price he is charging, what kind of cost he is operating on, what kind of product he is selling in the market. So every firm in that case they every firm utilizes this information. Every firm is going to utilize the information and try to beat its rivals in terms in making profit.

It will try to capture as much part of the market as possible to increase its profit or whatever its objective is in the market. The objective could be anything could be something other than profit maximization also but nevertheless the point here is in case of oligopoly the oligopolistic firm basically tries to be it is always abreast with the with information about its rivals what the rival is doing and it also knows that the rival also has keeps an keeps a watch on the firm itself and they are also going to change keep changing their price and quantity decision according to the actions of the of its other rivals.

So all the firms take this information into account and they decide on the price and quantity. So as a result in case of oligopoly determining the price and quantity becomes a little more tricky. It becomes a little more complicated than in the case of other market structures where it is a simple profit maximization of MC is equal to MR and one gets the combination of P and Q if the demand curve is known. But in case of oligopoly it is not as simple as that. So let us proceed and see.

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So before we move on we have defined oligopolistic market as having few firms. Now the question is how few is few? Is it like 3 firms, 4 firms, 6 firms, 10 firms? So how few is few is produced by top few firms in the industry. So this is a typical measure, is the concentration ratio which is the percentage of a market's total output produced by its four largest firms.

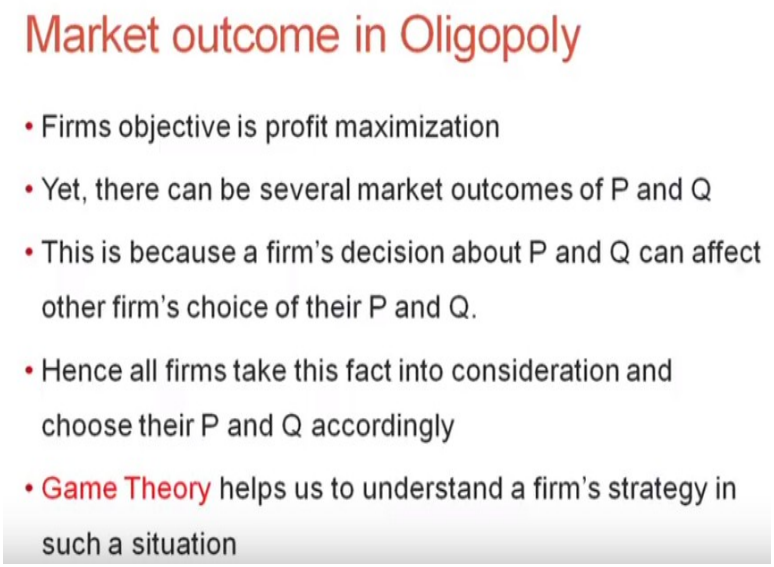
Now the rule of thumb is low concentration implies this concentration ratio 0% to 50% which happens in the case of say perfect competition to oligopoly. It can it this could be applicable, 0% or close to 0% would be a perfect competition and 50% it could be a oligopoly also. It could be oligopoly also where the 50% is maybe shared by 2 or 3 firms and the rest 50% is basically produced by whole lot of other firms. So that could also be a oligopoly.

And medium concentration is basically 50% to 80% which is likely to be an oligopoly if the top 4 largest firms are producing 50% to as high as 80% of the total output of the industry this market is likely to be an oligopoly and very high concentration is 80% to 100% which could range from oligopoly to monopoly.

So if it is like 100% then it is almost like a monopoly and the monopoly that we have defined in the case of when we discussed monopoly we said it is a single firm and if it is a one firm monopoly then it that case one firm concentration ratio of 100% implies a monopoly. That is it implies that 100% of output of the market is being produced by one single firm in the market.

So this is how this this is to just to give you an idea about how few is few. Now moving on to the market outcome in case of oligopoly. So the next question once we have understood what a oligopoly market structure basically looks like then the next obvious interest is to know what is the market outcome. What is the price and the Q that in case of oligopoly?

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So as in other cases we are going to assume that the firm's objective is profit maximization. So yet there can be several market outcomes of P and Q. This is because a firm's decision about P and Q can affect other firm's choice of their P and Q. Hence all firms take this fact into consideration and choose their P and Q accordingly and game theory helps us to understand a firm's strategy in such a situation.

Now going to again to explain what we mean by each of these points in the in this slide like as I already said in the other kinds of market structures there is a profit maximization rule of equating marginal revenue to marginal cost and you get a combination of P and Q whether it is perfect competition, whether it is monopoly, whether it is monopolistic competition.

But in case of oligopoly, MR is equal to MC if a firm just sticks to that principle and decides its P and Q. It also knows that once it decides its P and Q its rivals are going to see the P and Q that

this firm has settled for and they may decide their P and Q in such a way to capture as much part of the market as possible. This could be one example.

Say for example one firm decides that I am going to charge a price of 100 Rs and I am producing say 100 units and once this firm has decided this is the combination that I am going to stick to immediately the other firm whose objective is also to profit maximize he may see that if he reduces his price from 100 to say 80 he is going to get a lot of customers or lot of buyers are going to move from the first firm to this firm.

So it may then revive its price and accordingly and produce an output accordingly. So this is just an example because so this is so now the first firm he also knows that the second firm may behave this way and knowing this he has to decide how much price and price to charge and how much output to produce.

So hence although the objective may be profit maximization although we are going to discuss later that oligopoly firms may not always go for profit maximization in the short run and or profit maximization for the time being and they may decide to capture the market first and then go for profit maximization later once they have consolidated their position in the market.

So oligopoly firms, for oligopoly firms it is not so easy to decide what P to charge and output to produce. So that basically means that the decision making process of an oligopoly firm it involves making strategies, it involves making strategies taking into account the actions of the rivals.

So in that case it is very similar to game to a game like for example if you are playing chess or you are playing cards what move you are going to make on the chess board; you not only make your move but you also think about the following moves that your rival may make in response to your move and what you are going to do then once the rival makes that move.

So taking into consideration all the series of moves that are possible from your rivals eventually, you make your strategy on how to start your game or what to do, what your moves should be. So it is so in case of oligopoly also the decision making process is very similar to a game. The decision making process becomes very similar to a game and we are going to use the game

theory to understand a firm's strategy in such a situation. So we are going to do game theory later.

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So from our understanding of market module so far one might think that the potential for profit would be high as firms can fix a price higher than perfect competition or monopolistic competition since there are only few firms in the industry. Firms will be able to maintain profit greater than 0 in the long run as barrier to entry is high.

So these points I am making on the basis of what we understand about market structures so far. Yet firms keep watch on their rival's actions like raising or lowering price or output and plan their strategy accordingly. So this is our understanding from our this is our understanding about market models and we know that since there are only a few firms and they.

So basically the firms are probably facing negatively sloping demand curve we are going to develop look at the demand curve in details in the next following module and firms will be able to earn profit and probably they will be able to continue earning profit in the long run unlike monopolistic competition because there is barrier to entry is high. They know that more firms are not going to enter the market; so, that they know.

So yet although they are earning profits they are going to keep on keep a watch on their rival's actions like raising or lowering prices or output and accordingly they are going to make their strategy of whether to raise price or lower price or raise the quantity of output that they are producing or lower the quantity of output so these decisions will be dependent on the decision of their rivals also. So we will be doing or developing models to understand these actions more clearly in the following modules. Thank you.

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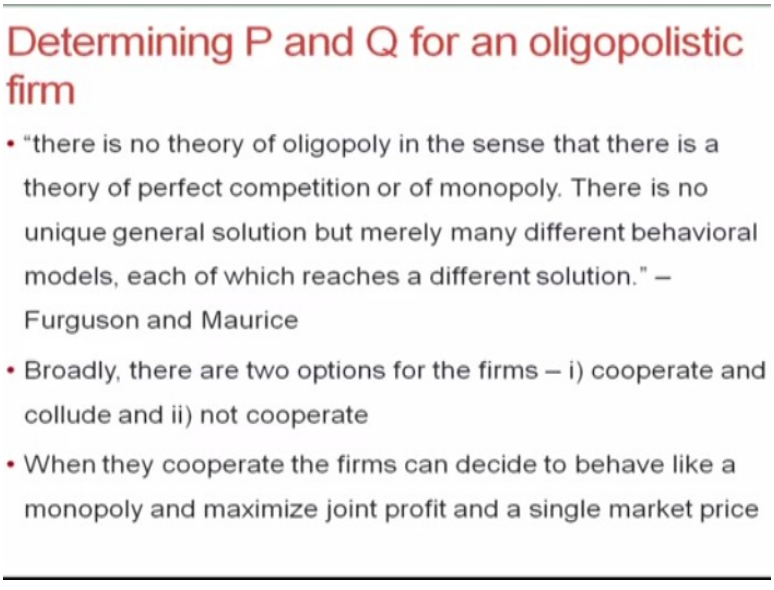
Indian Institute of Technology-Kharagpur

Lecture - 35

Oligopoly - Determining P and Q

Welcome back to our discussion on oligopoly. We have developed our understanding about what an oligopoly market structure looks like and what kind of products they might be selling and how the firm's decision of determining P and Q are dependent on the rivals' decision also. So we have had a preliminary discussion on this but we have not looked at any model yet. So that is what we are going to do in this module.

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So determining P and Q for an oligopolistic firm. So I quote Furguson and Maurice by saying that there is no theory of oligopoly in the sense that there is a theory of perfect competition or of monopoly. There is no unique general solution but merely many different behavioral models, each of which reaches a different solution.

So this is this summarizes our discussion from the previous module where we said that basically the oligopolistic firm has to make its P and Q choices depending on the behaviour or depending on the actions of its rivals; what kind of price the rival is charging, what kind of output it is supplying in the market, what is it likely to do in the future.

So these are the things that each of the firms keep into keep, each of the firms they keep into consideration and decide what P and Q to charge and as I also said in the previous class that it is basically like a strategy planning for the firms. They have to plan a strategy of how to behave in this kind of market where there are only few other players in the market and they have to stay in business and they have to earn a profit and they have to preferably do better than their rivals.

So there could be various like in case of a game, like in case of say a game of chess you know that there could be various starting moves one could think of. Those of you who play chess or play say cards they know that there could be various starting moves and also depending on whether the it is you are playing with white or black and accordingly you are going to make your moves and you are also going to decide that if my move is this way this is what these are the likely reactions of my competitor or the other player whom with whom I am playing the game of chess and you are going to decide accordingly that if that is the fall in the next move then I am going to do this in my third move.

So these are the things, these are the strategies that a player plans or if he is not the you know like opener in the game he may observe what the rival has done or what the competitor what has been the move of the competitor and accordingly he decides how or plans how the game is going to follow eventually.

So similarly in case of oligopoly also one can imagine various situations, various strategies of the firms to do well in business and accordingly one can imagine various reactions of the rival firms

and hence P and Q are going to vary from situation to situation. So we are going to discuss a few situations here.

So very broadly speaking there are 2 options for the firms, there are 2 options for the firms. So there are only 4 or 5 firms in the market so or imagine the case of 2 firms say for example Pepsi and Coke. Both of them are catering to the major soft drink demand of the country. Most of the people like they are either buying different products of coke or they are buying different products of Pepsi.

So basically major portion of the as we discussed about concentration ratio so probably more than 80% of soft drinks is being produced by only 2 firms in the market. So the concentration ratio is very high and they are like a oligopoly. They are sharing the market and so there are 2 options for the firms.

One option is they can collude or they can cooperate or they can say that see the two of us are only catering to the market. There are no other rivals here. Probably no one is going to enter the market because it is not easy because we have already created or established a reputation in this market and there are the consumers are we have more or less divided the consumers so let us collude, let us cooperate, and let us charge a common price and let us split up the market amongst us.

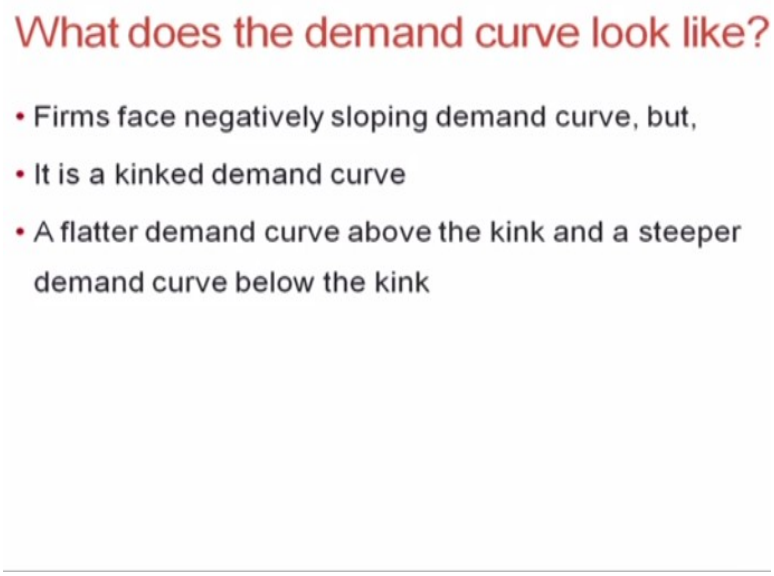
So this is one option that the firms have and the other option is they might decide not to cooperate and because there could be various reasons of not cooperating. First they do not trust each other so not they would not cooperate or they have a lot of confidence in themselves that they will be doing much better if they do not cooperate.

So whatever may be the reasons the other option for the firms is to not cooperate. So we are going to see so when the when they cooperate the firms can decide to behave like a monopoly and maximize joint profit and a single market price.

So when they cooperate they may decide they may say that let us behave like a monopoly. Let us get together. Let us behave like a monopoly. Let us charge a common price and let us split up the total output amongst ourselves which will probably be dependent on the cost structure of each of

the firms. So if they have very similar cost then probably it will be 50%, 50% of the output. So that situation we are going to discuss later.

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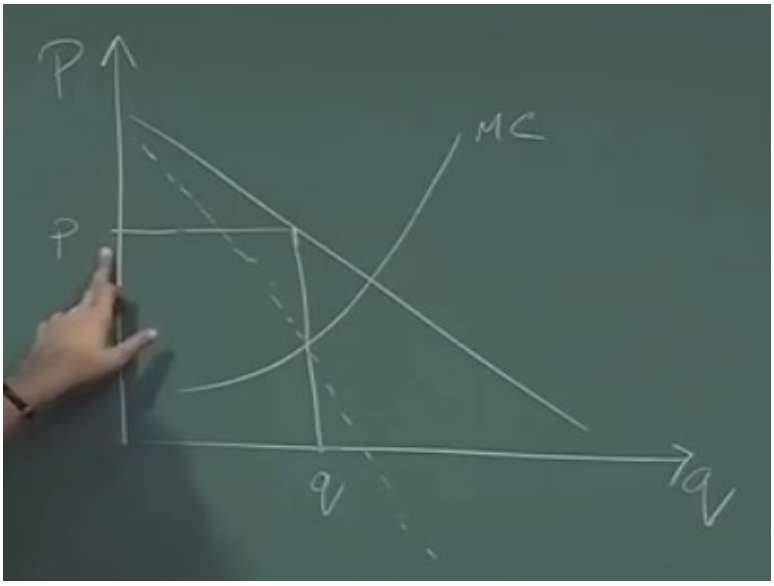


Now let us move on to the situation where they do not cooperate or the situation where a firm behaves individually. It makes its choices individually and then decides what price to charge and what quantity to produce. So when we try to do that first we need to develop the demand curve. So our question is what does the demand curve look like?

So the firms will be facing negatively sloping demand curve because they this is a market structure which lies between say monopolistic competition and monopoly both of them it could be product differentiation, it is very unique it is even if it is not unique then also the barrier to entry is very high so the demand curve is negatively sloping.

But it is a kinked demand curve. I am going to explain on board why it is a kinked demand curve. A flatter demand curve above the kink and a steeper demand curve below the kink. This is how the demand curve looks like for a oligopoly firm when they are not cooperating or when they are deciding P and Q on their own. So let us let me draw try to draw the demand curve what it looks like.

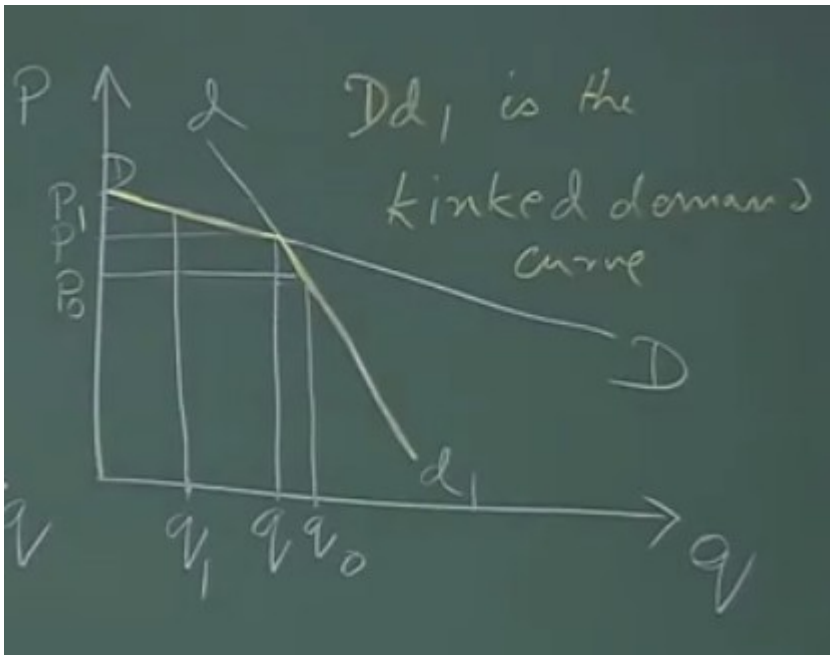
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So this is for a firm and so this is for a firm and it has a so you probably know the exercise already. So this is a firm and say it has started with this demand curve. It is its demand curve that the firm is facing and it is the marginal cost curve and the firm has it is a profit maximizer so obviously it has made MC is equal to MR and it has decided to produce q and it is decided to charge a price of P.

Now imagine that this is the situation in the market. This is the equilibrium situation where we start from. Now the so the prevailing price in the market is say P. Now what happens? So the prevailing price in the market is P. Now what happens if the firm decides to reduce the P. If the firm decides to so say imagine a situation where the firm is or even it is possible not to draw okay let it be here let me draw so this is P this is q and say this is the equilibrium P in the market.

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Now so if this is the P in the market and say the firm was selling a quantity of amount q and there are a few more rivals in the market who are keeping a close watch on this firm. Now the assumption here is suppose the firm decides I am going to reduce the price. Suppose the firm decides I am going to reduce the price. In that case what is going to happen?

So if it reduces the price to P 0 then how much is the quantity going to change? Now what is going to the assumption here is as soon as the firm reduces its price to P 0 the other firms are going to follow suit why? Because they know that since if they are operating in the same market and when they have this price that I have started with.

So say this is the equilibrium situation. Like this is the market at rest and it is a this is the prevailing price in the market. So everyone is charging around P. Now this firm decides I am going to reduce the price to P 0. So as soon as it reduces the price to P 0 what will happen is other firms in the market are also going to reduce their price to P 0. So all the firms will see that if they if since each of them faces a negatively sloping demand curve and one firm reduces the price so what will happen is they know that one firm has reduced the price so most of the market or most of the consumers they are going to rush to this firm and because there has been a price drop and hence the rivals to stop that from happening they are also going to reduce their price because they do not want to lose their customers to this firm.

So they are also going to reduce the price. Hence the firm which had hoped to increase its output because of by reducing price cannot increase it too much. So the increase in output will be very little say q to q 1 because all the firms' other firms have also reduced their price. So the demand curve for this firm now looks like this which is a steep demand curve which means that if the firm is to reduce the price, quantity, rise in quantity is not going to be much.

On the other hand, if this firm decides that I am going to raise the price due to whatever reason say the cost has increased. So he decides I am going to raise the price. So if he raises the price to P 1 what will happen is since it is a negatively sloping demand curve law of demand price increase associated with drop in quantity.

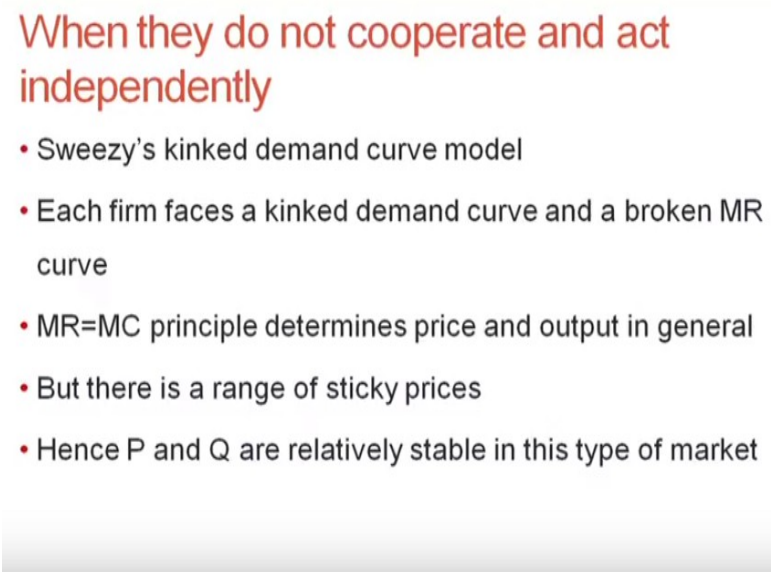
So since that is going to happen so all the other firms in the industry are not going to raise their price. They will think that okay fine this firm has raised its price so now the product of this firm is costlier so hopefully many of the consumers are going to come to us. So they are not going to do anything. They are not going to raise their price and as a result this firm is going to lose lot of customers. Hence the drop in quantity is going to be very big.

So when he raises the price, the drop in quantity is very big. Say let me call this q 0 to match with the P 0 and this is the q 1 to match with P 1. So here the drop in quantity is very high as a result in this portion the demand curve is flat. So above P the demand curve is very flat and below P the demand curve is steep.

So this is one demand curve and small d is another demand curve. So if the market starts from a stable price quantity combination in that case the demand curve facing each of the firm is a kinked demand curve. So it is a kinked demand curve. It looks like this, the yellow demand curve D, capital D d 1, d 1 is the kinked demand curve.

So capital D d 1 is the kinked demand curve that each of the firms is facing. That is it would be it would not like to raise the price but it has to follow its rivals if the price falls. So if the its rivals also they reduce their price it would probably follow suit but if the rivals are raising the price it will be reluctant to raise the price. Hence a flatter demand curve above the kink and a so a flatter demand curve above the kink and a steeper demand curve below the kink. So that is how the demand curve looks like.

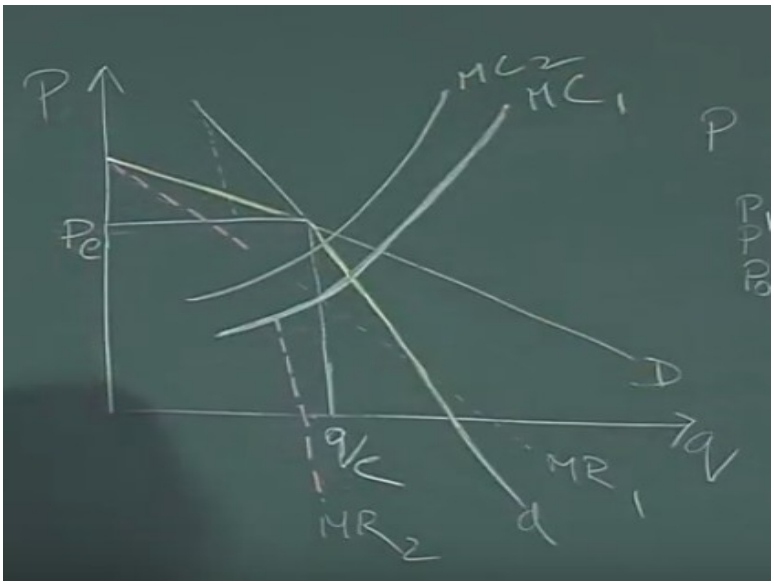
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So what happens when they do not cooperate and act independently? So this is called the Sweezy's kinked demand curve model, Sweezy's kinked demand curve model. So each firm faces a kinked demand curve and a broken market revenue curve. I have not yet drawn the market revenue curve, I am going to do that but let me first go through the slide and then explain the points that we have in the slide.

So once we know the marginal revenue curve and cost curve is given so MR is equal to MC principle determines price and output in general. But there is a range of sticky prices. Hence P and Q are relatively stable in this type of market. What does this mean? So this has just summarized the model. Let me draw what this looks like. So we have understood what the kinked demand curve looks like. Let me draw the marginal revenue curve here. Let me draw the marginal revenue curve.

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So I have drawn the 2 demand curves. One is the capital D other is the small d and from here I have the kinked demand curve. This is the kinked demand curve. So this is the kinked demand curve and this is the so this is the marginal cost curve. So this is where the so this was the equilibrium price, this was the equilibrium price and from the first demand curve capital D so this is where the market was stable.

Now what does the demand curve, what does the marginal revenue curve actually look like? Now the marginal revenue curve till P, this is P; so till P the marginal revenue curve follows the flat demand curve. The marginal revenue curve of the flat demand curve and then it is the marginal revenue of the steeper demand curve. So this is here. So this is the marginal, the pink marginal revenue is the marginal revenue curve. So and let me draw the marginal cost as green. So this is the this was the marginal cost of the so this was the marginal cost of the firm.

So this was the marginal cost curve of the firm and it started from P and the pink line is the pink dotted line is the marginal revenue curve. So marginal revenue as you will see it has a discontinuity between when the demand curve shifts from capital D to small d. So there is a discontinuity in the marginal revenue curve and this is the space where it is the marginal revenue curve has a discontinuity because of the kink in the demand curve.

Now the next question is if MR is equal to MC? MR is equal to MC basically had decided the equilibrium output and equilibrium quantity so let me call them q e and P e. So this is where the

market was stable. Now what happens say for example if the marginal now say the market started from here. Now what happens if the marginal cost of this firm goes up? So there is a firm, due to whatever reason its marginal cost has gone up.

So in that case the marginal cost is now say from MC 1 it has increased to MC 2. So what is it going to do? Is it going to so if it has increased from here so MC is equal to MR that would in other situations had this not been a kinked demand curve, had this been a say monopolistic competition firm what would it do?

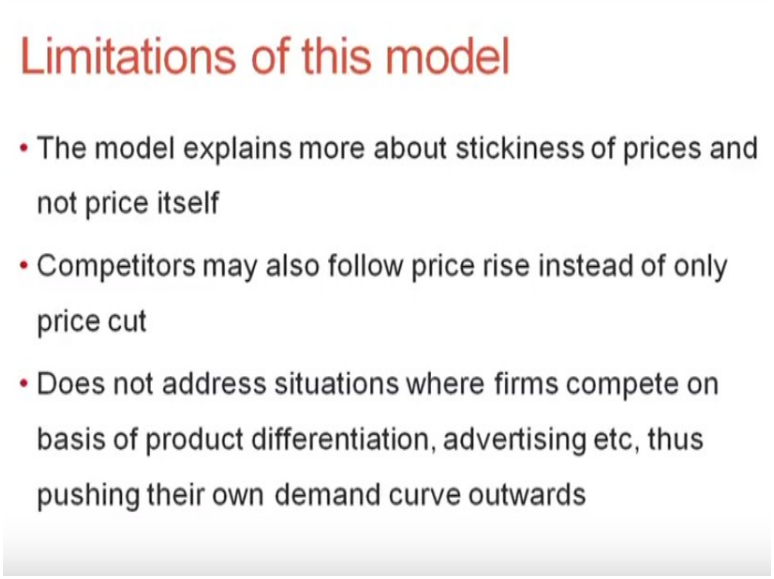
It would in general make MC is equal to MR and in that case price would go up because it would follow the MR 1 the it would be following the marginal revenue curve as continuity of this line MR 1. So in that case it would be charging a higher price and producing a lesser amount of output but in this situation since there is a kink in the demand curve because of this logic that if he increases the price he is going to lose a whole lot of output so (he is going to) he is not going to increase the price in this region.

So for a long period of time he is going to try not to raise his price. So he is not going to raise his price because he is going to lose a lot of customers which means that if marginal cost shifts up and down in this region he will continue to charge a price of P e and produce a quantity of q e for the fear of losing customers. So hence P and q are relatively stable in this type of market.

So what does that mean? That means that because of this kinked demand curve the firms are reluctant to change their prices and they try to stick to the prices that they have been charging and produce the same amount of output for a whole lot of for a for quite a range of marginal revenue and marginal cost. So that is what they are going to do for quite a range of marginal revenue quite a range of marginal cost even if marginal cost goes up a little bit they are they will be reluctant to change their prices.

Hence price is called sticky in this situation. So it is a sticky price situation which is happening here and the firm is not changing the firms are not changing their prices and hence this kind of kinked demand curve gives the oligopoly market a very relatively stable kind of price quantity combination where the firms do not change their prices so easily.

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But this is only one of the models of oligopoly and what are the limitations of this model? Now the limitations of this model is are that the model explains more about stickiness of prices and not price itself. So it talks about why price is fixed at P e but it does not say too much about why how prices will go above P e or how prices can fall below P e. So the model explains more about stickiness of prices and not price itself.

And competitors may also follow price rise instead of only price cut. This has been actually empirically proven that it is not always true that if the firm raises its price above P e other firms will not follow suit. That is actually not true because one can imagine in situations where there is an industry level change in cost situation say for example all the firms are having to pay a higher having to pay higher taxes or they are having to pay higher price for a certain input.

So in that case marginal cost is going to go up for everyone and if one firm takes the chance of actually raising the price other firms will be too relieved to come out of this sticky situation and charge a higher price like the other like this firm. So that happens all the time in the market and one could actually see different forms of price signaling happening in the market where different firms indicate that costs are going up and we need to increase our prices and that is kind of signal which happens so that is proof that it is not always that competitors do not follow price rise.

The third point is it does not address situations where firms compete on basis of product differentiation, advertising, etc. thus pushing their own demand curve outwards. So in this

situation again we have assumed that this demand curve is fixed for the firm. But what the model basically ignores is it is possible for the firms to product differentiate. They could try to advertise like the kind of product differentiation we discussed in case of monopolistic competition that they hold true in this case also and it is possible for the firms to go for various forms of product differentiation and thus capture a high share of the market and basically push their demand curve to the right and hence when marginal cost goes up they can actually charge a different price. Those kind of situations can also be imagined.

So that basically ends our discussion on the Sweezy's model of the kinked demand curve which is a very simple model of non-cooperating firms in oligopoly and next we are going to discuss about collusion and if that is possible among oligopoly firms when we take up we use the game theory we apply game theory to understanding the strategies of firms in responding to their rival's moves. Thank you.

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Lecture - 36

Oligopoly - P and Q Outcomes with Example

Hello and welcome back to our discussion on oligopoly. We had introduced the oligopoly market structure. We have talked about the about why the oligopoly market outcome is a little more complicated than monopoly or perfect competition because in the latter 2 cases even in case of monopolistic competition also the firms have to make their own decisions irrespective of what the other firms are doing in the industry or in the market.

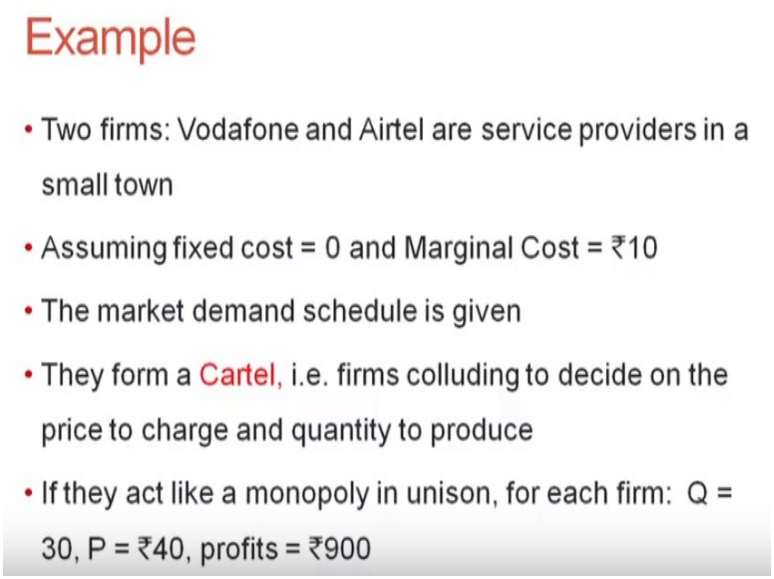
So there the output that they are going to produce, the price that they are going to charge and the profit that they are likely to make are solely dependent on their own cost conditions and nothing else. But in case of oligopoly we have seen that the firms since there are only a few firms and entry and exit is difficult so there are only a few firms in the market and lot of market power could be exercised but at the same time the firms are going to look at the price and quantity of its rivals and act accordingly or accordingly they are going to make their decisions of what price to charge and what output to produce. So we have looked at the kinked demand curve where we saw that there is a range where there is the prices are sticky and the output is also sticky.

So that kind of gives a stability to a oligopolistic market structure that for quite a range of marginal cost quite a range between which if the cost fluctuates for any firm it is not going to change its price or output. So that kind of imparts a stability to the oligopolistic market structures

where the firms would not like to move away from the price that they are already charging. So now in this module we are going to take an example.

We are really we are going to look at demand curve. We are going to have some cost structure and we are going to see that if a oligopoly firm has a demand curve like this and the cost structure and how is it likely to behave. What could be the price and what could be the output in such a situation where the firm actually takes into account the decision of its rival. So let us look at the example.

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Now the example that we are considering, there are 2 firms. They are Vodafone and Airtel. They are service providers in a small town. So they are they are the only 2 firms who have network and they provide services in that small town and we are assuming for simplicity a fixed cost equal to 0 and marginal cost equals Rs. 10.

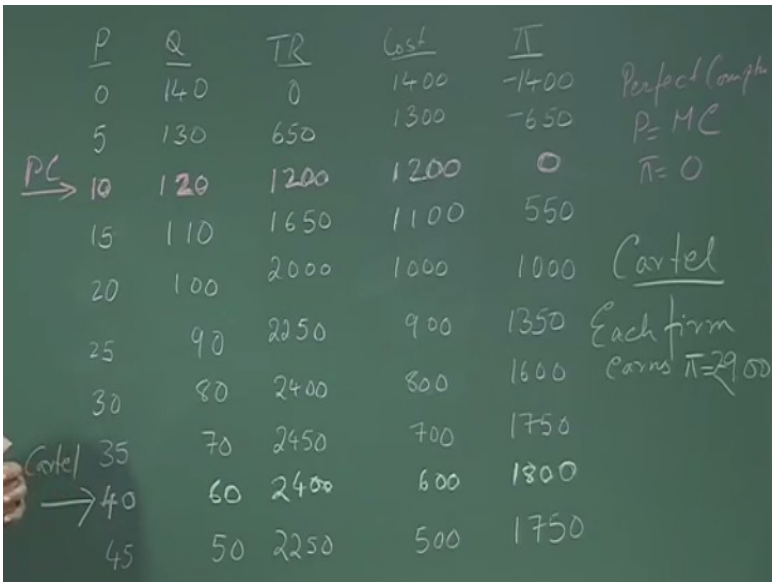
Now the market demand schedule is given. I am going to write down the market demand schedule on the board and then once the market demand schedule is given the firms have so here this is a special case of oligopoly where there are instead of few firms there are actually 2 firms.

So this situation is called duopoly where there are 2 firms. So we are taking a very simple example where there are 2 firms competing for the same market and they have very similar costs and how are they going to share the market between themselves.

So as I said in the previous modules that the firms have 2 choices. The one the first choice for the firm is to cooperate. The first choice is that one of the choices is that the firms could decide that let us work in unison and decide what price to charge and what output to produce. So that will sort of give us some clarity and we need not worry about the other firm.

So this is one option that they could try. The other option is not to cooperate and act independently and try to capture the as much of the market or make as much profit as possible in that market. So let us see what happens if they form a Cartel. Cartel means that is the firms that is firms colluding to decide on the price to charge and quantity to produce. That is what we mean by Cartel and here we will see that in when they act as Cartel they can decide to act as a monopoly. So let me first write down the demand schedule and then from there we can see.

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So, so price and quantity 0, 5, 10, 20, 30, sorry 30, 35, 40, 45. These are the prices that are possible and quantity is say 140, 130, 120, 110, 100, 90, 80, 15, 110, 20, 100, 25, 90, 30, 80. So this is the price schedule and accordingly these are the output schedule. So you can imagine the

output to be the number of people they are serving, number of customers, it can be imagined that way.

So this is the price and demand schedule. So let us write down the total revenue in each of these cases is 0, 650, 1200, 1650, 2000, 2250, 2400, 2450, 2400, 2250. So these are the total revenue. Cost is since marginal cost is fixed for every unit of output so this is 1400, this is 1300, then 1200, then 1100, 1000, 900, 800.

So these are the total cost and so profit as is equal to total revenue minus total cost is equal to -1400, -650, 0, 550, 1000, 1350, 1600, 1750, 1800, and 1750. So these are the, this is the cost. This is the revenue, this is cost and this is profit. So from here first would you be able to tell from here what is the what is likely to be the perfectly competitive output and what is likely to be the monopoly output.

So from here one can clearly see so in case of perfect competition what happens in case of competition? In case of perfect competition price is equal to MC and profit is equal to 0. Price is equal to MC and profit is equal to 0. So this basically gives us our competitive output. So this is our competitive output and this is our competitive profit.

So the pink line this is our perfectly competitive output. So if the firms are to behave as competitive firms, if they are to behave as competitive firms and take the price as given so in that in the market what they would do is they would produce the output so that the profit is equal to 0 so that their total revenue is equal to total cost or price is equal to their marginal cost and this is the competitive output.

So what would happen, now we said that the firms will decide to act like a monopoly. What if they collude? Now if the firms collude they are going to say that since monopoly gives us the maximum profit let us get together. We are the only 2 firms in this town. Let us get together, act as a monopoly and we are going to charge the price in such a way that it maximizes our joint profit.

So the total profit is going to be joint and our costs are same so we are just going to split the profit into half between the 2 of us and that is the price that we are going to charge and that is the output that we are going to produce.

So what they do is they make MC is equal to marginal revenue and accordingly from the demand curve they choose that level of price which they are going to decide the output on the basis of marginal cost equals marginal revenue and accordingly and from there they are going to trace back the output the price from the demand diagram from the demand curve and that is the price that they are going to charge.

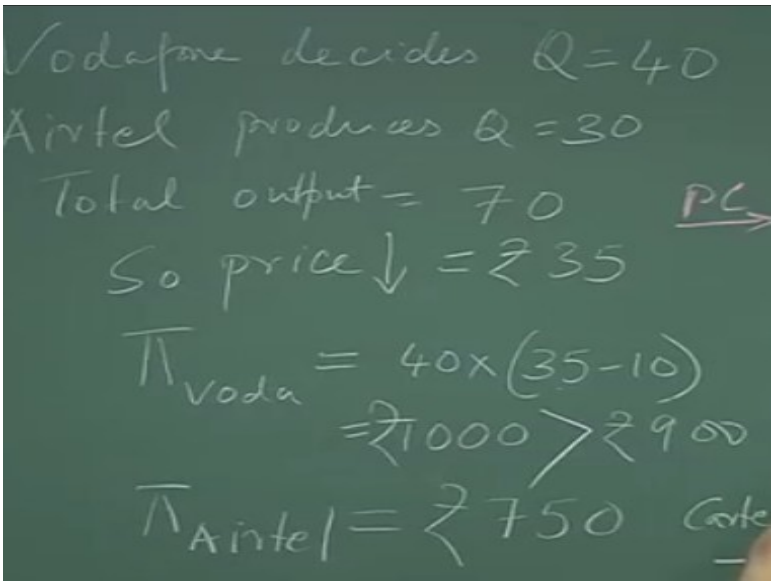
So that is going to give them the maximum revenue. So monopoly gives them the highest revenue so highest profit. So where is the highest profit in this schedule? The highest profit lies here 1800. I hope you are able to see it. So this is the highest profit that is being earned here, this is the highest profit that is being earned here at a output level of 60 and a price level of 40. So this is the monopoly output.

So had there been only one firm in this market that firm would have produced 60 units of output and charged a price of 40 Rs and gained a profit of 1800 Rs. Now since there are 2 firms in this market they are going to split this amongst themselves. So this is the Cartel situation. This is what they are going to do. So each of the firms is going to produce 30 units of output and they are going to charge a charge a price of 40 Rs and accordingly they are each of the firms is now going to get a so in case of Cartel each firm earns profit is equal to 900 Rs.

So each firm earns a profit equal to 900 Rs so the total profit in the market is 1800 Rs. So that is what they are going to do and this is what we see in the slide also. In the slide if they act like a monopoly in unison for each firm output is equal to 30 and price is equal to 40 Rs. So profit is equal to 900 Rs for each of the firms.

Now next thing to think about is each of the firm wants to maximize its profit. It has its own self-interest in mind. So what happens so the next the firm is he going to stop at producing 30 units? Is he going to stop at producing 30 units?

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Let us see what happens when one of the firms say for example Vodafone decides say Vodafone now decides that let me break the agreement, let me break the agreement without informing the other firm, without informing Airtel let me break the agreement and let me produce Q equals 40 instead of 30. So it decides I am going to produce 40 units instead of 30 units and in that case and the prevailing price in the market is what?

So as soon as he produces 40 units Airtel assuming that Airtel so Vodafone thinks that if it increases its output to 40 units Airtel will continue will continue to produce or let me just make it shorter; Airtel produces Q is equal to 30. So total output in the market is equal to 70 units now. So total output in the market is 70 units now and price immediately then falls to so price immediately falls to 35 Rs.

So price immediately falls to 35 Rs. So as soon as the price falls to 35 Rs what happens to Vodafone's profit? So in this situation if Vodafone decides that I am going to increase the output then profit of Vodafone profit of Vodafone is equal to 40 into 35 minus 10; cost, price, and output is equal to 1000 Rs which is more than 900 Rs.

So when they decide, when they collude Vodafone gets a profit of 900 Rs but if he breaks the agreement, produces another 10 more units in that case his profit is 1000 Rs as against 900 Rs so there is every incentive for Vodafone to break the agreement because all of them are profit maximizers and he would like to maximize his profit.

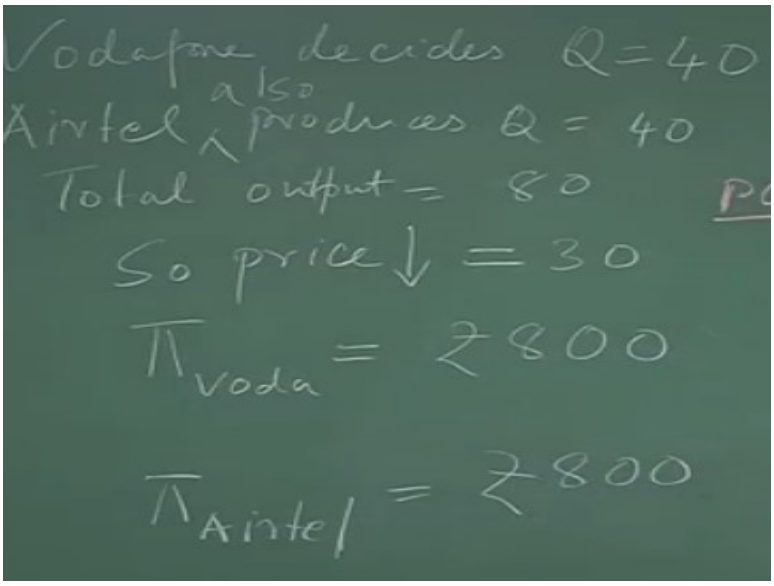
So he is going to break the agreement or Vodafone is going to break the agreement and charge a higher price (charge) or break the agreement and produce a higher amount of output thus bringing down the price in the market and hence getting a profit which is higher than the Cartel situation.

But now what happens and in this case what if Airtel continues to do the same; the profit of Airtel is, profit of Airtel falls, profit of Airtel falls to 750 Rs. So the profit of Airtel falls to 750 Rs. So that is what happens. So when 35, 70 profit is 1750. So when the one who is breaking the agreement he gets 1000 Rs out of this profit and the other one who sticks to the agreement he gets a much lesser profit than 900; he gets a lesser profit of 750 Rs.

So in that case so now the Airtel also knows if there is a collusion between the 2 firms and there is no way of legally enforcing this agreement in that case each of the firm has enough incentive to break the agreement. So each of the firm knows this also. So the Airtel will know that it makes sense for Vodafone to break the agreement and produce a higher amount of output.

So that is what Airtel is also going to do or to think about it in a different way is independently also the Airtel also thinks in the same lines as Vodafone and thinks that if I break the agreement I am going to get a higher return. I am going to get a higher profit. So let me break the agreement. So as a result what happens is each of the firms breaks the agreement and each of the firms ends up producing 40 units of output.

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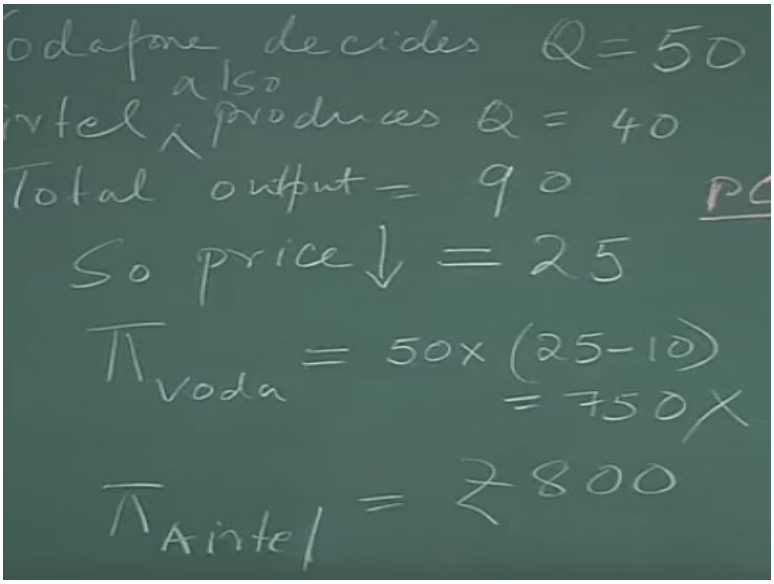
So each of the firms now, so each of the firms now break the agreement and Airtel also produces, now Airtel also produces Q is equal to 40. So total output in the market now becomes 80, even higher. Price now falls to 30 and now the profit falls to so each of the firms now this calculating in similar way. So profit of Vodafone now reduces to 800 Rs and for Airtel also profit reduces to 800 Rs.

So they each of them is going to break the agreement and each of them is now going to produce higher amount of output because they do not trust each other. Each of them is going to produce a higher amount of output. So firstly there is distrust amongst the firms that the other is going to probably break the agreement. Second is they are acting in self-interest and they are going to do whatever is going to make them best, make them the best of in this situation.

So this is what they are going to do and they will end up each getting a profit of 800 Rs which is now less than the Cartel outcome of 900 Rs. So what we see is even if the firms decide to collude and they agree to produce a certain amount of output which makes their profit highest in the joint profit highest, because of their self-interest getting priority they end up producing a output level which is actually much more than the monopolistic level of output and the result is that each of them ends up getting a profit which is less than the monopoly profit which is less than the Cartel outcome of profit.

So this is what happens and next another interesting thing to see would be are they going to keep on increasing the output so 40 is something we tried. Now Vodafone is it going to increase the output to 50. Is he going to increase it to 50? So what happens if Vodafone increases the output to 50?

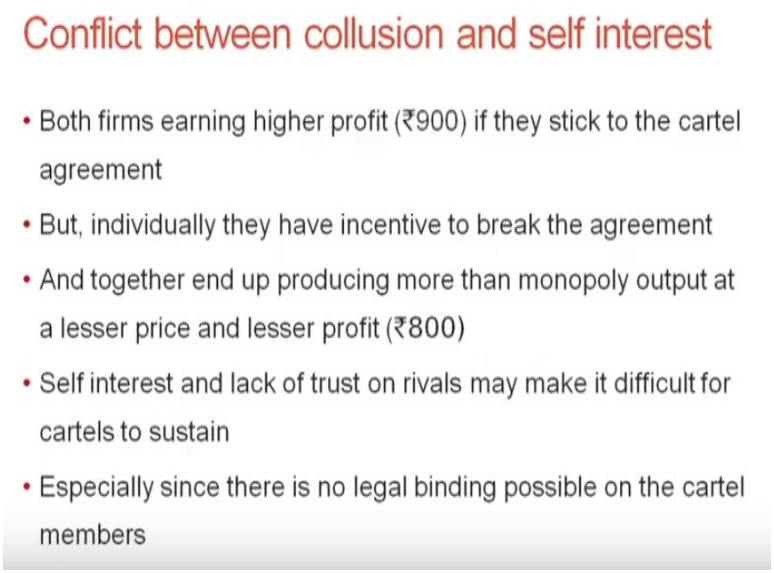
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So going by the same argument if the Vodafone increases its price to or output to 50 then this is 50 and total output then becomes 90. So price will be 25, price will be 25 and Vodafone profit will reduce to 50 into 25 minus 10 equals 750. So in that case Vodafone if Vodafone decides to increase its output even further to 50 units in that case its profit is going to fall from 40 units.

So in that case it is not going to increasing his output and it just going to continue to produce 40 units. So both the so this is the Vodafone is not going to choose so neither of the firms is going to increase the output even further beyond 40 units and they will settle at producing 40 units and together they are going to produce 80 units in the market.

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So to summarize there is conflict between collusion and self-interest. So although collusion leads to a higher profit for individual firms acting on self-interest leads the firms to such a situation where they end up getting a profit which is less than collusion outcome. So both firms earning higher profits if they stick to the Cartel agreement but individually they have incentive to break the agreement and together they end up producing more than monopoly output at a lesser price and lesser profit.

Self-interest and lack of trust on rivals may make it difficult for Cartels to sustain. Especially since there is no legal binding possible on the Cartel members. So as you may already know that Cartel is Cartel formation is not legal for the benefit of the consumers because if the firms get together and if there is legally if it is Cartel is enforceable then they will end up producing monopoly output at monopoly price which is not at all a desirable situation for the customers so for the consumers so this is something which is not legally approved.

So because since it is not legally enforceable there is every incentive for the firms to default on the agreement and they work on their they basically act on their own self-interest and they end up producing an output level which is higher than the monopoly output level which is which in a sense actually is beneficial for the consumers.

So this was an example that illustrated how a Cartel formation or how collusion is not so easy in case of oligopoly. In the following modules we are going to talk more about these. Thank you.

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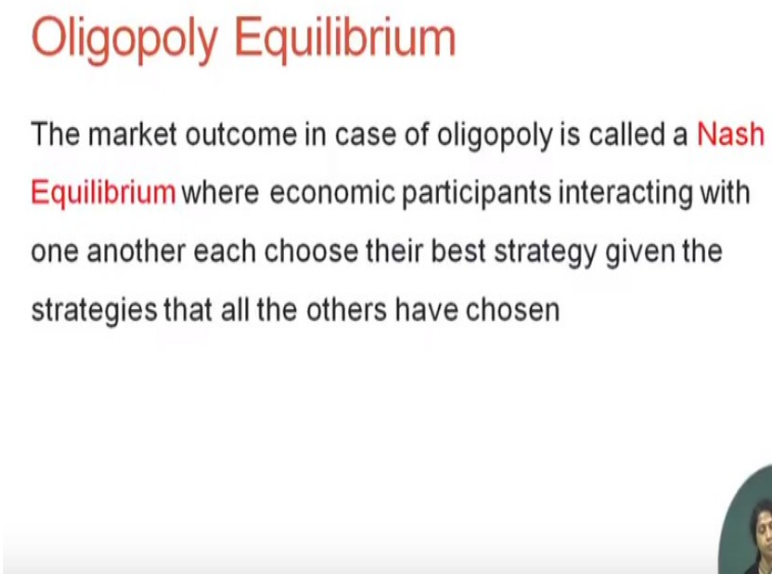
Lecture - 37

Difference between Market Outcomes

Welcome back to our discussion on oligopoly. We in the previous module we took an example and we saw what are the different outcomes if the firms behave in different ways. So if the firm behaved like a perfectly competitive firm they would be earning zero profit. Then we went on to see what would be the monopoly output in case of a situation where we have the demand schedule, we have the cost, and we have the profits and we clearly saw what the monopoly outcome was and we discussed why it is difficult for oligopoly firms to settle at the monopoly output and monopoly price in an oligopolistic market.

When there are more than one firms there each firm acting on its own self-interest they end up producing more output than the monopoly output. So this is what we saw in the earlier module and this is a very short module where we are going to just extend that discussion and see compare the different outputs.

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So oligopoly equilibrium is the market outcome in case of oligopoly is called a Nash equilibrium where economic participants interacting with one another each choose their best strategy given the strategies that all the others have chosen. So the market equilibrium in case of so as we saw in the earlier example we took the earlier example and in the earlier example we saw that the firms do not settle at the monopoly outcome. Although the monopoly outcome gives them the higher level of profit or the highest level of profit, they do not stick to the monopoly outcome and what they do is they move on to produce more output ending up at a market equilibrium where from where they do not have any incentive to move away from.

So they end up each of the firms ended up producing 40 units of output instead of 30 units of output and each of the firms ended up getting a profit of Rs 800 against the monopoly profit level of Rs 900 for each of the firms had they stuck to the monopoly output of total 60 units and 30 units by each of the firms.

So this situation or this outcome this equilibrium is called a Nash equilibrium where the participants basically when they interact with each other they know that each of the firms they know what the rival is probably going to do or what the rival is going to do and accordingly they change their decision output producing decision like in the example that we took earlier.

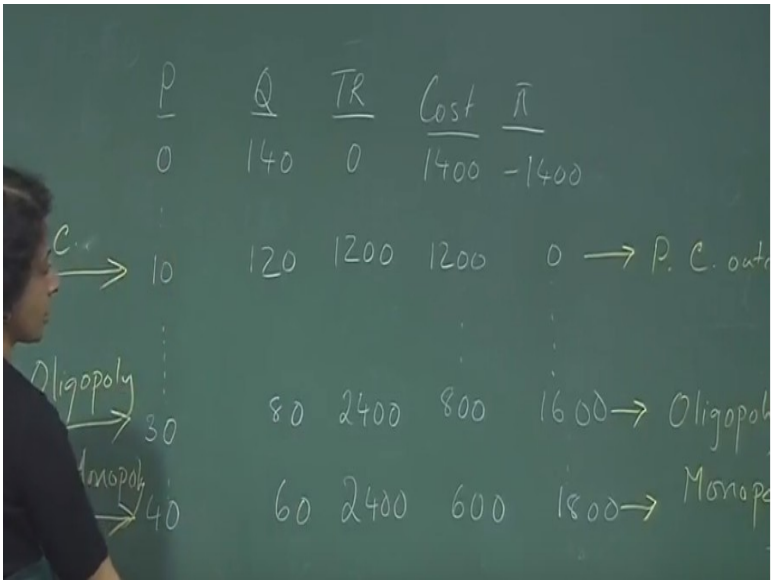
Each of the firms although they agree amongst themselves that each of us is going to produce 30 units yet each of the firms once they know that the other firm is going to produce 30 units it is

best for them to produce 40 units and secondly worse is if the firm distrust the rival or distrust the other firm and thinks that the other firm is going to break the agreement then also it makes sense for this firm to change its output level from 30 to 40 units.

So given the situation or given the strategy that has been chosen by the by its rival it is in the best interest of the firm to produce 40 units. Hence production or output of 40 units by each of the firms in the in our example was a Nash equilibrium and neither of the firms is going to move away from that output level of 40 units.

So this is called Nash equilibrium where economic participants interact with one another and they choose their best strategy given the strategies that all the others have chosen.

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So from our discussion so far we know that the market outcomes are such that competitive outcome we already saw in our example if you have if you have taken down the examples if you remember the so I am just going to write the so we had a price series here, a output series here. The first point was 0, 140, 0, 1400, and -1400. So then we had then 10, 120, 1200, 1200, and 0. Then 30, 80, 2400, 800, and 1600 and 40, 60, 2400, 600, and 1800.

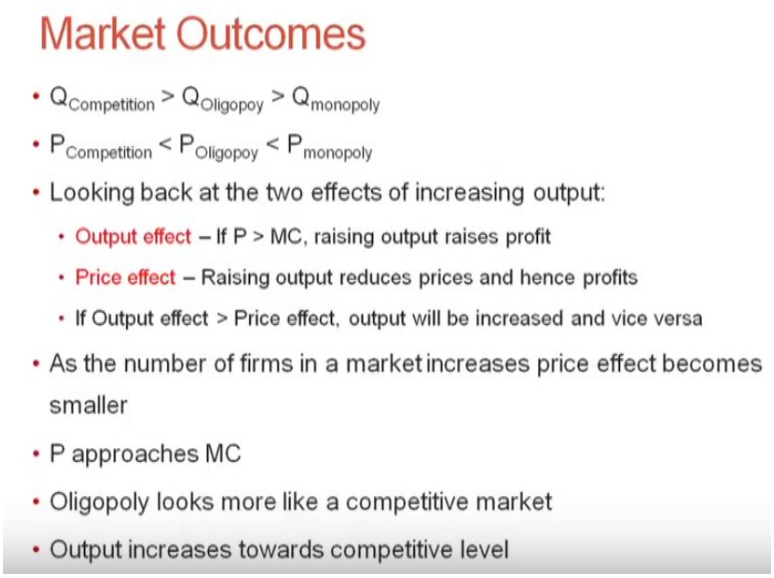
So I have not filled up the other figures but these are just for interest and we saw this was the perfect competition this was the perfectly competitive outcome. Our oligopolistic outcome came

out to be here. This was oligopoly and the final one this one was so this one was monopoly price. This one was oligopoly outcome and this was perfectly competitive outcome.

So as we see here the output is output in case of competition is highest 120 is more than 80 which is the oligopoly outcome and monopoly outcome is output is 60 which is lowest and the price is just the reverse and similarly in case of profit also competitive profit is 0. Oligopoly profit is 1600 and monopoly profit is 1800.

So these are the situations which so these are the differences in outcome between different when the firms behave as a as competitive firms or monopoly firms or oligopoly firms.

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So price in case of competition is lowest. In case of oligopoly is slightly higher and in case of monopoly it is highest. Now just why does that happen. So it is basically what we mean here is in each of these firms or each of these market structures of competition, oligopoly, and monopoly the difference is very broadly speaking the difference is in the number of sellers who are operating in the market.

So the number of sellers who can actually be present in the market that is that is a that results from the nature of the output that is being produced in these kinds of markets which we discussed initially when we talked about market structure but finally what price to charge and

what output to produce that basically comes from the interaction between all these firms in the market.

So higher the number of firms there is more competition in the market and price is lower and output is higher and profit is lower. So that is where it boils down to at the end of the day. So that is where it boils down to and although we have theoretically developed various models of competition, monopoly, monopolistic competition, and oligopoly and we have tried to make distinctions between the different kinds of market structures and hence develop theoretical models. We have drawn diagrams with demand curve, marginal cost etc. and seen what the firm's equilibrium output is going to be.

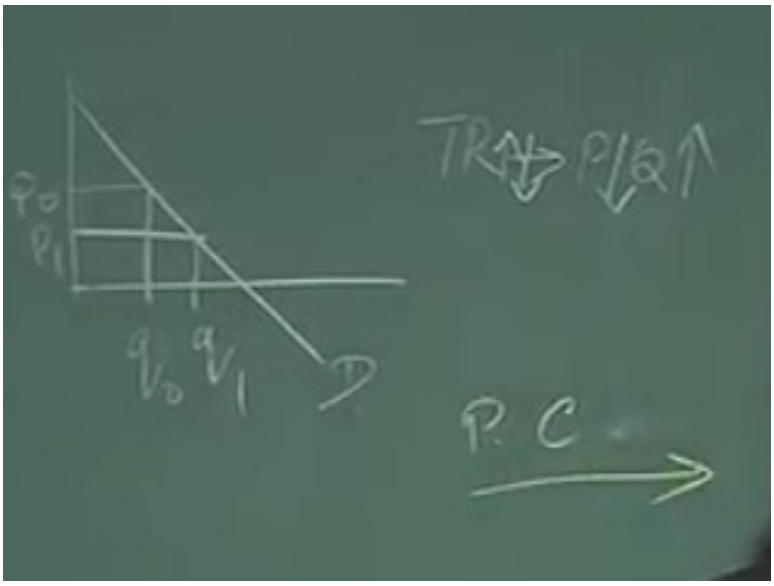
But in reality in the real world there is the demarcation between the market structure for different firms is quite grey. It is not in black and white that one firm is really a oligopolistic firm or is it a monopolistic competition or even if there are few firms more than one firm in the market yet they can behave like a monopoly. So all these sorts of situations are possible in a market. So finally it all boils down to how many firms are operating in the market and how they can decide their strategy, what is possible for them to do and what is not possible for them to do.

So to look at the outcomes in a different way, let us look back at the 2 effects of increasing output. If you may remember when we had introduced the negatively sloping demand curve in case of monopoly if you remember in case of perfect competition the demand curve facing the firm is horizontal. So the firm knows that it can sell as many units of output as possible at that given price.

But in case of the monopoly and monopolistic competition and any firm which had some amount of market power was facing a negatively sloping demand curve and when we introduced the negatively sloping demand curve we discussed about the 2 effects. We said there was this output effect and there was this price effect. So let me let us see what they were.

So looking back at the 2 effects of increasing output, one was the output effect that is if price is more than marginal cost raising output raises profit. Price effect was raising output reduces prices and hence profits. So if output effect is more than price effect, output will be increased and vice versa.

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So basically going back to the price effect and output effect if you may if you remember we discussed that when the firm facing a negatively sloping demand curve or a firm facing a negatively sloping demand curve and if this is the so for a firm facing a negatively sloping demand curve if the firm increases output there are 2 effects.

One is the increase in output from q 0 to q 1 which raises the total revenue by increasing the Q component here. But at the same time when the Q component increases price falls to P 1. So price falls to P 1. So TR when P when q increases there is a increasing impact on TR because of Q rising and as Q rises P falls to P 1 and hence there is a downward impact on revenue and hence profit also. So both these effects work in work together.

So there is an output effect and there is a price effect. In what situation is output going to be increased? Now output is going to be increased if the output effect is stronger than the price effect. So output will be increased or the firm it would be the interest of the firm to raise the output if this effect is more than this effect. So if the output increase effect is more then it makes sense for the firm to raise the output.

So keeping that in mind in what situation do the firms actually increase the output? So as the number of firms in a market increases, price effect becomes smaller. So as the number of firms in

a market increases price effect becomes smaller and hence P approaches marginal cost. As more and more firms enter the market so the output effect becomes stronger so it makes sense for the firms to increase so as more and more firms enter the market the output the amount of output goes up and it makes sense for the firms to increase output and price gradually approaches marginal cost.

So when that happens as price moves more and more towards marginal cost and output keeps on increasing because more firms have entered the market oligopoly looks more like a competitive market. So oligopoly then starts looking like a competitive market. So output increases towards competitive level. So when more and more firms enter the market then the output becomes bigger and bigger and price comes closer and closer to marginal cost and hence the oligopoly market outcome looks moves more towards competitive outcome.

So that is what happens and so another way of thinking about it is imagine that there were instead of 2 firms there were 3 firms. If there were 3 firms, you would see that if you could do it as a exercise actually that instead of 2 firms had there been 3 firms what would they do in that case. If had there been 3 firms so the monopoly output would be 1800 and they would be splitting it up amongst themselves so they would be producing 20 units each.

So you can keep on increasing the number of units and each firm when they break the promise and they increase the output they end up producing output which is more than 80 units. So as more and more firms enter this market this quantity is going to move say when it when from instead of 1 firm when there are 2 firms there is 80 units.

As they move as more and more firms enter the market this output keeps on increasing and in the extreme situation if there are whole lot of firms in the market they might end up producing almost close to this level of output. So that is the intuition behind it and that is the intuition behind it that in case of an oligopolistic market framework as more and more firms enter the market the divide between oligopoly and monopoly that becomes broader, wider and oligopoly market starts looking more and more like a competitive market.

Now this is this outcome could be compared or this outcome could be compared to the case of trade between countries to the benefits of trade between countries. How so? Say for example

there is a closed economy, there is a closed economy which has only a few manufacturers. Let us take the example of the car manufacturing industry. So let us take the example of cars and say there is say each of the country say taking the example of say US.

Now imagine that US has only 2 firms which manufacture cars. One is the Ford other is the General Motors. Now Ford and GM they together so they form a duopoly within the market setup in the US and they are catering to all the car requirements of the country. In that case they will end up somewhere here. They will end up producing number of cars which is similar to or commensurate with this kind of outcomes, slightly more than monopoly but less than competitive output.

Again, say Japan, Japan has 2 car manufacturers of its own say Toyota and Honda. So they also in their country if they are not allowed to trade. If Japan is does not trade, they also end up with only 2 firms within their country and they also produce less amount of output and again say Germany has BMW and Mercedes and they do not trade with any other country.

Now instead imagine that now these countries open up so each of these country say US now has instead of 2 car manufacturing companies they now have 6 car manufacturing companies. So they now have 2 of their own, they have 2 Japanese companies and 2 German companies who manufacture cars for their domestic market.

So as soon as the oligopoly market becomes bigger with instead of 2 firms they now have 6 firms in the market and they again they behave like a oligopoly and they are going to behave and make their decisions exactly like this and they will also not be probably be able to collude and they are not going to settle at this equilibrium but they will end up producing instead of this equilibrium their output will be bigger because each of the firm defaults on the agreement and each of the firm ends up producing more.

So the net result is the total output in the market goes up many folds. As the total output in the market goes up many folds' price approaches this competitive price and output approaches this competitive output. It may not be reaching the competitive price and output is unrealistic but at least it will be close. It will be much away from the monopoly equilibrium and it will be closer to competitive equilibrium.

So hence we see from these illustrations why it is important for a market to have a large number of firms. Why it is important for the government to see to it that there are no artificial entry restrictions or entry barriers for different companies. So to create a competitive setup a competitive environment in any market is desirable because as the market becomes larger and larger with large number of producers the final outcomes comes closer and closer to the competitive outcome which is the most efficient and which has the highest amount of welfare maximization possible in an economy.

So as you have more and more firms entering the market price goes down for the customers, for the consumers and quantity goes up for the consumers so they have a whole lot of choices available in the market so hence market outcome in case of a market outcome a perfectly competitive outcome, perfectly is something perfectly competitive although is unrealistic but at least competitive outcome is something which the markets and the government needs to need to strive for because of the betterment of consumption for all the consumers and it improves competition and it improves efficiency of production also because as price gets closer and closer to marginal cost the cost of producing the last unit of output becomes closer and closer to the price that the consumers are willing to pay for that unit of output which is an efficient outcome.

So that is a discussion about the comparison between oligopoly and other kinds of market structures and finally we have seen how the different market structures are very similar to each other.

The only difference being the change in the number of producers operating in the market and in the following module we are going to discuss about game theory because so far in case of oligopoly we have said that it is more about strategy of the firms deciding how to act depending on how the rivals are going to respond. So we are going to try to understand the decision making process of different firms through game theory in our next module. Thank you.

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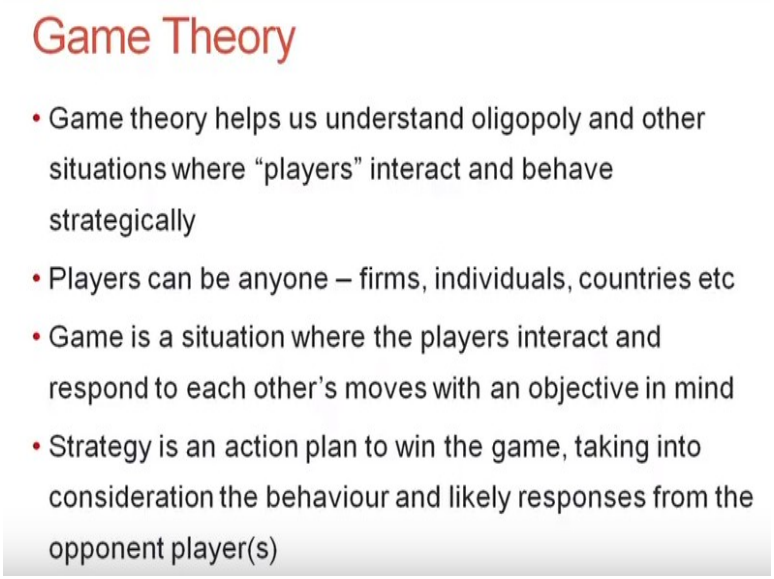
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Lecture - 38

Oligopoly – Game Theory

Welcome back to our discussion on oligopoly. In this module we are going to talk about game theory. Game theory is a very important tool in understanding how different individuals or different decision makers how they make decisions or how they make strategy depending on the moves when they are specially involved in interaction with other decision makers and they try to they have some objective in mind and they make decisions to attain that objective and how do they take into account the decision of the other stakeholders there and game theory is a very interesting and important tool which helps us in doing that.

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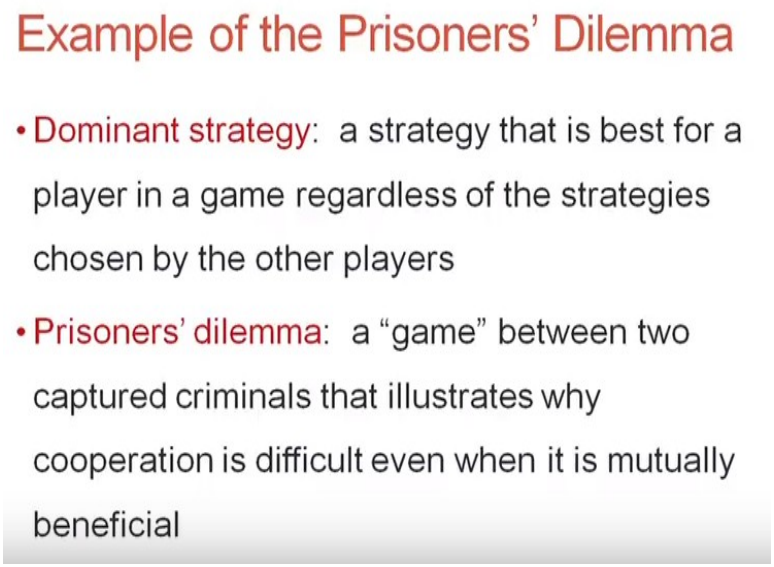
So game theory helps us understand oligopoly and other situations where players interact and behave strategically. The reason that we are doing game theory in our discussion of oligopoly is because as we have I have iterated a number of times that oligopoly is a situation where each of the players in the market or each of the producers in the market continuously keeps a watch on what the rivals are doing and accordingly they act on their self-interest and they take into account all information they have or understanding they have of their rivals and decide on the price to charge and output to produce in the market and so since it involves a strategy making process for the firms, game theory is an important tool which will help us understand the minds of the oligopoly players better.

So players can be anyone. Players could be firms, it could be individuals, it could be countries etc. So we are going to do some game theory examples in the following modules and we are going to see that how this tool can be applied not only in case of economics and not only in case of understanding firms but it can be actually applied to whole lot of situations.

So game is a situation where the players interact and respond to each other's moves with an objective in mind and strategy is an action plan to win the game, taking into consideration the behavior and likely responses from the opponent player or players. So game is a situation where they interact and respond to each other's moves and strategy is how they make decisions.

Strategy is what the players in any situation, players in a game or say the producers in the game of competing for maximum profit in a market setup how they plan strategies or what is their action plan, what is their decision making process so that they may emerge as winners in the game.

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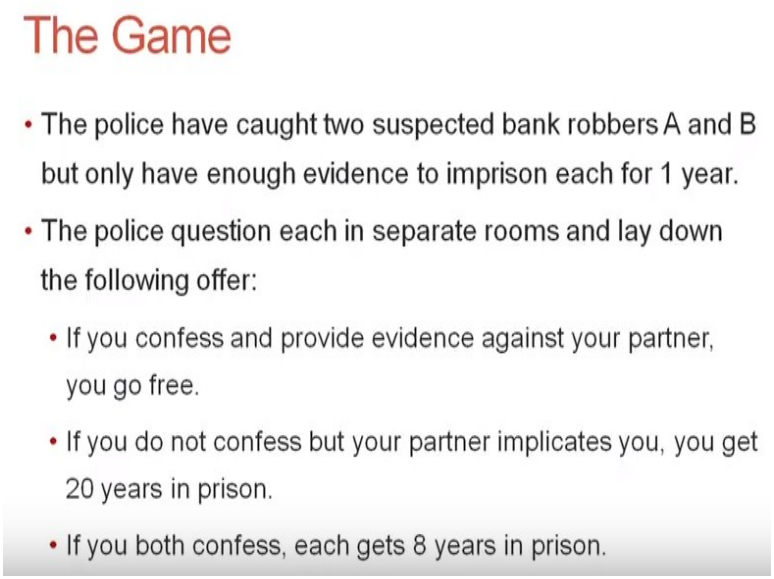
So this is called strategy and to understand that first let us take a very traditional example that is taken in case of game theory which is the prisoner's dilemma. So let me explain the game and so in case of a game there is let us these are some this is a definition which one it would be helpful if we understood this definition is what is a dominant strategy? So we know what a strategy is and what is a dominant strategy?

A dominant strategy is a strategy that is best for a player in a game regardless of the strategies chosen by the other players. So it is a strategy that is best for a player in a game regardless of the strategies chosen by the other players. And the prisoner's dilemma that is the game that we are going to discuss in this module is a game between 2 captured criminals that illustrates why cooperation is difficult even when it is mutually beneficial.

So why cooperation is difficult even when it is mutually beneficial is something we already saw through the example of the 2 firms Vodafone and Airtel that we took in the earlier module and

we saw why there is mutual distrust and there is among the 2 firms they act on their own self-interest and they end up producing higher amount of output than the monopoly outcome and they end up getting a lower amount of profit. So this we already have seen in the case of the market outcome and this prisoner's dilemma game is going to illustrate why it is difficult to collude. So let us go to the game.

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Now what is a game? The game is the police have caught 2 suspected bank robbers A and B but only have enough evidence to imprison each for 1 year. So these are hardcore criminals and they have been caught but the police does not have enough evidence. So the police questions each in separate rooms and lays down the following offer.

If you confess and provide evidence against your partner, you go free. If you do not confess but your partner implicates you, you get 20 years in prison. If you both confess, each gets 8 years in prison. So what happens here is these are 2 criminals who have, the police have kind of idea that they have committed a whole lot of crimes and but they do not have enough evidence to implicate them.

They do not have enough evidence to put them behind bars and the evidence that they have is only going to put them behind bars for only 1 year and obviously these two are partners in crime

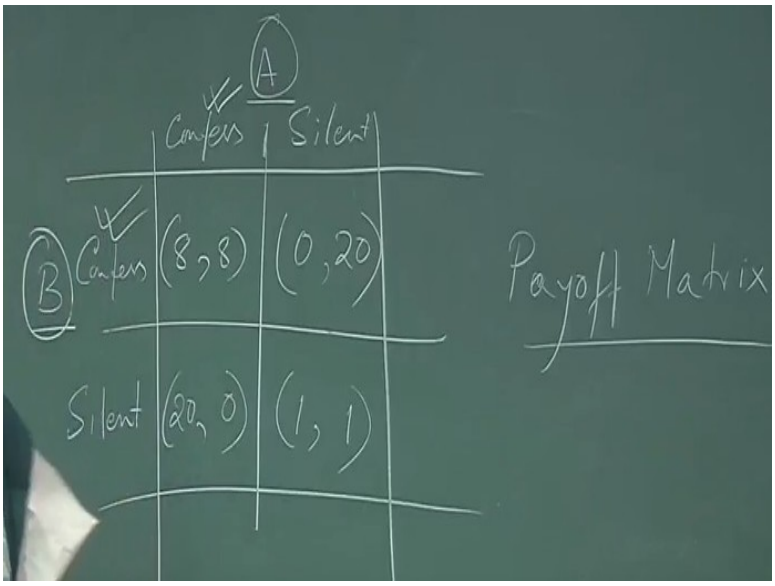
so the police knows that they have enough evidence against each other. They know what the other has done and probably if they help the police they will be able to provide enough information or evidence to the police.

So what the police does is they the police puts each of them in 2 different cells. They the police puts each of them in 2 different cells with no contact between each other and this is the offer that the police gives to each of them. To both A and B the police says that look if this is the deal what evidence we have is going to put you behind bars for 1 year.

Now if you confess and you provide enough evidence to against your partner so that your partner is implicated, in that case we are going to let you go free. So you are free. You do not have to serve any sentence and your partner goes behind bars for 20 years. But if you do not confess but your partner confesses and partner gives us enough evidence against you then you go behind bars for 20 years and he goes scot free and if both of you end up confessing, if you have also confessed and the other have also confessed and give evidence against each other, each of you is going to go behind bars for 8 years each.

So this is what the police offers to the 2 convicts A and B. Now what are they going to do? What is in their best interest? So from the game itself as you have already probably understood that it makes sense for them both of them to keep quiet. Had there been a probably a cell phone to talk to each other probably they would really agree that let us both of us let us be quiet although that is also not I do not know if you put yourself in the shoes of those convicts would you be trusting your partner or not. Is he going to tell on me? Should I keep quiet or should I talk? So this decision is what they have to make and this is the game and let us see what it looks like.

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So there is prisoner A and there is prisoner B. So they are they are convicts okay. So and so they have A and B. So they have choices, A can A has the choice to confess or keep silent. Similarly, B has the choice to confess or keep silent. Now what happens if both confess? According to what the police has offered to both of them if both confess then each gets a jail term of 8 years.

Now if A remains silent and B confesses then A gets 20 years in prison and B gets away scot free. So since B has confessed, so police will let him go without any punishment and A goes behind bars for 20 years. What happens when A confesses and B is silent? Just the opposite happens and B goes to jail for 20 years and A goes scot free.

And now when both A and B if they keep silent then both go behind bars for 1 year each. So from here it seems that it is probably in their joint best interest to keep silent. Now is it what they are going to do? Let us see. Now A is completely cut off from B and he cannot communicate with him so he just has to keep thinking what is probably B doing, what could probably B do and whatever B's action is, against that what action should I take? This is the way he is thinking.

Now A knows that B has 2 choices, confess or silent. So A thinks if B confesses what is the best strategy for what is the best thing for me to do given that B confesses. Now given that B confesses A's payoff is now when I say payoff let me a quick diversion this is called a payoff matrix in game theory. This is called a payoff matrix. This is called a payoff matrix in game theory where we have the payoffs for different players in each of the quadrants.

So now coming back to this game now that A knows that B has 2 choices either to confess or to keep silent. What happens if B confesses? If B confesses then A knows that if I confess when B has confessed, I am going to get 8 years in prison. If B confesses and I keep silent then I get 20 years in prison. So this is worse than this outcome so it makes sense for me to confess if B confesses.

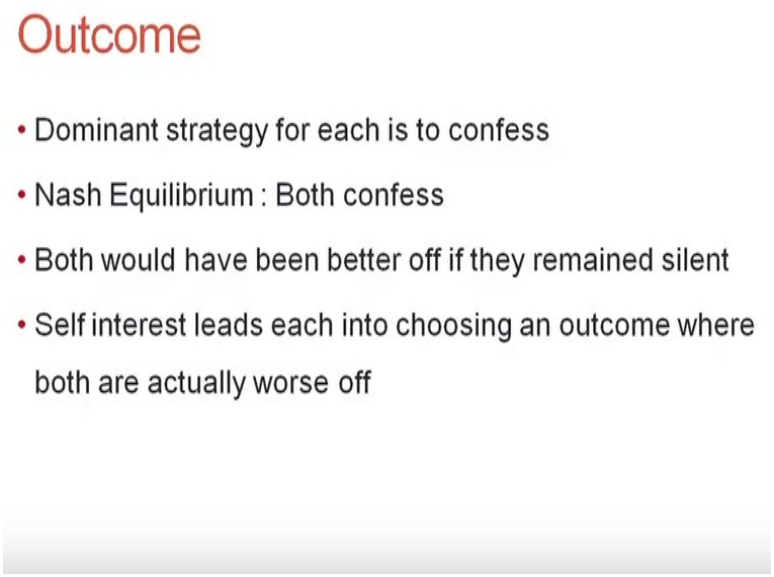
But what happens if B keeps silent? If B decides to keep silent what should I do? So A again looks at his own payoffs and sees that if B keeps silent I get 0 years in prison if I confess and if B keeps silent I get 1 year in prison if I keep silent. So here among the 2 payoffs, I am worse off if I keep silent and I am better off if I confess.

So in the situation where B keeps silent, then also it makes sense for me to confess. So in either case no matter what my rival does or what my partner in crime does it makes sense for me to confess. So A ends up confessing. Now coming to B, how does B think? B thinks in exactly the similar way. B thinks in exactly the similar way.

So B thinks that B is going to look at the different payoffs and B also thinks exactly the similar way that A has 2 choices, confess or silent. So if A confesses then if I confess I get 8 years. If I keep silent I get 20 years. If A keeps silent then if I confess, I get 0 years. If I keep silent I get 1 year. So for B also it makes sense to sorry so in that case for B also it makes sense to confess in both the situations.

So no matter what its rival does it makes sense for B to confess and both of them end up confessing as a result of which both of them end up in this box. So both of them end up confessing and both of them end up with a jail term of 8 years which is worse than this situation where they would have got a jail term of only 1 year each. So this is what happens in case of the prisoner's dilemma.

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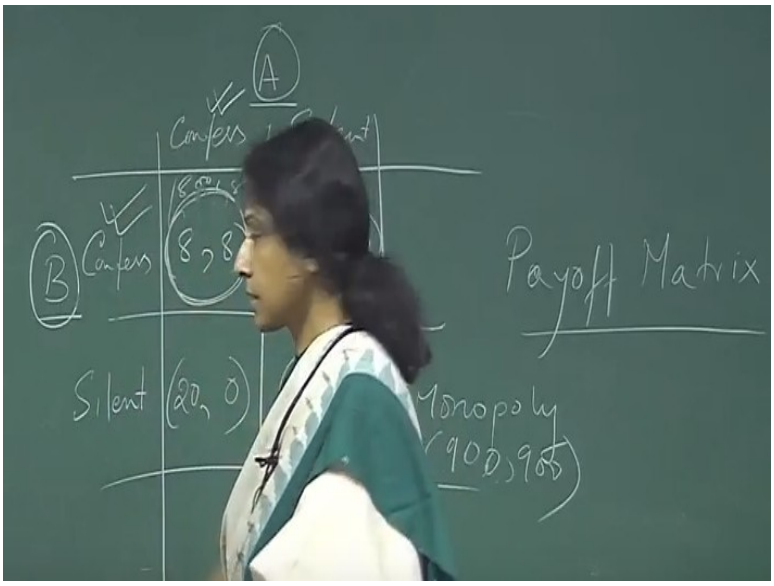


So the dominant strategy for each is to confess and Nash equilibrium is as per our definition of Nash equilibrium what happens in case of Nash equilibrium both the firms so each for the dominant strategy for each firm is to confess because that is if it confesses in both the situation confessing is the better choice, better strategy for A.

So that is the dominant strategy for each of the firms and Nash equilibrium is where both of them confess. Both would have been better off if they remained silent. Self-interest leads each into choosing an outcome where both are actually worse off. So self-interest leads each into choosing an outcome where both are actually worse off.

So this is the so this is the game theory in short and this as you can already probably relate to the oligopoly market outcome that we have already discussed this is the situation of the monopoly. This is the situation of the monopoly where had they cooperated then each would be earning a monopoly profit of 900 each.

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So the payoff would be 900 each but they cannot or each of them firstly they probably do not trust each other that the other is going to stick to the monopoly agreed monopoly output of 30 units so each of them is going to default and produce a higher amount of output and as a result of which both of them end up in this box which is the oligopoly outcome where each of them is going to get a profit of 800 each, so each of them is going to get a profit of 800 each.

So this is so it is very the oligopoly outcome is very similar to our outcome in the case of the prisoner's dilemma and this is the kind of dilemma that the different firms are going to go through when they compete with each other in a oligopolistic setup and they end up at a equilibrium, market equilibrium which is not the monopoly outcome and thankfully for the consumers because they get to pay a lower price and they get a higher amount of output and as the number of firms keep on increasing this payoff goes down and down and ultimately they are they the price is very close to marginal cost and output is very close to competitive output.

So this was our this was an introduction to the game theory. In the following modules we are going to take some examples of game theory and see what happens in different situations in a game theory and we are also going to look at repetitive repeated games what happens if the prisoners are repeated repeatedly caught.

So this could be one single situation where they are caught off guard and police have been able to catch them and put them into cells before they had any chance to communicate with each

other. Also going back to this example imagine that they somehow, A and B somehow know that the police is going to pressurize them to own up and the police is probably going to pressurize them to give enough evidence of talk about the crime and give evidence to the police and before going before getting separated both of them agree that I am not going to talk.

This is what they agree and then they are separated and then the police lays out this offer to each of them. In that situation is the outcome going to be different? Is the outcome going to be different? It is unlikely that the outcome is going to be different because each of the prisoners know that by any chance if the partner defaults then this defaults or if the partner cracks up and talks then he has to go to prison for 20 years.

So the amount of punishment that he is going to get if he keeps silent and his partner talks is so big that he would not take the chance and he is probably going to confess. So even if they have an agreement like in case of Cartel where even if they have an agreement of producing monopoly output they end up producing higher output.

Similarly, in this case also even if the partners have discussed amongst themselves not to talk, probably they are going to talk because firstly there is distrust they do not trust their partner fully and secondly they are going to act on their own self-interest. So self-interest comes foremost and it makes sense for them to confess and this is going to be the Nash equilibrium in both the situations whether the two the two criminals get to communicate with each other before talking, before being taken away, before being separated or they do not get to communicate with each other before being separated.

So this is what the game theory game looks like and we are going to discuss a few more games in the following modules. Thank you.

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Lecture - 39

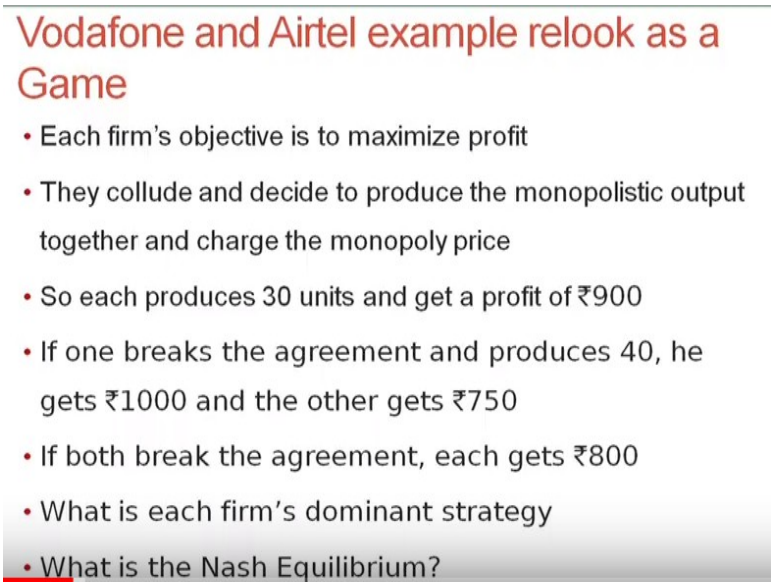
Oligopoly – Game Theory (Contd.)

Hello and welcome back to our discussion on oligopoly. We have looked at possible oligopoly model where we have looked at the traditional marginal cost, marginal revenue curves, what profit maximization would mean to an oligopolistic firm and we have looked at kinked demand curve and then we went on to understand a little about game theory because we said that in case of oligopoly it is more like a game for every individual firm where the firm has to keep not only a constant watch on its own demand and cost but also a constant watch on the price and quantity movements of its rivals.

And that is the reason that it is more like a game where the oligopolistic firm has to constantly make its own strategy or action plan to respond against different moves of its rivals. So we looked at the typical very traditional game of prisoner's dilemma which shows that why the 2 prisoners or why 2 individuals can settle on a less than optimum situation because of self-interest, they are driven by self-interest and so the choices that they make ultimately put them in a situation where they are not earning the maximum that they could have.

So this we started with game theory. Let us take a few more examples and see what are the situations, how the situations can change. Is there any possibility that the firms could actually stay put at a monopolistic situation. Is it possible for the firms to successfully collude? So we will see take a few examples and look at various possibilities.

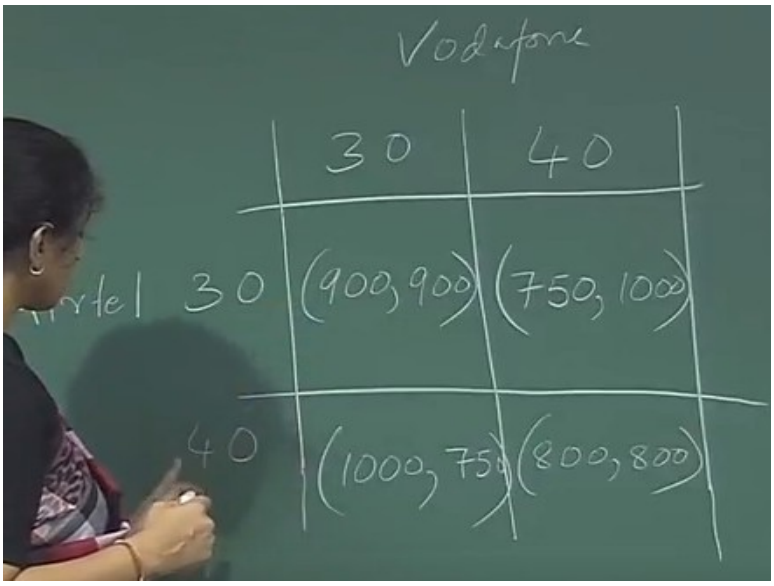
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So going back to our Vodafone and Airtel example, we look at the example, we relook at the example as a game. So now to quickly go through the example once more. Each firm's objective is to maximize profit. They collude and decide to produce the monopolistic output together and charge the monopoly price.

So each produces 30 units and gets a profit of 900 Rs. If one breaks the agreement and produces 40, he gets 1000 Rs and the other gets 750. If both break the agreement, each gets 800 Rs. So what is each firm's dominant strategy and what is the Nash Equilibrium? That is what we are interested in and let us look at the same problem as a game and try to draw the payoff matrix for this example.

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So we have Vodafone here and we have Airtel here and what are the choices that they can make? So they have colluded and they have decided they know, each of them knows that if they stick to the monopoly price and if they stick to the monopoly quantity of 30 units each that is the total output of 60 units in the market in that case each can earn a profit of 900 Rs.

So one option for them is to stick to the agreement and keep producing 30 units each. So that is one of the options; keep producing 30 units each. Now the other option is since price is now fixed each of the firms could be tempted to produce a little more. So if he produces a little more than he gets a bigger share of the market and ends up getting a higher profit.

So that is another choice. Say for example they can choose to produce 40 units each. So the other option is to produce 40 units each and then what is the, what does the payoff matrix look like then? So if each of them produces 30 units then the payoff matrix has profit of 900 for each firm.

If each of them ends up producing 40 units, both are producing 40 units so that the total output in the economy, in the market goes up to 80 units, in that case profit drops to 800 and 800 each. Now if Vodafone decides that I am going to produce 40 units and Airtel decides I am going to produce, Airtel continues to produce 30 units in that case Airtel is going to get a profit of 750 because now it is producing less and Vodafone is producing more and hence Vodafone gets a profit of 1000 Rs.

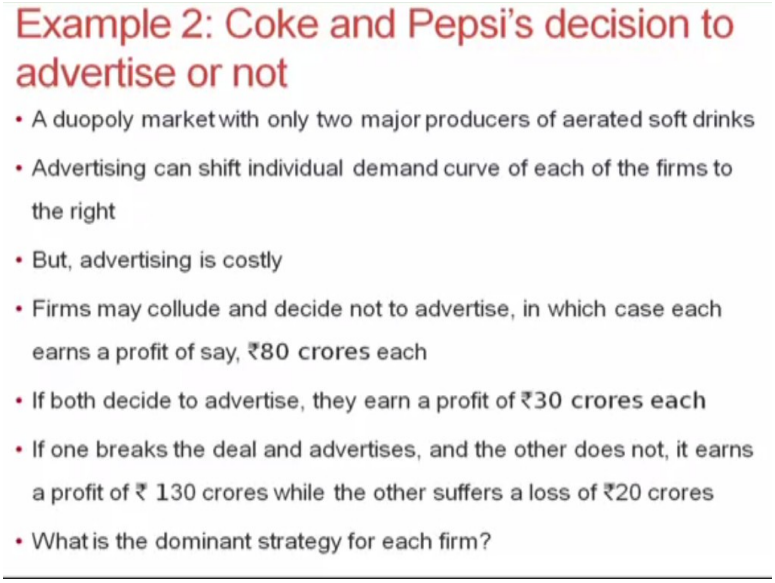
And the opposite happens when Airtel basically defaults on the agreement and produce 40 units and Vodafone sticks to the agreement and produces 30 units. In that case Airtel is going to get 1000 Rs and Vodafone gets 750. So this is what the payoff matrix actually looks like. So if we try to think about the decision making process of each of the firms in the framework of a game then the payoff matrix looks like this.

So since this is the payoff matrix how do they decide? So Vodafone decides that I have a choice either to produce 30 or 40. What should I do if Airtel has produced 30? So if Airtel has produced 30 it makes sense for me to produce 40 because in that case I will be getting a higher profit. What happens if Airtel has produced 40?

What happens if I cannot trust Airtel and Airtel actually ends up producing more. So what happens in that case, what should I do, what should my strategy be? So in case of, so the Vodafone now sees, looks at the payoff in both the cases and sees that if Airtel produces 40 I will be worse off if I continue to produce 30 and I should better be producing 40 so that I am at least I am at least getting 800 Rs which is same as what Airtel is going to get.

So in both the situations no matter what the decision of Airtel is, it makes sense for Vodafone to produce 40 units and same is the case for Airtel. Hence 40 units is the dominant strategy for each of the firms. So each of the firms it makes sense to produce 40 units and so this ends up becoming the Nash Equilibrium for both. So this is the Nash Equilibrium for both the firms. So this is how we look at the decision making in the framework of a game.

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Let us move on to another example, let us take up another example where basically it is a duopoly situation, it is a duopoly market with only 2 major producers of aerated soft drinks. That is we have the Coke and the Pepsi. They are the only ones who are supplying aerated soft drinks to this market and they have to make a, take a decision of whether to advertise or not.

So we have already spoken a quite a lot about advertisement, advertising, the pros and cons of advertising from the point of view of the firm also and we have seen that it is possible for the firms to attract a larger amount of customers to themselves by just advertise through just by advertising.

So it is a kind of product differentiation a notional differentiation that the firms are able to create in the minds of the customers. So now the but at the same time advertising can turn out to be really costly for the both the firms so it is the decision that how much I am going to gain by advertising because my demand is going to go up against how much am I going to lose because my cost is also going to go up because I will be spending so much money on advertising.

So each of the firms knows that they would have been better off if none of them advertised. So if none of the firms advertise in a market then no one needs to spend anything, so the demand is basically continues to be the same for everyone so there is no possibility of one firm poaching on the customers of other firms just to advertisement.

So this is the decision that the since there are only 2 firms in the industry it is one could imagine that the 2 firms would collude or cooperate and say that see it is costly for both of us so let us not advertise. So let us see what happens. So advertising can shift individual demand curve of each of the firms to the right but advertising is costly.

Firms may collude and decide not to advertise in which case each earns a profit of say 80 crore rupees each. So let us assume that this is the amount they earn each of the firm is earning if they are not advertising at all. Now if both decide to advertise, they earn a profit of Rs 30 crores each.

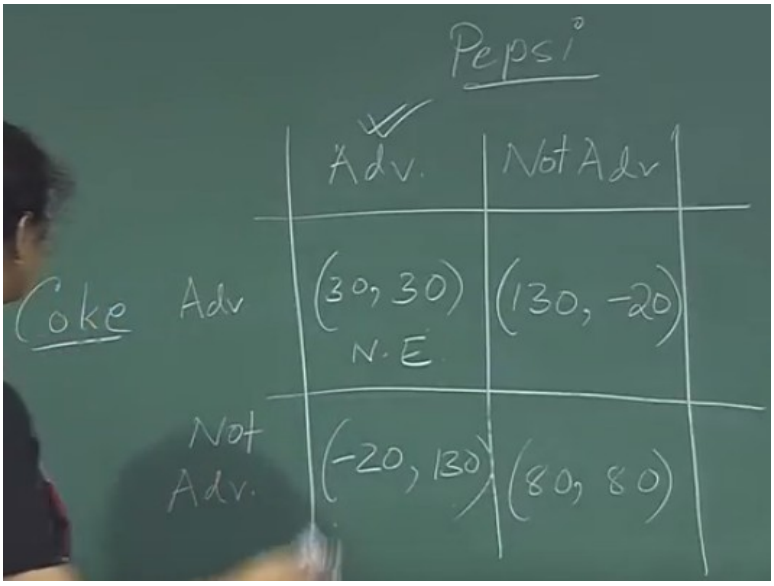
Why has the profit gone down? Because it is costly for both the firms and since both the firms are advertising so the impact on demand for both the firms is probably cancelling out and the net result is that both are incurring higher cost without much increase in demand for both the firms. Now what happens if they decide that let us advertise and then one firm basically breaks the agreement and goes on to advertise.

In that case what is going to happen in the market is although the firm which is actually breaking the agreement his demand is going to go up because he has advertised and the other firm has not advertised so he gets to he successfully is able to attract a lot of customers of the other firm which is not advertising.

So in that case since the other firm's customer base falls that is his demand curve shrinks and the demand curve of the firm that is advertising shifts to the right the firm the advertising firm gets a high profit of say 30 crores, while the firm which has not advertised he suffers a loss of say 20 crores, I am sorry it is it would not be 30, it should be 130 because now the profit is very high; he, it is earning a high profit.

So these are the so if one breaks the so if one firm breaks the deal and advertises and the other does not, it earns a profit of 130 crores while the other suffers a loss of 20 crores. So what is the dominant strategy for each firm in this case? So again let us look at the payoff matrix for each of the firms.

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So we have Pepsi here and say these are the 2 players and their options are to advertise and no advertise. Again here Coke also has similar options of advertise, not advertise. So these are the options that they have and as per our payoff matrix so if they if both advertises then each gets 30, 30. They do not advertise then each gets 80 and 80.

If Pepsi does not advertise and Coke advertises then Coke gets a very high profit of 130 and Pepsi has a loss of Pepsi incurs a loss of 20 crores, so we denote it as -20 and in case where Pepsi advertises and Coke does not advertise it is just the opposite. So it is -20 for Coke and 130 for Pepsi.

So again so it is now you probably know you have probably understood the exercise how we are going to decide what is the dominant strategy for both and we see that the dominant strategy for Pepsi is to advertise because when Coke advertises Pepsi's payoff is more here in case of advertising; when Coke does not advertise, then also Pepsi's payoff, Pepsi's payoff is more in case of advertising.

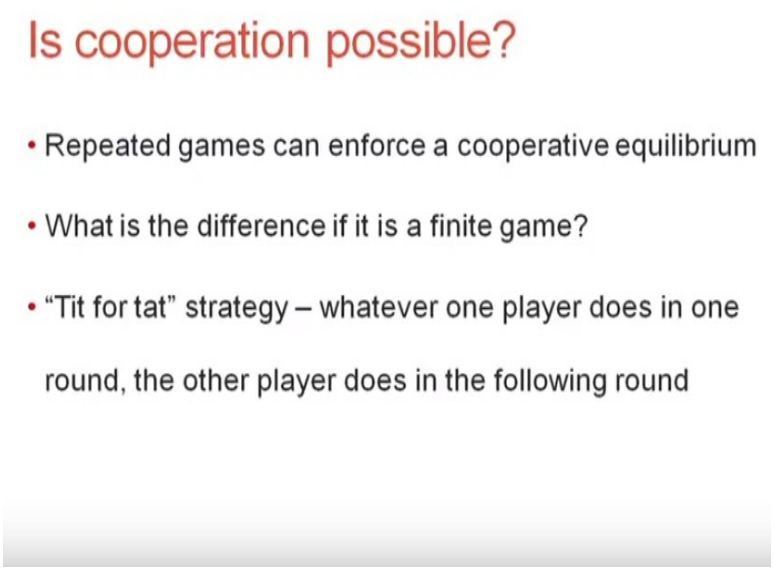
So advertising is the dominant strategy for Pepsi and the same for Coke also when Pepsi advertises Coke's payoff is 30 and in case of not advertising it is sorry and Pepsi not advertising it is -20. So in case of Pepsi also it is for Coke also it is advertising and in case of Pepsi also it is advertising.

So what they end up is this is the dominant strategy and this becomes their Nash Equilibrium. So this becomes their Nash Equilibrium. So although had they stuck to the agreement of not advertising each would be better off by getting a profit of 80 and 80, but since each of them acts on their self-interest and each of them is not being able to trust the other completely they end up in this box where they end up with a profit which is much lower than the collusion situation where they successfully collude.

So this is the example of Coke and Pepsi's decision. Now the question is does that mean that cooperation is never possible? Does that mean that cooperation is never possible because had that been the case then probably we would not be having, we would not be needing a law to stop Cartelization.

So Cartel's successfully sometimes run and that is the reason that the competition law is always watchful of firms if they are having any collusion and charging a monopoly price or quantity in the market. So there is a possibility of collusion and let us look at a few situations where cooperation may be possible.

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Now repeated games can enforce a cooperative equilibrium. How so? So let us take the same example and look at what happens. So let us go back to the same example. Now what happens

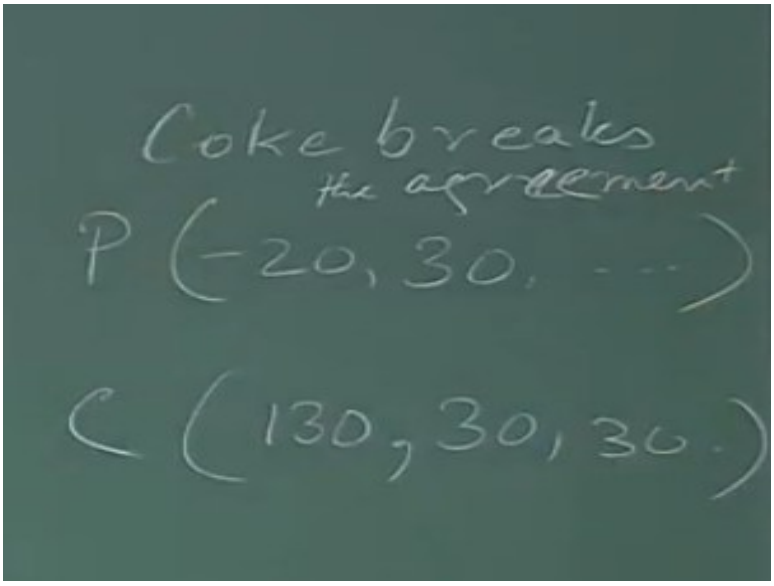
this is a one-time situation. So this is a one-time situation of Pepsi and Coke deciding or not deciding to advertise and then in this situation this is the payoff that we are this is the equilibrium that we are reaching.

Now what happens if it is a repeated game which means that this is a decision or this is a choice that Pepsi and Coke has to make every year. Say for example every year they have to make this decision whether to launch a new advertisement, new campaign or not. So this is a repeated game and in that case if it is a repeated game, then what is possible?

So in a repeated game now the Pepsi and Coke both know that if I break the agreement in the following period, if I break the agreement my rival is also going to break the agreement. So in that case I am worse off later. So although so let me give an example.

Say for example Pepsi and Coke they, say for example Pepsi tells Coke that let us collude, let us decide not to advertise but if you advertise, if you break the agreement then in the following all the periods I am going to advertise. So if this is a threat that one firm gives to the other in that case cooperation is possible, how so?

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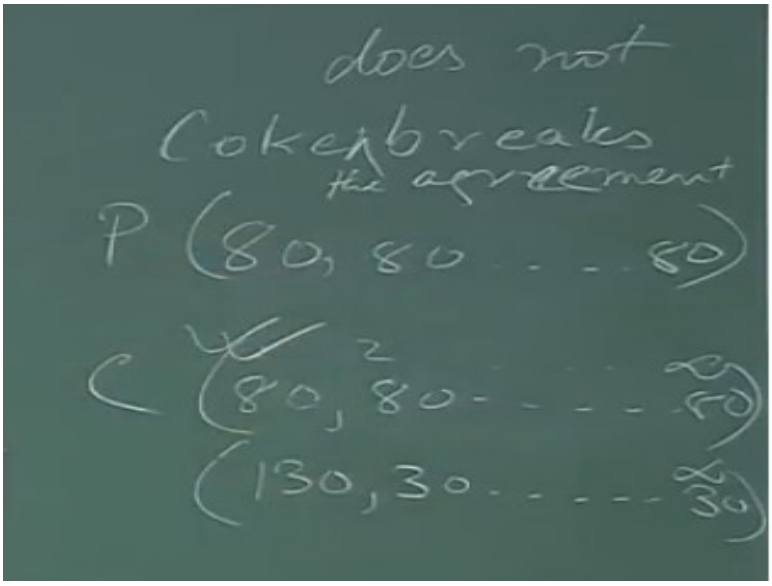


So let us look at their payoffs. Now say for example Pepsi gives the threat and both decide that we are not going to advertise, so for Pepsi and Coke. Now Pepsi has given Coke this threat. Now Coke if it decides that if I break the agreement then in the first period I am going to earn in the first period I am going to earn 130 crores and Pepsi is going to make a loss of 20 crores. In the following periods so once Pepsi suffers this loss Pepsi is never ever going to not advertise.

So Pepsi is going to advertise in all the following periods because that is what Pepsi has threatened to do. So in that case in all the following periods Pepsi will be earning a profit of 30 crores and Coke will also have to earn a profit of 30 crores because then it does not have any other because it knows that Pepsi is going to advertise in the following period so Coke is also going to advertise in the following period and all the consequent periods they are going to get a payoff of 30 each.

So this is the situation when Coke decides to break the agreement. Now what is the payoff if Coke does not break the agreement? So this is where Coke breaks the agreement. So what happens if Coke does not break the agreement? So let me write it here itself.

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So when Coke does not break the agreement, in that case each earns 80 crores in all the periods. So in that case, so in all the periods, in all the periods both are going to earn 80 each. So obviously to Coke this is a better situation than 130. So this is a better situation than this. This is a better situation than this because earning 130 in the first period and 30 in all the consecutive periods compared to earning 80 in all the periods is always so this is always a better choice for Coke, so Coke is going to stick to the agreement.

So this is where cooperation is possible and why is cooperation possible here? For Pepsi this is a worse situation. For Pepsi this is a worse situation because for Pepsi it earns a loss in the first period. So one thing that needs to be kept in mind is the objective of the firm is to maximize its own profit.

The objective of the firm is not to harm its harm its rivals or not to create a situation where the rival gets a earns a loss. So that is not the objective of the firm and since the objective of the firm is to maximize its own profit so none of the firms is going to break the agreement. So in case of repeated games, repeated games can enforce a cooperative equilibrium.

So even going back to the so this is although this is a situation of threat where actually I said Pepsi imposes a threat on Coke that I am going to always advertise later but that is not always necessary because sometimes the players learn through the game. Say for example in the case of prisoner's dilemma it is possible if they are repeatedly getting caught and repeatedly the same

thing is happening probably they will learn over the overtime during the repetition of the games that it is better not to talk. So repeated games can enforce a cooperative equilibrium.

Now what is the difference if it is a finite game? So here I said it goes on till so they know that forever from now we are going to earn this profit of where no one advertises. But what happens if say for example in the same situation where Pepsi has given a threat that I am going to advertise in the later periods and all of a sudden both of the firms come to know that in 10 years' time the government is going to pass a law that no one can advertise aerated soft drinks.

The advertisement of or promotion of aerated soft drinks is banned. So they know advertising will be banned from on the 10th year. Say note know that advertising will be banned on the after the 10th year. So till the 10th year from this is the 10th year and beyond that advertising will be banned. So in that case what are they going to do?

So in that case Pepsi just go to the situation of the 10th year now Coke knows that in the 11th year Pepsi cannot exercise the threat because advertising will be banned in any case. So to the in the 10th year Coke is going to break the agreement. So Coke is going to break the agreement in the 10th year and Pepsi also knows the same.

Pepsi knows that Coke is going to break the agreement in the 10th year. So Pepsi is also going to advertise in the 10th year. Now come back a year earlier in the 9th year Pepsi knows that Coke is going to break the agreement in the following year so and Coke also knows that he is it is going to break the agreement in the following year so they both advertise in the 9th year.

So this thinking or making or strategizing in this way or planning in this way, each of the firms come back a year earlier and they know that in the following period the agreement will be broken and the knowledge that the agreement will be broken at a later year basically breaks down the agreement in the very first year.

So (in the) from the very first year itself both are going to advertise and or this threat does not hold if there is a finite time for which the threat is applicable. So in that case both are going to advertise from the very first period for the 10 years and after that in any case there is no advertising. Okay another strategy is tit for tat strategy. Whatever one player does in one round, the other player does in the following round.

So this is also a situation where cooperation is possible because the firm learns from the reaction of its rival. So the year that it has advertised following year the rival is going to advertise. If it does not the other also does the same. So it knows that it can trust the other firm to do exactly the same thing that it does in the first year so in that case cooperation is possible because in that case cooperation in case of cooperation both of them are better off.

So this was about game theory. In the following module we are going to look at different ways of pricing in case of an oligopoly firm. Thank you.

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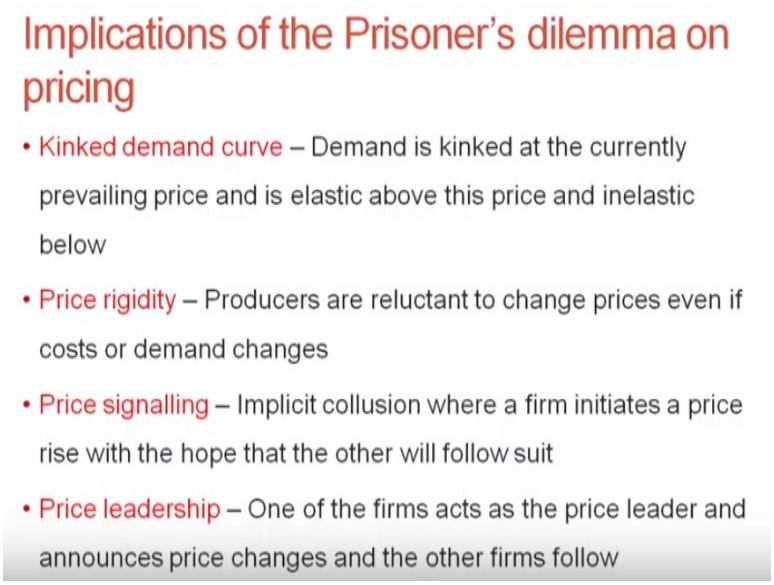
Lecture - 40

Oligopoly Pricing

Hello and welcome back to our discussion on oligopoly. We have looked at possible oligopolistic decision making process through game theory and in this module we are going to look at pricing in case of oligopoly. So if oligopolistic firm behaviour is ruled majorly by the moves and actions of the rivals then how do the oligopolistic firms decide what prices to charge and what output to produce.

So there are certain implications of the prisoner's dilemma on pricing. So we have seen already with lot of examples that because the prisoner's dilemma exists that is the dilemma whether to collude whether to trust the other players and stick to the agreement or whether to go by one's own self-interest this is the dilemma that all the firms are facing and that is the reason that pricing choices could be different.

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So what are the implications of this dilemma on pricing in case of an oligopoly firm? First is the kinked demand curve. So summarizing implications of the prisoner's dilemma is the first thing is the kinked demand curve which we looked at a few modules back. So the demand is kinked at the currently prevailing price and is elastic above this price and inelastic below.

So the demand is kinked at the currently prevailing price because here is the dilemma for the firms whether to change the price or not and we have seen that the firms they know that if they would like to increase the price the other firms are probably not going to follow suit in that case this particular firm is going to lose all its customers to the other firms.

Hence the demand curve is flat or highly elastic above this price. That is if the price is to be increased the laws of demand for this firm is going to be really huge but on the other hand if the firm is to reduce the price then it knows that probably it is not going to get a huge increase in output or a huge increase in demand for its product because other firms are also going to follow suit in this in reducing the price and hence the impact on quantity or the impact the output effect that we say that when there is a price decline the output is going to increase.

That does not happen and the firm kind of faces an inelastic demand curve below this price. So it is all but one thing which needs to be kept in mind is the firm faces this kinked demand curve because it perceives the actions of its rivals in this way. The firm perceives that the rivals are

probably not, probably not going to increase the price if it increases the price and the other firms are probably going to reduce the price if the firm reduces its price.

So acting on this speculation the firm kind of faces a kinked demand curve. So this kinked demand curve results in a price rigidity. We have seen that the kinked demand curve has a marginal revenue curve which has a break in the in between. So it has a break where the kinked happens and in this break and in this range we have shown through diagram shown through theoretical modeling that if the marginal cost fluctuates within that range if the marginal cost is to go up in that range the firm is probably not going to increase its price.

So the firm is sticky or rigid about changing its price for a certain range of costs. It is probably going to take the cost on itself and not pass it on to the consumers for the fear that it is going to lose its consumers to the other firms. So this is called price rigidity which the producers are reluctant to change prices even if costs or demand changes.

Another 2 more things that can happen because of prisoner's dilemma on pricing so since all the firms are acting with this prisoner's dilemma in oligopolistic setup it is possible for the firms to go for tacit collusion. So that is possible and 2 examples are one is price signaling and other is called the price leadership. So basically as we have been repeatedly saying while discussing oligopoly that the oligopolistic firm keeps a watch on the price movements of its rivals.

But at the same time explicitly colluding with the firms and deciding on a monopoly output and monopoly price is not a proof by law. So they cannot do that and it is obviously not legally enforceable also. But there are situations where the firms signal to each other that I would so although the firms are firms fear the fear that they are going to lose consumers when the price when they increase the price yet they would like to signal (to its) to their, to the other competitors in the market that I am going to increase the price and so do you?

So this is the kind of price signaling so the firms might feel that if I increase the price it is possible for the firms to increase their other firms to increase their price as well. So this is called price signaling where it is an implicit collusion where a firm initiates a price rise with the hope that the other firm will follow suit.

So this is a very common phenomenon in an oligopolistic market structure and it happens it is a kind of tacit agreement and price signaling can happen in a lot of ways say for example in the airlines industry it may happen that one of the airlines is contemplating a rise in its prices in one of the routes and it raises the price but keeps a watch on the other competitors if they are following suit.

Now one possibility is that the other firms emboldened by the price rise of this one firm is by price rise of one of its competitors they also increase their price. So if they increase their price so the objective of the first firm increasing their increasing its price is fulfilled. Now what happens if the other firms do not increase their price?

If the other firms do not increase their price so usually what happens in case of the airlines industry they are going to try this price increase during say the weekend. So during the weekend when the fare fares are low they are going to try this and if they see that the other competitors in the market are not increasing their price they will immediately come back to the lower their price and come back to the original price.

Now then again in the following weekend it will again increase the price so that is a very clear signal to the competitors that you have to increase your price otherwise I may go for deep discounting and actually I may end up getting your customers. So this is the kind of price signaling which happens in the market but one thing that needs to be kept in mind here is price signaling is not always about illegal collusion or the firms going for monopoly pricing.

It could have a economic rationale behind it. Say for example this has been very citing the example of Professor George Hay of Institute of Competition Law where he says he cites the example of 2 petrol pumps. Say imagine that there are 2 petrol pumps on a highway and there are no petrol pumps for miles altogether. So all the cars and vehicles which are passing through the highway they are dependent on only these 2 petrol pumps. So it is an oligopolistic market in this entire locality.

So for quite a few kilometers these are the 2 petrol pumps in the market, the only 2 petrol pumps in the market who are splitting the market between themselves. So now imagine that each of the petrol pumps is charging a price of say 68 Rs per litre of petrol.

Now one of the petrol pumps, now each of the firms they know, each of the petrol pumps obviously they are splitting the market they are in the same region same locality so their cost structures are also probably the same and they are just splitting the market between themselves. So may be the equal number of customers are almost going to each of the petrol pumps.

Now they know each of the firms each of the pumps they know that they could act as a monopoly if they would if they want to because in that entire area there is no other petrol pump and the all the cars are completely dependent on these two.

So one of the petrol pumps knows that it is possible to charge a price of 72 Rs per litre of petrol instead of 68 and it would really like to do so because it knows that if I increase the price to this much we can I can have monopoly profit and basically and if the other pump also follows the same price then both the firms together we are going to get a monopoly profit and we are going to split the number of customers equally amongst ourselves.

But there is no way it can communicate this directly to the other pump. That is illegal, it cannot do that. So what it does is it goes for price signaling in the market in the sense that it is going to increase its price on its own without making any without informing or sitting face to face with the other pump and deciding on the price it just takes the chance to increase its price.

Now is it not risky? Yes, it is risky for this firm for this pump say let us call this pump A. Now it is risky for pump A because as soon as it increases the price it will lose a whole lot of vehicles who will immediately go to pump B if pump B continues to charge 68 instead of 72 as the pump A is charging. So that is a huge risk that pump A is taking.

Now from the point of view of pump B it is very similar to the Pepsi and Coke example that we took. Now what are the choices in front of pump B? Now pump B knows that if it does not follow the rise in price by pump A so for a brief period probably for 5 days it is going to get the major of the vehicles on the highway.

But after that probably the other pump is going to go back to its lower price of 68, will continue to charge the lower price, will continue to charge the same price as pump B and they will go back to earning the profit that they were earning earlier. So and the other option to pump B is to

since A has already raised its price let me also raise my price and it knows in that case it is going to get a monopoly profit which will be higher than the profit that it was earning earlier.

So in that case both A and B continue to earn a higher profit for all the periods to come. In the first case where pump B decides not to raise its price it earns a higher profit only for a very brief period. So in that case it makes sense for pump B to raise its price also and both the firms have been able to raise their price successfully and continue to earn a higher profit.

So this is where this is a form of price signaling where actually firm A or the petrol pump A has not communicated anything to the other pump, it has acted on its own, taken a chance taken a risk, acted on its own and the other pump just followed it. So this is a price signaling that can happen in the market.

Another form of tacit collusion is price leadership. Price leadership, one of the firms acts as the price leader and announces price changes and the other firms follow. So what happens in the case of a kinked demand curve we said that there is a range between in range where the firms are wary of changing the price although the costs may increase or the demand may have come down. So they are reluctant to increase the price.

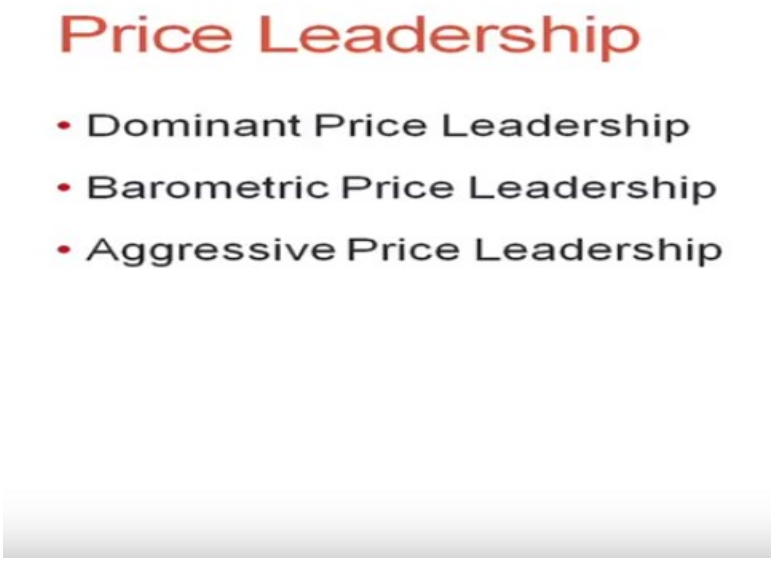
So they fear increasing the price. But it is not always clear to the firms where exactly is the end of the kink or end of the break in the marginal revenue and where it is safe to increase the price and the other firms are going to increase the price. So in that case in an industry or in a market what usually happens is the firms basically follow a price leader.

They basically follow a price leader so if one of the firms has increased its price, so everyone follows suit because they know that the situations are such that it makes sense to increase the price and for one thing in case of price leadership in most circumstances the unless it is a very aggressive price leader or a very aggressive firm which has decided to cut prices to emerge as the dominant player in the market in most other situations the firms are going to increase the price only when the costs have increased or demand has gone down and which is going to probably affect the entire industry as a whole in the entire market as a whole and that is where one firm is going to take the initiative of increasing the price and the other firms basically follow.

So one typical example is in the case of the banks. Banks have this prime lending rate. A prime lending rate is the rate which the banks charge to their the most established biggest corporate clients. Now this prime lending rate is dependent on the money market in movements in the money market of the economy and is dependent on all the rates of interest, in the, all the rates of interest are basically correlated to this rate of interest in the in banking.

Now the banks again since they could oligopolistic banks few banks who are lending it is possible that they would be reluctant to change this lending, prime lending rate because they do not want to lose these big corporate customers. So there could be a price leader who understand understands the changes in the money market and knows that the rates are going to be high for some periods to come because of changes in the economy and it takes initiatives to increase this lending rate and all the other banks follow. So this is a example of price leadership.

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Now to talk a little more in details, price leadership can be of 3 types. One is the dominant price leadership. Now dominant price leadership happens when there is one single large firm in the market. It is a dominant player in the market and it basically it is almost like a monopoly may not be a monopoly but it is almost like a monopoly and it is a very big firm and since it is a big firm catering to it has economies of scale and probably since it is a big firm which has emerged as a dominant player it has better and more efficient cost conditions and it decides its prices on the

basis of its own cost and demand in the market and the price that it charges since it is the dominant player in the market since it is a one large firm in this market and the price that it charges the other firms just have to follow this price because it this big firm is catering to the major part of the market so the firm the price that this firm is charging the other smaller firms have to follow this price and adjust their output and output accordingly because the other firms are just catering to the residual demand in this market.

But this is a dominant price leadership and the price leader here almost acts as a monopoly and it does not is it is not concerned about the cost or demand conditions of the other smaller firms in the market. So it is almost acting like a monopoly and it is not behaving like it probably does not have any prisoner's dilemma. So it is not going to consider the actions or reactions of the other firms in the industry and it is just going to change its price depending on its own cost conditions and demand and other firms are going to just follow suit and they are behaving like an oligopolistic like oligopolistic firm.

The second kind of price leadership is barometric price leadership. Now barometric price leadership can happen when there is a barometer firm. What do we mean by barometer firm? Basically this is a firm which probably gauges the changes in individual demand and cost first in the market. It is the first to understand changes in demand and cost conditions in the market and hence it basically changes the price and probably the other firms have accepted the role of this.

So this barometric price leadership this barometer firm may not be a dominant player in the firm. It may not a very large firm in the market yet it is possible for it to behave as the price leader because probably it has a history of judging the movements in the market, judging the movements of demand and cost in the market better than the other firms in the market and the other firms basically trust the price movement the price the price set by this firm and they also follow suit.

So barometric price leadership is another form of price leadership and third is the aggressive price leadership. Now in case of aggressive price leadership it is possible for a small firm, a small entrant in the market also to go for aggressive pricing that is it tries to establish its foothold in the market and accordingly it charges a very low price a competitive price and other firms are

forced to follow it because otherwise their customers are going to be taken away by this very aggressive pricing policy of this firm.

So that is a aggressive price leadership which can also happen in a market. So these are the different ways in which pricing could be influenced in a oligopolistic market framework and some of it is allowed by law because they are going by their economic rational of setting a price but collusion any explicit collusion amongst the firms to behave as a monopoly is not approved by law and that is something which the firms cannot do.

They cannot do this and that is the reason that basically collusion or Cartel formation they are intrinsically they are unsustainable. They cannot sustain for long because the firms do not trust each other and there they eventually at least one of the firms they it goes by self-interest and breaks the agreement which leads to breaking of the entire Cartel. Thank you.

Foundation Course in Managerial Economics

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Lecture - 41

Public Policy in Oligopoly

Hello and welcome back to the last module of this entire course and last module of our discussion on oligopoly. So we have so far discussed about situations where the oligopolistic firm does not go by its economic rationale of equating marginal revenue to marginal cost and ends up charging a price depending on what the other rivals are charging in the market.

We have looked at the kinked demand curve. We have looked at price leadership, price signaling etc. So we have seen that the oligopolistic market is somewhat different from the other kinds of markets where decision making process is solely dependent on economic rationale of equating marginal revenue to marginal cost if it is profit maximizing objective of the firm.

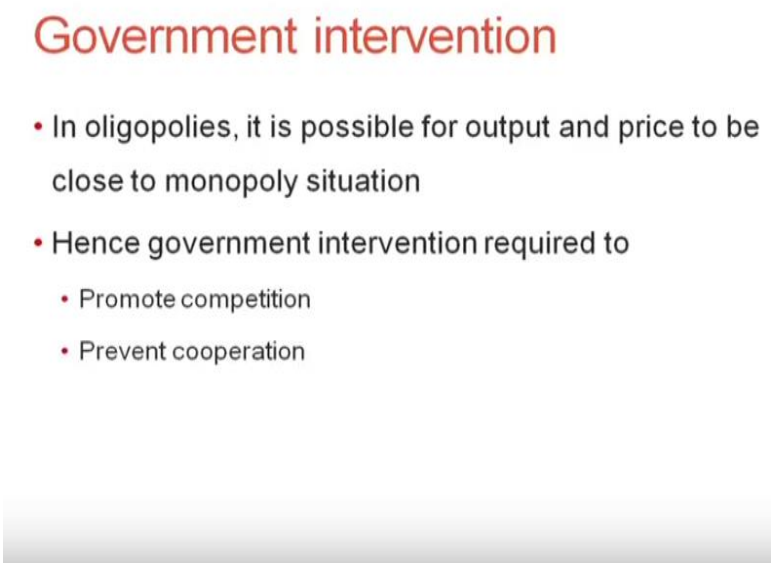
So in case of oligopoly, lot of subjectivity is involved in deciding what price to charge and what output to produce and we are going to see that in an oligopolistic market it is possible for the government to intervene. We have looked at government intervention in all kinds of market.

In perfect competition it is not required because that is the most efficient output that any market should strive to attend and in rest of all the markets in monopoly we have seen that there is requirement for government intervention so that the government can force the monopolist to produce a higher amount of output at a lower price.

In case of monopolistic competition, we have seen that since the firms although the firms are charging a price which is higher than marginal cost and but still the firms are charging a price which is equal to their average cost and their economic profit is equal to zero in the long run hence the government can cannot do anything much.

But in case of oligopoly we have seen that if there are few firms in the market there is a possibility of the firms colluding with each other and charging a monopoly price and producing a monopoly level of output. So government intervention is possible in case of a oligopolistic market so let us look at how the government intervenes.

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So in oligopolies it is possible for output and price to be close to the monopoly situation hence government intervention is required to promote competition and to prevent cooperation. So when we say to promote competition what we basically mean is in any oligopolistic firm we have said that oligopolistic market is defined with high entry barriers and the reason that the firms are behaving this way is because there are few firms in the industry and each firm can keep a track of the other firms and that is the reason that their decision making gets complicated and the resultant the market outcome is not exactly competitive outcome and competitive outcome or higher output with lower prices can only happen when a oligopolistic market moves more and more towards having more and more number of firms in the market.

And that is possible only if the barriers to entry are low and sometimes barrier to entry could be created by the firms themselves in the market. Barriers to entry can come from a lot of other things say information mismatch, lack of infrastructure; lot of things can happen and here is where the government can intervene to see to it that there are no artificial barriers to entry that have been created in the market by the existing firm so that other firms cannot enter the market.

So that if the existing firms say for example if the existing firms decide to go for predatory pricing that is they charge a price which is so low that it keeps all rivals out of the market. So that is a it is creating a artificial barrier to entry by charging a price which is so low that the other entrance will be deterred by this price to enter in the market.

So this is where the government can step in and prevent the firm from doing so and as a result remove this barrier to entry so that more firms can enter the market and raise the amount of output in the market and so this is a promoting competition and another is prevent cooperation.

That is if the firms cooperate amongst each other that is although that is good for the consumers it is not good, it is, if the sorry, if they cooperate, the firms cooperate with each other it is good for the firms because they end up getting monopoly profit but it is not good for the consumers because they have to charge as they have to pay a higher price and they get lesser choices, lesser amount of output in the market.

So preventing cooperation is another thing which the government tries to do through competition laws like the government is going to say stop firms from entering into mergers where there is a possibility that the firm will end up being the only sole player in the market. So these are preventing cooperation preventing Cartel formation, preventing price fixing; these are the things that the government likes would like to do so that the existing firms do not charge high prices and produce monopoly output.

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So which are the laws around the globe which uphold competition and forbid collusion like the earliest is probably the Sherman Antitrust Act of 1890 which forbids collusion between competitors. We have Clayton Antitrust Act again from US in 1914. These are the acts the competition laws of US under these acts and the Clayton Antitrust Act strengthened the rights of individuals damaged by anticompetitive arrangements between firms.

In India we have the competition law which is relatively newer than in the developed countries' laws. These laws were existing long back in the developed countries of US most European countries but in India the competition law came into being in 2002 after abolishing the monopolies and monopolistic and restrictive trade practices act the MRTP Act was abolished and we had the competition law in price in India.

And in China it is the NDRC. Even in China many countries, many developing countries are gradually recognizing the importance of creating a competitive business environment in the economy and hence having these laws in place which are which have the objective of creating a competitive environment, creating a level playing field for all the firms in the economy.

Now one important thing in case of competition law why was the Monopolies and Restrictive Trade Practices Act abolished and in place we have the competition law because in case of MRTP monopoly itself was considered as illegal and monopoly itself only the public sector firm could be a monopoly and no other firm could be a monopoly because monopoly per se was

considered to be bad but as we have seen during the our discussion of natural monopoly in this course that sometimes it is beneficial for the consumers to have one single firm catering to the needs of all the consumers because the firm is able to reap the benefits of economies of scale and pass it on to the consumers.

So monopoly per se is not bad. What is bad what is considered bad by the competition law is if a dominant player, if a monopoly player is abusing its position of dominance and charging or charging a higher price to the consumers or charging or producing lesser output in the market or in any other way abusing its position as a dominant player in the market.

So one example is the competition law where the real estate giant of DLF was tried under this law where one of their apartment building's Resident Flat Owners' Association, the Belaire Owners' Association they filed a law suit against DLF for abuse of dominance and they actually won the case and Supreme Court also upheld the decision of the competition commission of India and DLF was charged as was found guilty of abuse of dominance.

Now here what kind of abuse was done here basically DLF being a dominant player in this market they had gone ahead not honoring the agreements that was signed between the flat owners and DLF and eventually the flat owners had to face a lot of loses. They got the flat's, the timely delivery of the flats was not done and whole range of similar reasons were given against DLF and DLF could, it was thought that DLF could afford to do so because it was a dominant player in the market.

And but one thing which is of importance or which is of relevance here for this course is when we say that a firm is a dominant player or when we say that like in the previous module we said there is a dominant firm which is the price leader or say in the example of the 2 petrol pumps that we talked about on the highway there are only 2 petrol pumps and they are almost behaving like monopoly because there are only 2 firms instead of one and for a very huge area there is no other petrol pump.

So that does not mean that petrol pump itself is a monopolistic or oligopolistic market. So basically this kind of puts things in perspective in the sense that throughout the entire course we have kept talking about market and we have kept talking about product. Now the questions is

when we look at competition when we looked at look at pricing's decision for different firms the question is how are we to define if it is a monopoly market, if it is competitive market, if it is a monopolistic competition, if it is an oligopoly.

So basically it means that all of this distinction depends on what is the exact nature of the market that the firms are operating in and what is the relevant product that the firms are selling. So in that context we can only say if a firm is operating as a monopolist or operating as a oligopolist. Say for example had these petrol pumps been in a very crowded middle of the city. Then probably it is a monopolistic competition. They are all very similar, lot of petrol pumps around and the buyers have can choose between the petrol pumps. So say if one petrol pump it provides a free car wash, it is basically product differentiating and most of the drivers would probably like to take their cars there. So it is a monopolistic competition.

But the same petrol pump if it is in the middle of nowhere, it is a huge area, completely remote area where there is no petrol pump inside for long distances and cars are, long distance cars are travelling on the highway, there the petrol pump can afford to behave like a monopoly and charge a very high price.

So all of this is in the context of what the market looks like and what is the product that it is selling. So coming back to the example of DLF that I gave when the case was tried in court the first thing that the court needed to decide was so it had to be so if the Belaire Association or the flat owners' association was accusing DLF of abusing its dominant position, first it had to be proved if DLF at all had a dominant position and if dominant position then in what market, in which product?

So then that had to be decided by the court of law first. So the court basically said it is selling these high-end apartments to a particular category of customers who can afford to buy such expensive, luxurious apartments. So that is the product that they have defined first. Second is which is more important in this case is it has to be defined or decided if DLF is the dominant player in the market.

So what do we mean by dominant player and if you may remember we had discussed about concentration ratio. We said that what percentage of the output is produced by the 4 major

players in the market. So it seemed very easy then but now the question is what product are they selling and so it is it the market for only apartments? Is it the market for both commercial and residential apartments? Is it the market for only high high-end, luxurious residential apartments?

So that has to be narrowed down first. Secondly as in the case of the petrol pump, geographic location also matters in case of the market. Now in case of DLF, DLF probably argued that we have we are in real estate business all over the country. So that is the market we are operating in and so the amount the percentage share in the market that we hold is much less than what it is nowhere close to being the dominant player because there are so many big real estate giants all over India.

But here the question is DLF is selling these apartments in Gurgaon which has a specific characteristic of its own, it is close to Delhi and it is a very major upcoming official corporate hub and it has its own clientele or own requirement of residential apartments and so the market in Gurgaon cannot be compared to market in any other place.

For example, a person, a consumer or a customer who is deciding to purchase a apartment in Gurgaon is not going to, is unlikely to consider an apartment anywhere else say in Bangalore or Calcutta or Mumbai as a substitute. Probably he is considering apartments only in Gurgaon. So in for this case DLF is the dominant player only in Gurgaon or the market that is to be considered for this case for Gurgaon for DLF is Gurgaon and no other place.

So this is what is the reason that I went into the details of this case is basically it is not to talk about the case but to bring to your understanding that definition of the market and definition of the market as we narrow down the product, as we narrow down the market the real existence of the firm as a monopoly or an oligopoly or a monopolistic competition that becomes more and more clear.

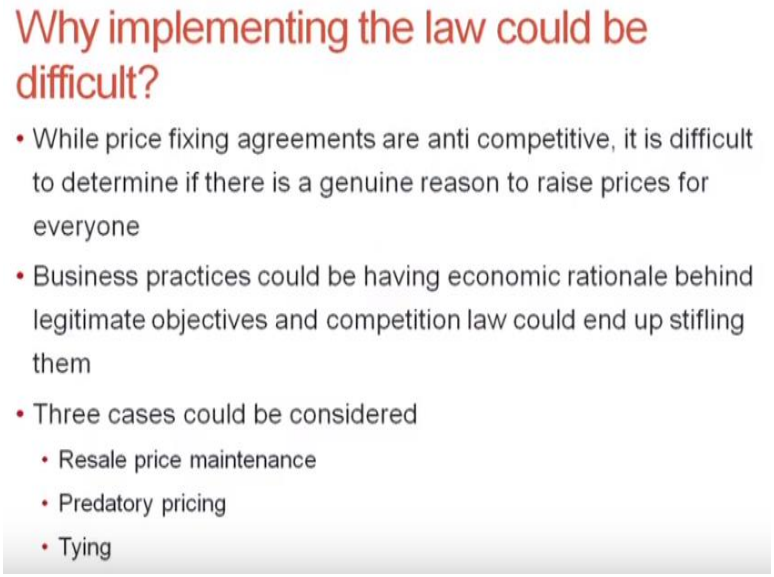
So the firm, the individual firms know this. So even a single firm which maybe producing multiple types of products yet it could be a monopolist in producing one product and it could be oligopolist in producing some other product and it could be into monopolistic competition in producing yet other products. So this is something I wanted to highlight.

Second is, so we have all these laws in place but at the same time there have been lot of dispute all around the globe about implementation of these laws because often it becomes very difficult to differentiate between what is legal and what is economically or what is rational as an economic entity to do.

So every firm exists as a rational economic agent which makes his choices depending on some very legitimate objective of say profit maximization and it takes certain decision depending on that and it could be probably shown through the theoretical models that it was the obvious choice for the firm to do but in the eyes of the law it could be interpreted as anti-competitive.

So this is a conflict that exists all the time and hence each of the cases the cases none of the cases if you go through the different kinds of cases that have been tried all over the world, none of the cases are exactly similar to each other.

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So why implementing the law could be difficult while price fixing agreements are anti-competitive, it is difficult to determine if there is a genuine reason to raise prices for everyone. So we have seen that there is a kinked demand curve and there is a range in which firms are reluctant to raise prices. They are afraid of raising prices because they do not want to lose customers.

On the other hand, it is possible for the firms to collude and raise the prices to a monopolistic level. So if the if a firm is a oligopolist and it is in that range where it is scared of increasing the prices yet its costs have gone up and it increases the price and similarly the same situation happens all over the market and all the firms increase their price, is that price fixing; is it possible that the firms have all colluded and they have decided we are going to increase the price.

So these are certain very delicate situations where it is not easy to determine whether something anti-competitive or whether something illegal has happened or is it just a natural consequence of the economic circumstances. So second is business practices could be having economic rationale behind legitimate objectives and competition law could end up stifling them.

So this is one argument that the economist or this is one thing that the economist keep on reminding that too much implementation of the law or too much legal restrictions on the firms could make them lose the breathing space that they have to behave or to flourish as or to thrive as businesses and that could be harmful for the entire economy as a whole.

Now three cases could be considered. One is resale price maintenance. So we are taking three examples where this kind of situation can arise which probably in eyes of law it could be termed as anti-competitive yet it could be explained to have its own economic rationale behind it. Now three cases to be considered; one is called the resale price maintenance.

Now often it happens that it can be seen that the manufacturer sets a lower limit to prices for the retailers. That is the retailers cannot charge or cannot provide a discount below a certain level to its customers. Now this could be considered as anti-competitive among the retailers because firstly manufacturers do not have manufacturers are not going to gain by imposing a lower price restriction on the retailers.

All that it is probably doing is restricting competition amongst retailers and preventing the consumers from actually gaining from competition between retailers and finally getting the product at a lower price. But yet manufacturers could be doing that and the one reason behind this is probably the manufacturer is trying to prevent the discount retailers from free riding on services of the full service retailers.

So let me give an example. Quite a few years back I mean it has been more than probably a decade or so or even more that, the Bosch speakers were introduced in India. The Bosch speakers came to India and we had retailers selling the Bosch speakers.

Now at that time when the speakers were launched in India there were exclusive showrooms which had very the entire the these showrooms used to have special rooms to display the sound, effectiveness of sound of Bosch speakers because that is how the audio facilities inside the or the infrastructure within the showrooms were such that they would highlight the special features of the Bosch sound systems.

So obviously these retailers were had spent a lot of money to display this product to the customers who are new to this product. Now Bosch had now these showrooms the Bosch would be honoring the warranties of these products sold in these showrooms only and not by the discount retailers.

Now what is the reason behind it? The reason being that it is possible that say for example me as a customer I might have gone to one of the full service showrooms got a full test of the sound system the sounds the speakers and identified what I exactly wanted and then gone to a discount retailer and said that okay I would like to purchase this and give me at a lower price.

So in that case why would the full service retailer spent so much effort and money on selling the Bosch speakers in a specialized showroom like his. So these are the reasons that resale price maintenance could be controlled by the manufacturers. Another example is predatory pricing. Predatory pricing as I already said that it is possible for certain firms to cut the prices so low that other firms do not enter the, so the other firms cannot enter the market.

Now predatory pricing is also argued against by economist who say that it really does not make sense for a firm to take so much of risk because if one is going for predatory pricing this firm is probably cutting its price below its cost and incurring a loss to drive out the rivals or stop the entry of other rivals in the firm.

Why would the firm take such a risk because it is it cannot continue to stay at this low price? It has to increase its price eventually and in that case whenever it increases its price the other firms' other rivals who have probably been waiting for this firm to drive out the competitors they will

jump right back into the market and take the advantage of this predatory pricing of the first firm. So this predatory pricing is also something which the economists do not think is actually possible.

Third is tying. Tying is something where a firm or a company may actually tie a weak product with a strong product and sell it in the market. This is also considered anti-competitive but again say for example when Microsoft would add a browser to its operating system. So this is something which is considered as anti-competitive because a firm could be it is possible for a firm to tie a weak product with its strong product but here again the argument which the economists put forth is the consumers would not be willing to pay a price which is higher than the total price that they would be paying individually for the 2 products. So there is and tying is something where the different firms actually price discriminate. Tying is a is a form of price discrimination and it is and price discrimination is not illegal. So tying is again something which is argued against by economists.

So coming back so that basically ends our discussion on oligopoly and we have seen that an oligopoly is probably the most realistic of the market situations and we have seen that different firms it is the all the in spite of all the theoretical models that we have gone into, which clears our understanding about how what is the economic rationale behind or what is the logic behind firms to charge a certain price when they do so; but at the same time the firms also need to take into account the decision of all the other firms or the other players in the market to decide on their price and output. Thank you.

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